

ZlibValidation

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Chapter 1

ZlibValidation

1.1 Description

Command line tool to validate standard cell libraries in .lib format.

1.2 Usage

```
ZlibValidation
Usage: ./zlibvalidation [OPTIONS] [SUBCOMMAND]

Options:
  -h,--help          Print this help message and exit
  -v,--version        Display program version information and exit

Subcommands:
  parse              Parse the Liberty file and write JSON to a file
  mono               Check the monotonicity of timing arc values
  compare            Compare the comparison library against the reference library and report differences
  supercell          Generate supercells for the given Liberty file
  zlibboost          ZlibBoost - Multi-threaded Library Processing Tool
  clear              Clear the log, JSON, map, markdown files in this directory
  verilog            Generate Verilog file for given Liberty file
  spice              Generate SPICE file for given Liberty file
  func               Check functional equivalence of two Liberty files or Verilog files
```

1.3 Development Diary

1.3.1 2025-01-27

- 创建了一个C++空项目，使用CMake管理编译链，并手动添加了外部库 libsi2dr_liberty.a。
- 添加了 CLI11 库以处理命令行参数解析。
- 添加了 spdlog 日志库，实现高效的日志格式化输出。

1.3.2 2025-01-28

- 添加 `nlohmann/json` 库用于 JSON 数据存储和检索。
- 在 `version.h` 中存储项目作者、版本、构建日期等信息，由 CMake 自动维护。
- 简化了 `--version` 参数解析的代码实现，并添加了 `--mode` 选项以选择工作模式。
- 使用 `spdlog` 库的日志记录器将 debug 级别及以上的日志输出到文件，将 info 级别及以上的日志输出到控制台。
- 将库文件顶层封装为 `LibFile` 对象，提供了读取、解析和修改库文件的方法，包含名称和 PVT 信息作为公共属性。
 - 但仍然使用过程化方法循环迭代读取库文件名称和 PVT 信息。
- 按照 C++ 标准库、第三方库和本地库的顺序重新排列了 `#include` 语句。

1.3.3 2025-02-01

- 使用面向对象编程 (OOP) 重构了原先用 C 语言循环遍历的代码，将简单属性的读取封装为 `LibAttribute` 类，
 - 并将函数返回的 `char *` 类型的属性值转换为 `std::string` 类型，便于外部使用。
- 将属性迭代过程封装为 `AttributesIterator` 类，在析构函数中处理迭代退出，实现资源获取即初始化 (RAII)。

1.3.4 2025-02-10

- 将组的迭代过程封装为 `GroupsIterator` 类，同样在析构函数中处理迭代退出。
- 封装了 `LibGroup` 类，提供了读取组名、类别、属性、子组的方法。
- 将类的定义和实现分离，将类的实现放在 `src` 目录下，将类的定义放在 `include` 目录下，修改了 `CMakeLists.txt` 自动收集文件，便于模块化编译。

1.3.5 2025-02-11

- 类的头文件改为 `hpp` 后缀，修改了 `CMakeLists.txt` 中的文件收集规则，CMake 编译时间戳更新。
- 为 `LibFile` 类添加了输出同名 json 格式文件的方法 `LibFile::writeJsonToFile`，目前能输出 PVT、`cell_name`、`cell_footprint`、`cell_area` 信息。
- [x] voltage 浮点数误差未解决。

1.3.6 2025-02-14

- 在 `json_utils.cpp` 中，实现了以 JSON 数据结构循环迭代存储库 (lib) 信息的功能，具体包括 `generateCellJson`, `generatePinJson`, `generatePowerJson`, `generateLutJson` 等函数，并添加了同名的 `hpp` 头文件。
- [x] 时序 (timing) 信息解析功能待完成。
- 新增辅助函数 `parseStringToVector`，该函数可将以逗号分隔的字符串解析为浮点数向量，以便于读取 Look Up Table (LUT)。
- 为 `LibFile` 对象添加了私有属性 `lib_json_`，用于存储库 (lib) 的 JSON 对象。
- 封装了用于迭代复杂属性值的 `ValuesIterator` 类。
- 修改了 `LibAttribute::isComplex()` 函数的返回值类型，由原类型变更为 `bool` 类型。
- 重写了 `LibGroup::getName()` 函数，使其直接返回 `std::string` 类型的组名字。
- 重构了迭代器使用代码。移除了中间变量类似 `si2drGroupsIdT sub_groups` 的声明步骤。直接使用 `lib_group.getGroups()` 的返回值进行初始化 `GroupsIterator`，提升了代码的可读性和简洁性。

1.3.7 2025-02-15

- 实现 `generateTimingJson` 函数，用于全面解析时序信息。

1.3.8 2025-02-17

- 简化了迭代器在外部的调用流程，删除所有迭代器的 `begin()` 方法，改为在构造函数内部初始化。

1.3.9 2025-02-18

- 新建 `LibFile::mono()` 方法，准备实现库文件输出引脚的 `values` 单调递增性检查。
- 修改了 `setupLogger()` 函数，将日志文件名可作为参数传入，便于根据不同工作模式切换日志文件。

1.3.10 2025-02-19

- 修改了 `mode == "mono"` 情况下的逻辑，使用 C++17 引入的 `std::filesystem::exists()` 函数检查同名 JSON 文件是否存在，如果 JSON 文件不存在，则先调用库文件解析功能，生成 JSON 文件，然后再执行单调性检查。
- 实现了 `LibFile::mono()` 方法，检查 `cell_rise`、`cell_fall`、`rise_transition`、`fall_transition` 这四种 timing 信息，针对其 LUT 数据结构，检查其 `value` 值在表格的每一行是否保持单调递增。

1.3.11 2025-02-20

- `LibFile::mono()` 方法增加结果统计功能，在程序运行的最后输出单调性检查中 passed (通过) 和 failed (失败) 的 cell 数量，输出格式参考 `liberate_lv` 工具的风格。
- 以 `sl018_ff_3.96_-40.lib` 为例，与 `liberate_lv` 工具对比测试，结果均为 205 out of 559 cells failed。
- 对于非单调信息警告，增加输出 when 信息，方便用户查看具体位置。

1.3.12 2025-02-25

- 重构命令行参数解析，使用 `CLI11` 库的子命令，在每个回调函数中实现 `parse`、`mono` 功能。

1.3.13 2025-02-26

- 子命令可自定义输出文件名、日志文件名，并设置了相应的默认值。
- 改为在 `Logger` 初始化时打印版本信息，避免在 `-h`，`-v` 时多余打印。
- `LibFile::mono()` 方法增加对 `input_slew` 的单调性检查，如果在输入时有指定，就检查 value 矩阵的每一列是否单调。经过测试，`tcbn65lpbc.lib` 输出与 `liberate_lv` 工具一致。

1.3.14 2025-02-27

- 多线程并行尝试，采用 GDB 调试：
 - `si2dr_liberty` 库在解析 `Liberty` 文件时，可能使用了共享的数据结构（例如哈希表、字符串表）来存储解析结果或中间数据。
 - 在多线程并行解析多个文件时，不同的线程同时调用 `si2dr_liberty` 库的函数，并发地访问和操作这些共享数据结构。
 - `si2dr_liberty` 库可能没有采取足够的线程同步措施来保护这些共享数据结构，导致了数据竞争。
 - 数据竞争最终导致了内存损坏，使得 `strcmp` 函数在后续操作中访问了无效内存，触发了段错误。

1.3.15 2025-02-28

- 修改 `LibFile` 的构造函数与析构函数，`read` 方法改为私有，移入 `parse` 方法。将 `si2dr` 初始化与销毁放在 `parse` 方法中，使得 `mono` 方法与 `si2dr` 库解耦，从而实现多线程并行验证单调性。
- [x] 顶层 `si2drPIGetGroups` 未退出的 bug: `WARNING: si2drPIQuit: GetGroups called 1 more times than IterQuit`，内存泄漏？
- 学到了可以在 `CMakeLists.txt` 中使用 `set(CMAKE_BUILD_TYPE Release)` 减少编译器优化程度，配合 GDB 等调试工具检查堆栈信息。

1.3.16 2025-03-01

- `sub_groups_iter` 是在每次 `for` 循环的迭代中声明的局部变量，在每次迭代结束后其作用域结束，析构函数被调用，释放相关资源。所以 `si2drPIQuit` 未发现内存泄漏。
- 顶层 `group_iter` 的析构函数发生在 `parse` 方法结束时，所以 `parse` 方法末尾的 `si2drPIQuit` 检测到了顶层 `groups` 未退出的情况。
- 综上，`si2drPIQuit` 的警告是由于当时顶层 `groups` 未退出导致的，在 `parse` 方法结束后能正确释放，不会造成实际上的内存泄漏。
- [] 顺序执行 `parse`，`log` 输出仍存在问题：解析第二第三个库的时候会多余打印之前库的PVT信息，时间上也比单独解析一个库的总是要慢 (`parse` 6s 左右，`mono` 1s)。并行执行 `mono`，`log` 输出会集中到最后一个库的文件。
- 给 `LibFile` 类的私有属性添加了初始值，避免了未初始化的问题，吗？

1.3.17 2025-03-07

- 实现了 `compare` 子命令的解析，能够对两个库文件依次进行解析，并生成对应的 JSON 文件。
- [] 仍然存在日志重复输出的问题，多文件处理时数据可能会相互干扰。

1.3.18 2025-03-10

- 新增 `supercell` 子命令和 `LibFile::supercell` 方法，用于生成超级单元以供验证。
- 集成了 `zlibboost` 子命令，支持调用开源库特征化工具 `zlibboost`。已与开发者取得联系。
- 调整了子命令输出文件的位置为当前目录，避免文件混淆。
- 添加了 `clear` 子命令，方便清理生成的 JSON 文件和日志文件。

1.3.19 2025-03-14

- 添加了 `tabulate/table` 库，通过 CMake FetchContent 进行管理，用于生成格式化的表格输出。
- 新建了 `compare.cpp` 和 `compare.hpp` 文件，创建了 `LibraryComparator` 类，用于比较两个库文件的差异。目前实现了读取参考库 JSON 文件的功能。

1.3.20 2025-03-15

- 使用 `std::filesystem::path` 重构了 `LibFile` 类的文件路径管理，方便文件路径的拼接和处理。`LibFile` 类添加了成员变量 `basename_`、`filename_`、`libname_`、`jsonname_` 和 `loggername_`，用于存储文件名、库名、JSON 文件名和日志文件名。
- 通过作用域 (scope) 管理 `LibFile::parse` 方法中的顶层迭代器的生命周期，提前结束并调用 `si2drPIQuit` 保证资源释放，解决了分析多个库时日志重复输出和数据保存错误的问题。
- `clear` 子命令增加了删除 `.map` 和 `.md` 文件的功能。
- 实现了 `mono` 子命令的多线程并行。首先检查是否是多文件输入，如果是，就顺序检查是否准备好了每个文件的 JSON 文件，否则顺序进行多文件库解析。在所有文件的 JSON 准备就绪后，根据文件数量创建线程池，每个线程负责一个库的单调性检查。
- 重构了 `spdlog` 日志初始化，返回 `logger` 对象而不是设置全局默认 `logger`。保留名为 `APP_NAME` 的全局 `logger`，用于输出总体提示信息。
- 为 `LibFile` 类添加了独立的成员变量 `logger`，用于记录每个库文件的日志信息。
- 完成了 `supercell` 子命令的多线程并行，处理逻辑与 `mono` 子命令类似。
- `LibraryComparator::generateReport()` 方法测试了 `tabulate` 库的 markdown 表格输出功能，完善了报告的头部信息输出，包括参考库、比较库、相对容差、软件信息、报告生成时间和图例说明。

1.3.21 2025-03-16

- 针对 `compare` 子命令，增加了对报告文件名的检查，确保其以 `.md` 或 `.txt` 结尾。若不符合，则给出警告并自动添加 `.md` 后缀。
- 进一步完善了 `LibraryComparator::generateReport()` 方法。现在，该方法能够遍历比较库中的所有 `cell`，并在参考库中查找对应的 `cell`。如果找到，则调用 `LibraryComparator::compareCell()` 方法进行比较；否则，会记录 `cell` 未找到的警告信息。
- 新增了 `LibraryComparator::compareCell()` 方法，用于比较两个库中同名 `cell` 的输出引脚。该方法会遍历比较库 `cell` 的每个输出引脚，在参考库 `cell` 中寻找同名引脚。如果找到，则调用 `comparePin` 函数比较引脚；如果未找到，则记录警告信息。如果比较库 `cell` 没有输出引脚，则会记录一条信息级别的日志。
- 新增了 `LibraryComparator::comparePin()` 方法，用于比较两个库中同名 `cell` 的同名引脚。它会遍历比较库 `pin` 的每个 `timing arc`，并在参考库 `pin` 中查找相同类型的 `timing arc`。如果找到，则调用 `compareTimingArc` 函数进行比较；如果未找到，则记录警告信息。如果比较库 `pin` 没有 `timing arc`，则记录一条信息级别的日志。

- 修改了 `LibraryComparator::compareTimingArc()` 方法，用于比较两个 timing arc JSON 对象。该方法首先记录正在比较的 timing arc 的类型，然后提取 "related_pin" 属性。接着，它遍历预定义的 timing arc 名称列表 ("cell_rise", "cell_fall", "rise_transition", "fall_transition")，并在比较库中查找这些 timing arc。如果找到，则在参考库中查找对应的 timing arc，并调用 `compareLut` 方法进行比较。如果参考库中未找到对应的 timing arc，则记录警告信息。
- 修改了 `LibraryComparator::compareLut()` 方法（原 `compareValue` 方法，已更正命名），用于比较两个 JSON 对象中具有相同名称的 value。它首先提取两个 JSON 对象中的 "index_1" 和 "index_2" 数组，如果这两个数组不相等，则记录错误信息并返回。如果索引匹配，则比较 LUT 中的实际 value 值，并根据相对容差 (reltol) 和绝对容差 (abstol) 标记差异。
- 重新设计了层级比较方法，现在可以逐层级地传入 `cell_name`、`pin_name`、`timing_type`、`related_pin` 和 `arc_name`，以便在最内层制表或提示时能够准确输出层级信息。
- 修复了 `LibraryComparator::compareValue()` 方法的命名错误，已更正为 `LibraryComparator::compareLut`
- 已经可以输出表格，数据数量和商用工具结果一致，但格式还需要进一步调整。
- 新增了 `abstol` 参数，默认值为 0.002ns，用于设置绝对容差，与库验证工具保持一致。

1.3.22 2025-03-17

- 新增 `verilog` 和 `spice` 子命令，用于生成 Verilog 和 SPICE 代码。
- 完善 `LibraryComparator` 报告生成：
 - 为每个 cell 增加表头，仅在存在表格数据时输出表格，避免冗余输出。
 - 增加 cell 结果统计功能，输出结果统计表格，目前仅支持 Timing 中 Delay 的比较。

1.3.23 2025-03-18

- 优化统计表格，包含 | Cell Name | Data Type | Failed Count | Avg Diff | Avg Diff% | Max Diff | Max Diff% | Outliers |。
- 优化报告对比表格，包含 | Pin Name | Reference | Comparison | Diff | Diff % | Type | Arc Name | Row # | Index_1 | Column # | Index_2 | Note |
- 报告表格的 stdout 输出增加表头加粗、黄色的格式化。
- 增加 `si2drSimpleAttrGetBooleanValue` 封装，用于获取布尔类型的属性值，以判断引脚是否是时钟引脚。
- `LibFile::mono()` 增加对 `min_pulse_width` 的单调性检查：
 - 针对包含 "input_pins" 的 cell，遍历每个输入引脚。
 - 如果引脚是时钟引脚，则检查其 "timing_arcs"。

- 针对 "timing_type" 为 "min_pulse_width" 的 timing arc, 对 "rise_constraint" 和 "fall_constraint" 调用 checkTimingArcMonotonicity 进行单调性检查。
- checkTimingArcMonotonicity 函数日志区分有无 when 信息, 输出不同日志。
- checkTimingArcMonotonicity 核心比较功能取消了等于的情况, 相等情况默认为通过。
- checkTimingArcMonotonicity 核心比较功能增加了判断 related_pin 是否等于当前 pin, 不等于则跳过。最终改为: 如果矩阵中当前的值小于前一个值, 并且当前引脚不是相关引脚, 或者当前值和前一个值都为零, 那么就认为这些值不是单调递增的。

1.3.24 2025-03-19

- 完善 supercell 方法, 如果有有时钟信号引脚, 链式长度被设为 1, 再生成超级单元。
- 解决 voltage 浮点数误差问题, `std::round(voltage_ * 100) / 100.0` 保留两位小数。

1.3.25 2025-03-21

- 新增 func 子命令, 用于检查库或者 Verilog 文件的逻辑等价性。

1.3.26 2025-03-24

- 调研了 C++ 操作 Verilog 相关的库, 找到了一个实用的 Verilog 库: `slang`, 它可以解析出抽象语法树 (AST)。已修改 CMakeLists 文件, 将 slang 集成到项目中, 目前正在研究其 API。

1.3.27 2025-03-25

- 新增 `verilog_utils.cpp` 和 `verilog_utils.hpp` 文件, 并创建了 `VerilogVisitor` 类, 用于自定义遍历 Verilog AST。
- 验证了 slang 对不同类型逻辑单元的解析能力:
 - 简单组合逻辑 (如 AND2D0)、复杂时序逻辑 (如 CMPE42D1) 和基本时序逻辑 (如 DFQD1) 均能被正确解析。
 - 用户自定义原语 (UDP), 例如 `tsmc_dff`, 会被错误地识别为模块实例化, 并产生 "Invalid instance declaration" 警告。
 - 成功提取了模块名、端口方向 (input、output) 及名称。
 - 能够解析命名端口连接的层次化实例化, 并获取模块名、实例名称和端口映射关系。
 - 能够解析门级原语实例化, 并获取门类型、实例名称、输出端口和输入端口。

1.3.28 2025-03-26

- 进一步完善了对 UDP (如 `tsmc_dff`) 端口映射关系的解析, 但其内部 `Entry` 尚未处理。
- 尝试了通过继承 `slang::syntax::SyntaxRewriter` 类创建自定义类 `CellExtractor`, 实现了只保留目标模块、删除其他模块, 并将修改后的语法树另存为新文件。使用 `slang::syntax::SyntaxPrinter::printFile(*tree)` 方法可以将 Verilog 代码输出到文件。
- 创建了 `CellPrinter` 类, 继承自 `slang::syntax::SyntaxVisitor` 类。当访问到目标模块时, 使用 `module.toString()` 方法也能将 Verilog 代码输出到文件, 但发现输出结果中空行会被删除。
- 修改了 `LibFile::supercell()` 方法, 该方法可以根据输入的 `cell_names` 生成指定的 `supercell`, 并在遇到未找到的单元时提示警告信息。
- 优化了时钟引脚的处理方式, 现在时钟引脚不再被视为 `supercell` 的输入引脚, 而是仅记录为时序单元, 并跳过存入 `input_pins` 集合的步骤。
- 引入了 Doxygen 文档生成工具, 用于可视化分析项目, 并生成了 Doxygen HTML 文档和 LaTeX 参考手册。
- [x] 尚未解决手册中文显示不正确的问题, 可能需要自定义 LaTeX 头文件。

1.3.29 2025-03-27

- **重大失误!** 误操作 Git 分支, 导致部分代码修改丢失。务必牢记: **** 及时提交代码! ****
- 实现了 `LibFile::verilog` 方法的核心流程:
 - 首先调用 `this->supercell()` 生成超级单元。
 - 读取 `.map` 文件, 提取单元映射关系到 `std::pair<std::string, std::string>` 中 (原始单元名 -> 超级单元名/模块名)。
 - 从 `lib_json_` 中提取单元的输入/输出端口信息。
 - 根据是否存在时钟引脚, 确定 Verilog 模块中 `instance` 的生成数量 (时序逻辑为 1, 组合逻辑等于 `chain length`)。
 - 通过字符串操作生成 ANSI 风格的端口列表, 并与模块名组合成完整的 `fullModuleText`。
 - 使用 `slang::syntax::SyntaxTree::fromText` 从 `fullModuleText` 生成语法树。
 - 将语法树传递给 `ModuleRewriter` 类, 使用其 `transform` 方法遍历模块成员, 并通过 `handle()` 方法进行处理。
 - 最后, 使用 `slang::syntax::SyntaxPrinter::printFile(*tree)` 将处理后的语法树输出到文件。
- `LibFile::verilog` 目前能够正确输出模块名和 ANSI 风格的端口声明。

1.3.30 2025-03-28

- 实现了在模块成员列表中插入新节点的功能。具体而言，在 `ModuleRewriter::handle(const slang::syntax::ModuleDeclarationSyntax &module)` 方法中，通过 `insertAtBack(module.members, newDataNode)` 函数，成功地在模块成员的末尾插入了链式单元所需的中间变量声明，例如 `wire OP_i;`。
- 成功地在模块内部添加了关键连接网络变量。通过解析 `CELLNAME__X#__CRITICALPORT__OUTPUTPORT` 格式的字符串，提取出关键输入端口 (critical input port) 和输出端口 (OUTPUTPORT)，用于后续的实例化端口连接。
- 增加了用于处理多输出接口的中间网络变量 `P__UNUSEOUTPUTPORT`，用于连接不需要考察的输出端口。
- 成功地在模块内添加了模块实例化所需的模块名和实例名称，完成了端口映射关系的连接处理。
- 验证了代码对简单组合逻辑单元（如 `INVD0` 反相器）和复杂组合逻辑单元（如 `FA1D0` 全加器）的输出能力，结果均符合预期。
- 为 `ModuleRewriter` 类添加了私有属性 `std::map<std::string, std::string> portInfoMap_`，用于存储端口映射关系，将端口名映射到其方向 (input/output)。
- 完成了实例化端口连接的细节处理：
 - 如果是关键输入端口且为第一个实例，则直接连接到输入端口；否则，连接到 `OP_(i-1)` 这样的中间网络变量。
 - 如果是关键输出端口，且为最后一个实例，则连接到输出端口；否则，连接到 `OP_i` 中间网络变量。
 - 如果是其他输出端口且不是最后一个实例，则连接到 `P_i__portName` 这样的中间网络变量；否则，连接名等于端口名。
 - 其余输入端口，连接名等于端口名。
- 文档方面，GitHub 远程仓库已开放为公开访问，并配置了 GitHub Pages。可以通过 <https://cedar17.github.io/ZlibValidation/> 访问项目文档，该页面包含 Doxygen 生成的 HTML 文档和仓库地址的链接。
- 配置了 GitHub Actions，实现了 Doxygen 文档和 Graphviz 图的自动化构建，以及 LaTeX 参考手册的自动编译（暂不支持中文）。每次在 `dev` 和 `main` 分支的提交都会触发文档和参考手册的生成，并发布到 `gh-pages` 分支。
- 将第三方库放置到 `include_3rd_party` 目录下，方便管理。修改了 `CMakeLists` 文件，将第三方库的头文件路径添加到 `include_directories` 中。修改了 `Doxyfile` 文件，将第三方库的头文件路径添加到 `INPUT` 中，便于文档生成工具理解第三方库函数、类的关系及使用方法。

1.3.31 2025-03-29

- 修复了 Doxygen 生成 LaTeX 参考手册时中文显示不正确的问题。通过修改 Doxyfile 文件，添加 `EXTRA_PACKAGES = ctex` 选项，并指定 `lualatex` 作为编译器，成功解决了中文显示问题。同时，从 `include_3rd_party` 目录中移除了 `Allsyntax.h` 文件，避免了由于 LaTeX 文档过大导致的 "TeX capacity exceeded, sorry [main memory size=5000000]" 内存不足问题。
- 实现了 Verilog 文件的顶层模块生成功能，采用非 ANSI 风格的端口声明方式，格式与 `liberate_lv` 工具生成的完全一致。实现过程如下：
 - 首先将各模块的 Verilog 代码存放到临时文件中。
 - 在处理模块的过程中，使用 `std::vector<std::string>` 收集每个模块的 `input_pins` 和 `output_pins`。
 - 遍历所有模块名称列表，拼接模块名和引脚名，生成顶层模块的 `all_input_ports` 和 `all_output_ports`。
 - 使用字符串拼接方式构建 `validate_top` 模块的完整代码。
 - 最后将临时文件内容与顶层模块代码合并，输出到最终的 Verilog 文件中。
- 更改 `deploy-docs.yml`，修复了 GitHub Action 生成的 LaTeX 参考手册中的图像显示问题。具体步骤如下：
 - 设置系统时区为 Asia/Shanghai (东八区)。
 - 安装 miniconda 并更新 conda。
 - 通过 conda 安装 1.9.5 版本的 Doxygen，并安装 graphviz 以解决依赖关系，确保每个 dot 图都能正确编译出 pdf 文件。
 - 使用 `doxygen` 命令生成 HTML 文档和 tex 文件。
 - 采用 action marketplace 提供的 `xu-cheng/texlive-action@v2` 安装全量的 `texlive2024`。
 - 进入 `./doc_doxygen/latex` 目录，运行 `make` 命令使用 `lualatex` 编译 tex 文件，生成 pdf 文件。
 - 使用 `JamesIves/github-pages-deploy-action@v4` 将生成的 pdf 文件上传到 GitHub Pages 发布，并保留 `README.md` 和 `index.html` 文件。

1.3.32 2025-03-30

- 完成了 `LibFile::logic(const std::string &cell_name)` 方法，该方法返回指定 `cell_name` 所有输出引脚对应的逻辑表达式，以 `std::map<std::string, std::string>` 的形式返回，存储输出引脚名称到其逻辑表达式字符串的映射关系。
- 实现了 `func` 子命令的基本框架，用于检查库或 Verilog 文件的逻辑等价性。
 - 完善了 `funcLibFile` 函数，能够处理 Liberty 文件和 Verilog 文件作为参考和比较对象。
 - 增加了文件类型判断，根据文件扩展名选择不同的处理方式。

- 实现了对 .lib 文件的解析，并提取指定 cell 的逻辑表达式。
- 目前仅支持 Liberty 文件的逻辑表达式提取，Verilog 文件的逻辑表达式提取功能尚未完成 (TODO)。
- 增加了对 report 文件名的检查，确保其以 .md 或 .txt 结尾。若不符合，则给出警告并自动添加 .md 后缀。
- 实现了从 Liberty 文件中提取逻辑表达式的功能，使用了 `LibFile::logic` 方法。
- 增加了日志输出，记录每个 cell 的逻辑表达式提取和比较过程。
- 逻辑比较部分目前为空，待实现 (TODO)。

1.3.33 2025-03-31

- 实现了从 Verilog 结构网表中提取逻辑表达式的功能，使用了 `LogicExtractor` 类。
 - 设计了如何使用 Slang 库从 Verilog 结构网表中提取逻辑表达式的方案：
 - * 使用 Slang 解析 Verilog 文件获取语法树 (AST)。
 - * 遍历 AST 定位目标模块。
 - * 在目标模块内构建网表的内部表示，识别输入、输出、线网及门级连接关系。
 - * 对每个输出端口，递归反向追踪门和线网直至到达主输入端口。
 - * 在回溯过程中根据遇到的门类型构建逻辑表达式。
 - * 应用记忆化技术避免对相同线网/信号的重复计算。
 - 创建了 `LogicExtractor` 类，用于从 Verilog 代码中提取逻辑表达式。
 - * `LogicExtractor` 类使用 Slang 库解析 Verilog 文件，并构建目标模块的网表表示。
 - * 实现了 `handle(const slang::syntax::ModuleDeclarationSyntax &module)` 方法，用于定位目标模块，并初始化内部数据结构。
 - * 实现了 `handle(const slang::syntax::PortDeclarationSyntax &portDecl)` 方法，用于提取端口信息，包括端口方向和名称。
 - * 实现了 `handle(const slang::syntax::NetDeclarationSyntax &netDecl)` 方法，用于提取线网信息。
 - * 实现了 `handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst)` 方法，用于提取门级单元信息，包括门类型、输入和输出信号。
 - * 实现了 `getLogicExpressions()` 方法，用于根据提取的网表信息，递归地推导每个输出端口的逻辑表达式。
 - * 实现了 `deriveLogicRecursive(const std::string &signalName)` 方法，用于递归地推导指定信号的逻辑表达式。
 - * 实现了 `formatExpression(const GateInfo &gateInfo, const std::vector<std::string> &inputExprs)` 方法，用于根据门类型和输入表达式，生成逻辑表达式字符串。
 - 实现了 `extractAndPrintNetlistInfo(const std::string &verilog_file, const std::string &cell)` 函数，用于提取并打印网表信息，包括输入、输出、线网和门级单元。

- 实现了 `extractLogicFromVerilog(const std::string &verilog_file, const std::string &cell)` 函数，用于从 Verilog 文件中提取指定 cell 的逻辑表达式，并以 `std::map<std::string, std::string>` 的形式返回。
- 为了更好地组织代码结构，将 `LogicExtractor` 类从 `verilog_utils.hpp` 和 `verilog_utils.cpp` 中分离出来，并创建了独立的 `LogicExtractor.hpp` 和 `LogicExtractor.cpp` 文件。同时，统一了所有 `hpp` 和 `cpp` 文件的命名规范，使其与对应的类名保持一致。

Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

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Chapter 3

Hierarchical Index

3.1 Class Hierarchy

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Chapter 4

Class Index

4.1 Class List

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File Index

5.1 File List

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Chapter 6

Namespace Documentation

6.1 slang Namespace Reference

Namespaces

- namespace [parsing](#)
- namespace [syntax](#)

6.2 slang::parsing Namespace Reference

6.3 slang::syntax Namespace Reference

Namespaces

- namespace [detail](#)

Classes

- struct [ConstTokenOrSyntax](#)
A token or a constant syntax node.
- class [DeferredSourceRange](#)
- struct [PtrTokenOrSyntax](#)
A token pointer or a syntax node.
- class [SeparatedSyntaxList](#)
- class [SyntaxList](#)
A syntax node that represents a list of child syntax nodes.

- class [SyntaxListBase](#)
A base class for syntax nodes that represent a list of items.
- class [SyntaxNode](#)
Base class for all syntax nodes.
- class [SyntaxPrinter](#)
- class [SyntaxRewriter](#)
- class [SyntaxTree](#)
- class [SyntaxVisitor](#)
- class [TokenList](#)
A syntax node that represents a list of child tokens.
- struct [TokenOrSyntax](#)
A token or a syntax node.

Functions

- SLANG_EXPORT [SyntaxNode](#) * [clone](#) (const [SyntaxNode](#) &node, BumpAllocator &alloc)
Performs a shallow clone of the given syntax node.
- SLANG_EXPORT [SyntaxNode](#) * [deepClone](#) (const [SyntaxNode](#) &node, BumpAllocator &alloc)
Performs a deep clone of the given syntax node.
- template<std::derived_from< [SyntaxNode](#) > T>
T * [clone](#) (const T &node, BumpAllocator &alloc)
Performs a shallow clone of the given syntax node.
- template<std::derived_from< [SyntaxNode](#) > T>
T * [deepClone](#) (const T &node, BumpAllocator &alloc)
Performs a deep clone of the given syntax node.
- template<typename T >
[SyntaxList](#)< T > * [deepClone](#) (const [SyntaxList](#)< T > &node, BumpAllocator &alloc)
- template<typename T >
[SeparatedSyntaxList](#)< T > * [deepClone](#) (const [SeparatedSyntaxList](#)< T > &node, BumpAllocator &alloc)
- [TokenList](#) * [deepClone](#) (const [TokenList](#) &node, BumpAllocator &alloc)

6.3.1 Function Documentation

6.3.1.1 clone() [1/2]

```
SLANG_EXPORT SyntaxNode * slang::syntax::clone (
    const SyntaxNode & node,
    BumpAllocator & alloc )
```

Performs a shallow clone of the given syntax node.

All members will be simply copied to the new instance. The instance will be allocated with the provided allocator. Here is the caller graph for this function:



6.3.1.2 clone() [2/2]

```
template<std::derived_from< SyntaxNode > T>
T * slang::syntax::clone (
    const T & node,
    BumpAllocator & alloc )
```

Performs a shallow clone of the given syntax node.

All members will be simply copied to the new instance. The instance will be allocated with the provided allocator.

Definition at line 233 of file [SyntaxNode.h](#).

Here is the call graph for this function:

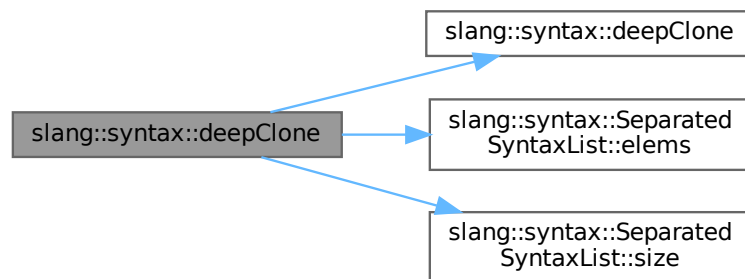


6.3.1.3 deepClone() [1/5]

```
template<typename T >
SeparatedSyntaxList< T > * slang::syntax::deepClone (
    const SeparatedSyntaxList< T > & node,
    BumpAllocator & alloc )
```

Definition at line 477 of file [SyntaxNode.h](#).

Here is the call graph for this function:



6.3.1.4 deepClone() [2/5]

```
template<typename T >
SyntaxList< T > * slang::syntax::deepClone (
    const SyntaxList< T > & node,
    BumpAllocator & alloc )
```

Definition at line 343 of file [SyntaxNode.h](#).

Here is the call graph for this function:



6.3.1.5 deepClone() [3/5]

```
SLANG_EXPORT SyntaxNode * slang::syntax::deepClone (
    const SyntaxNode & node,
    BumpAllocator & alloc )
```

Performs a deep clone of the given syntax node.

All members will be cloned recursively to create a complete new copy of the syntax tree. All cloned instances will be allocated with the provided allocator. Here is the caller graph for this function:

**6.3.1.6 deepClone()** [4/5]

```
template<std::derived_from< SyntaxNode > T>
T * slang::syntax::deepClone (
    const T & node,
    BumpAllocator & alloc )
```

Performs a deep clone of the given syntax node.

All members will be cloned recursively to create a complete new copy of the syntax tree. All cloned instances will be allocated with the provided allocator.

Definition at line 243 of file [SyntaxNode.h](#).

Here is the call graph for this function:



6.3.1.7 deepClone() [5/5]

```
TokenList * slang::syntax::deepClone (
    const TokenList & node,
    BumpAllocator & alloc ) [inline]
```

Definition at line 488 of file [SyntaxNode.h](#).

6.4 slang::syntax::detail Namespace Reference

Classes

- struct [ChangeCollection](#)
- struct [RemoveChange](#)
- struct [ReplaceChange](#)
- struct [SyntaxChange](#)

Typedefs

- using [InsertChangeMap](#) = flat_hash_map< const [SyntaxNode](#) *, std::vector< [SyntaxChange](#) > >
- using [ModifyChangeMap](#) = flat_hash_map< const [SyntaxNode](#) *, std::variant< [RemoveChange](#), [ReplaceChange](#) > >
- using [ListChangeMap](#) = flat_hash_map< const [SyntaxNode](#) *, std::vector< [SyntaxChange](#) > >

Functions

- SLANG_EXPORT std::shared_ptr< [SyntaxTree](#) > [transformTree](#) (BumpAllocator &&alloc, const std::shared_ptr< [SyntaxTree](#) > &tree, const [ChangeCollection](#) &commits, const std::vector< std::shared_ptr< [SyntaxTree](#) > > &tempTrees, const SourceLibrary *library)

6.4.1 Typedef Documentation

6.4.1.1 InsertChangeMap

```
using slang::syntax::detail::InsertChangeMap = typedef flat_hash_map<const SyntaxNode*, std::vector<SyntaxChange> >
```

Definition at line 68 of file [SyntaxVisitor.h](#).

6.4.1.2 ListChangeMap

```
using slang::syntax::detail::ListChangeMap = typedef flat_hash_map<const SyntaxNode*, std::vector<SyntaxChange>  
>
```

Definition at line 70 of file [SyntaxVisitor.h](#).

6.4.1.3 ModifyChangeMap

```
using slang::syntax::detail::ModifyChangeMap = typedef flat_hash_map<const SyntaxNode*, std::variant<RemoveChange,  
ReplaceChange> >
```

Definition at line 69 of file [SyntaxVisitor.h](#).

6.4.2 Function Documentation

6.4.2.1 transformTree()

```
SLANG_EXPORT std::shared_ptr< SyntaxTree > slang::syntax::detail::transformTree (  
    BumpAllocator && alloc,  
    const std::shared_ptr< SyntaxTree > & tree,  
    const ChangeCollection & commits,  
    const std::vector< std::shared_ptr< SyntaxTree > > & tempTrees,  
    const SourceLibrary * library )
```

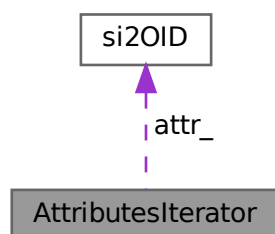

Chapter 7

Class Documentation

7.1 AttributesIterator Class Reference

```
#include <Iterators.hpp>
```

Collaboration diagram for AttributesIterator:



Public Member Functions

- [AttributesIterator](#) ([si2drAttrslidT](#) attrs, [si2drErrorT](#) &err)
- [~AttributesIterator](#) ()
- void [next](#) ()
- bool [end](#) ()
- [LibAttribute](#) [get](#) ()

Private Attributes

- [si2drAttrIdT](#) *attrs_*
- [si2drAttrIdT](#) *attr_*
- [si2drErrorT](#) & *err_*

7.1.1 Detailed Description

Definition at line 23 of file [lterators.hpp](#).

7.1.2 Constructor & Destructor Documentation

7.1.2.1 AttributesIterator()

```
AttributesIterator::AttributesIterator (
    si2drAttrIdT attrs,
    si2drErrorT & err )
```

Definition at line 13 of file [lterators.cpp](#).

7.1.2.2 ~AttributesIterator()

```
AttributesIterator::~AttributesIterator ( )
```

Definition at line 17 of file [lterators.cpp](#).

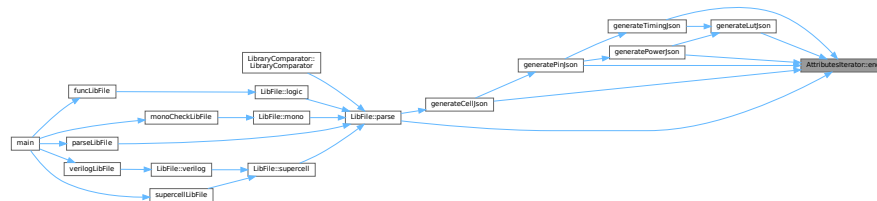
7.1.3 Member Function Documentation

7.1.3.1 end()

```
bool AttributesIterator::end ( )
```

Definition at line 20 of file [l iterators.cpp](#).

Here is the caller graph for this function:

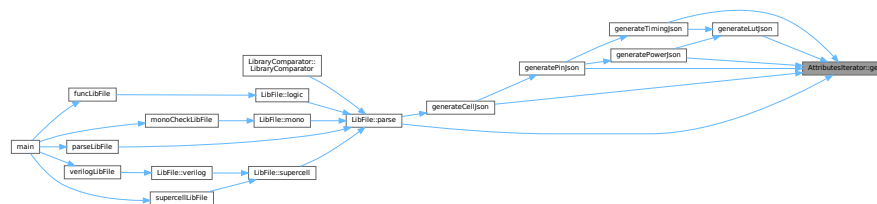


7.1.3.2 get()

```
LibAttribute AttributesIterator::get ( )
```

Definition at line 22 of file [l iterators.cpp](#).

Here is the caller graph for this function:

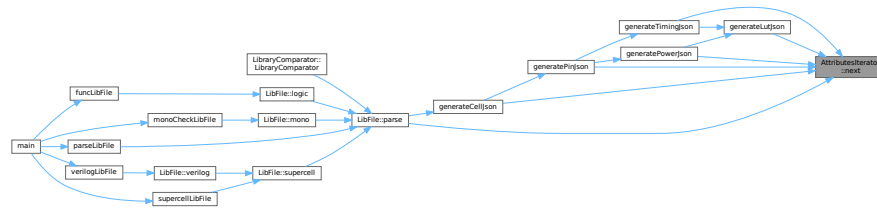


7.1.3.3 next()

```
void AttributesIterator::next ( )
```

Definition at line 19 of file [l iterators.cpp](#).

Here is the caller graph for this function:



7.1.4 Member Data Documentation

7.1.4.1 attr_

`si2drAttrIdT` `AttributesIterator::attr_` [private]

Definition at line 33 of file [lterators.hpp](#).

7.1.4.2 attrs_

`si2drAttrsIdT` `AttributesIterator::attrs_` [private]

Definition at line 32 of file [lterators.hpp](#).

7.1.4.3 err_

`si2drErrorT&` `AttributesIterator::err_` [private]

Definition at line 34 of file [lterators.hpp](#).

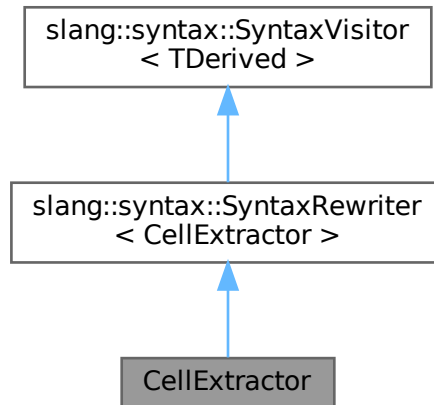
The documentation for this class was generated from the following files:

- `include/lterators.hpp`
- `src/lterators.cpp`

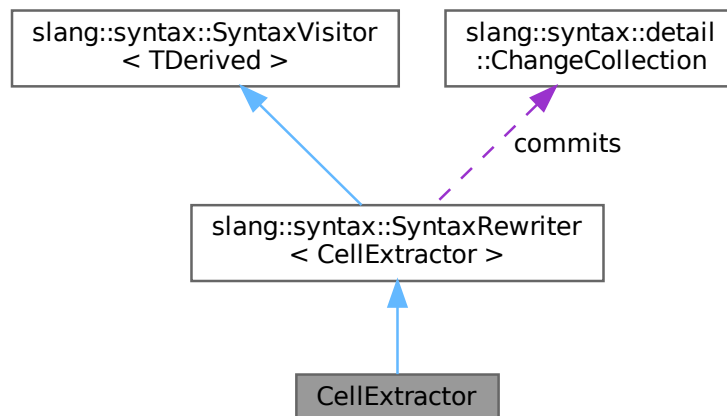
7.2 CellExtractor Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for CellExtractor:



Collaboration diagram for CellExtractor:



Public Member Functions

- `CellExtractor` (const std::string &targetCell)
- void `handle` (const slang::syntax::ModuleDeclarationSyntax &module)
- bool `foundTargetCell` () const

Private Attributes

- `const std::string & targetCell_`
- `bool foundTarget_`

Additional Inherited Members

7.2.1 Detailed Description

Definition at line 39 of file [verilog_utils.hpp](#).

7.2.2 Constructor & Destructor Documentation

7.2.2.1 CellExtractor()

```
CellExtractor::CellExtractor (  
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 41 of file [verilog_utils.hpp](#).

7.2.3 Member Function Documentation

7.2.3.1 foundTargetCell()

```
bool CellExtractor::foundTargetCell ( ) const
```

Definition at line 364 of file [verilog_utils.cpp](#).

Here is the caller graph for this function:

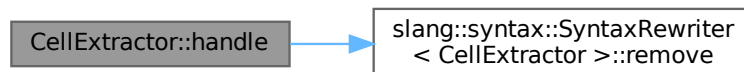


7.2.3.2 handle()

```
void CellExtractor::handle (  
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 348 of file [verilog_utils.cpp](#).

Here is the call graph for this function:



7.2.4 Member Data Documentation

7.2.4.1 foundTarget_

```
bool CellExtractor::foundTarget_ [private]
```

Definition at line 48 of file [verilog_utils.hpp](#).

7.2.4.2 targetCell_

```
const std::string& CellExtractor::targetCell_ [private]
```

Definition at line 47 of file [verilog_utils.hpp](#).

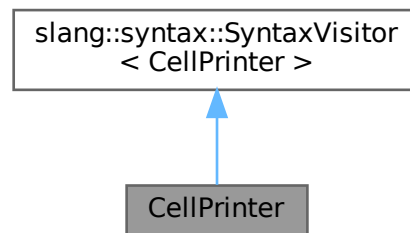
The documentation for this class was generated from the following files:

- [include/verilog_utils.hpp](#)
- [src/verilog_utils.cpp](#)

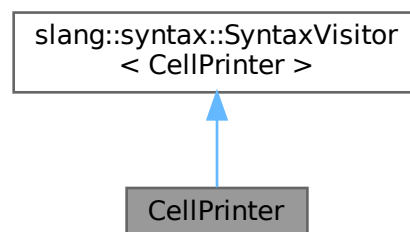
7.3 CellPrinter Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for CellPrinter:



Collaboration diagram for CellPrinter:



Public Member Functions

- [CellPrinter](#) (const std::string &targetCell, std::ostream &out)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)

Private Attributes

- const std::string & [targetCell_](#)
- std::ostream & [out_](#)
- bool [foundTarget_](#)

7.3.1 Detailed Description

Definition at line 52 of file [verilog_utils.hpp](#).

7.3.2 Constructor & Destructor Documentation

7.3.2.1 CellPrinter()

```
CellPrinter::CellPrinter (
    const std::string & targetCell,
    std::ostream & out ) [inline], [explicit]
```

Definition at line 54 of file [verilog_utils.hpp](#).

7.3.3 Member Function Documentation

7.3.3.1 handle()

```
void CellPrinter::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 367 of file [verilog_utils.cpp](#).

Here is the call graph for this function:



7.3.4 Member Data Documentation

7.3.4.1 foundTarget_

```
bool CellPrinter::foundTarget_ [private]
```

Definition at line 61 of file [verilog_utils.hpp](#).

7.3.4.2 out_

```
std::ostream& CellPrinter::out_ [private]
```

Definition at line 60 of file [verilog_utils.hpp](#).

7.3.4.3 targetCell_

```
const std::string& CellPrinter::targetCell_ [private]
```

Definition at line 59 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- include/[verilog_utils.hpp](#)
- src/[verilog_utils.cpp](#)

7.4 slang::syntax::detail::ChangeCollection Struct Reference

```
#include <SyntaxVisitor.h>
```

Public Member Functions

- void [clear](#) ()
- bool [empty](#) () const

Public Attributes

- [InsertChangeMap](#) [insertBefore](#)
- [InsertChangeMap](#) [insertAfter](#)
- [ModifyChangeMap](#) [removeOrReplace](#)
- [ListChangeMap](#) [listInsertAtFront](#)
- [ListChangeMap](#) [listInsertAtBack](#)

7.4.1 Detailed Description

Definition at line 72 of file [SyntaxVisitor.h](#).

7.4.2 Member Function Documentation

7.4.2.1 clear()

```
void slang::syntax::detail::ChangeCollection::clear ( ) [inline]
```

Definition at line 79 of file [SyntaxVisitor.h](#).

Here is the caller graph for this function:



7.4.2.2 empty()

```
bool slang::syntax::detail::ChangeCollection::empty ( ) const [inline]
```

Definition at line 87 of file [SyntaxVisitor.h](#).

Here is the caller graph for this function:



7.4.3 Member Data Documentation

7.4.3.1 insertAfter

[InsertChangeMap](#) slang::syntax::detail::ChangeCollection::insertAfter

Definition at line 74 of file [SyntaxVisitor.h](#).

7.4.3.2 insertBefore

[InsertChangeMap](#) slang::syntax::detail::ChangeCollection::insertBefore

Definition at line 73 of file [SyntaxVisitor.h](#).

7.4.3.3 listInsertAtBack

[ListChangeMap](#) slang::syntax::detail::ChangeCollection::listInsertAtBack

Definition at line 77 of file [SyntaxVisitor.h](#).

7.4.3.4 listInsertAtFront

[ListChangeMap](#) slang::syntax::detail::ChangeCollection::listInsertAtFront

Definition at line 76 of file [SyntaxVisitor.h](#).

7.4.3.5 removeOrReplace

[ModifyChangeMap](#) slang::syntax::detail::ChangeCollection::removeOrReplace

Definition at line 75 of file [SyntaxVisitor.h](#).

The documentation for this struct was generated from the following file:

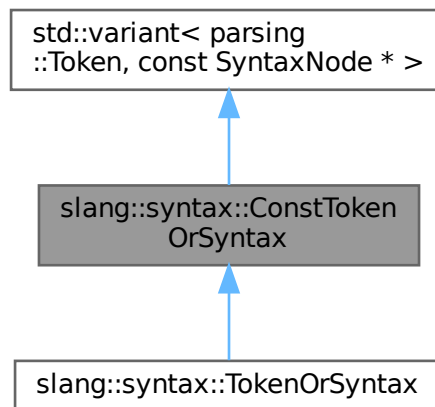
- `include_3rd_party/SyntaxVisitor.h`

7.5 slang::syntax::ConstTokenOrSyntax Struct Reference

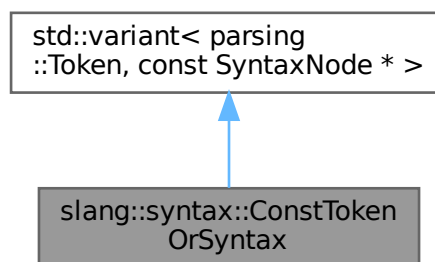
A token or a constant syntax node.

```
#include <SyntaxNode.h>
```

Inheritance diagram for slang::syntax::ConstTokenOrSyntax:



Collaboration diagram for slang::syntax::ConstTokenOrSyntax:



Public Types

- using [Base](#) = std::variant< parsing::Token, const [SyntaxNode](#) * >

Public Member Functions

- [ConstTokenOrSyntax](#) (parsing::Token [token](#))
- [ConstTokenOrSyntax](#) (const [SyntaxNode](#) *[node](#))
- [ConstTokenOrSyntax](#) (nullptr_t)
- bool [isToken](#) () const
- bool [isNode](#) () const
- parsing::Token [token](#) () const
- const [SyntaxNode](#) * [node](#) () const
- SourceRange [range](#) () const

Gets the source range for the token or syntax node.

7.5.1 Detailed Description

A token or a constant syntax node.

Definition at line 50 of file [SyntaxNode.h](#).

7.5.2 Member Typedef Documentation

7.5.2.1 Base

```
using slang::syntax::ConstTokenOrSyntax::Base = std::variant<parsing::Token, const SyntaxNode*>
```

Definition at line 51 of file [SyntaxNode.h](#).

7.5.3 Constructor & Destructor Documentation

7.5.3.1 ConstTokenOrSyntax() ^[1/3]

```
slang::syntax::ConstTokenOrSyntax::ConstTokenOrSyntax (
    parsing::Token token ) [inline]
```

Definition at line 52 of file [SyntaxNode.h](#).

7.5.3.2 ConstTokenOrSyntax() [2/3]

```
slang::syntax::ConstTokenOrSyntax::ConstTokenOrSyntax (  
    const SyntaxNode * node ) [inline]
```

Definition at line 53 of file [SyntaxNode.h](#).

7.5.3.3 ConstTokenOrSyntax() [3/3]

```
slang::syntax::ConstTokenOrSyntax::ConstTokenOrSyntax (  
    nullptr_t ) [inline]
```

Definition at line 54 of file [SyntaxNode.h](#).

7.5.4 Member Function Documentation

7.5.4.1 isNode()

```
bool slang::syntax::ConstTokenOrSyntax::isNode ( ) const [inline]
```

Returns

true if the object is a syntax node.

Definition at line 60 of file [SyntaxNode.h](#).

7.5.4.2 isToken()

```
bool slang::syntax::ConstTokenOrSyntax::isToken ( ) const [inline]
```

Returns

true if the object is a token.

Definition at line 57 of file [SyntaxNode.h](#).

7.5.4.3 node()

```
const SyntaxNode * slang::syntax::ConstTokenOrSyntax::node ( ) const [inline]
```

Gets access to the object as a syntax node (throwing an exception if it's not actually a syntax node).

Definition at line 68 of file [SyntaxNode.h](#).

7.5.4.4 range()

```
SourceRange slang::syntax::ConstTokenOrSyntax::range ( ) const
```

Gets the source range for the token or syntax node.

7.5.4.5 token()

```
parsing::Token slang::syntax::ConstTokenOrSyntax::token ( ) const [inline]
```

Gets access to the object as a token (throwing an exception if it's not actually a token).

Definition at line 64 of file [SyntaxNode.h](#).

The documentation for this struct was generated from the following file:

- include_3rd_party/[SyntaxNode.h](#)

7.6 slang::syntax::DeferredSourceRange Class Reference

```
#include <SyntaxNode.h>
```

Public Member Functions

- [DeferredSourceRange](#) ()=default
- [DeferredSourceRange](#) (SourceRange range)
- [DeferredSourceRange](#) (const [SyntaxNode](#) &node)
- [DeferredSourceRange](#) (const [SyntaxNode](#) &node, SourceRange range)
- SourceRange [get](#) () const
- SourceRange [operator*](#) () const
- const [SyntaxNode](#) * [syntax](#) () const

Private Attributes

- SourceRange [range](#)
- PointerIntPair< const [SyntaxNode](#) *, 1, 1, bool > [node](#)

7.6.1 Detailed Description

Represents a source range or a way to get one by materializing it from a syntax node. This exists to avoid computing the source range of a node unless it's actually needed.

Definition at line [250](#) of file [SyntaxNode.h](#).

7.6.2 Constructor & Destructor Documentation

7.6.2.1 DeferredSourceRange() [1/4]

```
slang::syntax::DeferredSourceRange::DeferredSourceRange ( ) [default]
```

7.6.2.2 DeferredSourceRange() [2/4]

```
slang::syntax::DeferredSourceRange::DeferredSourceRange (
    SourceRange range ) [inline]
```

Definition at line [253](#) of file [SyntaxNode.h](#).

7.6.2.3 DeferredSourceRange() [3/4]

```
slang::syntax::DeferredSourceRange::DeferredSourceRange (
    const SyntaxNode & node ) [inline]
```

Definition at line [254](#) of file [SyntaxNode.h](#).

7.6.2.4 DeferredSourceRange() [4/4]

```
slang::syntax::DeferredSourceRange::DeferredSourceRange (
    const SyntaxNode & node,
    SourceRange range ) [inline]
```

Definition at line [255](#) of file [SyntaxNode.h](#).

7.6.3 Member Function Documentation

7.6.3.1 get()

```
SourceRange slang::syntax::DeferredSourceRange::get ( ) const [inline]
```

Definition at line [258](#) of file [SyntaxNode.h](#).

Here is the caller graph for this function:



7.6.3.2 operator*()

```
SourceRange slang::syntax::DeferredSourceRange::operator* ( ) const [inline]
```

Definition at line [267](#) of file [SyntaxNode.h](#).

Here is the call graph for this function:



7.6.3.3 syntax()

```
const SyntaxNode * slang::syntax::DeferredSourceRange::syntax ( ) const [inline]
```

Definition at line 269 of file [SyntaxNode.h](#).

7.6.4 Member Data Documentation

7.6.4.1 node

```
PointerIntPair<const SyntaxNode*, 1, 1, bool> slang::syntax::DeferredSourceRange::node [mutable], [private]
```

Definition at line 273 of file [SyntaxNode.h](#).

7.6.4.2 range

```
SourceRange slang::syntax::DeferredSourceRange::range [mutable], [private]
```

Definition at line 272 of file [SyntaxNode.h](#).

The documentation for this class was generated from the following file:

- include_3rd_party/[SyntaxNode.h](#)

7.7 GateInfo Struct Reference

```
#include <LogicExtractor.hpp>
```

Public Attributes

- slang::parsing::TokenKind [kind](#)
- std::string [gateTypeName](#)
- std::vector< std::string > [inputSignals](#)
- std::string [outputSignal](#)

7.7.1 Detailed Description

Definition at line 22 of file [LogicExtractor.hpp](#).

7.7.2 Member Data Documentation

7.7.2.1 gateTypeName

```
std::string GateInfo::gateTypeName
```

Definition at line 24 of file [LogicExtractor.hpp](#).

7.7.2.2 inputSignals

```
std::vector<std::string> GateInfo::inputSignals
```

Definition at line 25 of file [LogicExtractor.hpp](#).

7.7.2.3 kind

```
slang::parsing::TokenKind GateInfo::kind
```

Definition at line 23 of file [LogicExtractor.hpp](#).

7.7.2.4 outputSignal

```
std::string GateInfo::outputSignal
```

Definition at line 26 of file [LogicExtractor.hpp](#).

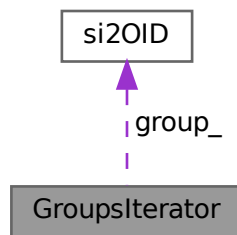
The documentation for this struct was generated from the following file:

- include/[LogicExtractor.hpp](#)

7.8 GroupsIterator Class Reference

```
#include <Iterators.hpp>
```

Collaboration diagram for GroupsIterator:



Public Member Functions

- `GroupsIterator` (`si2drGroupsIdT` groups, `si2drErrorT` &err)
- `~GroupsIterator` ()
- void `next` ()
- bool `end` ()
- `LibGroup` `get` ()

Private Attributes

- `si2drGroupsIdT` groups_
- `si2drGroupIdT` group_
- `si2drErrorT` & err_

7.8.1 Detailed Description

Definition at line 9 of file `Iterators.hpp`.

7.8.2 Constructor & Destructor Documentation

7.8.2.1 GroupsIterator()

```
GroupsIterator::GroupsIterator (
    si2drGroupsIdT groups,
    si2drErrorT & err )
```

Definition at line 3 of file [lterators.cpp](#).

7.8.2.2 ~GroupsIterator()

```
GroupsIterator::~GroupsIterator ( )
```

Definition at line 6 of file [lterators.cpp](#).

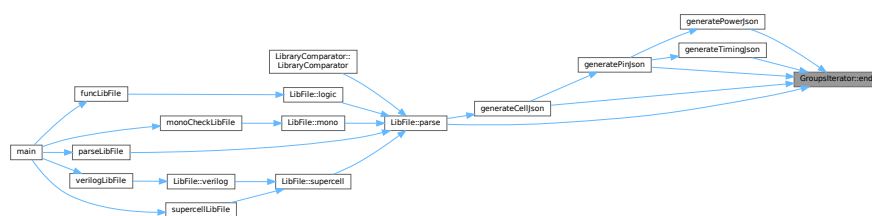
7.8.3 Member Function Documentation

7.8.3.1 end()

```
bool GroupsIterator::end ( )
```

Definition at line 9 of file [lterators.cpp](#).

Here is the caller graph for this function:

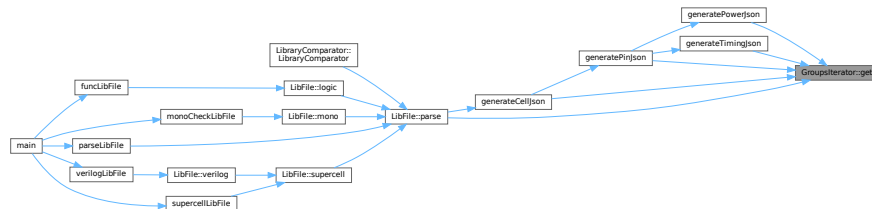


7.8.3.2 get()

`LibGroup GroupsIterator::get ( )`

Definition at line 11 of file [Iterators.cpp](#).

Here is the caller graph for this function:

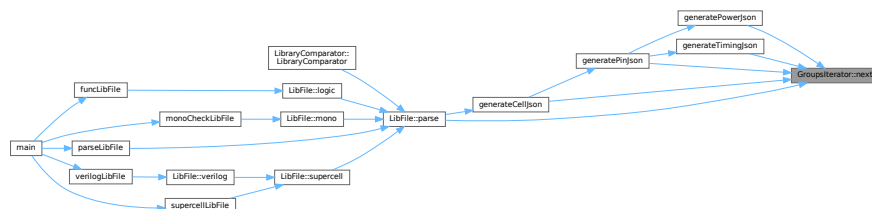


7.8.3.3 next()

`void GroupsIterator::next ( )`

Definition at line 8 of file [Iterators.cpp](#).

Here is the caller graph for this function:



7.8.4 Member Data Documentation

7.8.4.1 err_

`si2drErrorT& GroupsIterator::err_ [private]`

Definition at line 20 of file [Iterators.hpp](#).

7.8.4.2 group_

`si2drGroupIdT GroupsIterator::group_ [private]`

Definition at line 19 of file [lterators.hpp](#).

7.8.4.3 groups_

`si2drGroupsIdT GroupsIterator::groups_ [private]`

Definition at line 18 of file [lterators.hpp](#).

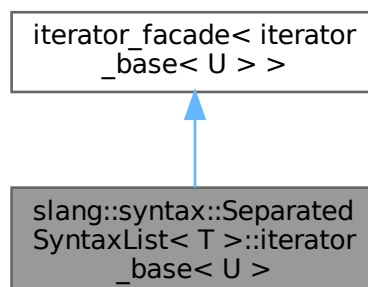
The documentation for this class was generated from the following files:

- [include/lterators.hpp](#)
- [src/lterators.cpp](#)

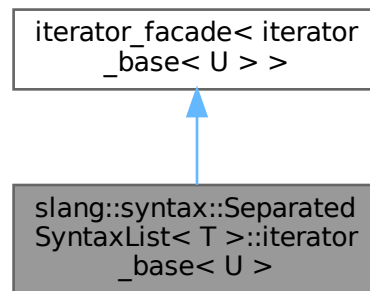
7.9 slang::syntax::SeparatedSyntaxList< T >::iterator_base< U > > Class Template Reference

```
#include <SyntaxNode.h>
```

Inheritance diagram for slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >:



Collaboration diagram for slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >:



Public Types

- using `ParentList` = `std::conditional_t< std::is_const_v< std::remove_pointer_t< U > >, const SeparatedSyntaxList, SeparatedSyntaxList >`

Public Member Functions

- `iterator_base` ()
- `iterator_base` (`ParentList` &`list`, `size_t` `index`)
- `U` `dereference` () const
- `bool` `equals` (const `iterator_base` &`other`) const
- `ptrdiff_t` `distance_to` (const `iterator_base` &`other`) const
- `void` `advance` (`ptrdiff_t` `n`)

Private Attributes

- `ParentList` * `list`
- `size_t` `index`

7.9.1 Detailed Description

```

template<typename T>
template<typename U>
class slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >

```

An iterator that will iterate over just the nodes (and skip the delimiters) in the parent `SeparatedSyntaxList`.

Definition at line 386 of file `SyntaxNode.h`.

7.9.2 Member Typedef Documentation

7.9.2.1 ParentList

```
template<typename T >
template<typename U >
using slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >::ParentList = std::conditional_t<std::is_↵
const_v<std::remove_pointer_t<U> >, const SeparatedSyntaxList, SeparatedSyntaxList>
```

Definition at line 388 of file [SyntaxNode.h](#).

7.9.3 Constructor & Destructor Documentation

7.9.3.1 iterator_base() [1/2]

```
template<typename T >
template<typename U >
slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >::iterator_base ( ) [inline]
```

Definition at line 391 of file [SyntaxNode.h](#).

7.9.3.2 iterator_base() [2/2]

```
template<typename T >
template<typename U >
slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >::iterator_base (
    ParentList & list,
    size_t index ) [inline]
```

Definition at line 392 of file [SyntaxNode.h](#).

7.9.4 Member Function Documentation

7.9.4.1 advance()

```
template<typename T >
template<typename U >
void slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >::advance (
    ptrdiff_t n ) [inline]
```

Definition at line 404 of file [SyntaxNode.h](#).

7.9.4.2 dereference()

```
template<typename T >
template<typename U >
U slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >::dereference ( ) const [inline]
```

Definition at line 394 of file [SyntaxNode.h](#).

7.9.4.3 distance_to()

```
template<typename T >
template<typename U >
ptrdiff_t slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >::distance_to (
    const iterator_base< U > & other ) const [inline]
```

Definition at line 400 of file [SyntaxNode.h](#).

7.9.4.4 equals()

```
template<typename T >
template<typename U >
bool slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >::equals (
    const iterator_base< U > & other ) const [inline]
```

Definition at line 396 of file [SyntaxNode.h](#).

7.9.5 Member Data Documentation

7.9.5.1 index

```
template<typename T >
template<typename U >
size_t slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >::index [private]
```

Definition at line 408 of file [SyntaxNode.h](#).

7.9.5.2 list

```
template<typename T >
template<typename U >
ParentList* slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >::list [private]
```

Definition at line 407 of file [SyntaxNode.h](#).

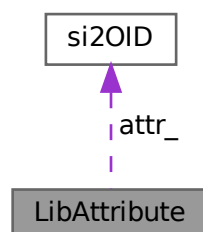
The documentation for this class was generated from the following file:

- [include_3rd_party/SyntaxNode.h](#)

7.10 LibAttribute Class Reference

```
#include <LibAttribute.hpp>
```

Collaboration diagram for LibAttribute:



Public Member Functions

- [LibAttribute](#) ([si2drAttrIdT](#) attr, [si2drErrorT](#) &err)
- [~LibAttribute](#) ()
- [std::string](#) [getName](#) ()
- [bool](#) [isComplex](#) ()
- [si2drValuesIdT](#) [getValues](#) ()
- [long int](#) [getInt](#) ()
- [double](#) [getFloat](#) ()
- [std::string](#) [getString](#) ()
- [bool](#) [getBoolean](#) ()

Private Attributes

- [si2drAttrIdT](#) attr_
- [si2drErrorT](#) & err_

7.10.1 Detailed Description

Definition at line 8 of file [LibAttribute.hpp](#).

7.10.2 Constructor & Destructor Documentation

7.10.2.1 LibAttribute()

```
LibAttribute::LibAttribute (  
    si2drAttrIdT attr,  
    si2drErrorT & err )
```

Definition at line 3 of file [LibAttribute.cpp](#).

7.10.2.2 ~LibAttribute()

```
LibAttribute::~LibAttribute ( )
```

Definition at line 5 of file [LibAttribute.cpp](#).

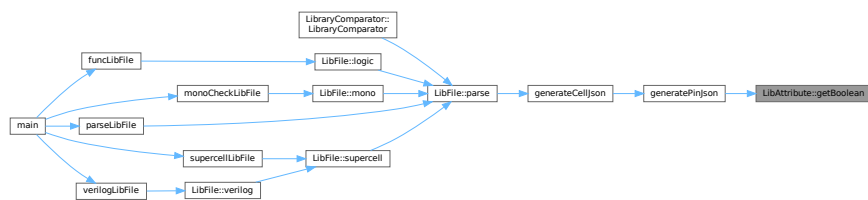
7.10.3 Member Function Documentation

7.10.3.1 getBoolean()

```
bool LibAttribute::getBoolean ( )
```

Definition at line 25 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

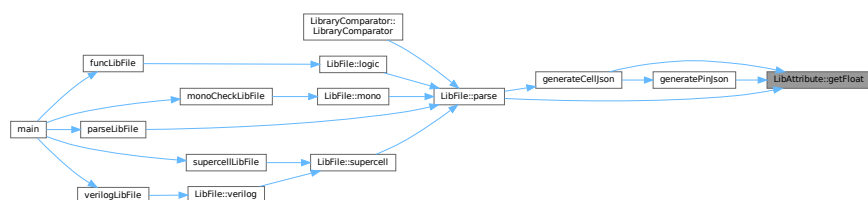


7.10.3.2 getFloat()

```
double LibAttribute::getFloat ( )
```

Definition at line 18 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

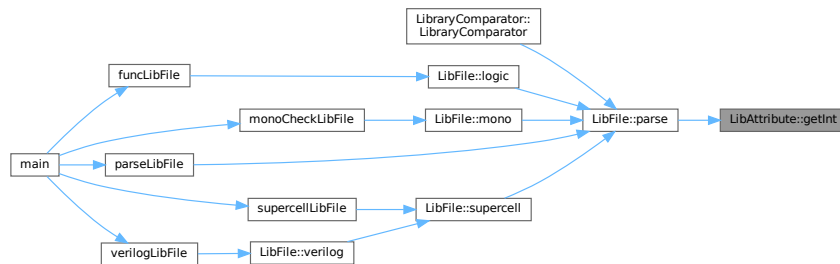


7.10.3.3 getInt()

```
long int LibAttribute::getInt ( )
```

Definition at line 16 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

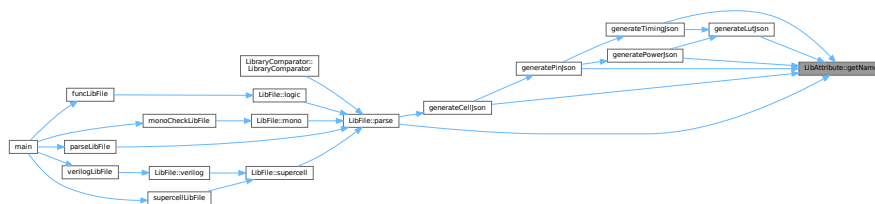


7.10.3.4 getName()

```
std::string LibAttribute::getName ( )
```

Definition at line 7 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

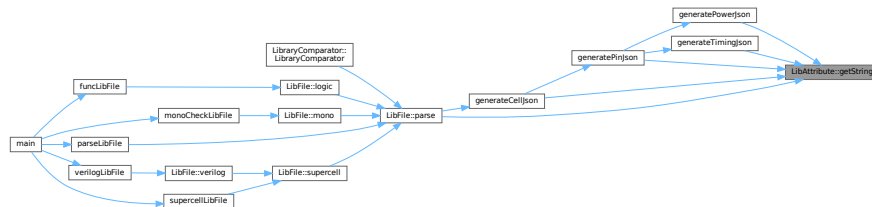


7.10.3.5 getString()

```
std::string LibAttribute::getString ( )
```

Definition at line 20 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:

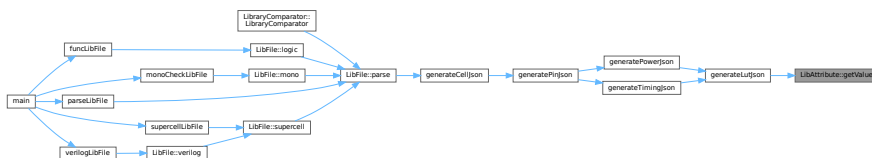


7.10.3.6 getValues()

```
si2drValuesIdT LibAttribute::getValues ( )
```

Definition at line 14 of file [LibAttribute.cpp](#).

Here is the caller graph for this function:



7.10.3.7 isComplex()

```
bool LibAttribute::isComplex ( )
```

Definition at line 12 of file [LibAttribute.cpp](#).

7.10.4 Member Data Documentation

7.10.4.1 attr_

`si2drAttrIdT` `LibAttribute::attr_` [private]

Definition at line 21 of file [LibAttribute.hpp](#).

7.10.4.2 err_

`si2drErrorT&` `LibAttribute::err_` [private]

Definition at line 22 of file [LibAttribute.hpp](#).

The documentation for this class was generated from the following files:

- [include/LibAttribute.hpp](#)
- [src/LibAttribute.cpp](#)

7.11 liberty_value_data Struct Reference

```
#include <si2dr_liberty.h>
```

Public Attributes

- `int` [dimensions](#)
- `int *` [dim_sizes](#)
- `LONG_DOUBLE **` [index_info](#)
- `LONG_DOUBLE *` [values](#)

7.11.1 Detailed Description

Definition at line 612 of file [si2dr_liberty.h](#).

7.11.2 Member Data Documentation

7.11.2.1 dim_sizes

```
int* liberty_value_data::dim_sizes
```

Definition at line 615 of file [si2dr_liberty.h](#).

7.11.2.2 dimensions

```
int liberty_value_data::dimensions
```

Definition at line 614 of file [si2dr_liberty.h](#).

7.11.2.3 index_info

```
LONG_DOUBLE** liberty_value_data::index_info
```

Definition at line 617 of file [si2dr_liberty.h](#).

7.11.2.4 values

```
LONG_DOUBLE* liberty_value_data::values
```

Definition at line 621 of file [si2dr_liberty.h](#).

The documentation for this struct was generated from the following file:

- [include_3rd_party/si2dr_liberty.h](#)

7.12 LibFile Class Reference

```
#include <LibFile.hpp>
```

Public Member Functions

- `LibFile` (const std::string &filepath, const std::string &loggername)
Constructs a `LibFile` object with specified file path and logger name.
- `~LibFile` ()
- void `writeJsonToFile` ()
Writes the JSON data stored in the object to a file.
- void `parse` ()
Parses the Liberty file and extracts library information into a JSON structure.
- void `modify` ()
- void `mono` (const bool is_slew)
Validates that lookup tables are monotonically increasing with respect to the output load.
- void `supercell` (const int chain_length, const std::vector< std::string > &cell_names)
Creates supercells based on standard cells in the library file.
- void `verilog` (const int chain_length, const std::vector< std::string > &cell_names)
Generates Verilog files for cell validation based on the library file.
- std::map< std::string, std::string > `logic` (const std::string &cell_name)

Public Attributes

- std::shared_ptr< spdlog::logger > `logger_`
- std::filesystem::path `filepath_`
- std::string `basename_`
- std::string `filename_`
- std::string `libname_` = ""
- std::string `jsonname_` = ""
- std::string `loggername_` = ""
- `json lib_json_` = json::object()

Private Member Functions

- void `read` ()
Reads the Liberty file specified by the `filename_` member variable.
- bool `checkTimingArcMonotonicity` (const json &cell, const json &pin, const json &arc, const std::string &type, bool is_slew)
Checks if the values in a timing arc are monotonically increasing.

Private Attributes

- `si2drErrorT err_`
- int `process_` = 0
- float `voltage_` = 0.0
- int `temperature_` = 0

7.12.1 Detailed Description

Definition at line 13 of file [LibFile.hpp](#).

7.12.2 Constructor & Destructor Documentation

7.12.2.1 LibFile()

```
LibFile::LibFile (
    const std::string & filepath,
    const std::string & loggername )
```

Constructs a [LibFile](#) object with specified file path and logger name.

This constructor initializes a [LibFile](#) object with the given file path and creates a logger with the specified name. It sets up:

- Two logging sinks: a colored console sink and a file sink
- File path information including filename, basename, and a corresponding JSON filename
- Log levels (info for console, debug for file)

Parameters

<i>filepath</i>	The path to the file to be processed
<i>loggername</i>	The name for the logger and the log file

Definition at line 28 of file [LibFile.cpp](#).

7.12.2.2 ~LibFile()

```
LibFile::~LibFile ( )
```

Definition at line 48 of file [LibFile.cpp](#).

7.12.3 Member Function Documentation

7.12.3.1 checkTimingArcMonotonicity()

```
bool LibFile::checkTimingArcMonotonicity (
    const json & cell,
    const json & pin,
    const json & arc,
    const std::string & timing_arc_name,
    bool is_slew ) [private]
```

Checks if the values in a timing arc are monotonically increasing.

This function examines the values in the specified timing arc to ensure they are monotonically increasing. It checks monotonicity in two dimensions:

1. Across rows (by output load capacitance)
2. Across columns (by input slew, only if is_slew is true)

The function logs detailed information about any non-monotonic values found, including cell name, pin names, and conditional statements (when clause) if present.

Parameters

<i>cell</i>	The JSON object representing the cell being checked
<i>pin</i>	The JSON object representing the pin being checked
<i>arc</i>	The JSON object representing the timing arc being checked
<i>timing_arc_name</i>	The name of the timing arc to check (e.g., "cell_rise", "cell_fall")
<i>is_slew</i>	Boolean flag indicating whether to check monotonicity across input slew values

Returns

true if all values in the timing arc are monotonic, false otherwise

Exceptions

<i>None, but</i>	logs errors or warnings for invalid data formats or non-monotonic values
------------------	--

Definition at line 218 of file [LibFile.cpp](#).

Here is the caller graph for this function:

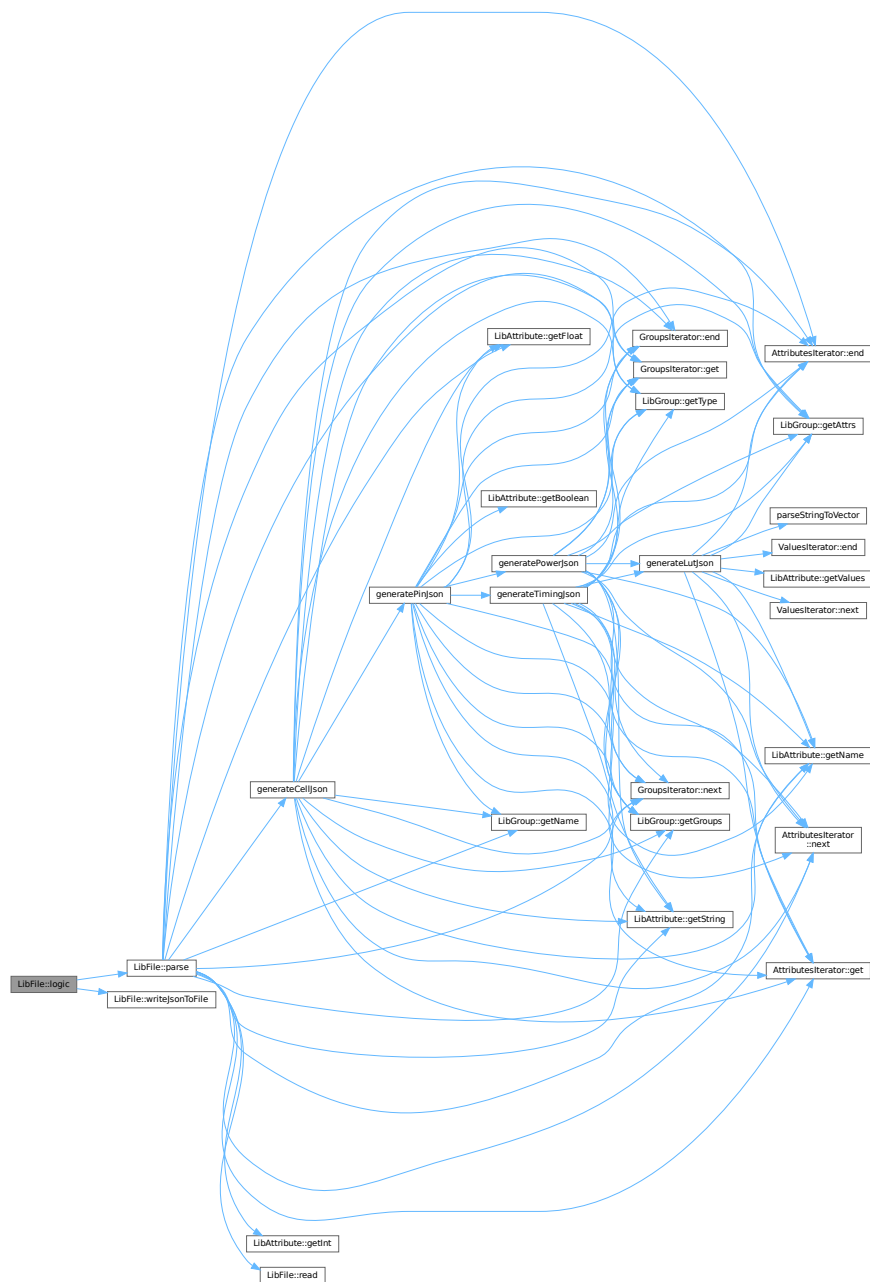


7.12.3.2 logic()

```
std::map< std::string, std::string > LibFile::logic (  
    const std::string & cell_name )
```

Definition at line 891 of file [LibFile.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.12.3.3 `modify()`

```
void LibFile::modify ( )
```

Definition at line 195 of file [LibFile.cpp](#).

7.12.3.4 `mono()`

```
void LibFile::mono (
    const bool is_slew )
```

Validates that lookup tables are monotonically increasing with respect to the output load.

This function checks the following timing data in the current library to ensure monotonicity:

- `cell_rise` and `retaining_rise`
- `cell_fall` and `retaining_fall`
- `rise_transition` and `retain_rise_slew`
- `fall_transition` and `retain_fall_slew`
- `min_pulse_width` constraints (`rise_constraint`, `fall_constraint`)

The function first checks if a parsed JSON representation exists. If not, it parses the Liberty file and creates the JSON file. It then evaluates each timing arc in all cells and reports the pass/fail status at the end.

Parameters

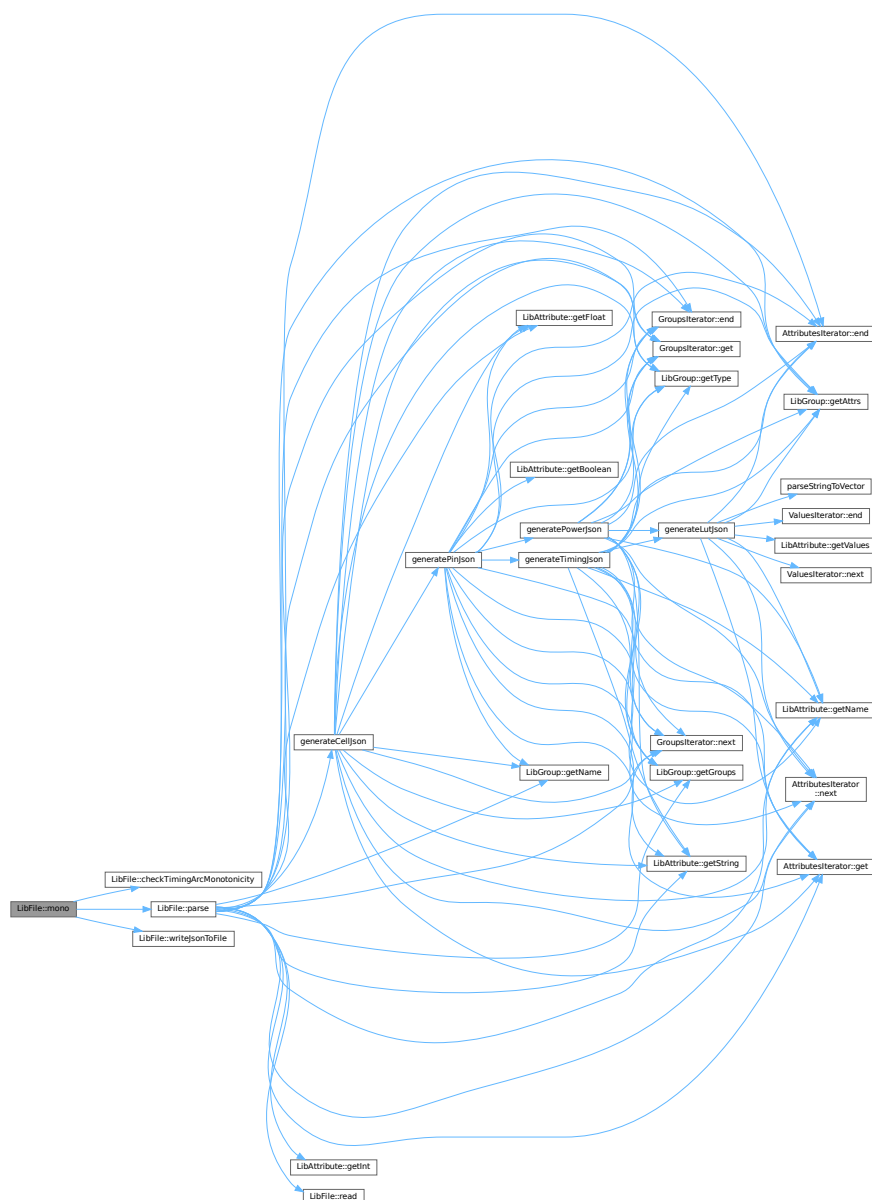
<code><i>is_slew</i></code>	Boolean flag indicating whether to check slew values rather than delay values
-----------------------------	---

The function outputs:

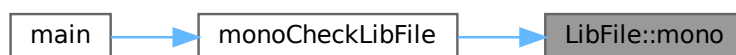
- Number of cells that passed/failed the monotonicity check
- List of cells that failed the check (if any)

Definition at line 344 of file [LibFile.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.12.3.5 parse()

```
void LibFile::parse ( )
```

Parses the Liberty file and extracts library information into a JSON structure.

This method handles the parsing of a Liberty file by:

1. Initializing the SI2 parser interface error handler
2. Reading the Liberty file contents
3. Extracting the library name
4. Processing second-level groups including:
 - Cell definitions, which are added to the JSON structure
 - Operating conditions, extracting PVT (Process, Voltage, Temperature) values

The parsed data is stored in the `lib_json_` member variable, with the following structure:

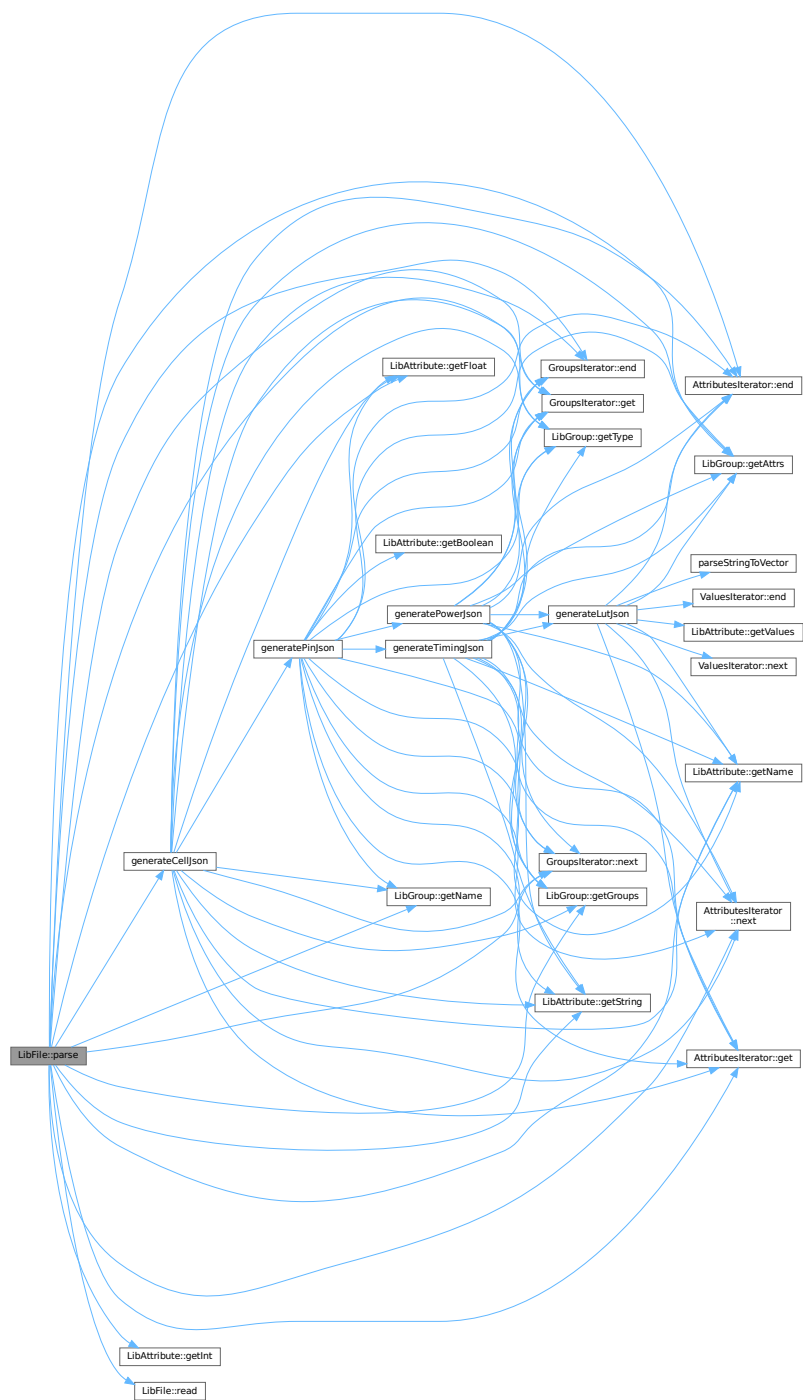
- "library_name": Name of the library
- "cells": Array of cell definitions
- "process": Process value
- "voltage": Voltage value (rounded to 2 decimal places)
- "temperature": Temperature value

Note

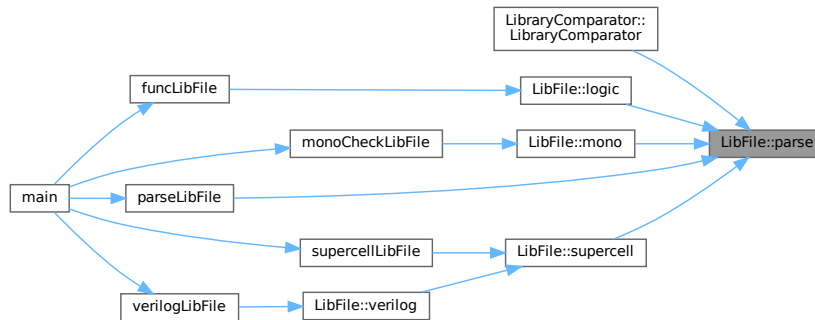
This method creates local scopes to ensure proper cleanup of SI2 Iterators.

Definition at line [129](#) of file [LibFile.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.12.3.6 read()

```
void LibFile::read ( ) [private]
```

Reads the Liberty file specified by the filename_ member variable.

This function attempts to read a Liberty file and logs the process. It measures the time taken to read the file and logs any errors encountered during the read operation. If an error occurs, the function logs the error and terminates the program.

- Logs the start of the read operation.
- Measures the duration of the read operation.
- Checks for specific errors (invalid name or syntax error) and logs them.
- Terminates the program with specific exit codes if errors are detected.
- Logs the completion of the read operation and the time taken.

Note

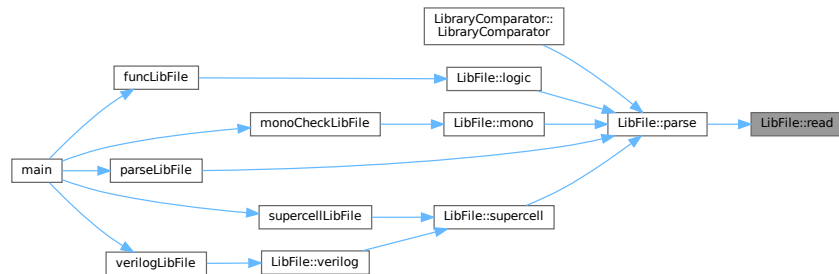
This function uses the spdlog library for logging and the si2drReadLibertyFile function for reading the Liberty file.

Exceptions

<i>This</i>	function will terminate the program with exit codes 301 or 401 if specific errors are encountered.
-------------	--

Definition at line 90 of file [LibFile.cpp](#).

Here is the caller graph for this function:



7.12.3.7 supercell()

```

void LibFile::supercell (
    const int chain_length,
    const std::vector< std::string > & cell_names )

```

Creates supercells based on standard cells in the library file.

This method creates supercells by combining standard cells in chains. For each input-output pin pair of a cell, it creates a supercell entry with naming format: <cellname>**X**<chain_length><input_pin>↔__<output_pin>. The results are written to a .map file.

For sequential cells (with clock pins), the chain length is always set to 1 regardless of the requested chain length.

Parameters

<i>chain_length</i>	The length of the chain for combinational cells
<i>cell_names</i>	Vector of specific cell names to process. If empty, all cells will be processed.

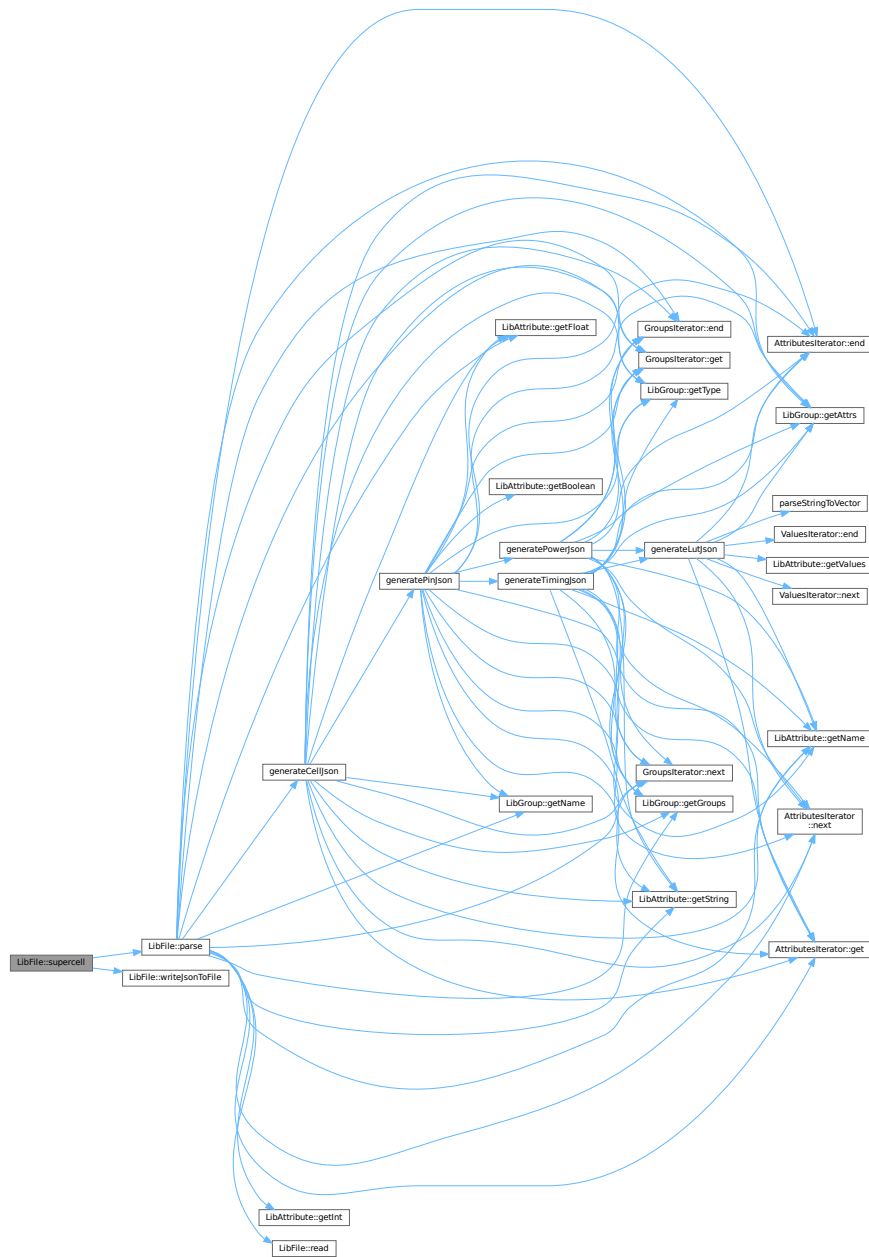
Note

Before creating supercells, this method checks for the existence of a JSON representation of the Liberty file and parses it if not found.

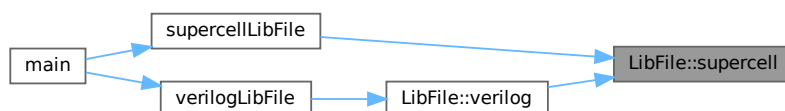
The method will report any requested cells that were not found in the library.

Definition at line 473 of file [LibFile.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.12.3.8 verilog()

```
void LibFile::verilog (
    const int chain_length,
    const std::vector< std::string > & cell_names )
```

Generates Verilog files for cell validation based on the library file.

This function creates Verilog modules for each cell in the specified chain and generates a top-level module that connects them all together. The process involves:

1. Creating supercells based on the specified chain length and cell names
2. Reading the mapping file to identify supercell name relationships
3. Generating individual Verilog modules for each supercell, handling both sequential and combinational cells differently
4. Creating ANSI-style port definitions for each module
5. Using the slang library to build proper syntax trees for Verilog modules
6. Creating a top-level module that instantiates all the individual modules
7. Writing the complete Verilog output to the output file

Sequential cells are instantiated once, while combinational cells are instantiated multiple times based on the specified chain length.

Parameters

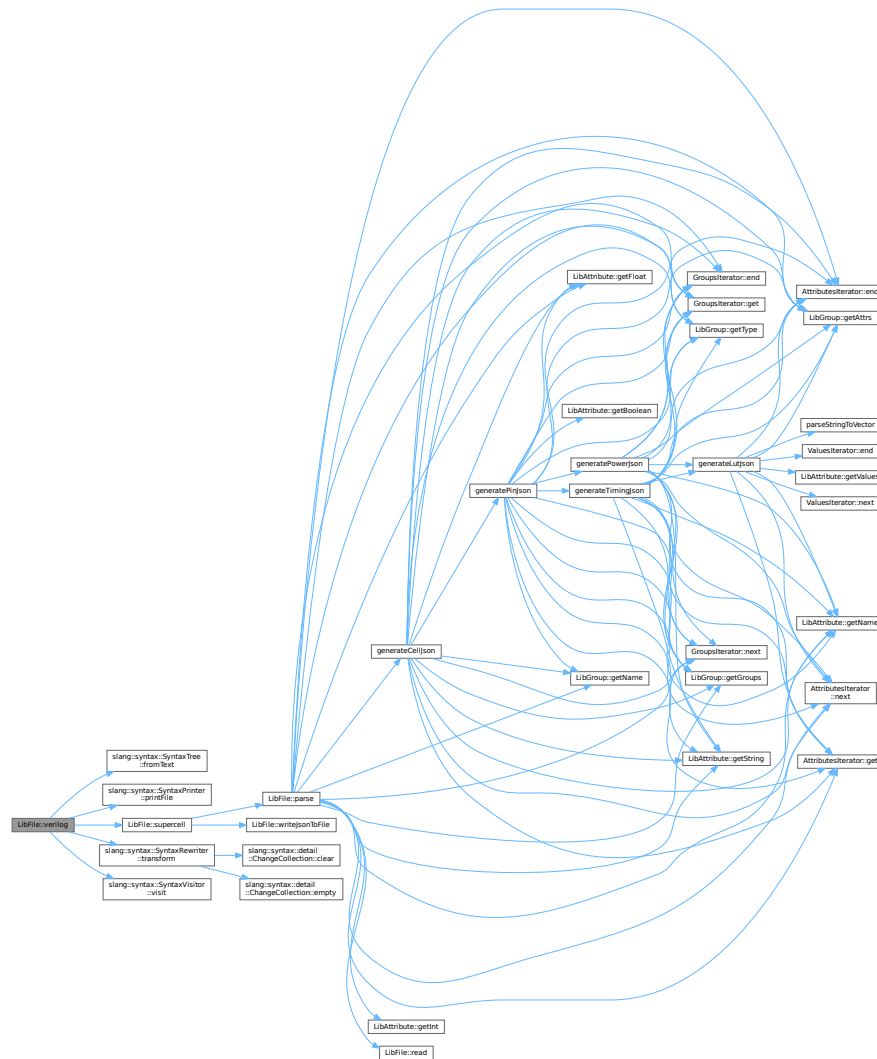
<i>chain_length</i>	The number of times to chain combinational cells together
<i>cell_names</i>	Vector of cell names to include in the Verilog generation

Note

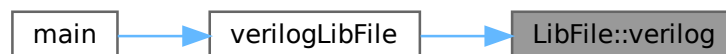
The function writes a temporary file during processing that might not be removed when the function completes (commented out cleanup code).

Definition at line 627 of file [LibFile.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.12.3.9 writeJsonToFile()

```
void LibFile::writeJsonToFile ( )
```

Writes the JSON data stored in the object to a file.

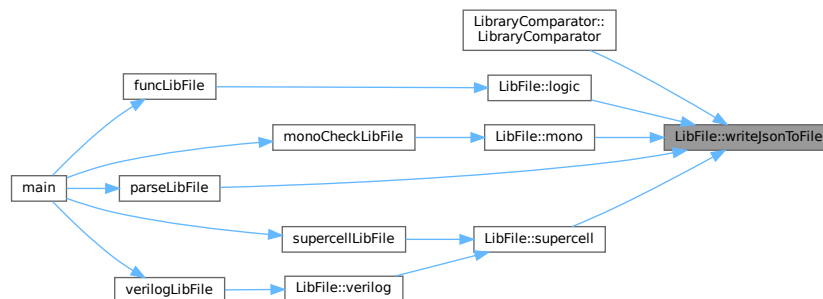
This method opens the file specified by `jsonname_` and writes the content of `lib_json_` with an indentation of 2 spaces. If the file cannot be opened, an error message is logged. Upon successful writing, an informational message is logged.

Exceptions

<i>None</i>	directly, but <code>std::ofstream</code> operations may throw exceptions
-------------	--

Definition at line 59 of file [LibFile.cpp](#).

Here is the caller graph for this function:



7.12.4 Member Data Documentation

7.12.4.1 `basename_`

`std::string LibFile::basename_`

Definition at line 19 of file [LibFile.hpp](#).

7.12.4.2 `err_`

`si2drErrorT LibFile::err_` [private]

Definition at line 34 of file [LibFile.hpp](#).

7.12.4.3 filename_

```
std::string LibFile::filename_
```

Definition at line 20 of file [LibFile.hpp](#).

7.12.4.4 filepath_

```
std::filesystem::path LibFile::filepath_
```

Definition at line 18 of file [LibFile.hpp](#).

7.12.4.5 jsonname_

```
std::string LibFile::jsonname_ = ""
```

Definition at line 22 of file [LibFile.hpp](#).

7.12.4.6 lib_json_

```
json LibFile::lib_json_ = json::object()
```

Definition at line 24 of file [LibFile.hpp](#).

7.12.4.7 libname_

```
std::string LibFile::libname_ = ""
```

Definition at line 21 of file [LibFile.hpp](#).

7.12.4.8 logger_

```
std::shared_ptr<spdlog::logger> LibFile::logger_
```

Definition at line 17 of file [LibFile.hpp](#).

7.12.4.9 loggername_

```
std::string LibFile::loggername_ = ""
```

Definition at line 23 of file [LibFile.hpp](#).

7.12.4.10 process_

```
int LibFile::process_ = 0 [private]
```

Definition at line 35 of file [LibFile.hpp](#).

7.12.4.11 temperature_

```
int LibFile::temperature_ = 0 [private]
```

Definition at line 37 of file [LibFile.hpp](#).

7.12.4.12 voltage_

```
float LibFile::voltage_ = 0.0 [private]
```

Definition at line 36 of file [LibFile.hpp](#).

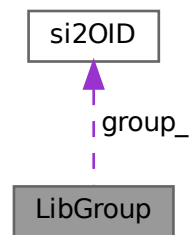
The documentation for this class was generated from the following files:

- [include/LibFile.hpp](#)
- [src/LibFile.cpp](#)

7.13 LibGroup Class Reference

```
#include <LibGroup.hpp>
```

Collaboration diagram for LibGroup:



Public Member Functions

- `LibGroup` (`si2drGroupIdT` group, `si2drErrorT` &err)
- `~LibGroup` ()
- `std::string` `getName` ()
- `std::string` `getType` ()
- `si2drAttrIdT` `getAttrs` ()
- `si2drGroupIdT` `getGroups` ()

Private Attributes

- `si2drGroupIdT` `group_`
- `si2drErrorT` & `err_`

7.13.1 Detailed Description

Definition at line 8 of file `LibGroup.hpp`.

7.13.2 Constructor & Destructor Documentation

7.13.2.1 LibGroup()

```
LibGroup::LibGroup (
    si2drGroupIdT group,
    si2drErrorT & err )
```

Definition at line 3 of file [LibGroup.cpp](#).

7.13.2.2 ~LibGroup()

```
LibGroup::~~LibGroup ( )
```

Definition at line 5 of file [LibGroup.cpp](#).

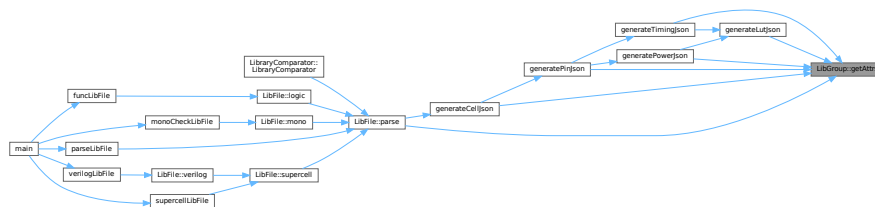
7.13.3 Member Function Documentation

7.13.3.1 getAttrs()

```
si2drAttrsIdT LibGroup::getAttrs ( )
```

Definition at line 19 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

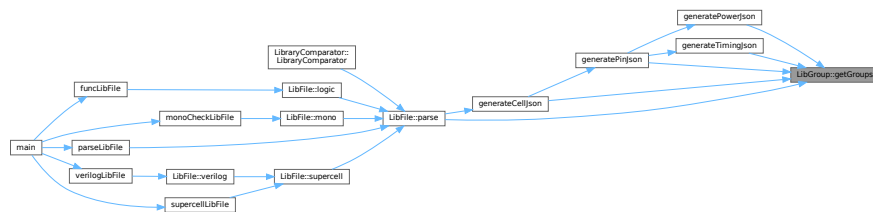


7.13.3.2 getGroups()

```
si2drGroupsIdT LibGroup::getGroups ( )
```

Definition at line 21 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

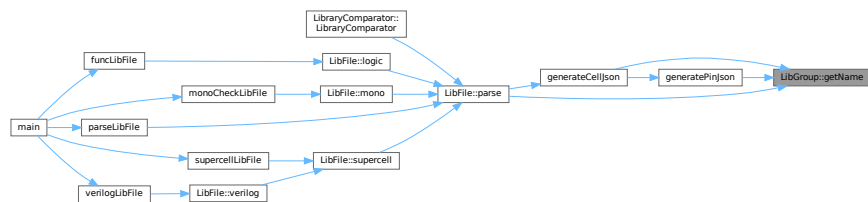


7.13.3.3 getName()

```
std::string LibGroup::getName ( )
```

Definition at line 7 of file [LibGroup.cpp](#).

Here is the caller graph for this function:

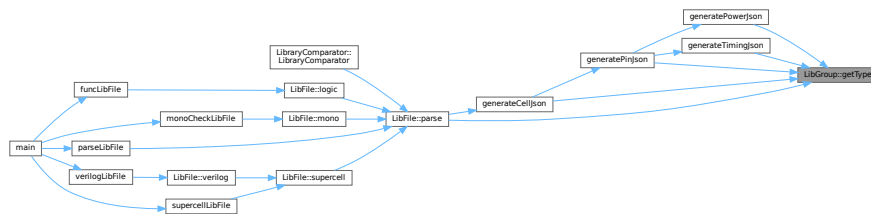


7.13.3.4 getType()

```
std::string LibGroup::getType ( )
```

Definition at line 14 of file [LibGroup.cpp](#).

Here is the caller graph for this function:



7.13.4 Member Data Documentation

7.13.4.1 err_

```
si2drErrorT& LibGroup::err_ [private]
```

Definition at line 19 of file [LibGroup.hpp](#).

7.13.4.2 group_

```
si2drGroupIdT LibGroup::group_ [private]
```

Definition at line 18 of file [LibGroup.hpp](#).

The documentation for this class was generated from the following files:

- [include/LibGroup.hpp](#)
- [src/LibGroup.cpp](#)

7.14 LibraryComparator Class Reference

```
#include <LibraryComparator.hpp>
```

Public Member Functions

- [LibraryComparator](#) ([LibFile](#) &ref_libfile, [LibFile](#) &comp_libfile, double reltol, double abstol)
Constructor for the [LibraryComparator](#) class.
- void [generateReport](#) (const std::string &output_file)
Generates a comparison report between the reference and comparison libraries.

Public Attributes

- std::filesystem::path [ref_lib_path_](#)
- std::filesystem::path [comp_lib_path_](#)

Private Member Functions

- void [compareCell](#) (const std::string &cell_name, const [json](#) &ref_cell, const [json](#) &comp_cell, Table &table)
Compares the output pins of a cell between a reference JSON and a comparison JSON.
- void [comparePin](#) (const std::string &cell_name, const std::string &pin_name, const [json](#) &ref_pin, const [json](#) &comp_pin, Table &table)
Compares the timing arcs of a given pin between a reference JSON and a comparison JSON.
- void [compareTimingArc](#) (const std::string &cell_name, const std::string &pin_name, const std::string &timing_type, const [json](#) &ref_timing_arc, const [json](#) &comp_timing_arc, Table &table)
Compares a timing arc between two JSON objects.
- void [compareLut](#) (const std::string &cell_name, const std::string &pin_name, const std::string &timing_type, const std::string &related_pin, const std::string &arc_name, const [json](#) &ref_lut, const [json](#) &comp_lut, Table &table)
Compares lookup tables (LUTs) between reference and comparison libraries for a specific timing arc.

Private Attributes

- [json](#) [ref_json_](#)
- [json](#) [comp_json_](#)
- double [reltol_](#)
- double [abstol_](#)

7.14.1 Detailed Description

Definition at line 12 of file [LibraryComparator.hpp](#).

7.14.2 Constructor & Destructor Documentation

7.14.2.1 LibraryComparator()

```
LibraryComparator::LibraryComparator (
    LibFile & ref_libfile,
    LibFile & comp_libfile,
    double reltol,
    double abstol )
```

Constructor for the [LibraryComparator](#) class.

Initializes a [LibraryComparator](#) object by loading JSON data from reference and comparison library files. If JSON files don't exist, parses the library files first and creates the JSON files.

The constructor performs the following steps:

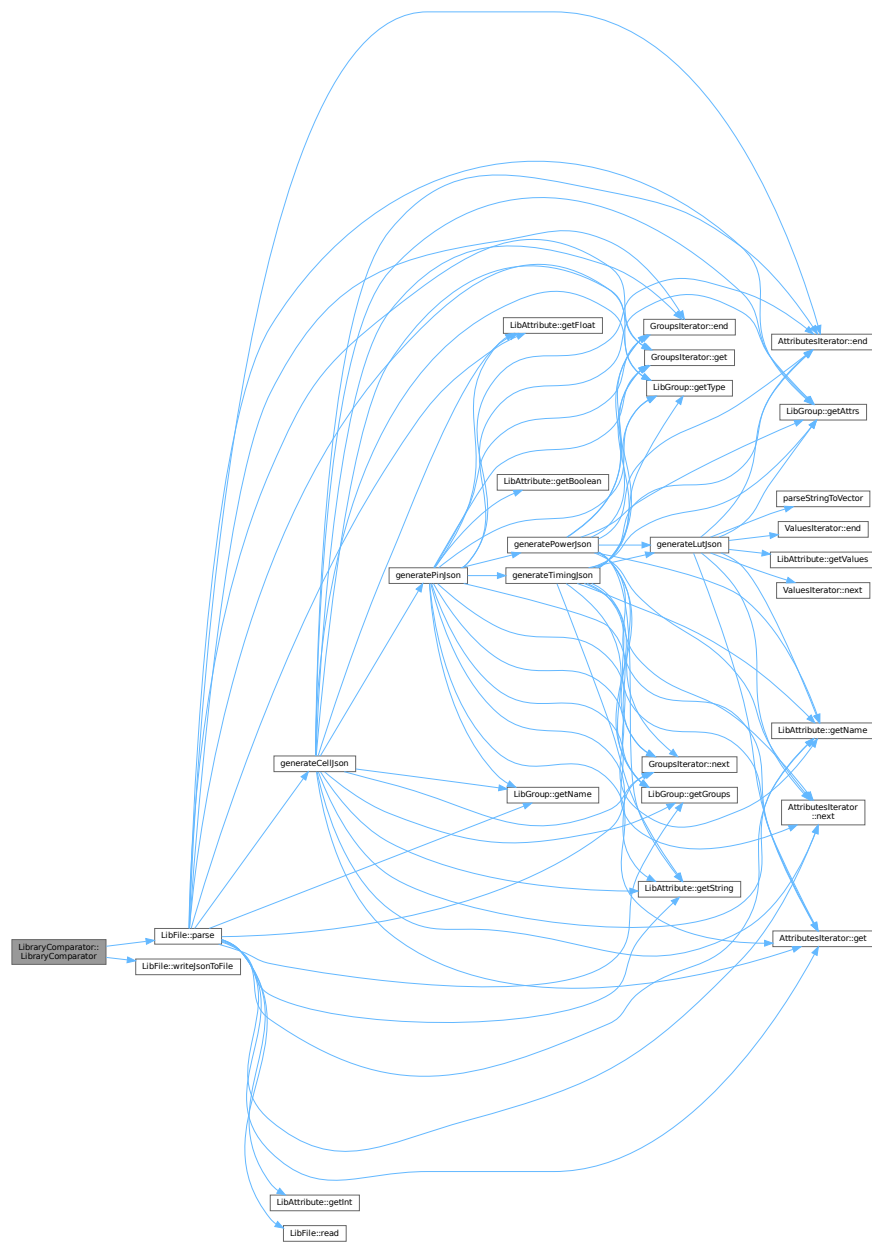
1. Looks for reference JSON file, parses library file if not found
2. Loads reference JSON data into `ref_json_` member
3. Looks for comparison JSON file, parses library file if not found
4. Loads comparison JSON data into `comp_json_` member
5. Stores file paths and initializes tolerance values for comparison

Parameters

<i>ref_libfile</i>	Reference library file object
<i>comp_libfile</i>	Comparison library file object
<i>reltol</i>	Relative tolerance for numerical comparisons (default defined elsewhere)
<i>abstol</i>	Absolute tolerance for numerical comparisons (default defined elsewhere)

Definition at line 31 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:



7.14.3 Member Function Documentation

7.14.3.1 compareCell()

```
void LibraryComparator::compareCell (
    const std::string & cell_name,
```

```
const json & ref_cell,
const json & comp_cell,
Table & table ) [private]
```

Compares the output pins of a cell between a reference JSON and a comparison JSON.

This function iterates through the output pins of the comparison cell and checks if each pin exists in the reference cell. If a pin is found in both cells, it calls the comparePin function to compare the details of the pin. If a pin is not found in the reference cell, a warning is logged. If the comparison cell does not contain any output pins, an informational message is logged.

Parameters

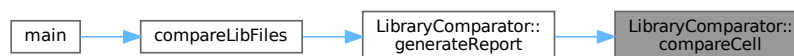
<i>cell_name</i>	The name of the cell being compared.
<i>ref_cell</i>	The reference JSON object containing the cell data.
<i>comp_cell</i>	The comparison JSON object containing the cell data.
<i>table</i>	The table object where the comparison results are stored.

Definition at line 308 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.14.3.2 compareLut()

```
void LibraryComparator::compareLut (
    const std::string & cell_name,
    const std::string & pin_name,
```

```

const std::string & timing_type,
const std::string & related_pin,
const std::string & arc_name,
const json & ref_lut,
const json & comp_lut,
Table & table ) [private]

```

Compares lookup tables (LUTs) between reference and comparison libraries for a specific timing arc.

This function compares the lookup table values between reference and comparison libraries for a given timing arc, checking the consistency of indices and values. It verifies that the indices match between libraries and then compares each value in the LUT against relative and absolute tolerance thresholds. Any discrepancies that exceed the tolerance limits are recorded in the provided table.

Parameters

<i>cell_name</i>	The name of the cell containing the LUT.
<i>pin_name</i>	The name of the pin associated with the LUT.
<i>timing_type</i>	The timing type of the arc (e.g., "rise_transition", "fall_transition").
<i>related_pin</i>	The related pin for the timing arc.
<i>arc_name</i>	The name of the specific timing arc being compared.
<i>ref_lut</i>	The lookup table from the reference library (in JSON format).
<i>comp_lut</i>	The lookup table from the comparison library (in JSON format).
<i>table</i>	Table object to store comparison results for any mismatches found.

Note

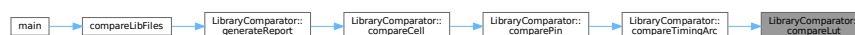
The function logs various information/warning/error messages:

- Logs index mismatches as warnings
- Logs value mismatches as debug messages
- Logs format errors in LUT structure as errors
- Records values exceeding tolerance in the provided table

The comparison uses both relative tolerance (`reltol_`) and absolute tolerance (`abstol_`) to determine if values differ significantly.

Definition at line 107 of file [LibraryComparator.cpp](#).

Here is the caller graph for this function:



7.14.3.3 comparePin()

```
void LibraryComparator::comparePin (
    const std::string & cell_name,
    const std::string & pin_name,
    const json & ref_pin,
    const json & comp_pin,
    Table & table ) [private]
```

Compares the timing arcs of a given pin between a reference JSON and a comparison JSON.

This function iterates through the timing arcs of the comparison pin and looks for matching timing types in the reference pin. If a matching timing type is found, it calls the compareTimingArc function to compare the timing arcs. If a timing type is not found in the reference JSON, a warning is logged.

Parameters

<i>cell_name</i>	The name of the cell containing the pin.
<i>pin_name</i>	The name of the pin to compare.
<i>ref_pin</i>	The reference JSON object containing the pin's data.
<i>comp_pin</i>	The comparison JSON object containing the pin's data.
<i>table</i>	A reference to a Table object where comparison results are stored.

Definition at line 269 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.14.3.4 compareTimingArc()

```
void LibraryComparator::compareTimingArc (
    const std::string & cell_name,
    const std::string & pin_name,
    const std::string & timing_type,
    const json & ref_timing_arc,
    const json & comp_timing_arc,
    Table & table ) [private]
```

Compares a timing arc between two JSON objects.

This function compares a specific timing arc (e.g., cell_rise, cell_fall, rise_transition, fall_transition) between a reference JSON object and a comparison JSON object. It extracts the related pin from the reference timing arc and iterates through a predefined list of arc names. If an arc name is found in the comparison timing arc, it checks if the same arc name exists in the reference timing arc. If both exist, it calls the compareLut function to compare the LUT data associated with the arc. If the arc name is found in the comparison JSON but not in the reference JSON, a warning message is logged.

Parameters

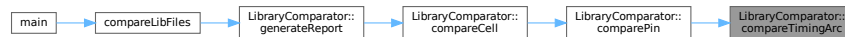
<i>cell_name</i>	The name of the cell.
<i>pin_name</i>	The name of the pin.
<i>timing_type</i>	The type of timing (e.g., "combinational", "sequential").
<i>ref_timing_arc</i>	The reference JSON object containing the timing arc information.
<i>comp_timing_arc</i>	The comparison JSON object containing the timing arc information.
<i>table</i>	The table to store comparison results.

Definition at line 229 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.14.3.5 generateReport()

```
void LibraryComparator::generateReport (
    const std::string & output_file )
```

Generates a comparison report between the reference and comparison libraries.

This function generates a detailed comparison report between the reference library and the comparison library. The report includes metadata such as the reference and comparison library paths, absolute and relative tolerances, and the application details. It also includes a legend explaining various symbols used in the report.

The function iterates through each cell in the comparison library, compares it with the corresponding cell in the reference library, and generates a table of differences. If there are any differences, it calculates summary statistics such as the average and maximum differences, and the number of outliers. The results are written to the specified output file in either Markdown or plain text format.

Parameters

<i>output_file</i>	The path to the output file where the report will be written.
--------------------	---

Definition at line 350 of file [LibraryComparator.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.14.4 Member Data Documentation

7.14.4.1 abstol_

```
double LibraryComparator::abstol_ [private]
```

Definition at line 23 of file [LibraryComparator.hpp](#).

7.14.4.2 comp_json_

```
json LibraryComparator::comp_json_ [private]
```

Definition at line 21 of file [LibraryComparator.hpp](#).

7.14.4.3 comp_lib_path_

```
std::filesystem::path LibraryComparator::comp_lib_path_
```

Definition at line 16 of file [LibraryComparator.hpp](#).

7.14.4.4 ref_json_

```
json LibraryComparator::ref_json_ [private]
```

Definition at line 20 of file [LibraryComparator.hpp](#).

7.14.4.5 ref_lib_path_

```
std::filesystem::path LibraryComparator::ref_lib_path_
```

Definition at line 15 of file [LibraryComparator.hpp](#).

7.14.4.6 reltol_

```
double LibraryComparator::reltol_ [private]
```

Definition at line 22 of file [LibraryComparator.hpp](#).

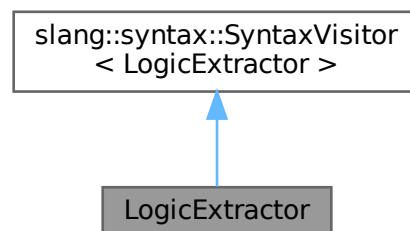
The documentation for this class was generated from the following files:

- include/[LibraryComparator.hpp](#)
- src/[LibraryComparator.cpp](#)

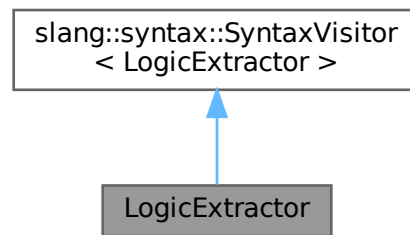
7.15 LogicExtractor Class Reference

```
#include <LogicExtractor.hpp>
```

Inheritance diagram for LogicExtractor:



Collaboration diagram for LogicExtractor:



Public Member Functions

- `LogicExtractor` (const std::string &targetCell)
- void `handle` (const slang::syntax::ModuleDeclarationSyntax &module)
- void `handle` (const slang::syntax::PortDeclarationSyntax &portDecl)
- void `handle` (const slang::syntax::NonAnsiPortListSyntax &portList)
- void `handle` (const slang::syntax::NetDeclarationSyntax &netDecl)
- void `handle` (const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst)
- const std::unordered_map< std::string, `GateInfo` > & `getExtractedGates` () const
- const std::unordered_set< std::string > & `getPrimaryInputs` () const
- const std::unordered_set< std::string > & `getPrimaryOutputs` () const
- const std::unordered_set< std::string > & `getInternalWires` () const
- std::map< std::string, std::string > `getLogicExpressions` ()

Private Member Functions

- std::string `deriveLogicRecursive` (const std::string &signalName)
- std::string `formatExpression` (const `GateInfo` &gateInfo, const std::vector< std::string > &inputExprs)

Private Attributes

- const std::string & `targetCell_`
- bool `inTargetModule_`
- bool `parsingComplete_`
- std::unordered_set< std::string > `primaryInputs_`
- std::unordered_set< std::string > `primaryOutputs_`
- std::unordered_set< std::string > `internalWires_`
- std::unordered_map< std::string, std::string > `portDirections_`
- std::unordered_map< std::string, `GateInfo` > `gateOutputDrivers_`
- std::unordered_map< std::string, std::string > `logicCache_`

7.15.1 Detailed Description

Definition at line 30 of file [LogicExtractor.hpp](#).

7.15.2 Constructor & Destructor Documentation

7.15.2.1 LogicExtractor()

```
LogicExtractor::LogicExtractor (
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 33 of file [LogicExtractor.hpp](#).

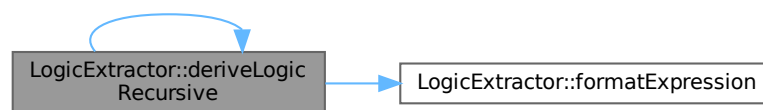
7.15.3 Member Function Documentation

7.15.3.1 deriveLogicRecursive()

```
std::string LogicExtractor::deriveLogicRecursive (
    const std::string & signalName ) [private]
```

Definition at line 380 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.15.3.2 formatExpression()

```
std::string LogicExtractor::formatExpression (
    const GateInfo & gateInfo,
    const std::vector< std::string > & inputExprs ) [private]
```

Definition at line 437 of file [LogicExtractor.cpp](#).

Here is the caller graph for this function:



7.15.3.3 getExtractedGates()

```
const std::unordered_map< std::string, GateInfo > & LogicExtractor::getExtractedGates ( ) const [inline]
```

Definition at line 50 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:

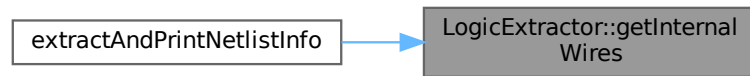


7.15.3.4 getInternalWires()

```
const std::unordered_set< std::string > & LogicExtractor::getInternalWires ( ) const [inline]
```

Definition at line 55 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:



7.15.3.5 getLogicExpressions()

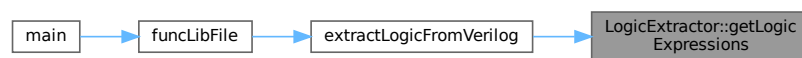
```
std::map< std::string, std::string > LogicExtractor::getLogicExpressions ( )
```

Definition at line 350 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:

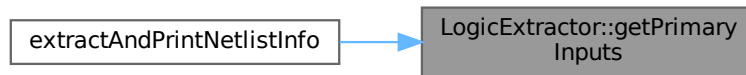


7.15.3.6 getPrimaryInputs()

```
const std::unordered_set< std::string > & LogicExtractor::getPrimaryInputs ( ) const [inline]
```

Definition at line 53 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:

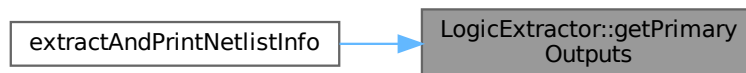


7.15.3.7 getPrimaryOutputs()

```
const std::unordered_set< std::string > & LogicExtractor::getPrimaryOutputs ( ) const [inline]
```

Definition at line 54 of file [LogicExtractor.hpp](#).

Here is the caller graph for this function:

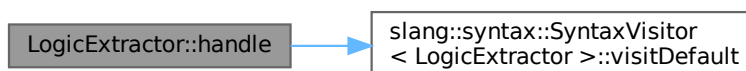


7.15.3.8 handle() [1/5]

```
void LogicExtractor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 5 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



7.15.3.9 handle() [2/5]

```
void LogicExtractor::handle (
    const slang::syntax::NetDeclarationSyntax & netDecl )
```

Definition at line 201 of file [LogicExtractor.cpp](#).

7.15.3.10 handle() [3/5]

```
void LogicExtractor::handle (
    const slang::syntax::NonAnsiPortListSyntax & portList )
```

Definition at line 112 of file [LogicExtractor.cpp](#).

7.15.3.11 handle() [4/5]

```
void LogicExtractor::handle (
    const slang::syntax::PortDeclarationSyntax & portDecl )
```

Definition at line 49 of file [LogicExtractor.cpp](#).

7.15.3.12 handle() [5/5]

```
void LogicExtractor::handle (
    const slang::syntax::PrimitiveInstantiationSyntax & primitiveInst )
```

Definition at line 221 of file [LogicExtractor.cpp](#).

7.15.4 Member Data Documentation

7.15.4.1 gateOutputDrivers_

```
std::unordered_map<std::string, GateInfo> LogicExtractor::gateOutputDrivers_ [private]
```

Definition at line 76 of file [LogicExtractor.hpp](#).

7.15.4.2 inTargetModule_

```
bool LogicExtractor::inTargetModule_ [private]
```

Definition at line 63 of file [LogicExtractor.hpp](#).

7.15.4.3 internalWires_

```
std::unordered_set<std::string> LogicExtractor::internalWires_ [private]
```

Definition at line 69 of file [LogicExtractor.hpp](#).

7.15.4.4 logicCache_

```
std::unordered_map<std::string, std::string> LogicExtractor::logicCache_ [private]
```

Definition at line 79 of file [LogicExtractor.hpp](#).

7.15.4.5 parsingComplete_

```
bool LogicExtractor::parsingComplete_ [private]
```

Definition at line 64 of file [LogicExtractor.hpp](#).

7.15.4.6 portDirections_

```
std::unordered_map<std::string, std::string> LogicExtractor::portDirections_ [private]
```

Definition at line 73 of file [LogicExtractor.hpp](#).

7.15.4.7 primaryInputs_

```
std::unordered_set<std::string> LogicExtractor::primaryInputs_ [private]
```

Definition at line 67 of file [LogicExtractor.hpp](#).

7.15.4.8 primaryOutputs_

```
std::unordered_set<std::string> LogicExtractor::primaryOutputs_ [private]
```

Definition at line 68 of file [LogicExtractor.hpp](#).

7.15.4.9 targetCell_

```
const std::string& LogicExtractor::targetCell_ [private]
```

Definition at line 62 of file [LogicExtractor.hpp](#).

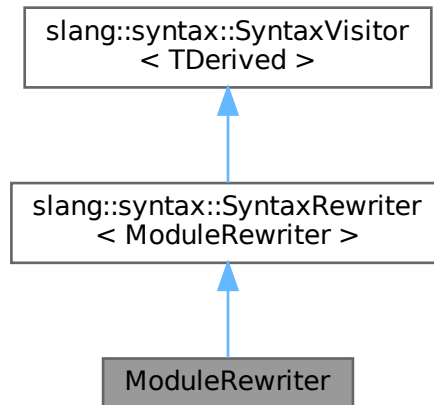
The documentation for this class was generated from the following files:

- [include/LogicExtractor.hpp](#)
- [src/LogicExtractor.cpp](#)

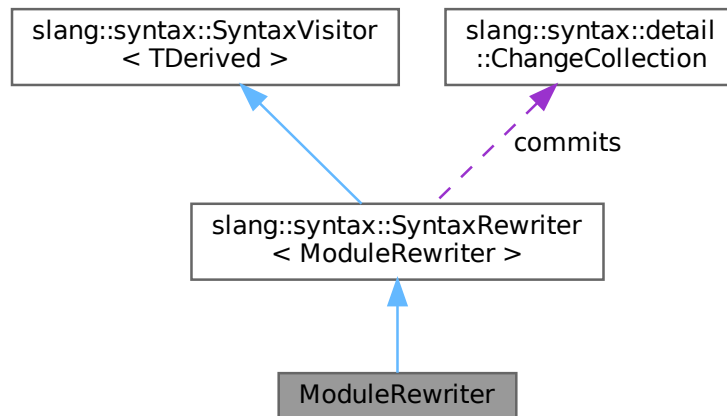
7.16 ModuleRewriter Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for ModuleRewriter:



Collaboration diagram for ModuleRewriter:



Public Member Functions

- [ModuleRewriter](#) (const std::vector< std::string > &inputPins, const std::vector< std::string > &outputPins, const std::pair< std::string, std::string > &supercell_entry, int instance_count, std::shared_ptr< spdlog::logger > logger)

- void [handle](#) (const [slang::syntax::SyntaxNode](#) &node)
Processes a syntax node in the module's AST.
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)

Public Attributes

- std::shared_ptr< spdlog::logger > [logger_](#)

Private Attributes

- const std::vector< std::string > & [inputPins_](#)
- const std::vector< std::string > & [outputPins_](#)
- const std::string [cellName_](#)
- const std::string [moduleName_](#)
- std::map< std::string, std::string > [portInfoMap_](#)
- int [depth_](#)
- int [instance_count_](#)

Additional Inherited Members

7.16.1 Detailed Description

Definition at line 65 of file [verilog_utils.hpp](#).

7.16.2 Constructor & Destructor Documentation

7.16.2.1 ModuleRewriter()

```
ModuleRewriter::ModuleRewriter (
    const std::vector< std::string > & inputPins,
    const std::vector< std::string > & outputPins,
    const std::pair< std::string, std::string > & supercell_entry,
    int instance_count,
    std::shared_ptr< spdlog::logger > logger ) [inline], [explicit]
```

Definition at line 67 of file [verilog_utils.hpp](#).

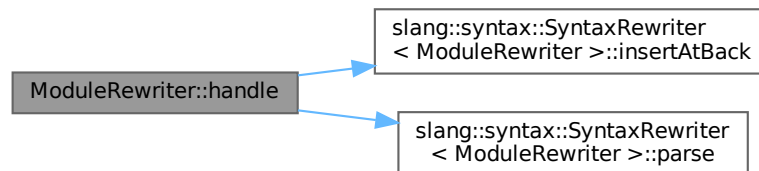
7.16.3 Member Function Documentation

7.16.3.1 `handle()` [1/2]

```
void ModuleRewriter::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 406 of file [verilog_utils.cpp](#).

Here is the call graph for this function:



7.16.3.2 `handle()` [2/2]

```
void ModuleRewriter::handle (
    const slang::syntax::SyntaxNode & node )
```

Processes a syntax node in the module's AST.

This method is called for each syntax node during the traversal of the AST. It logs debug information about the current node and continues processing its child nodes by recursively calling the appropriate visit methods.

Parameters

<i>node</i>	The syntax node to process
-------------	----------------------------

Definition at line 396 of file [verilog_utils.cpp](#).

Here is the call graph for this function:



7.16.4 Member Data Documentation

7.16.4.1 cellName_

```
const std::string ModuleRewriter::cellName_ [private]
```

Definition at line 81 of file [verilog_utils.hpp](#).

7.16.4.2 depth_

```
int ModuleRewriter::depth_ [private]
```

Definition at line 84 of file [verilog_utils.hpp](#).

7.16.4.3 inputPins_

```
const std::vector<std::string>& ModuleRewriter::inputPins_ [private]
```

Definition at line 79 of file [verilog_utils.hpp](#).

7.16.4.4 instance_count_

```
int ModuleRewriter::instance_count_ [private]
```

Definition at line 85 of file [verilog_utils.hpp](#).

7.16.4.5 logger_

```
std::shared_ptr<spdlog::logger> ModuleRewriter::logger_
```

Definition at line 74 of file [verilog_utils.hpp](#).

7.16.4.6 moduleName_

```
const std::string ModuleRewriter::moduleName_ [private]
```

Definition at line 82 of file [verilog_utils.hpp](#).

7.16.4.7 outputPins_

```
const std::vector<std::string>& ModuleRewriter::outputPins_ [private]
```

Definition at line 80 of file [verilog_utils.hpp](#).

7.16.4.8 portInfoMap_

```
std::map<std::string, std::string> ModuleRewriter::portInfoMap_ [private]
```

Definition at line 83 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

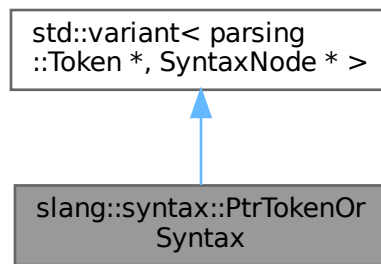
- include/[verilog_utils.hpp](#)
- src/[verilog_utils.cpp](#)

7.17 slang::syntax::PtrTokenOrSyntax Struct Reference

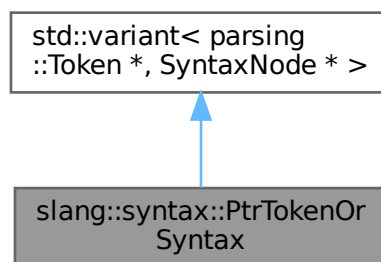
A token pointer or a syntax node.

```
#include <SyntaxNode.h>
```

Inheritance diagram for slang::syntax::PtrTokenOrSyntax:



Collaboration diagram for slang::syntax::PtrTokenOrSyntax:



Public Types

- using [Base](#) = std::variant< parsing::Token *, [SyntaxNode](#) * >

Public Member Functions

- [PtrTokenOrSyntax](#) (parsing::Token **token*)
- [PtrTokenOrSyntax](#) (SyntaxNode **node*)
- [PtrTokenOrSyntax](#) (nullptr_t)
- bool [isToken](#) () const
- bool [isNode](#) () const
- parsing::Token * [token](#) () const
- SyntaxNode * [node](#) () const
- SourceRange [range](#) () const

7.17.1 Detailed Description

A token pointer or a syntax node.

Definition at line 24 of file [SyntaxNode.h](#).

7.17.2 Member Typedef Documentation

7.17.2.1 Base

```
using slang::syntax::PtrTokenOrSyntax::Base = std::variant<parsing::Token*, SyntaxNode*>
```

Definition at line 25 of file [SyntaxNode.h](#).

7.17.3 Constructor & Destructor Documentation

7.17.3.1 PtrTokenOrSyntax() [1/3]

```
slang::syntax::PtrTokenOrSyntax::PtrTokenOrSyntax (
    parsing::Token * token ) [inline]
```

Definition at line 26 of file [SyntaxNode.h](#).

7.17.3.2 PtrTokenOrSyntax() [2/3]

```
slang::syntax::PtrTokenOrSyntax::PtrTokenOrSyntax (  
    SyntaxNode * node ) [inline]
```

Definition at line 27 of file [SyntaxNode.h](#).

7.17.3.3 PtrTokenOrSyntax() [3/3]

```
slang::syntax::PtrTokenOrSyntax::PtrTokenOrSyntax (  
    nullptr_t ) [inline]
```

Definition at line 28 of file [SyntaxNode.h](#).

7.17.4 Member Function Documentation

7.17.4.1 isNode()

```
bool slang::syntax::PtrTokenOrSyntax::isNode ( ) const [inline]
```

Returns

true if the object is a syntax node.

Definition at line 34 of file [SyntaxNode.h](#).

7.17.4.2 isToken()

```
bool slang::syntax::PtrTokenOrSyntax::isToken ( ) const [inline]
```

Returns

true if the object is a token.

Definition at line 31 of file [SyntaxNode.h](#).

7.17.4.3 node()

```
SyntaxNode * slang::syntax::PtrTokenOrSyntax::node ( ) const [inline]
```

Gets access to the object as a syntax node (throwing an exception if it's not actually a syntax node).

Definition at line 42 of file [SyntaxNode.h](#).

Here is the caller graph for this function:



7.17.4.4 range()

```
SourceRange slang::syntax::PtrTokenOrSyntax::range ( ) const
```

Gets the source range for the token or syntax node or NoLocation if it points to nullptr.

7.17.4.5 token()

```
parsing::Token * slang::syntax::PtrTokenOrSyntax::token ( ) const [inline]
```

Gets access to the object as a token (throwing an exception if it's not actually a token).

Definition at line 38 of file [SyntaxNode.h](#).

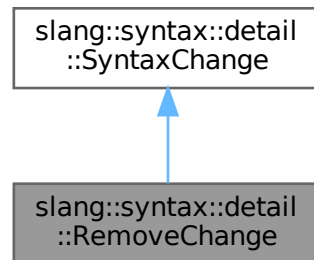
The documentation for this struct was generated from the following file:

- `include_3rd_party/SyntaxNode.h`

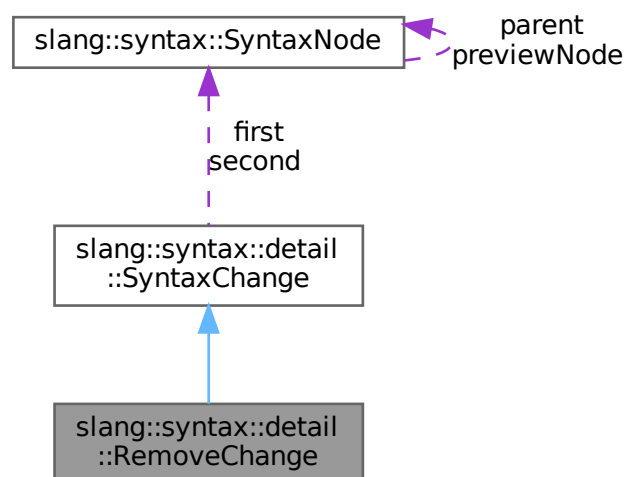
7.18 slang::syntax::detail::RemoveChange Struct Reference

```
#include <SyntaxVisitor.h>
```

Inheritance diagram for slang::syntax::detail::RemoveChange:



Collaboration diagram for slang::syntax::detail::RemoveChange:



Additional Inherited Members

7.18.1 Detailed Description

Definition at line 65 of file [SyntaxVisitor.h](#).

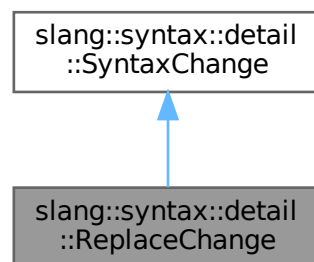
The documentation for this struct was generated from the following file:

- `include_3rd_party/SyntaxVisitor.h`

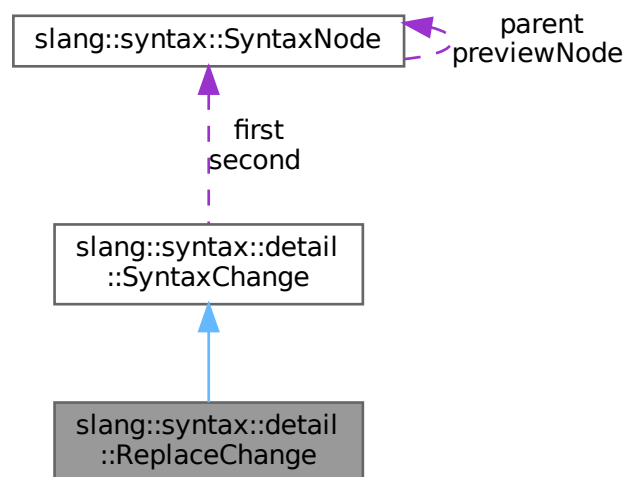
7.19 slang::syntax::detail::ReplaceChange Struct Reference

```
#include <SyntaxVisitor.h>
```

Inheritance diagram for slang::syntax::detail::ReplaceChange:



Collaboration diagram for slang::syntax::detail::ReplaceChange:



Additional Inherited Members

7.19.1 Detailed Description

Definition at line 66 of file [SyntaxVisitor.h](#).

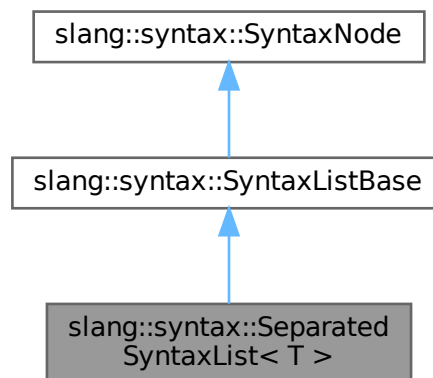
The documentation for this struct was generated from the following file:

- include_3rd_party/[SyntaxVisitor.h](#)

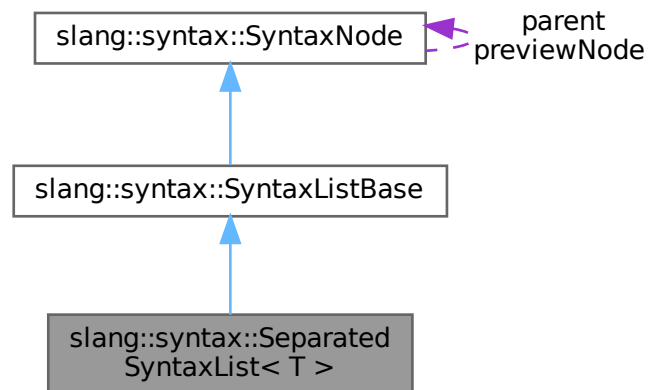
7.20 slang::syntax::SeparatedSyntaxList< T > Class Template Reference

```
#include <SyntaxNode.h>
```

Inheritance diagram for slang::syntax::SeparatedSyntaxList< T >:



Collaboration diagram for slang::syntax::SeparatedSyntaxList< T >:



Classes

- class [iterator_base](#)

Public Types

- using [iterator](#) = [iterator_base](#)< T * >
- using [const_iterator](#) = [iterator_base](#)< const T * >

Public Member Functions

- [SeparatedSyntaxList](#) (nullptr_t)
- [SeparatedSyntaxList](#) (std::span< [TokenOrSyntax](#) > [elements](#))
- std::span< const [ConstTokenOrSyntax](#) > [elems](#) () const
- std::span< [TokenOrSyntax](#) > [elems](#) ()
- bool [empty](#) () const
- size_t [size](#) () const noexcept
- const T * [operator\[\]](#) (size_t index) const
- T * [operator\[\]](#) (size_t index)
- [iterator](#) [begin](#) ()
- [iterator](#) [end](#) ()
- [const_iterator](#) [begin](#) () const
- [const_iterator](#) [end](#) () const
- [const_iterator](#) [cbegin](#) () const
- [const_iterator](#) [cend](#) () const

Private Member Functions

- [TokenOrSyntax getChild](#) (size_t index) final
Gets the child (token or node) at the given index.
- [ConstTokenOrSyntax getChild](#) (size_t index) const final
Gets the child (token or node) at the given index.
- [PtrTokenOrSyntax getChildPtr](#) (size_t index) final
- void [setChild](#) (size_t index, [TokenOrSyntax](#) child) final
Sets the child (token or node) at the given index.
- [SyntaxListBase](#) * [clone](#) (BumpAllocator &alloc) const final
Clones the list into a new node using the provided allocator.
- void [resetAll](#) (BumpAllocator &alloc, std::span< const [TokenOrSyntax](#) > children) final

Private Attributes

- std::span< [TokenOrSyntax](#) > [elements](#)

Additional Inherited Members

7.20.1 Detailed Description

```
template<typename T>
class slang::syntax::SeparatedSyntaxList< T >
```

A syntax node that represents a token-separated list of child syntax nodes. The stored children are assumed to alternate between delimiters (such as a comma) and nodes.

Definition at line 381 of file [SyntaxNode.h](#).

7.20.2 Member Typedef Documentation

7.20.2.1 const_iterator

```
template<typename T >
using slang::syntax::SeparatedSyntaxList< T >::const_iterator = iterator\_base<const T*>
```

Definition at line 412 of file [SyntaxNode.h](#).

7.20.2.2 iterator

```
template<typename T >
using slang::syntax::SeparatedSyntaxList< T >::iterator = iterator_base<T*>
```

Definition at line 411 of file [SyntaxNode.h](#).

7.20.3 Constructor & Destructor Documentation

7.20.3.1 SeparatedSyntaxList() [1/2]

```
template<typename T >
slang::syntax::SeparatedSyntaxList< T >::SeparatedSyntaxList (
    nullptr_t ) [inline]
```

Definition at line 414 of file [SyntaxNode.h](#).

7.20.3.2 SeparatedSyntaxList() [2/2]

```
template<typename T >
slang::syntax::SeparatedSyntaxList< T >::SeparatedSyntaxList (
    std::span< TokenOrSyntax > elements ) [inline]
```

Definition at line 415 of file [SyntaxNode.h](#).

7.20.4 Member Function Documentation

7.20.4.1 begin() [1/2]

```
template<typename T >
iterator slang::syntax::SeparatedSyntaxList< T >::begin ( ) [inline]
```

Definition at line 443 of file [SyntaxNode.h](#).

7.20.4.2 begin() [2/2]

```
template<typename T >
const_iterator slang::syntax::SeparatedSyntaxList< T >::begin ( ) const [inline]
```

Definition at line 445 of file [SyntaxNode.h](#).

7.20.4.3 cbegin()

```
template<typename T >
const_iterator slang::syntax::SeparatedSyntaxList< T >::cbegin ( ) const [inline]
```

Definition at line 447 of file [SyntaxNode.h](#).

7.20.4.4 cend()

```
template<typename T >
const_iterator slang::syntax::SeparatedSyntaxList< T >::cend ( ) const [inline]
```

Definition at line 448 of file [SyntaxNode.h](#).

7.20.4.5 clone()

```
template<typename T >
SyntaxListBase * slang::syntax::SeparatedSyntaxList< T >::clone (
    BumpAllocator & alloc ) const [inline], [final], [private], [virtual]
```

Clones the list into a new node using the provided allocator.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 461 of file [SyntaxNode.h](#).

7.20.4.6 elems() [1/2]

```
template<typename T >
std::span< TokenOrSyntax > slang::syntax::SeparatedSyntaxList< T >::elems ( ) [inline]
```

Returns

the elements of nodes in the list

Definition at line 425 of file [SyntaxNode.h](#).

7.20.4.7 elems() [2/2]

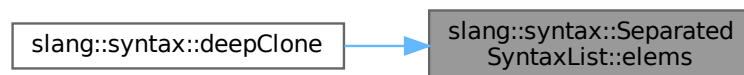
```
template<typename T >
std::span< const ConstTokenOrSyntax > slang::syntax::SeparatedSyntaxList< T >::elems ( ) const [inline]
```

Returns

the elements of nodes in the list

Definition at line 419 of file [SyntaxNode.h](#).

Here is the caller graph for this function:

**7.20.4.8 empty()**

```
template<typename T >
bool slang::syntax::SeparatedSyntaxList< T >::empty ( ) const [inline]
```

Returns

true if the list is empty, and false if it has elements.

Definition at line 428 of file [SyntaxNode.h](#).

7.20.4.9 end() [1/2]

```
template<typename T >
iterator slang::syntax::SeparatedSyntaxList< T >::end ( ) [inline]
```

Definition at line 444 of file [SyntaxNode.h](#).

7.20.4.10 end() [2/2]

```
template<typename T >
const_iterator slang::syntax::SeparatedSyntaxList< T >::end ( ) const [inline]
```

Definition at line 446 of file [SyntaxNode.h](#).

7.20.4.11 getChild() [1/2]

```
template<typename T >
ConstTokenOrSyntax slang::syntax::SeparatedSyntaxList< T >::getChild (
    size_t index ) const [inline], [final], [private], [virtual]
```

Gets the child (token or node) at the given index.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 452 of file [SyntaxNode.h](#).

7.20.4.12 getChild() [2/2]

```
template<typename T >
TokenOrSyntax slang::syntax::SeparatedSyntaxList< T >::getChild (
    size_t index ) [inline], [final], [private], [virtual]
```

Gets the child (token or node) at the given index.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 451 of file [SyntaxNode.h](#).

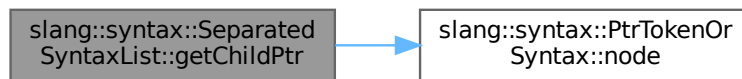
7.20.4.13 getChildPtr()

```
template<typename T >
PtrTokenOrSyntax slang::syntax::SeparatedSyntaxList< T >::getChildPtr (
    size_t index ) [inline], [final], [private], [virtual]
```

Implements [slang::syntax::SyntaxListBase](#).

Definition at line [453](#) of file [SyntaxNode.h](#).

Here is the call graph for this function:



7.20.4.14 operator[]() [1/2]

```
template<typename T >
T * slang::syntax::SeparatedSyntaxList< T >::operator[] (
    size_t index ) [inline]
```

Definition at line [438](#) of file [SyntaxNode.h](#).

7.20.4.15 operator[]() [2/2]

```
template<typename T >
const T * slang::syntax::SeparatedSyntaxList< T >::operator[] (
    size_t index ) const [inline]
```

Definition at line [433](#) of file [SyntaxNode.h](#).

7.20.4.16 resetAll()

```
template<typename T >
void slang::syntax::SeparatedSyntaxList< T >::resetAll (
    BumpAllocator & alloc,
    std::span< const TokenOrSyntax > children ) [inline], [final], [private], [virtual]
```

Overwrites all children with the new set of provided *children* (and making a copy with the provided allocator).

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 465 of file [SyntaxNode.h](#).

7.20.4.17 setChild()

```
template<typename T >
void slang::syntax::SeparatedSyntaxList< T >::setChild (
    size_t index,
    TokenOrSyntax child ) [inline], [final], [private], [virtual]
```

Sets the child (token or node) at the given index.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 459 of file [SyntaxNode.h](#).

7.20.4.18 size()

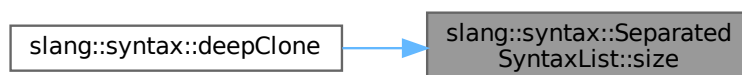
```
template<typename T >
size_t slang::syntax::SeparatedSyntaxList< T >::size ( ) const [inline], [noexcept]
```

Returns

the number of nodes in the list (doesn't include delimiter tokens in the count).

Definition at line 431 of file [SyntaxNode.h](#).

Here is the caller graph for this function:



7.20.5 Member Data Documentation

7.20.5.1 elements

```
template<typename T >
std::span<TokenOrSyntax> slang::syntax::SeparatedSyntaxList< T >::elements [private]
```

Definition at line 473 of file [SyntaxNode.h](#).

The documentation for this class was generated from the following file:

- include_3rd_party/[SyntaxNode.h](#)

7.21 si2drExprT Struct Reference

```
#include <si2dr_liberty.h>
```

Collaboration diagram for si2drExprT:



Public Attributes

- [si2drExprTypeT](#) type
- union {
 - [si2drInt32T](#) i
 - [si2drFloat64T](#) d
 - [si2drStringT](#) s
 - [si2drBooleanT](#) b
 } u
- [si2drValueTypeT](#) valuetype
- struct [si2drExprT](#) * left
- struct [si2drExprT](#) * right

7.21.1 Detailed Description

Definition at line 172 of file [si2dr_liberty.h](#).

7.21.2 Member Data Documentation

7.21.2.1 b

[si2drBooleanT](#) si2drExprT::b

Definition at line 180 of file [si2dr_liberty.h](#).

7.21.2.2 d

[si2drFloat64T](#) si2drExprT::d

Definition at line 178 of file [si2dr_liberty.h](#).

7.21.2.3 i

[si2drInt32T](#) si2drExprT::i

Definition at line 177 of file [si2dr_liberty.h](#).

7.21.2.4 left

struct [si2drExprT](#)* si2drExprT::left

Definition at line 185 of file [si2dr_liberty.h](#).

7.21.2.5 right

```
struct si2drExprT* si2drExprT::right
```

Definition at line [186](#) of file [si2dr_liberty.h](#).

7.21.2.6 s

```
si2drStringT si2drExprT::s
```

Definition at line [179](#) of file [si2dr_liberty.h](#).

7.21.2.7 type

```
si2drExprTypeT si2drExprT::type
```

Definition at line [174](#) of file [si2dr_liberty.h](#).

7.21.2.8

```
union { ... } si2drExprT::u
```

7.21.2.9 valuetype

```
si2drValueTypeT si2drExprT::valuetype
```

Definition at line [183](#) of file [si2dr_liberty.h](#).

The documentation for this struct was generated from the following file:

- `include_3rd_party/si2dr_liberty.h`

7.22 si2OID Struct Reference

```
#include <si2dr_liberty.h>
```

Public Attributes

- void * [v1](#)
- void * [v2](#)

7.22.1 Detailed Description

Definition at line [79](#) of file [si2dr_liberty.h](#).

7.22.2 Member Data Documentation

7.22.2.1 v1

```
void* si2OID::v1
```

Definition at line [79](#) of file [si2dr_liberty.h](#).

7.22.2.2 v2

```
void * si2OID::v2
```

Definition at line [79](#) of file [si2dr_liberty.h](#).

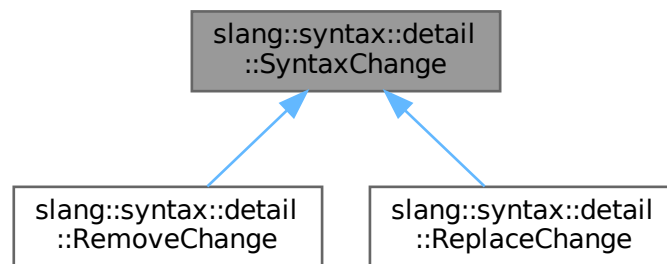
The documentation for this struct was generated from the following file:

- [include_3rd_party/si2dr_liberty.h](#)

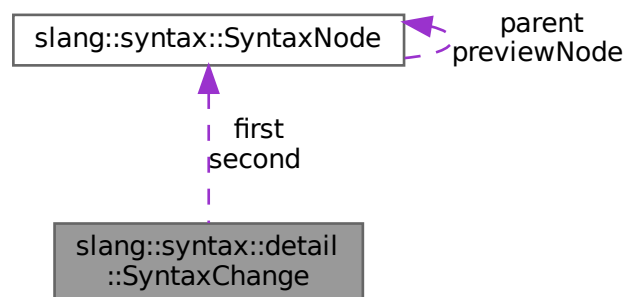
7.23 slang::syntax::detail::SyntaxChange Struct Reference

```
#include <SyntaxVisitor.h>
```

Inheritance diagram for slang::syntax::detail::SyntaxChange:



Collaboration diagram for slang::syntax::detail::SyntaxChange:



Public Attributes

- `const SyntaxNode * first = nullptr`
- `SyntaxNode * second = nullptr`
- `parsing::Token separator = {}`

7.23.1 Detailed Description

Definition at line 59 of file [SyntaxVisitor.h](#).

7.23.2 Member Data Documentation

7.23.2.1 first

```
const SyntaxNode* slang::syntax::detail::SyntaxChange::first = nullptr
```

Definition at line 60 of file [SyntaxVisitor.h](#).

7.23.2.2 second

```
SyntaxNode* slang::syntax::detail::SyntaxChange::second = nullptr
```

Definition at line 61 of file [SyntaxVisitor.h](#).

7.23.2.3 separator

```
parsing::Token slang::syntax::detail::SyntaxChange::separator = {}
```

Definition at line 62 of file [SyntaxVisitor.h](#).

The documentation for this struct was generated from the following file:

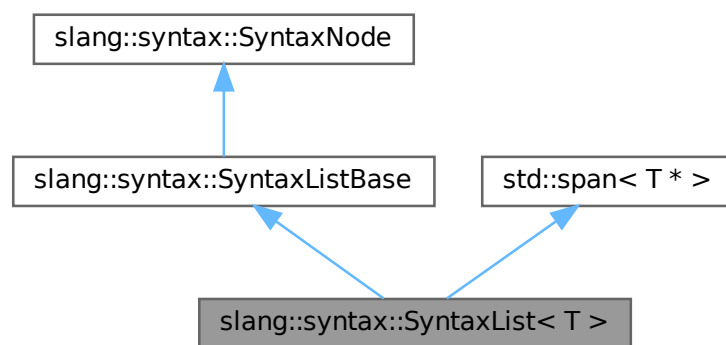
- include_3rd_party/[SyntaxVisitor.h](#)

7.24 slang::syntax::SyntaxList< T > Class Template Reference

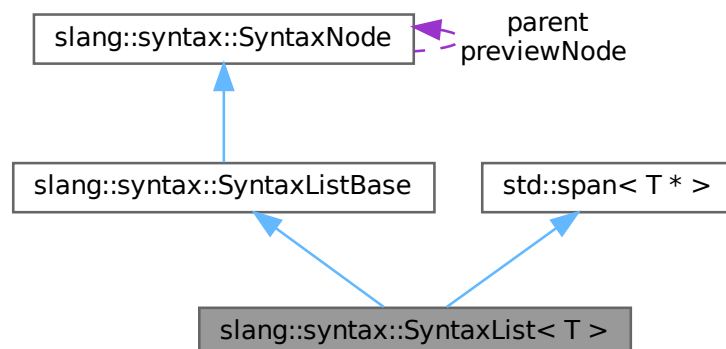
A syntax node that represents a list of child syntax nodes.

```
#include <SyntaxNode.h>
```

Inheritance diagram for slang::syntax::SyntaxList< T >:



Collaboration diagram for slang::syntax::SyntaxList< T >:



Public Member Functions

- [SyntaxList](#) (nullptr_t)
- [SyntaxList](#) (std::span< T * > elements)

Private Member Functions

- [TokenOrSyntax getChild](#) (size_t index) final
Gets the child (token or node) at the given index.
- [ConstTokenOrSyntax getChild](#) (size_t index) const final
Gets the child (token or node) at the given index.
- [PtrTokenOrSyntax getChildPtr](#) (size_t index) final
- void [setChild](#) (size_t index, [TokenOrSyntax](#) child) final
Sets the child (token or node) at the given index.
- [SyntaxListBase * clone](#) (BumpAllocator &alloc) const final
Clones the list into a new node using the provided allocator.
- void [resetAll](#) (BumpAllocator &alloc, std::span< const [TokenOrSyntax](#) > children) final

Additional Inherited Members

7.24.1 Detailed Description

```
template<typename T>
class slang::syntax::SyntaxList< T >
```

A syntax node that represents a list of child syntax nodes.

Definition at line 313 of file [SyntaxNode.h](#).

7.24.2 Constructor & Destructor Documentation

7.24.2.1 SyntaxList() ^[1/2]

```
template<typename T >
slang::syntax::SyntaxList< T >::SyntaxList (
    nullptr_t ) [inline]
```

Definition at line 315 of file [SyntaxNode.h](#).

7.24.2.2 SyntaxList() [2/2]

```
template<typename T >
slang::syntax::SyntaxList< T >::SyntaxList (
    std::span< T * > elements ) [inline]
```

Definition at line 316 of file [SyntaxNode.h](#).

7.24.3 Member Function Documentation

7.24.3.1 clone()

```
template<typename T >
SyntaxListBase * slang::syntax::SyntaxList< T >::clone (
    BumpAllocator & alloc ) const [inline], [final], [private], [virtual]
```

Clones the list into a new node using the provided allocator.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 328 of file [SyntaxNode.h](#).

7.24.3.2 getChild() [1/2]

```
template<typename T >
ConstTokenOrSyntax slang::syntax::SyntaxList< T >::getChild (
    size_t index ) const [inline], [final], [private], [virtual]
```

Gets the child (token or node) at the given index.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 321 of file [SyntaxNode.h](#).

7.24.3.3 getChild() [2/2]

```
template<typename T >
TokenOrSyntax slang::syntax::SyntaxList< T >::getChild (
    size_t index ) [inline], [final], [private], [virtual]
```

Gets the child (token or node) at the given index.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 320 of file [SyntaxNode.h](#).

7.24.3.4 getChildPtr()

```
template<typename T >
PtrTokenOrSyntax slang::syntax::SyntaxList< T >::getChildPtr (
    size_t index ) [inline], [final], [private], [virtual]
```

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 322 of file [SyntaxNode.h](#).

7.24.3.5 resetAll()

```
template<typename T >
void slang::syntax::SyntaxList< T >::resetAll (
    BumpAllocator & alloc,
    std::span< const TokenOrSyntax > children ) [inline], [final], [private], [virtual]
```

Overwrites all children with the new set of provided *children* (and making a copy with the provided allocator).

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 332 of file [SyntaxNode.h](#).

7.24.3.6 setChild()

```
template<typename T >
void slang::syntax::SyntaxList< T >::setChild (
    size_t index,
    TokenOrSyntax child ) [inline], [final], [private], [virtual]
```

Sets the child (token or node) at the given index.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 324 of file [SyntaxNode.h](#).

The documentation for this class was generated from the following file:

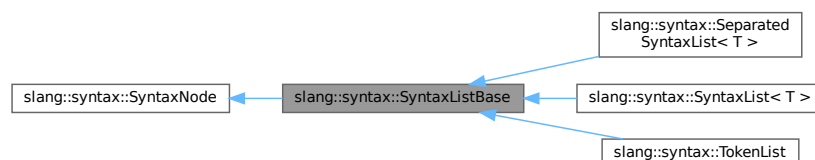
- [include_3rd_party/SyntaxNode.h](#)

7.25 slang::syntax::SyntaxListBase Class Reference

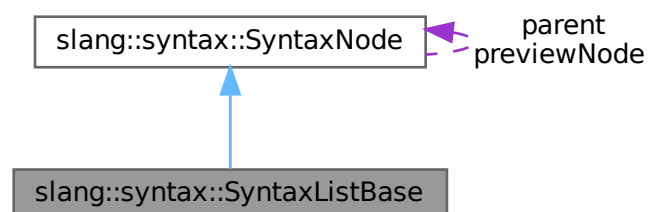
A base class for syntax nodes that represent a list of items.

```
#include <SyntaxNode.h>
```

Inheritance diagram for slang::syntax::SyntaxListBase:



Collaboration diagram for slang::syntax::SyntaxListBase:



Public Member Functions

- `size_t getChildCount () const`
Gets the number of child items in the node.
- `virtual TokenOrSyntax getChild (size_t index)=0`
Gets the child (token or node) at the given index.
- `virtual ConstTokenOrSyntax getChild (size_t index) const =0`
Gets the child (token or node) at the given index.
- `virtual PtrTokenOrSyntax getChildPtr (size_t index)=0`
- `virtual void setChild (size_t index, TokenOrSyntax child)=0`
Sets the child (token or node) at the given index.
- `virtual SyntaxListBase * clone (BumpAllocator &alloc) const =0`
Clones the list into a new node using the provided allocator.
- `virtual void resetAll (BumpAllocator &alloc, std::span< const TokenOrSyntax > children)=0`

Static Public Member Functions

- `static bool isKind (SyntaxKind kind)`
- `static bool isChildOptional (size_t index)`

Protected Member Functions

- `SyntaxListBase (SyntaxKind kind, size_t childCount)`

Protected Attributes

- `size_t childCount`

Additional Inherited Members

7.25.1 Detailed Description

A base class for syntax nodes that represent a list of items.

Definition at line 277 of file [SyntaxNode.h](#).

7.25.2 Constructor & Destructor Documentation

7.25.2.1 SyntaxListBase()

```
slang::syntax::SyntaxListBase::SyntaxListBase (
    SyntaxKind kind,
    size_t childCount ) [inline], [protected]
```

Definition at line 306 of file [SyntaxNode.h](#).

7.25.3 Member Function Documentation

7.25.3.1 clone()

```
virtual SyntaxListBase * slang::syntax::SyntaxListBase::clone (
    BumpAllocator & alloc ) const [pure virtual]
```

Clones the list into a new node using the provided allocator.

Implemented in [slang::syntax::SyntaxList< T >](#), [slang::syntax::TokenList](#), and [slang::syntax::SeparatedSyntaxList< T >](#).

7.25.3.2 getChild() [1/2]

```
virtual ConstTokenOrSyntax slang::syntax::SyntaxListBase::getChild (
    size_t index ) const [pure virtual]
```

Gets the child (token or node) at the given index.

Implemented in [slang::syntax::SyntaxList< T >](#), [slang::syntax::TokenList](#), and [slang::syntax::SeparatedSyntaxList< T >](#).

7.25.3.3 getChild() [2/2]

```
virtual TokenOrSyntax slang::syntax::SyntaxListBase::getChild (
    size_t index ) [pure virtual]
```

Gets the child (token or node) at the given index.

Implemented in [slang::syntax::SyntaxList< T >](#), [slang::syntax::TokenList](#), and [slang::syntax::SeparatedSyntaxList< T >](#).

7.25.3.4 getChildCount()

```
size_t slang::syntax::SyntaxListBase::getChildCount ( ) const [inline]
```

Gets the number of child items in the node.

Definition at line 280 of file [SyntaxNode.h](#).

7.25.3.5 getChildPtr()

```
virtual PtrTokenOrSyntax slang::syntax::SyntaxListBase::getChildPtr (
    size_t index ) [pure virtual]
```

Implemented in [slang::syntax::SyntaxList< T >](#), [slang::syntax::TokenList](#), and [slang::syntax::SeparatedSyntaxList< T >](#).

7.25.3.6 isChildOptional()

```
static bool slang::syntax::SyntaxListBase::isChildOptional (
    size_t index ) [static]
```

7.25.3.7 isKind()

```
static bool slang::syntax::SyntaxListBase::isKind (
    SyntaxKind kind ) [static]
```

7.25.3.8 resetAll()

```
virtual void slang::syntax::SyntaxListBase::resetAll (
    BumpAllocator & alloc,
    std::span< const TokenOrSyntax > children ) [pure virtual]
```

Overwrites all children with the new set of provided *children* (and making a copy with the provided allocator).

Implemented in [slang::syntax::SyntaxList< T >](#), [slang::syntax::TokenList](#), and [slang::syntax::SeparatedSyntaxList< T >](#).

7.25.3.9 setChild()

```
virtual void slang::syntax::SyntaxListBase::setChild (
    size_t index,
    TokenOrSyntax child ) [pure virtual]
```

Sets the child (token or node) at the given index.

Implemented in [slang::syntax::SyntaxList< T >](#), [slang::syntax::TokenList](#), and [slang::syntax::SeparatedSyntaxList< T >](#).

7.25.4 Member Data Documentation

7.25.4.1 childCount

```
size_t slang::syntax::SyntaxListBase::childCount [protected]
```

Definition at line 308 of file [SyntaxNode.h](#).

The documentation for this class was generated from the following file:

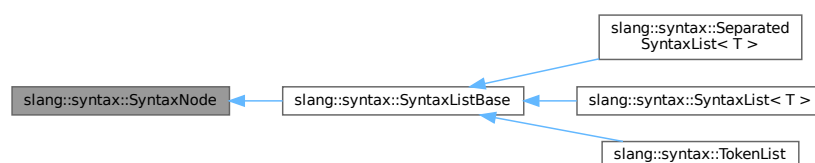
- include_3rd_party/[SyntaxNode.h](#)

7.26 slang::syntax::SyntaxNode Class Reference

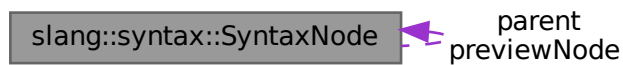
Base class for all syntax nodes.

```
#include <SyntaxNode.h>
```

Inheritance diagram for slang::syntax::SyntaxNode:



Collaboration diagram for slang::syntax::SyntaxNode:



Public Types

- using `Token` = `parsing::Token`

Public Member Functions

- `std::string toString () const`
Print the node and all of its children to a string.
- `Token getFirstToken () const`
Get the first leaf token in this subtree.
- `Token getLastToken () const`
Get the last leaf token in this subtree.
- `Token * getFirstTokenPtr ()`
Get the first leaf token as a mutable pointer in this subtree.
- `Token * getLastTokenPtr ()`
Get the last leaf token a mutable pointer in this subtree.
- `SourceRange sourceRange () const`
Get the source range of the node.
- `const SyntaxNode * childNode (size_t index) const`
- `SyntaxNode * childNode (size_t index)`
- `Token childToken (size_t index) const`
- `Token * childTokenPtr (size_t index)`
- `size_t getChildCount () const`
Gets the number of (direct) children underneath this node in the tree.
- `bool isEquivalentTo (const SyntaxNode &other) const`
- `template<typename T >`
`T & as ()`
- `template<typename T >`
`const T & as () const`
- `template<typename T >`
`T * as_if ()`

- `template<typename T >`
`const T * as\_if () const`
- `template<typename TVisitor , typename... Args>`
`decltype(auto) visit (TVisitor &visitor, Args &&... args)`
- `template<typename TVisitor , typename... Args>`
`decltype(auto) visit (TVisitor &visitor, Args &&... args) const`

Static Public Member Functions

- static bool [isKind](#) (SyntaxKind)
- static bool [isChildOptional](#) (size_t)

Public Attributes

- SyntaxKind [kind](#)
The kind of syntax node.
- [SyntaxNode](#) * [parent](#) = nullptr
- const [SyntaxNode](#) * [previewNode](#) = nullptr
An optional pointer to a syntax node that can be useful to know ahead of time when visiting this node.

Protected Member Functions

- [SyntaxNode](#) (SyntaxKind [kind](#))

Private Member Functions

- [ConstTokenOrSyntax](#) [getChild](#) (size_t index) const
- [TokenOrSyntax](#) [getChild](#) (size_t index)
- [PtrTokenOrSyntax](#) [getChildPtr](#) (size_t index)

7.26.1 Detailed Description

Base class for all syntax nodes.

Definition at line 90 of file [SyntaxNode.h](#).

7.26.2 Member Typedef Documentation

7.26.2.1 Token

```
using slang::syntax::SyntaxNode::Token = parsing::Token
```

Definition at line 92 of file [SyntaxNode.h](#).

7.26.3 Constructor & Destructor Documentation

7.26.3.1 SyntaxNode()

```
slang::syntax::SyntaxNode::SyntaxNode (  
    SyntaxKind kind ) [inline], [explicit], [protected]
```

Definition at line 207 of file [SyntaxNode.h](#).

7.26.4 Member Function Documentation

7.26.4.1 as() [1/2]

```
template<typename T >  
T & slang::syntax::SyntaxNode::as ( ) [inline]
```

Reinterprets this node as being of type T. In debug this will assert that the dynamic kind is appropriate for the specified static type.

Definition at line 158 of file [SyntaxNode.h](#).

7.26.4.2 as() [2/2]

```
template<typename T >  
const T & slang::syntax::SyntaxNode::as ( ) const [inline]
```

Reinterprets this node as being of type T. In debug this will assert that the dynamic kind is appropriate for the specified static type.

Definition at line 166 of file [SyntaxNode.h](#).

7.26.4.3 `as_if()` [1/2]

```
template<typename T >
T * slang::syntax::SyntaxNode::as_if ( ) [inline]
```

Definition at line 173 of file [SyntaxNode.h](#).

7.26.4.4 `as_if()` [2/2]

```
template<typename T >
const T * slang::syntax::SyntaxNode::as_if ( ) const [inline]
```

Definition at line 181 of file [SyntaxNode.h](#).

7.26.4.5 `childNode()` [1/2]

```
SyntaxNode * slang::syntax::SyntaxNode::childNode (
    size_t index )
```

Gets the child syntax node at the specified index. If the child at the given index is not a node (probably a token) then this returns null.

7.26.4.6 `childNode()` [2/2]

```
const SyntaxNode * slang::syntax::SyntaxNode::childNode (
    size_t index ) const
```

Gets the child syntax node at the specified index. If the child at the given index is not a node (probably a token) then this returns null.

7.26.4.7 `childToken()`

```
Token slang::syntax::SyntaxNode::childToken (
    size_t index ) const
```

Gets the child token at the specified index. If the child at the given index is not a token (probably a node) then this returns an empty Token.

7.26.4.8 childTokenPtr()

```
Token * slang::syntax::SyntaxNode::childTokenPtr (
    size_t index )
```

Gets a pointer to the child token at the specified index. If the child at the given index is not a token (probably a node) then this returns null.

7.26.4.9 getChild() [1/2]

```
TokenOrSyntax slang::syntax::SyntaxNode::getChild (
    size_t index ) [private]
```

7.26.4.10 getChild() [2/2]

```
ConstTokenOrSyntax slang::syntax::SyntaxNode::getChild (
    size_t index ) const [private]
```

7.26.4.11 getChildCount()

```
size_t slang::syntax::SyntaxNode::getChildCount ( ) const
```

Gets the number of (direct) children underneath this node in the tree.

7.26.4.12 getChildPtr()

```
PtrTokenOrSyntax slang::syntax::SyntaxNode::getChildPtr (
    size_t index ) [private]
```

7.26.4.13 getFirstToken()

```
Token slang::syntax::SyntaxNode::getFirstToken ( ) const
```

Get the first leaf token in this subtree.

7.26.4.14 `getFirstTokenPtr()`

```
Token * slang::syntax::SyntaxNode::getFirstTokenPtr ( )
```

Get the first leaf token as a mutable pointer in this subtree.

7.26.4.15 `getLastToken()`

```
Token slang::syntax::SyntaxNode::getLastToken ( ) const
```

Get the last leaf token in this subtree.

7.26.4.16 `getLastTokenPtr()`

```
Token * slang::syntax::SyntaxNode::getLastTokenPtr ( )
```

Get the last leaf token a mutable pointer in this subtree.

7.26.4.17 `isChildOptional()`

```
static bool slang::syntax::SyntaxNode::isChildOptional (
    size_t ) [inline], [static]
```

Derived nodes should implement this and return true if child at provided index is pointer not wrapped in `not_null`.

Definition at line 204 of file [SyntaxNode.h](#).

7.26.4.18 `isEquivalentTo()`

```
bool slang::syntax::SyntaxNode::isEquivalentTo (
    const SyntaxNode & other ) const
```

Returns true if this syntax node is "equivalent" to the other provided syntax node. Equivalence here is determined by the entire subtrees having the same kinds of syntax nodes in the same order and all leaf tokens having the same kinds and value text.

7.26.4.19 isKind()

```
static bool slang::syntax::SyntaxNode::isKind (
    SyntaxKind ) [inline], [static]
```

A base implementation of the method that checks correctness of dynamic casting. Derived nodes should reimplement this and return true if the provided syntax kind is compatible with the static type of the object.

Definition at line 200 of file [SyntaxNode.h](#).

7.26.4.20 sourceRange()

```
SourceRange slang::syntax::SyntaxNode::sourceRange ( ) const
```

Get the source range of the node.

7.26.4.21 toString()

```
std::string slang::syntax::SyntaxNode::toString ( ) const
```

Print the node and all of its children to a string.

7.26.4.22 visit() ^[1/2]

```
template<typename TVisitor , typename... Args>
decltype(auto) slang::syntax::SyntaxNode::visit (
    TVisitor & visitor,
    Args &&... args )
```

Applies a visitor object to this node by dispatching based on the dynamic kind. The given *args* are forwarded to the visitor.

7.26.4.23 visit() [2/2]

```
template<typename TVisitor , typename... Args>
decltype(auto) slang::syntax::SyntaxNode::visit (
    TVisitor & visitor,
    Args &&... args ) const
```

Applies a visitor object to this node by dispatching based on the dynamic kind. The given *args* are forwarded to the visitor.

7.26.5 Member Data Documentation

7.26.5.1 kind

```
SyntaxKind slang::syntax::SyntaxNode::kind
```

The kind of syntax node.

Definition at line 95 of file [SyntaxNode.h](#).

7.26.5.2 parent

```
SyntaxNode\* slang::syntax::SyntaxNode::parent = nullptr
```

The parent node of this syntax node. The root of the syntax tree does not have a parent (will be nullptr).

Definition at line 99 of file [SyntaxNode.h](#).

7.26.5.3 previewNode

```
const SyntaxNode\* slang::syntax::SyntaxNode::previewNode = nullptr
```

An optional pointer to a syntax node that can be useful to know ahead of time when visiting this node.

The node, if set, is underneath this node in the syntax tree.

For example, an enum declaration deep inside an expression tree needs to be known up front to add its members to its parent scope.

Definition at line 108 of file [SyntaxNode.h](#).

The documentation for this class was generated from the following file:

- [include_3rd_party/SyntaxNode.h](#)

7.27 slang::syntax::SyntaxPrinter Class Reference

```
#include <SyntaxPrinter.h>
```

Public Member Functions

- [SyntaxPrinter](#) ()=default
- [SyntaxPrinter](#) (const SourceManager &[sourceManager](#))
- [SyntaxPrinter](#) & [append](#) (std::string_view text)
- [SyntaxPrinter](#) & [print](#) (parsing::Trivia trivia)
- [SyntaxPrinter](#) & [print](#) (parsing::Token token)
- [SyntaxPrinter](#) & [print](#) (const [SyntaxNode](#) &node)
- [SyntaxPrinter](#) & [print](#) (const [SyntaxTree](#) &tree)
- [SyntaxPrinter](#) & [setIncludeTrivia](#) (bool include)
- [SyntaxPrinter](#) & [setIncludeMissing](#) (bool include)
- [SyntaxPrinter](#) & [setIncludeSkipped](#) (bool include)
- [SyntaxPrinter](#) & [setIncludeDirectives](#) (bool include)
- [SyntaxPrinter](#) & [setIncludePreprocessed](#) (bool include)
- [SyntaxPrinter](#) & [setIncludeComments](#) (bool include)
- [SyntaxPrinter](#) & [setSquashNewlines](#) (bool include)
- std::string [str](#) () const

Static Public Member Functions

- static std::string [printFile](#) (const [SyntaxTree](#) &tree)

Private Attributes

- std::string [buffer](#)
- const SourceManager * [sourceManager](#) = nullptr
- bool [includeTrivia](#) = true
- bool [includeMissing](#) = false
- bool [includeSkipped](#) = false
- bool [includeDirectives](#) = false
- bool [includePreprocessed](#) = true
- bool [includeComments](#) = true
- bool [squashNewlines](#) = true

7.27.1 Detailed Description

Provides support for printing tokens, trivia, or whole syntax trees back to source code.

Definition at line 25 of file [SyntaxPrinter.h](#).

7.27.2 Constructor & Destructor Documentation

7.27.2.1 SyntaxPrinter() [1/2]

```
slang::syntax::SyntaxPrinter::SyntaxPrinter ( ) [default]
```

7.27.2.2 SyntaxPrinter() [2/2]

```
slang::syntax::SyntaxPrinter::SyntaxPrinter (
    const SourceManager & sourceManager ) [explicit]
```

7.27.3 Member Function Documentation

7.27.3.1 append()

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::append (
    std::string_view text )
```

Append raw text to the buffer.

Returns

a reference to this object, to allow chaining additional method calls.

7.27.3.2 print() [1/4]

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::print (
    const SyntaxNode & node )
```

Print the provided *node* to the internal buffer.

Returns

a reference to this object, to allow chaining additional method calls.

7.27.3.3 print() [2/4]

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::print (
    const SyntaxTree & tree )
```

Print the provided *tree* to the internal buffer.

Returns

a reference to this object, to allow chaining additional method calls.

7.27.3.4 print() [3/4]

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::print (
    parsing::Token token )
```

Print the provided *token* to the internal buffer.

Returns

a reference to this object, to allow chaining additional method calls.

7.27.3.5 print() [4/4]

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::print (  
    parsing::Trivia trivia )
```

Print the provided *trivia* to the internal buffer.

Returns

a reference to this object, to allow chaining additional method calls.

7.27.3.6 printFile()

```
static std::string slang::syntax::SyntaxPrinter::printFile (  
    const SyntaxTree & tree ) [static]
```

A helper method that assists in printing an entire syntax tree back to source text. A [SyntaxPrinter](#) with useful defaults is constructed, the tree is printed, and the resulting text is returned. Here is the caller graph for this function:



7.27.3.7 setIncludeComments()

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::setIncludeComments (  
    bool include ) [inline]
```

Sets whether to include comments when printing syntax.

Returns

a reference to this object, to allow chaining additional method calls.

Definition at line 87 of file [SyntaxPrinter.h](#).

7.27.3.8 setIncludeDirectives()

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::setIncludeDirectives (  
    bool include ) [inline]
```

Sets whether to include preprocessor directives when printing syntax.

Returns

a reference to this object, to allow chaining additional method calls.

Definition at line 73 of file [SyntaxPrinter.h](#).

7.27.3.9 setIncludeMissing()

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::setIncludeMissing (  
    bool include ) [inline]
```

Sets whether to include missing (automatically inserted) nodes when printing syntax.

Returns

a reference to this object, to allow chaining additional method calls.

Definition at line 59 of file [SyntaxPrinter.h](#).

7.27.3.10 setIncludePreprocessed()

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::setIncludePreprocessed (  
    bool include ) [inline]
```

Sets whether to include preprocessor-expanded tokens when printing syntax.

Returns

a reference to this object, to allow chaining additional method calls.

Definition at line 80 of file [SyntaxPrinter.h](#).

7.27.3.11 setIncludeSkipped()

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::setIncludeSkipped (  
    bool include ) [inline]
```

Sets whether to include skipped (due to some sort of error) nodes when printing syntax.

Returns

a reference to this object, to allow chaining additional method calls.

Definition at line 66 of file [SyntaxPrinter.h](#).

7.27.3.12 setIncludeTrivia()

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::setIncludeTrivia (  
    bool include ) [inline]
```

Sets whether to include trivia when printing syntax.

Returns

a reference to this object, to allow chaining additional method calls.

Definition at line 52 of file [SyntaxPrinter.h](#).

7.27.3.13 setSquashNewlines()

```
SyntaxPrinter & slang::syntax::SyntaxPrinter::setSquashNewlines (  
    bool include ) [inline]
```

Sets whether to squash adjacent newlines down into one when printing syntax.

Returns

a reference to this object, to allow chaining additional method calls.

Definition at line 94 of file [SyntaxPrinter.h](#).

7.27.3.14 str()

```
std::string slang::syntax::SyntaxPrinter::str ( ) const [inline]
```

Returns

a copy of the internal text buffer.

Definition at line 100 of file [SyntaxPrinter.h](#).

7.27.4 Member Data Documentation

7.27.4.1 buffer

```
std::string slang::syntax::SyntaxPrinter::buffer [private]
```

Definition at line 108 of file [SyntaxPrinter.h](#).

7.27.4.2 includeComments

```
bool slang::syntax::SyntaxPrinter::includeComments = true [private]
```

Definition at line 115 of file [SyntaxPrinter.h](#).

7.27.4.3 includeDirectives

```
bool slang::syntax::SyntaxPrinter::includeDirectives = false [private]
```

Definition at line 113 of file [SyntaxPrinter.h](#).

7.27.4.4 includeMissing

```
bool slang::syntax::SyntaxPrinter::includeMissing = false [private]
```

Definition at line 111 of file [SyntaxPrinter.h](#).

7.27.4.5 includePreprocessed

```
bool slang::syntax::SyntaxPrinter::includePreprocessed = true [private]
```

Definition at line 114 of file [SyntaxPrinter.h](#).

7.27.4.6 includeSkipped

```
bool slang::syntax::SyntaxPrinter::includeSkipped = false [private]
```

Definition at line 112 of file [SyntaxPrinter.h](#).

7.27.4.7 includeTrivia

```
bool slang::syntax::SyntaxPrinter::includeTrivia = true [private]
```

Definition at line 110 of file [SyntaxPrinter.h](#).

7.27.4.8 sourceManager

```
const SourceManager* slang::syntax::SyntaxPrinter::sourceManager = nullptr [private]
```

Definition at line 109 of file [SyntaxPrinter.h](#).

7.27.4.9 squashNewlines

```
bool slang::syntax::SyntaxPrinter::squashNewlines = true [private]
```

Definition at line 116 of file [SyntaxPrinter.h](#).

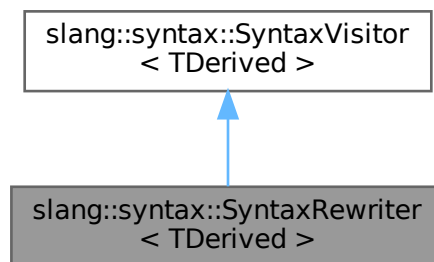
The documentation for this class was generated from the following file:

- include_3rd_party/[SyntaxPrinter.h](#)

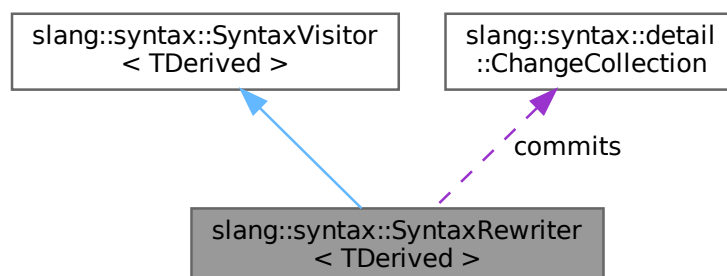
7.28 slang::syntax::SyntaxRewriter< TDerived > Class Template Reference

```
#include <SyntaxVisitor.h>
```

Inheritance diagram for slang::syntax::SyntaxRewriter< TDerived >:



Collaboration diagram for slang::syntax::SyntaxRewriter< TDerived >:



Public Member Functions

- [SyntaxRewriter](#) ()
- `std::shared_ptr< SyntaxTree > transform` (const `std::shared_ptr< SyntaxTree >` &tree, const `SourceLibrary *library=nullptr`)

Protected Types

- using [Token](#) = `parsing::Token`

Protected Member Functions

- [SyntaxNode](#) & [parse](#) (std::string_view text)
A helper for derived classes that parses some text into syntax nodes.
- void [remove](#) (const [SyntaxNode](#) &oldNode)
Register a removal for the given syntax node from the tree.
- void [replace](#) (const [SyntaxNode](#) &oldNode, [SyntaxNode](#) &newNode)
Replace the given oldNode with newNode in the rewritten tree.
- void [insertBefore](#) (const [SyntaxNode](#) &oldNode, [SyntaxNode](#) &newNode)
Insert newNode before oldNode in the rewritten tree.
- void [insertAfter](#) (const [SyntaxNode](#) &oldNode, [SyntaxNode](#) &newNode)
Insert newNode after oldNode in the rewritten tree.
- void [insertAtFront](#) (const [SyntaxListBase](#) &list, [SyntaxNode](#) &newNode, [Token](#) separator={})
Insert newNode at the front of list in the rewritten tree.
- void [insertAtBack](#) (const [SyntaxListBase](#) &list, [SyntaxNode](#) &newNode, [Token](#) separator={})
Insert newNode at the back of list in the rewritten tree.
- [Token](#) [makeToken](#) (parsing::TokenKind kind, std::string_view text, std::span< const parsing::Trivia > trivia={})
- [Token](#) [makeId](#) (std::string_view text, std::span< const parsing::Trivia > trivia={})
- [Token](#) [makeId](#) (std::string_view text, parsing::Trivia trivia)
- [Token](#) [makeComma](#) ()

Protected Attributes

- BumpAllocator [alloc](#)
- SyntaxFactory [factory](#)

Static Protected Attributes

- static const parsing::Trivia [SingleSpace](#) {parsing::TriviaKind::Whitespace, " "sv}

Private Attributes

- SourceManager * [sourceManager](#) = nullptr
- [detail::ChangeCollection](#) commits
- std::vector< std::shared_ptr< [SyntaxTree](#) > > [tempTrees](#)

7.28.1 Detailed Description

```
template<typename TDerived>
class slang::syntax::SyntaxRewriter< TDerived >
```

A helper class that assists in rewriting syntax trees – useful for automated refactoring tools.

Definition at line 102 of file [SyntaxVisitor.h](#).

7.28.2 Member Typedef Documentation

7.28.2.1 Token

```
template<typename TDerived >
using slang::syntax::SyntaxRewriter< TDerived >::Token = parsing::Token [protected]
```

Definition at line 131 of file [SyntaxVisitor.h](#).

7.28.3 Constructor & Destructor Documentation

7.28.3.1 SyntaxRewriter()

```
template<typename TDerived >
slang::syntax::SyntaxRewriter< TDerived >::SyntaxRewriter ( ) [inline]
```

Definition at line 104 of file [SyntaxVisitor.h](#).

7.28.4 Member Function Documentation

7.28.4.1 insertAfter()

```
template<typename TDerived >
void slang::syntax::SyntaxRewriter< TDerived >::insertAfter (
    const SyntaxNode & oldNode,
    SyntaxNode & newNode ) [inline], [protected]
```

Insert *newNode* after *oldNode* in the rewritten tree.

Definition at line 163 of file [SyntaxVisitor.h](#).

7.28.4.2 insertAtBack()

```
template<typename TDerived >
void slang::syntax::SyntaxRewriter< TDerived >::insertAtBack (
    const SyntaxListBase & list,
    SyntaxNode & newNode,
    Token separator = {} ) [inline], [protected]
```

Insert *newNode* at the back of *list* in the rewritten tree.

Definition at line 173 of file [SyntaxVisitor.h](#).

7.28.4.3 insertAtFront()

```
template<typename TDerived >
void slang::syntax::SyntaxRewriter< TDerived >::insertAtFront (
    const SyntaxListBase & list,
    SyntaxNode & newNode,
    Token separator = {} ) [inline], [protected]
```

Insert *newNode* at the front of *list* in the rewritten tree.

Definition at line 168 of file [SyntaxVisitor.h](#).

7.28.4.4 insertBefore()

```
template<typename TDerived >
void slang::syntax::SyntaxRewriter< TDerived >::insertBefore (
    const SyntaxNode & oldNode,
    SyntaxNode & newNode ) [inline], [protected]
```

Insert *newNode* before *oldNode* in the rewritten tree.

Definition at line 158 of file [SyntaxVisitor.h](#).

7.28.4.5 makeComma()

```
template<typename TDerived >
Token slang::syntax::SyntaxRewriter< TDerived >::makeComma ( ) [inline], [protected]
```

Definition at line 191 of file [SyntaxVisitor.h](#).

Here is the call graph for this function:



7.28.4.6 makeId() [1/2]

```
template<typename TDerived >
Token slang::syntax::SyntaxRewriter< TDerived >::makeId (
    std::string_view text,
    parsing::Trivia trivia ) [inline], [protected]
```

Definition at line 186 of file [SyntaxVisitor.h](#).

Here is the call graph for this function:



7.28.4.7 makeId() [2/2]

```

template<typename TDerived >
Token slang::syntax::SyntaxRewriter< TDerived >::makeId (
    std::string_view text,
    std::span< const parsing::Trivia > trivia = {} ) [inline], [protected]
  
```

Definition at line 182 of file [SyntaxVisitor.h](#).

Here is the caller graph for this function:



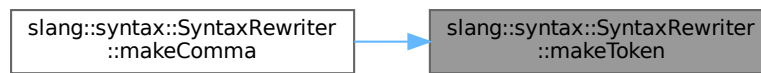
7.28.4.8 makeToken()

```

template<typename TDerived >
Token slang::syntax::SyntaxRewriter< TDerived >::makeToken (
    parsing::TokenKind kind,
    std::string_view text,
    std::span< const parsing::Trivia > trivia = {} ) [inline], [protected]
  
```

Definition at line 177 of file [SyntaxVisitor.h](#).

Here is the caller graph for this function:



7.28.4.9 parse()

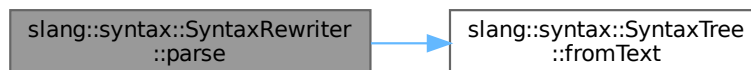
```

template<typename TDerived >
SyntaxNode & slang::syntax::SyntaxRewriter< TDerived >::parse (
    std::string_view text ) [inline], [protected]
  
```

A helper for derived classes that parses some text into syntax nodes.

Definition at line 134 of file [SyntaxVisitor.h](#).

Here is the call graph for this function:



7.28.4.10 remove()

```

template<typename TDerived >
void slang::syntax::SyntaxRewriter< TDerived >::remove (
    const SyntaxNode & oldNode ) [inline], [protected]
  
```

Register a removal for the given syntax node from the tree.

Definition at line 140 of file [SyntaxVisitor.h](#).

7.28.4.11 replace()

```
template<typename TDerived >
void slang::syntax::SyntaxRewriter< TDerived >::replace (
    const SyntaxNode & oldNode,
    SyntaxNode & newNode ) [inline], [protected]
```

Replace the given *oldNode* with *newNode* in the rewritten tree.

Definition at line 149 of file [SyntaxVisitor.h](#).

7.28.4.12 transform()

```
template<typename TDerived >
std::shared_ptr< SyntaxTree > slang::syntax::SyntaxRewriter< TDerived >::transform (
    const std::shared_ptr< SyntaxTree > & tree,
    const SourceLibrary * library = nullptr ) [inline]
```

Transforms the given syntax tree using the rewriter. The tree will be visited in order and each node will be given a chance to be rewritten (by providing [visit\(\)](#) implementations in the derived class).

Returns

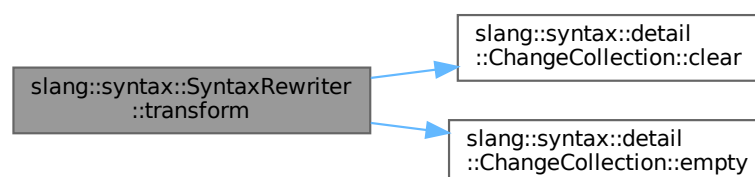
if no changes are requested, returns the original syntax tree. Otherwise, the changes are applied and the newly rewritten syntax tree is returned.

Note

lifetimes of new and old trees are independent from each other. If you shallow clone something from old tree into new one, you have to make sure that `shared_ptr` to original doesn't go out of scope too early.

Definition at line 116 of file [SyntaxVisitor.h](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.28.5 Member Data Documentation

7.28.5.1 alloc

```
template<typename TDerived >
BumpAllocator slang::syntax::SyntaxRewriter< TDerived >::alloc [protected]
```

Definition at line 193 of file [SyntaxVisitor.h](#).

7.28.5.2 commits

```
template<typename TDerived >
detail::ChangeCollection slang::syntax::SyntaxRewriter< TDerived >::commits [private]
```

Definition at line 200 of file [SyntaxVisitor.h](#).

7.28.5.3 factory

```
template<typename TDerived >
SyntaxFactory slang::syntax::SyntaxRewriter< TDerived >::factory [protected]
```

Definition at line 194 of file [SyntaxVisitor.h](#).

7.28.5.4 SingleSpace

```
template<typename TDerived >
const parsing::Trivia slang::syntax::SyntaxRewriter< TDerived >::SingleSpace {parsing::TriviaKind::Whitespace, "
"sv}  [static], [protected]
```

Definition at line 196 of file [SyntaxVisitor.h](#).

7.28.5.5 sourceManager

```
template<typename TDerived >
SourceManager* slang::syntax::SyntaxRewriter< TDerived >::sourceManager = nullptr  [private]
```

Definition at line 199 of file [SyntaxVisitor.h](#).

7.28.5.6 tempTrees

```
template<typename TDerived >
std::vector<std::shared_ptr<SyntaxTree> > slang::syntax::SyntaxRewriter< TDerived >::tempTrees  [private]
```

Definition at line 201 of file [SyntaxVisitor.h](#).

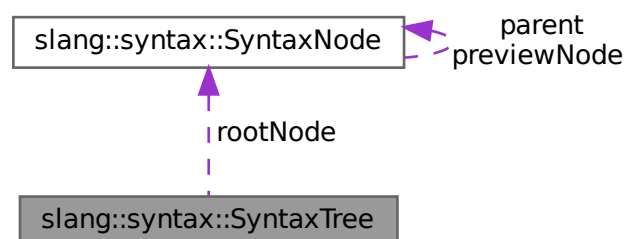
The documentation for this class was generated from the following file:

- [include_3rd_party/SyntaxVisitor.h](#)

7.29 slang::syntax::SyntaxTree Class Reference

```
#include <SyntaxTree.h>
```

Collaboration diagram for slang::syntax::SyntaxTree:



Public Types

- using [TreeOrError](#) = nonstd::expected< std::shared_ptr< [SyntaxTree](#) >, std::pair< std::error_code, std::string_view > >
- using [MacroList](#) = std::span< const DefineDirectiveSyntax *const >

Public Member Functions

- [SyntaxTree](#) ([SyntaxNode](#) *root, SourceManager &[sourceManager](#), BumpAllocator &&[alloc](#), const SourceLibrary *library, const std::shared_ptr< [SyntaxTree](#) > &parent=nullptr)
- [SyntaxTree](#) ([SyntaxTree](#) &&other)=default
- ~[SyntaxTree](#) ()
- Diagnostics & [diagnostics](#) ()
Gets any diagnostics generated while parsing.
- BumpAllocator & [allocator](#) ()
Gets the allocator containing the memory for the parse tree.
- SourceManager & [sourceManager](#) ()
Gets the source manager used to build the syntax tree.
- const SourceManager & [sourceManager](#) () const
Gets the source manager used to build the syntax tree.
- const SourceLibrary * [getSourceLibrary](#) () const
Gets the source library with which the syntax tree is associated.
- [SyntaxNode](#) & [root](#) ()
Gets the root of the syntax tree.
- const [SyntaxNode](#) & [root](#) () const
Gets the root of the syntax tree.
- const Bag & [options](#) () const
The options used to construct the syntax tree.
- const parsing::ParserMetadata & [getMetadata](#) () const
Gets various bits of metadata collected during parsing.
- [MacroList](#) [getDefinedMacros](#) () const
Gets the list of macros that were defined at the end of the loaded source file.

Static Public Member Functions

- static [TreeOrError](#) [fromFile](#) (std::string_view path)
- static [TreeOrError](#) [fromFile](#) (std::string_view path, SourceManager &[sourceManager](#), const Bag &[options](#)={})
- static [TreeOrError](#) [fromFiles](#) (std::span< const std::string_view > paths)
- static [TreeOrError](#) [fromFiles](#) (std::span< const std::string_view > paths, SourceManager &[sourceManager](#), const Bag &[options](#)={})

- static std::shared_ptr< [SyntaxTree](#) > [fromText](#) (std::string_view text, std::string_view name="source", std::string_view path="")
- static std::shared_ptr< [SyntaxTree](#) > [fromText](#) (std::string_view text, const Bag &[options](#), std::string_view name="source"sv, std::string_view path="")
- static std::shared_ptr< [SyntaxTree](#) > [fromText](#) (std::string_view text, SourceManager &[sourceManager](#), std::string_view name="source"sv, std::string_view path="", const Bag &[options](#)={}, const SourceLibrary *[library](#)=nullptr)
- static std::shared_ptr< [SyntaxTree](#) > [fromFileInMemory](#) (std::string_view text, SourceManager &[sourceManager](#), std::string_view name="source"sv, std::string_view path="", const Bag &[options](#)={})
- static std::shared_ptr< [SyntaxTree](#) > [fromBuffer](#) (const SourceBuffer &buffer, SourceManager &[sourceManager](#), const Bag &[options](#)={}, [MacroList](#) inheritedMacros={})
- static std::shared_ptr< [SyntaxTree](#) > [fromBuffers](#) (std::span< const SourceBuffer > buffers, SourceManager &[sourceManager](#), const Bag &[options](#)={}, [MacroList](#) inheritedMacros={})
- static std::shared_ptr< [SyntaxTree](#) > [fromLibraryMapFile](#) (std::string_view path, SourceManager &[sourceManager](#), const Bag &[options](#)={})
- static std::shared_ptr< [SyntaxTree](#) > [fromLibraryMapText](#) (std::string_view text, SourceManager &[sourceManager](#), std::string_view name="source"sv, std::string_view path="", const Bag &[options](#)={})
- static std::shared_ptr< [SyntaxTree](#) > [fromLibraryMapBuffer](#) (const SourceBuffer &buffer, SourceManager &[sourceManager](#), const Bag &[options](#)={})
- static SourceManager & [getDefaultSourceManager](#) ()

Public Attributes

- bool [isLibraryUnit](#) = false

Private Member Functions

- [SyntaxTree](#) ([SyntaxNode](#) *[root](#), const SourceLibrary *[library](#), SourceManager &[sourceManager](#), BumpAllocator &&[alloc](#), Diagnostics &&[diagnostics](#), parsing::ParserMetadata &&[metadata](#), std::vector< const DefinedDirectiveSyntax * > &&[macros](#), Bag [options](#))

Static Private Member Functions

- static std::shared_ptr< [SyntaxTree](#) > [create](#) (SourceManager &[sourceManager](#), std::span< const SourceBuffer > source, const Bag &[options](#), [MacroList](#) inheritedMacros, bool guess)

Private Attributes

- [SyntaxNode](#) * [rootNode](#)
- const SourceLibrary * [library](#)
- SourceManager & [sourceMan](#)
- BumpAllocator [alloc](#)
- Diagnostics [diagnosticsBuffer](#)
- Bag [options_](#)
- std::unique_ptr< parsing::ParserMetadata > [metadata](#)
- std::vector< const DefineDirectiveSyntax * > [macros](#)

7.29.1 Detailed Description

The [SyntaxTree](#) is the easiest way to interface with the lexer / preprocessor / parser stack. Give it some source text and it produces a parse tree.

The [SyntaxTree](#) object owns all of the memory for the parse tree, so it must live for as long as you need to access its syntax nodes.

Definition at line 39 of file [SyntaxTree.h](#).

7.29.2 Member Typedef Documentation

7.29.2.1 MacroList

```
using slang::syntax::SyntaxTree::MacroList = std::span<const DefineDirectiveSyntax* const>
```

Definition at line 43 of file [SyntaxTree.h](#).

7.29.2.2 TreeOrError

```
using slang::syntax::SyntaxTree::TreeOrError = nonstd::expected<std::shared_ptr<SyntaxTree>, std::pair<std::error_code, std::string_view> >
```

Definition at line 41 of file [SyntaxTree.h](#).

7.29.3 Constructor & Destructor Documentation

7.29.3.1 SyntaxTree() [1/3]

```
slang::syntax::SyntaxTree::SyntaxTree (
    SyntaxNode * root,
    SourceManager & sourceManager,
    BumpAllocator && alloc,
    const SourceLibrary * library,
    const std::shared_ptr< SyntaxTree > & parent = nullptr )
```

7.29.3.2 SyntaxTree() [2/3]

```
slang::syntax::SyntaxTree::SyntaxTree (
    SyntaxTree && other ) [default]
```

7.29.3.3 ~SyntaxTree()

```
slang::syntax::SyntaxTree::~~SyntaxTree ( )
```

7.29.3.4 SyntaxTree() [3/3]

```
slang::syntax::SyntaxTree::SyntaxTree (
    SyntaxNode * root,
    const SourceLibrary * library,
    SourceManager & sourceManager,
    BumpAllocator && alloc,
    Diagnostics && diagnostics,
    parsing::ParserMetadata && metadata,
    std::vector< const DefineDirectiveSyntax * > && macros,
    Bag options ) [private]
```

7.29.4 Member Function Documentation

7.29.4.1 allocator()

```
BumpAllocator & slang::syntax::SyntaxTree::allocator ( ) [inline]
```

Gets the allocator containing the memory for the parse tree.

Definition at line 187 of file [SyntaxTree.h](#).

7.29.4.2 create()

```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::create (
    SourceManager & sourceManager,
    std::span< const SourceBuffer > source,
    const Bag & options,
    MacroList inheritedMacros,
    bool guess ) [static], [private]
```

7.29.4.3 diagnostics()

```
Diagnostics & slang::syntax::SyntaxTree::diagnostics ( ) [inline]
```

Gets any diagnostics generated while parsing.

Definition at line 184 of file [SyntaxTree.h](#).

7.29.4.4 fromBuffer()

```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::fromBuffer (
    const SourceBuffer & buffer,
    SourceManager & sourceManager,
    const Bag & options = {},
    MacroList inheritedMacros = {} ) [static]
```

Creates a syntax tree from an already loaded source buffer. *buffer* is the loaded source buffer. *sourceManager* is the manager that owns the buffer. *options* is an optional bag of lexer, preprocessor, and parser options. *inheritedMacros* is a list of macros to predefine in the new syntax tree.

Returns

the created and parsed syntax tree.

7.29.4.5 fromBuffers()

```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::fromBuffers (
    std::span< const SourceBuffer > buffers,
    SourceManager & sourceManager,
    const Bag & options = {},
    MacroList inheritedMacros = {} ) [static]
```

Creates a syntax tree by concatenating several loaded source buffers. *buffers* is the list of buffers that should be concatenated to form the compilation unit to parse. *sourceManager* is the manager that owns the buffers. *options* is an optional bag of lexer, preprocessor, and parser options. *inheritedMacros* is a list of macros to predefine in the new syntax tree.

Returns

the created and parsed syntax tree.

7.29.4.6 fromFile() [1/2]

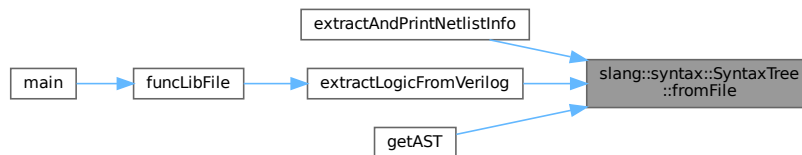
```
static TreeOrError slang::syntax::SyntaxTree::fromFile (
    std::string_view path ) [static]
```

Creates a syntax tree from a full compilation unit. *path* is the path to the source file on disk.

Returns

the created and parsed syntax tree.

Here is the caller graph for this function:



7.29.4.7 fromFile() [2/2]

```
static TreeOrError slang::syntax::SyntaxTree::fromFile (
    std::string_view path,
    SourceManager & sourceManager,
    const Bag & options = {} ) [static]
```

Creates a syntax tree from a full compilation unit. *path* is the path to the source file on disk. *sourceManager* is the manager that owns all of the loaded source code. *options* is an optional bag of lexer, preprocessor, and parser options.

Returns

the created and parsed syntax tree on success, or an OS error code if the file fails to load.

7.29.4.8 fromFileInMemory()

```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::fromFileInMemory (
    std::string_view text,
    SourceManager & sourceManager,
    std::string_view name = "source"sv,
    std::string_view path = "",
    const Bag & options = {} ) [static]
```

Creates a syntax tree from a full compilation unit already in memory. *text* is the actual source code text. *sourceManager* is the manager that owns all of the loaded source code. *name* is an optional name to give to the loaded source buffer. *path* is an optional path to give to the loaded source buffer. *options* is an optional bag of lexer, preprocessor, and parser options.

Returns

the created and parsed syntax tree.

7.29.4.9 fromFiles() [1/2]

```
static TreeOrError slang::syntax::SyntaxTree::fromFiles (
    std::span< const std::string_view > paths ) [static]
```

Creates a syntax tree by concatenating several files loaded from disk. *paths* is the list of paths to the source files on disk.

Returns

the created and parsed syntax tree on success, or an OS error code if the file fails to load.

7.29.4.10 fromFiles() [2/2]

```
static TreeOrError slang::syntax::SyntaxTree::fromFiles (
    std::span< const std::string_view > paths,
    SourceManager & sourceManager,
    const Bag & options = {} ) [static]
```

Creates a syntax tree by concatenating several files loaded from disk. *paths* is the list of paths to the source files on disk. *sourceManager* is the manager that owns all of the loaded source code. *options* is an optional bag of lexer, preprocessor, and parser options.

Returns

the created and parsed syntax tree on success, or an OS error code if the file fails to load.

7.29.4.11 fromLibraryMapBuffer()

```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::fromLibraryMapBuffer (
    const SourceBuffer & buffer,
    SourceManager & sourceManager,
    const Bag & options = {} ) [static]
```

Creates a syntax tree from a library map already loaded into a source buffer. *buffer* is the loaded source buffer. *sourceManager* is the manager that owns the buffer. *options* is an optional bag of lexer, preprocessor, and parser options.

Returns

the created and parsed syntax tree.

7.29.4.12 fromLibraryMapFile()

```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::fromLibraryMapFile (
    std::string_view path,
    SourceManager & sourceManager,
    const Bag & options = {} ) [static]
```

Creates a syntax tree from a library map file. *path* is the path to the source file on disk. *sourceManager* is the manager that owns all of the loaded source code. *options* is an optional bag of lexer, preprocessor, and parser options.

Returns

the created and parsed syntax tree.

7.29.4.13 fromLibraryMapText()

```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::fromLibraryMapText (
    std::string_view text,
    SourceManager & sourceManager,
    std::string_view name = "source"sv,
    std::string_view path = "",
    const Bag & options = {} ) [static]
```

Creates a syntax tree from a library map located in memory. *text* is the actual source code text. *sourceManager* is the manager that owns all of the loaded source code. *name* is an optional name to give to the loaded source buffer. *path* is an optional path to give to the loaded source buffer. *options* is an optional bag of lexer, preprocessor, and parser options.

Returns

the created and parsed syntax tree.

7.29.4.14 fromText() [1/3]

```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::fromText (
    std::string_view text,
    const Bag & options,
    std::string_view name = "source"sv,
    std::string_view path = "" ) [static]
```

Creates a syntax tree by guessing at what might be in the given source snippet. *text* is the actual source code text. *options* is a bag of lexer, preprocessor, and parser options. *name* is an optional name to give to the loaded source buffer. *path* is an optional path to give to the loaded source buffer.

Returns

the created and parsed syntax tree.

7.29.4.15 fromText() [2/3]

```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::fromText (
    std::string_view text,
    SourceManager & sourceManager,
    std::string_view name = "source"sv,
    std::string_view path = "",
    const Bag & options = {},
    const SourceLibrary * library = nullptr ) [static]
```

Creates a syntax tree by guessing at what might be in the given source snippet. *text* is the actual source code text. *sourceManager* is the manager that owns all of the loaded source code. *name* is an optional name to give to the loaded source buffer. *path* is an optional path to give to the loaded source buffer. *options* is an optional bag of lexer, preprocessor, and parser options. *library* the source library to associated with the parsed tree

Returns

the created and parsed syntax tree.

7.29.4.16 fromText() [3/3]

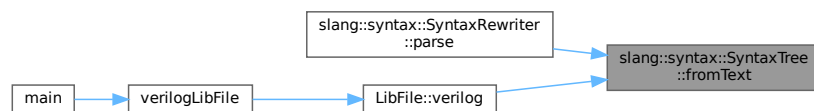
```
static std::shared_ptr< SyntaxTree > slang::syntax::SyntaxTree::fromText (
    std::string_view text,
    std::string_view name = "source",
    std::string_view path = "" ) [static]
```

Creates a syntax tree by guessing at what might be in the given source snippet. *text* is the actual source code text. *name* is an optional name to give to the loaded source buffer. *path* is an optional path to give to the loaded source buffer.

Returns

the created and parsed syntax tree.

Here is the caller graph for this function:

**7.29.4.17 getDefaultSourceManager()**

```
static SourceManager & slang::syntax::SyntaxTree::getDefaultSourceManager ( ) [static]
```

This is a shared default source manager for cases where the user doesn't care about managing the lifetime of loaded source. Note that all of the source loaded by this thing will live in memory for the lifetime of the process.

7.29.4.18 getDefinedMacros()

```
MacroList slang::syntax::SyntaxTree::getDefinedMacros ( ) const [inline]
```

Gets the list of macros that were defined at the end of the loaded source file.

Definition at line 211 of file [SyntaxTree.h](#).

7.29.4.19 getMetadata()

```
const parsing::ParserMetadata & slang::syntax::SyntaxTree::getMetadata ( ) const [inline]
```

Gets various bits of metadata collected during parsing.

Definition at line 208 of file [SyntaxTree.h](#).

7.29.4.20 getSourceLibrary()

```
const SourceLibrary * slang::syntax::SyntaxTree::getSourceLibrary ( ) const [inline]
```

Gets the source library with which the syntax tree is associated.

Definition at line 196 of file [SyntaxTree.h](#).

7.29.4.21 options()

```
const Bag & slang::syntax::SyntaxTree::options ( ) const [inline]
```

The options used to construct the syntax tree.

Definition at line 205 of file [SyntaxTree.h](#).

7.29.4.22 root() [1/2]

```
SyntaxNode & slang::syntax::SyntaxTree::root ( ) [inline]
```

Gets the root of the syntax tree.

Definition at line 199 of file [SyntaxTree.h](#).

7.29.4.23 root() [2/2]

```
const SyntaxNode & slang::syntax::SyntaxTree::root ( ) const [inline]
```

Gets the root of the syntax tree.

Definition at line 202 of file [SyntaxTree.h](#).

7.29.4.24 sourceManager() [1/2]

```
SourceManager & slang::syntax::SyntaxTree::sourceManager ( ) [inline]
```

Gets the source manager used to build the syntax tree.

Definition at line 190 of file [SyntaxTree.h](#).

7.29.4.25 sourceManager() [2/2]

```
const SourceManager & slang::syntax::SyntaxTree::sourceManager ( ) const [inline]
```

Gets the source manager used to build the syntax tree.

Definition at line 193 of file [SyntaxTree.h](#).

7.29.5 Member Data Documentation

7.29.5.1 alloc

```
BumpAllocator slang::syntax::SyntaxTree::alloc [private]
```

Definition at line 232 of file [SyntaxTree.h](#).

7.29.5.2 diagnosticsBuffer

```
Diagnostics slang::syntax::SyntaxTree::diagnosticsBuffer [private]
```

Definition at line 233 of file [SyntaxTree.h](#).

7.29.5.3 isLibraryUnit

```
bool slang::syntax::SyntaxTree::isLibraryUnit = false
```

Indicates whether this syntax tree represents a "library" compilation unit, which means that modules declared within it are not automatically instantiated.

Definition at line 47 of file [SyntaxTree.h](#).

7.29.5.4 library

```
const SourceLibrary* slang::syntax::SyntaxTree::library [private]
```

Definition at line 230 of file [SyntaxTree.h](#).

7.29.5.5 macros

```
std::vector<const DefineDirectiveSyntax*> slang::syntax::SyntaxTree::macros [private]
```

Definition at line 236 of file [SyntaxTree.h](#).

7.29.5.6 metadata

```
std::unique_ptr<parsing::ParserMetadata> slang::syntax::SyntaxTree::metadata [private]
```

Definition at line 235 of file [SyntaxTree.h](#).

7.29.5.7 options_

Bag slang::syntax::SyntaxTree::options_ [private]

Definition at line 234 of file [SyntaxTree.h](#).

7.29.5.8 rootNode

[SyntaxNode*](#) slang::syntax::SyntaxTree::rootNode [private]

Definition at line 229 of file [SyntaxTree.h](#).

7.29.5.9 sourceMan

SourceManager& slang::syntax::SyntaxTree::sourceMan [private]

Definition at line 231 of file [SyntaxTree.h](#).

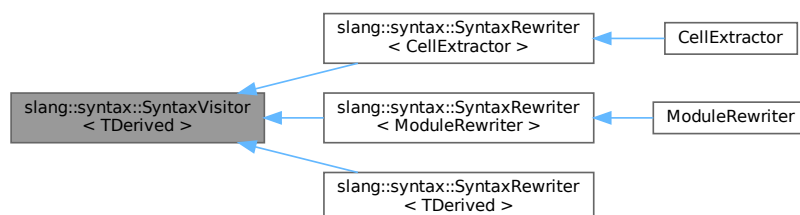
The documentation for this class was generated from the following file:

- include_3rd_party/[SyntaxTree.h](#)

7.30 slang::syntax::SyntaxVisitor< TDerived > Class Template Reference

```
#include <SyntaxVisitor.h>
```

Inheritance diagram for slang::syntax::SyntaxVisitor< TDerived >:



Public Member Functions

- `template<typename T >`
`void visit (T &&t)`
Visit the provided node, of static type T.
- `template<typename T >`
`void visitDefault (T &&node)`

Private Member Functions

- `void visitToken (parsing::Token)`

7.30.1 Detailed Description

```
template<typename TDerived>
class slang::syntax::SyntaxVisitor< TDerived >
```

Use this type as a base class for syntax tree visitors. It will default to traversing all children of each node. Add implementations for any specific node types you want to handle.

Definition at line 25 of file [SyntaxVisitor.h](#).

7.30.2 Member Function Documentation

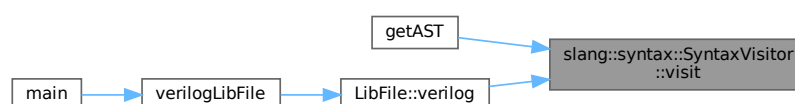
7.30.2.1 visit()

```
template<typename TDerived >
template<typename T >
void slang::syntax::SyntaxVisitor< TDerived >::visit (
    T && t ) [inline]
```

Visit the provided node, of static type T.

Definition at line 29 of file [SyntaxVisitor.h](#).

Here is the caller graph for this function:



7.30.2.2 visitDefault()

```
template<typename TDerived >
template<typename T >
void slang::syntax::SyntaxVisitor< TDerived >::visitDefault (
    T && node ) [inline]
```

The default handler invoked when no [visit\(\)](#) method is overridden for a particular type. Will visit all child nodes by default.

Definition at line 39 of file [SyntaxVisitor.h](#).

Here is the caller graph for this function:



7.30.2.3 visitToken()

```
template<typename TDerived >
void slang::syntax::SyntaxVisitor< TDerived >::visitToken (
    parsing::Token ) [inline], [private]
```

Definition at line 54 of file [SyntaxVisitor.h](#).

The documentation for this class was generated from the following file:

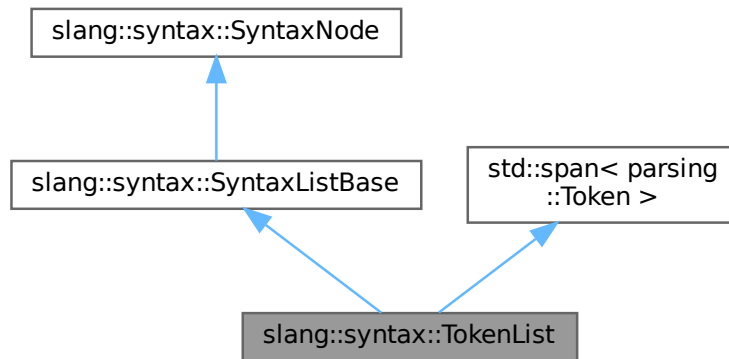
- include_3rd_party/[SyntaxVisitor.h](#)

7.31 slang::syntax::TokenList Class Reference

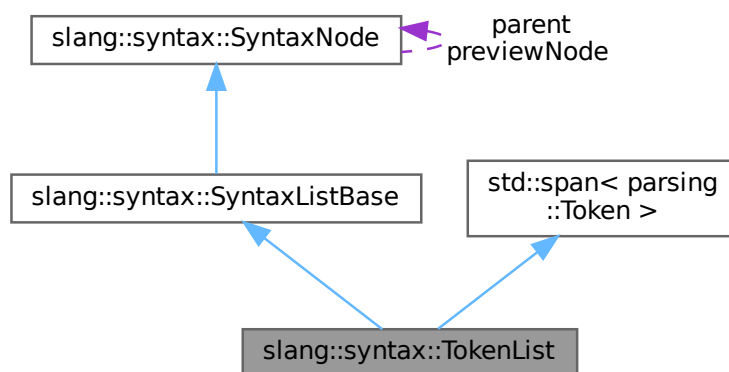
A syntax node that represents a list of child tokens.

```
#include <SyntaxNode.h>
```

Inheritance diagram for slang::syntax::TokenList:



Collaboration diagram for slang::syntax::TokenList:



Public Member Functions

- [TokenList](#) (nullptr_t)
- [TokenList](#) (std::span< [Token](#) > elements)

Private Member Functions

- [TokenOrSyntax getChild](#) (size_t index) final
Gets the child (token or node) at the given index.
- [ConstTokenOrSyntax getChild](#) (size_t index) const final
Gets the child (token or node) at the given index.
- [PtrTokenOrSyntax getChildPtr](#) (size_t index) final
- void [setChild](#) (size_t index, [TokenOrSyntax](#) child) final
Sets the child (token or node) at the given index.
- [SyntaxListBase * clone](#) (BumpAllocator &alloc) const final
Clones the list into a new node using the provided allocator.
- void [resetAll](#) (BumpAllocator &alloc, std::span< const [TokenOrSyntax](#) > children) final

Additional Inherited Members

7.31.1 Detailed Description

A syntax node that represents a list of child tokens.

Definition at line 352 of file [SyntaxNode.h](#).

7.31.2 Constructor & Destructor Documentation

7.31.2.1 TokenList() ^[1/2]

```
slang::syntax::TokenList::TokenList (
    nullptr_t ) [inline]
```

Definition at line 354 of file [SyntaxNode.h](#).

7.31.2.2 TokenList() ^[2/2]

```
slang::syntax::TokenList::TokenList (
    std::span< Token > elements ) [inline]
```

Definition at line 355 of file [SyntaxNode.h](#).

7.31.3 Member Function Documentation

7.31.3.1 clone()

```
SyntaxListBase * slang::syntax::TokenList::clone (  
    BumpAllocator & alloc ) const    [inline], [final], [private], [virtual]
```

Clones the list into a new node using the provided allocator.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 364 of file [SyntaxNode.h](#).

7.31.3.2 getChild() [1/2]

```
ConstTokenOrSyntax slang::syntax::TokenList::getChild (  
    size_t index ) const    [inline], [final], [private], [virtual]
```

Gets the child (token or node) at the given index.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 360 of file [SyntaxNode.h](#).

7.31.3.3 getChild() [2/2]

```
TokenOrSyntax slang::syntax::TokenList::getChild (  
    size_t index )    [inline], [final], [private], [virtual]
```

Gets the child (token or node) at the given index.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 359 of file [SyntaxNode.h](#).

7.31.3.4 getChildPtr()

```
PtrTokenOrSyntax slang::syntax::TokenList::getChildPtr (
    size_t index ) [inline], [final], [private], [virtual]
```

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 361 of file [SyntaxNode.h](#).

7.31.3.5 resetAll()

```
void slang::syntax::TokenList::resetAll (
    BumpAllocator & alloc,
    std::span< const TokenOrSyntax > children ) [inline], [final], [private], [virtual]
```

Overwrites all children with the new set of provided *children* (and making a copy with the provided allocator).

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 368 of file [SyntaxNode.h](#).

7.31.3.6 setChild()

```
void slang::syntax::TokenList::setChild (
    size_t index,
    TokenOrSyntax child ) [inline], [final], [private], [virtual]
```

Sets the child (token or node) at the given index.

Implements [slang::syntax::SyntaxListBase](#).

Definition at line 362 of file [SyntaxNode.h](#).

The documentation for this class was generated from the following file:

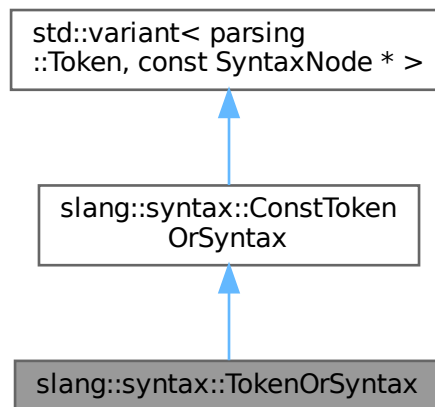
- [include_3rd_party/SyntaxNode.h](#)

7.32 slang::syntax::TokenOrSyntax Struct Reference

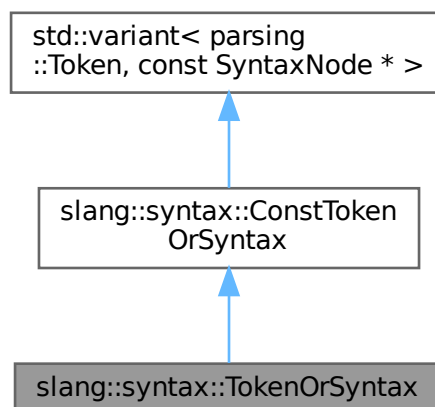
A token or a syntax node.

```
#include <SyntaxNode.h>
```

Inheritance diagram for slang::syntax::TokenOrSyntax:



Collaboration diagram for slang::syntax::TokenOrSyntax:



Public Member Functions

- [TokenOrSyntax](#) (parsing::Token *token*)
- [TokenOrSyntax](#) (SyntaxNode **node*)
- [TokenOrSyntax](#) (nullptr_t)
- [SyntaxNode](#) * *node* () const

Additional Inherited Members

7.32.1 Detailed Description

A token or a syntax node.

Definition at line 75 of file [SyntaxNode.h](#).

7.32.2 Constructor & Destructor Documentation

7.32.2.1 TokenOrSyntax() [1/3]

```
slang::syntax::TokenOrSyntax::TokenOrSyntax (  
    parsing::Token token ) [inline]
```

Definition at line 76 of file [SyntaxNode.h](#).

7.32.2.2 TokenOrSyntax() [2/3]

```
slang::syntax::TokenOrSyntax::TokenOrSyntax (  
    SyntaxNode * node ) [inline]
```

Definition at line 77 of file [SyntaxNode.h](#).

7.32.2.3 TokenOrSyntax() [3/3]

```
slang::syntax::TokenOrSyntax::TokenOrSyntax (  
    nullptr_t ) [inline]
```

Definition at line 78 of file [SyntaxNode.h](#).

7.32.3 Member Function Documentation

7.32.3.1 node()

```
SyntaxNode * slang::syntax::TokenOrSyntax::node ( ) const [inline]
```

Gets access to the object as a syntax node (throwing an exception if it's not actually a syntax node).

Definition at line 82 of file [SyntaxNode.h](#).

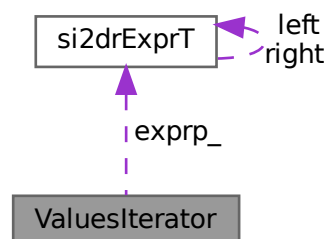
The documentation for this struct was generated from the following file:

- [include_3rd_party/SyntaxNode.h](#)

7.33 ValuesIterator Class Reference

```
#include <Iterators.hpp>
```

Collaboration diagram for ValuesIterator:



Public Member Functions

- [ValuesIterator](#) ([si2drValuesIdT](#) values, [si2drErrorT](#) &err)
- [~ValuesIterator](#) ()
- void [next](#) ()
- bool [end](#) ()

Public Attributes

- [si2drValuesIdT values_](#)
- [si2drValueTypeT vtype_](#)
- [si2drInt32T int_](#)
- [si2drFloat64T float_](#)
- [si2drStringT str_](#)
- [si2drBooleanT bool_](#)
- [si2drExprT * exprp_](#)
- [si2drErrorT & err_](#)

7.33.1 Detailed Description

Definition at line 37 of file [lterators.hpp](#).

7.33.2 Constructor & Destructor Documentation

7.33.2.1 ValuesIterator()

```
ValuesIterator::ValuesIterator (
    si2drValuesIdT values,
    si2drErrorT & err )
```

Definition at line 24 of file [lterators.cpp](#).

7.33.2.2 ~ValuesIterator()

```
ValuesIterator::~ValuesIterator ( )
```

Definition at line 27 of file [lterators.cpp](#).

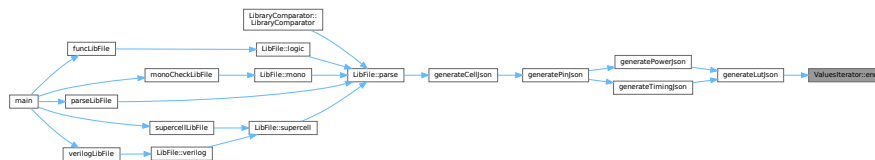
7.33.3 Member Function Documentation

7.33.3.1 end()

```
bool ValuesIterator::end ( )
```

Definition at line 32 of file [Iterators.cpp](#).

Here is the caller graph for this function:

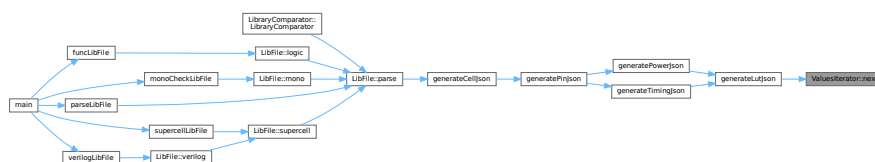


7.33.3.2 next()

```
void ValuesIterator::next ( )
```

Definition at line 29 of file [Iterators.cpp](#).

Here is the caller graph for this function:



7.33.4 Member Data Documentation

7.33.4.1 bool_

```
si2drBooleanT ValuesIterator::bool_
```

Definition at line 49 of file [Iterators.hpp](#).

7.33.4.2 err_

`si2drErrorT& ValuesIterator::err_`

Definition at line 51 of file [lterators.hpp](#).

7.33.4.3 exprp_

`si2drExprT* ValuesIterator::exprp_`

Definition at line 50 of file [lterators.hpp](#).

7.33.4.4 float_

`si2drFloat64T ValuesIterator::float_`

Definition at line 47 of file [lterators.hpp](#).

7.33.4.5 int_

`si2drInt32T ValuesIterator::int_`

Definition at line 46 of file [lterators.hpp](#).

7.33.4.6 str_

`si2drStringT ValuesIterator::str_`

Definition at line 48 of file [lterators.hpp](#).

7.33.4.7 values_

[si2drValuesIdT](#) ValuesIterator::values_

Definition at line 44 of file [lterators.hpp](#).

7.33.4.8 vtype_

[si2drValueTypeT](#) ValuesIterator::vtype_

Definition at line 45 of file [lterators.hpp](#).

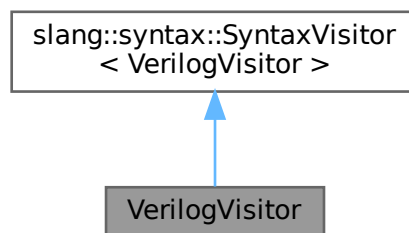
The documentation for this class was generated from the following files:

- include/[lterators.hpp](#)
- src/[lterators.cpp](#)

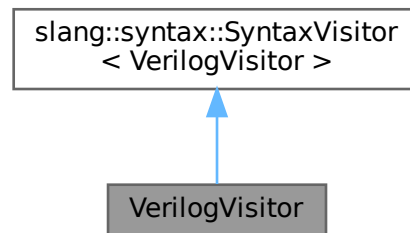
7.34 VerilogVisitor Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for VerilogVisitor:



Collaboration diagram for VerilogVisitor:



Public Member Functions

- `VerilogVisitor` (const std::string &targetCell)
- void `handle` (const slang::syntax::SyntaxNode &node)
- void `handle` (const slang::syntax::ModuleDeclarationSyntax &module)
- void `handle` (const slang::syntax::PortDeclarationSyntax &portDecl)
- void `handle` (const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst)
- void `handle` (const slang::syntax::SpecifyBlockSyntax &specifyBlock)

Private Attributes

- const std::string & `targetCell_`
- int `depth_`
- bool `inTargetModule_`

7.34.1 Detailed Description

Definition at line 21 of file [verilog_utils.hpp](#).

7.34.2 Constructor & Destructor Documentation

7.34.2.1 VerilogVisitor()

```
VerilogVisitor::VerilogVisitor (
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 23 of file [verilog_utils.hpp](#).

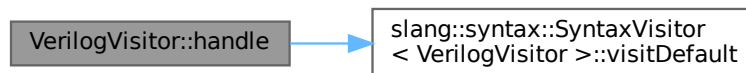
7.34.3 Member Function Documentation

7.34.3.1 handle() [1/5]

```
void VerilogVisitor::handle (
    const slang::syntax::HierarchyInstantiationSyntax & hierarchyInst )
```

Definition at line 78 of file [verilog_utils.cpp](#).

Here is the call graph for this function:

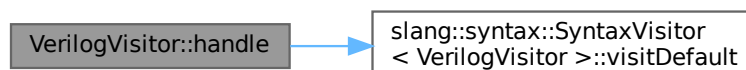


7.34.3.2 handle() [2/5]

```
void VerilogVisitor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 17 of file [verilog_utils.cpp](#).

Here is the call graph for this function:



7.34.3.3 handle() [3/5]

```
void VerilogVisitor::handle (
    const slang::syntax::PortDeclarationSyntax & portDecl )
```

Definition at line 46 of file [verilog_utils.cpp](#).

7.34.3.4 handle() [4/5]

```
void VerilogVisitor::handle (
    const slang::syntax::SpecifyBlockSyntax & specifyBlock )
```

Definition at line 258 of file [verilog_utils.cpp](#).

Here is the call graph for this function:

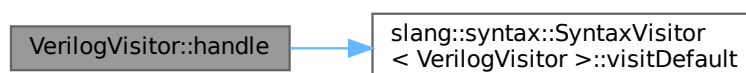


7.34.3.5 handle() [5/5]

```
void VerilogVisitor::handle (
    const slang::syntax::SyntaxNode & node )
```

Definition at line 6 of file [verilog_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



7.34.4 Member Data Documentation

7.34.4.1 depth_

```
int VerilogVisitor::depth_ [private]
```

Definition at line 34 of file [verilog_utils.hpp](#).

7.34.4.2 inTargetModule_

```
bool VerilogVisitor::inTargetModule_ [private]
```

Definition at line 35 of file [verilog_utils.hpp](#).

7.34.4.3 targetCell_

```
const std::string& VerilogVisitor::targetCell_ [private]
```

Definition at line 33 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- [include/verilog_utils.hpp](#)
- [src/verilog_utils.cpp](#)

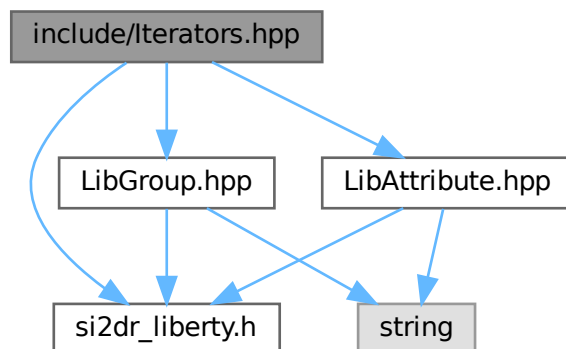
Chapter 8

File Documentation

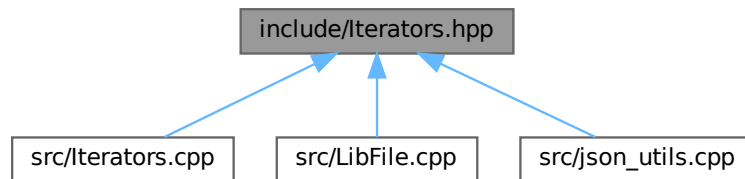
8.1 include/Iterators.hpp File Reference

```
#include "si2dr_liberty.h"  
#include "LibAttribute.hpp"  
#include "LibGroup.hpp"
```

Include dependency graph for Iterators.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [GroupsIterator](#)
- class [AttributesIterator](#)
- class [ValuesIterator](#)

8.2 Iterators.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef ITERATORS_H
00002 #define ITERATORS_H
00003
00004 #include "si2dr_liberty.h"
00005
00006 #include "LibAttribute.hpp"
00007 #include "LibGroup.hpp"
00008
00009 class GroupsIterator {
00010 public:
00011     GroupsIterator(si2drGroupsIdT groups, si2drErrorT &err);
00012     ~GroupsIterator();
00013     void next();
00014     bool end();
00015     LibGroup get();
00016
00017 private:
00018     si2drGroupsIdT groups_;
00019     si2drGroupIdT group_;
00020     si2drErrorT &err_;
00021 };
00022
00023 class AttributesIterator {
00024 public:
00025     AttributesIterator(si2drAttrsIdT attrs, si2drErrorT &err);
00026     ~AttributesIterator();
00027     void next();
00028     bool end();
00029     LibAttribute get();
00030
00031 private:
00032     si2drAttrsIdT attrs_;
  
```

```

00033     si2drAttrIdT attr_;
00034     si2drErrorT &err_;
00035 };
00036
00037 class ValuesIterator {
00038 public:
00039     ValuesIterator(si2drValuesIdT values, si2drErrorT &err);
00040     ~ValuesIterator();
00041     void next();
00042     bool end();
00043
00044     si2drValuesIdT values_;
00045     si2drValueTypeT vtype_;
00046     si2drInt32T int_;
00047     si2drFloat64T float_;
00048     si2drStringT str_;
00049     si2drBooleanT bool_;
00050     si2drExprT *exprp_;
00051     si2drErrorT &err_;
00052 };
00053
00054 #endif // ITERATORS_H

```

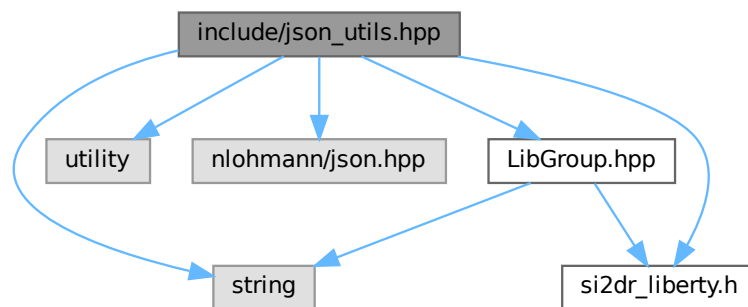
8.3 include/json_utils.hpp File Reference

```

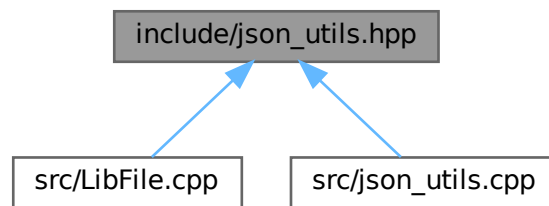
#include <string>
#include <utility>
#include "nlohmann/json.hpp"
#include "si2dr_liberty.h"
#include "LibGroup.hpp"

```

Include dependency graph for json_utils.hpp:



This graph shows which files directly or indirectly include this file:



Typedefs

- using `json` = `nlohmann::json`

Functions

- `json generateLutJson` (`LibGroup` &`lib_lut_group`, `si2drErrorT` &`err`)
Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.
- `json generatePowerJson` (`LibGroup` &`lib_power_group`, `si2drErrorT` &`err`)
Converts a Liberty power group into a JSON representation.
- `std::pair< std::string, json > generatePinJson` (`LibGroup` &`lib_pin_group`, `si2drErrorT` &`err`)
Generates a JSON representation of a Liberty pin group.
- `json generateCellJson` (`LibGroup` &`lib_cell_group`, `si2drErrorT` &`err`)
Generates a JSON representation of a cell from a [LibGroup](#) object.

8.3.1 Typedef Documentation

8.3.1.1 json

```
using json = nlohmann::json
```

Definition at line 12 of file [json_utils.hpp](#).

8.3.2 Function Documentation

8.3.2.1 generateCellJson()

```
json generateCellJson (  
    LibGroup & lib_cell_group,  
    si2drErrorT & err )
```

Generates a JSON representation of a cell from a [LibGroup](#) object.

This function processes a [LibGroup](#) representing a cell and converts it into a JSON object. It extracts the cell name and processes specific attributes such as area, cell_leakage_power, and cell_footprint. It also processes pins within the cell by categorizing them based on their direction (input, output, internal, inout).

Parameters

<i>lib_cell_group</i>	The LibGroup object representing the cell
<i>err</i>	Reference to a si2drErrorT object for error handling

Returns

json A JSON object containing the cell's information with the following structure:

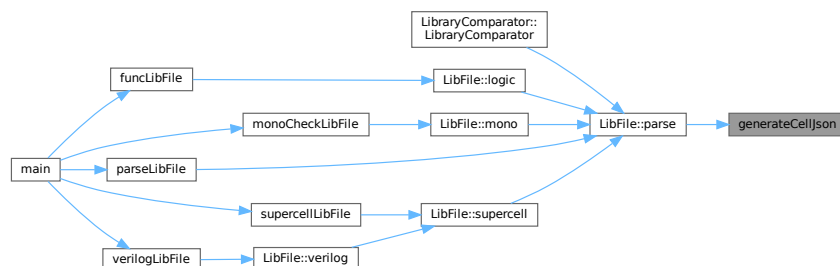
- "cell_name": string - name of the cell
- "area": float (optional) - cell area if present
- "cell_leakage_power": float (optional) - cell leakage power if present
- "cell_footprint": string (optional) - cell footprint if present
- "input_pins": array - list of input pins
- "output_pins": array - list of output pins
- "internal_pins": array - list of internal pins
- "inout_pins": array - list of inout pins

Definition at line 272 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.3.2.2 generateLutJson()

```
json generateLutJson (
    LibGroup & lib_lut_group,
    si2drErrorT & err )
```

Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.

This function iterates through the attributes of the provided [LibGroup](#) object representing a LUT and converts them into a JSON structure. It specifically handles the following attributes:

- "index_1": Converted to a vector and stored in the JSON
- "index_2": Converted to a vector and stored in the JSON
- "values": Each value is parsed into a vector and added to an array in the JSON

Any other attributes encountered will trigger a warning message.

Parameters

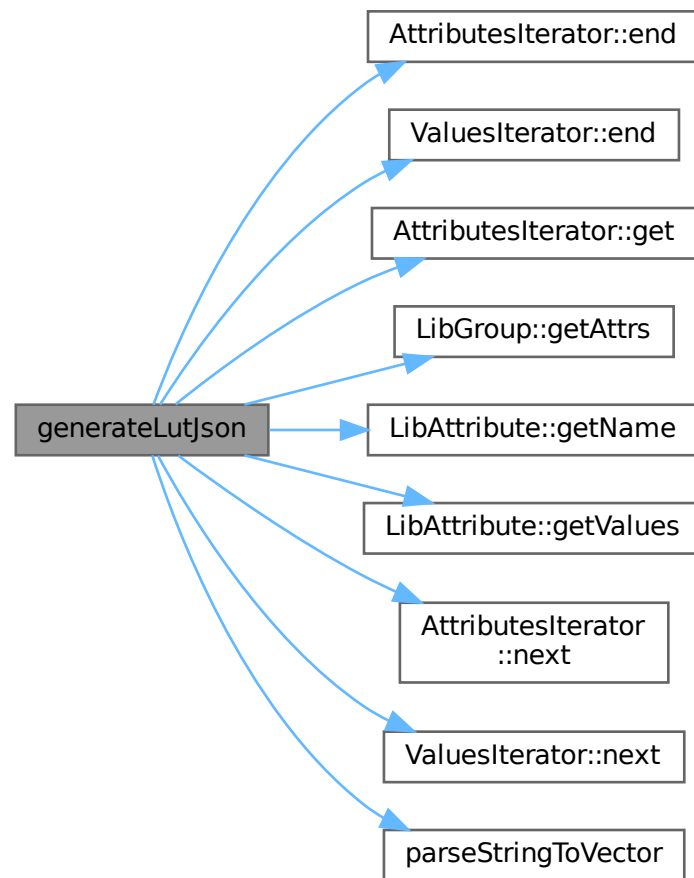
<i>lib_lut_group</i>	The LibGroup object containing the LUT data to be converted
<i>err</i>	Reference to an <code>si2drErrorT</code> object to track any errors during processing

Returns

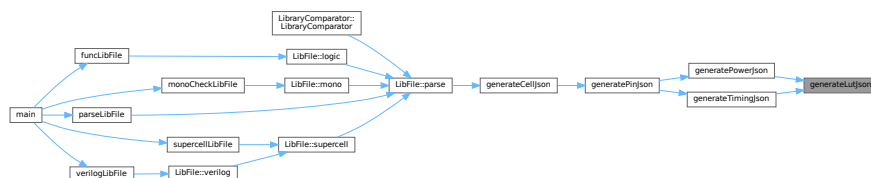
json A JSON object representing the LUT data with keys for "index_1", "index_2", and "values"

Definition at line 62 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.3.2.3 generatePinJson()

```
std::pair< std::string, json > generatePinJson (
```



```
LibGroup & lib_pin_group,  
si2drErrorT & err )
```

Generates a JSON representation of a Liberty pin group.

This function traverses a Liberty pin group, extracts relevant attributes and sub-groups, and converts them into a JSON object. It also determines the pin's direction.

Processes the following pin attributes:

- direction
- max_transition, capacitance, rise_capacitance, fall_capacitance, max_capacitance (float types)
- function, power_down_function, related_ground_pin, related_power_pin, three_state (string types)
- clock (boolean type)

Handles the following sub-groups:

- internal_power: Converted using [generatePowerJson\(\)](#)
- timing: Converted using [generateTimingJson\(\)](#)

Parameters

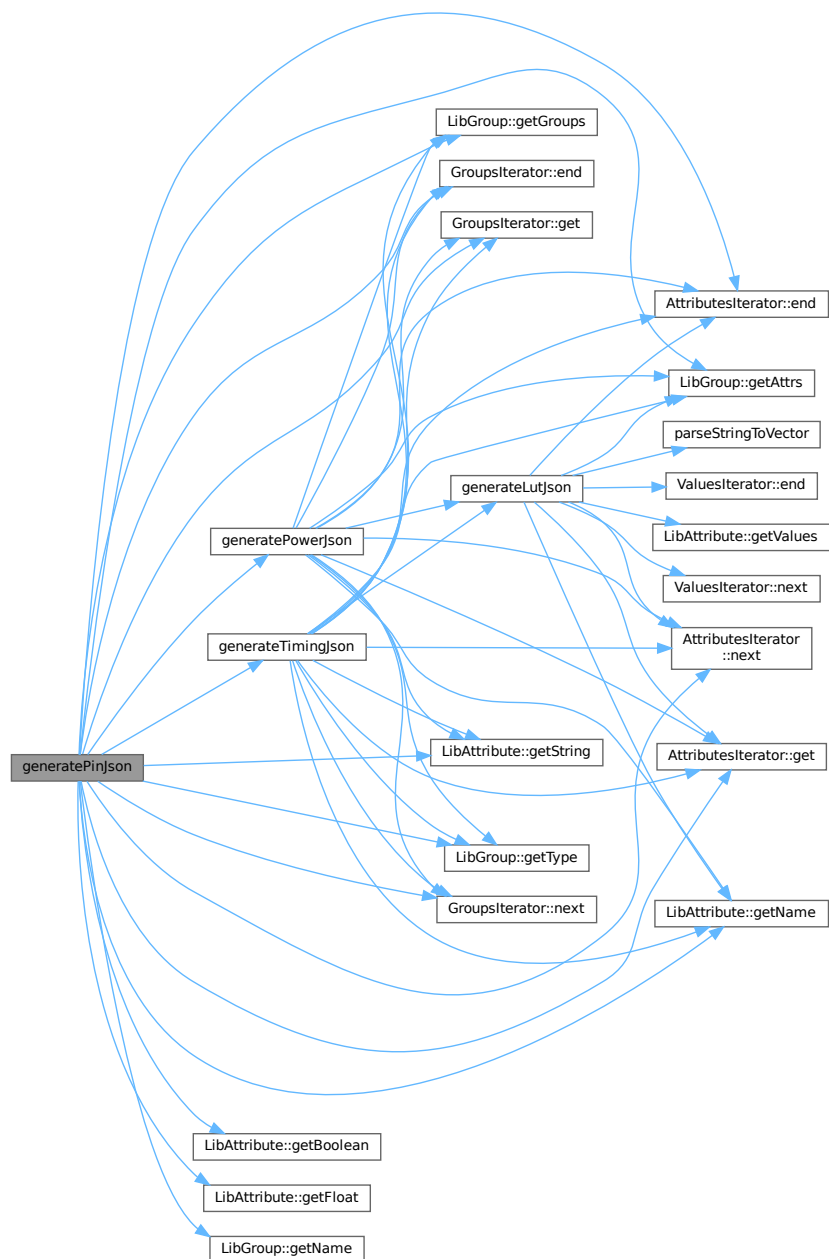
<i>lib_pin_group</i>	The Liberty pin group to process
<i>err</i>	Reference to an error object for tracking Liberty API errors

Returns

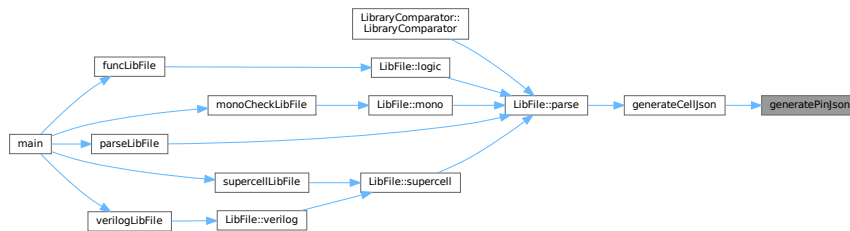
A pair containing the pin direction (string) and the JSON representation of the pin

Definition at line 206 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.3.2.4 generatePowerJson()

```

json generatePowerJson (
    LibGroup & lib_power_group,
    si2drErrorT & err )

```

Converts a Liberty power group into a JSON representation.

This function processes a Liberty power group and converts its attributes and subgroups into a JSON object. It handles attributes like "when", "related_pin", and "related_pg_pin", as well as "rise_power" and "fall_power" subgroups.

Parameters

<i>lib_power_group</i>	The Liberty power group to convert
<i>err</i>	Reference to an si2drErrorT object for error handling

Returns

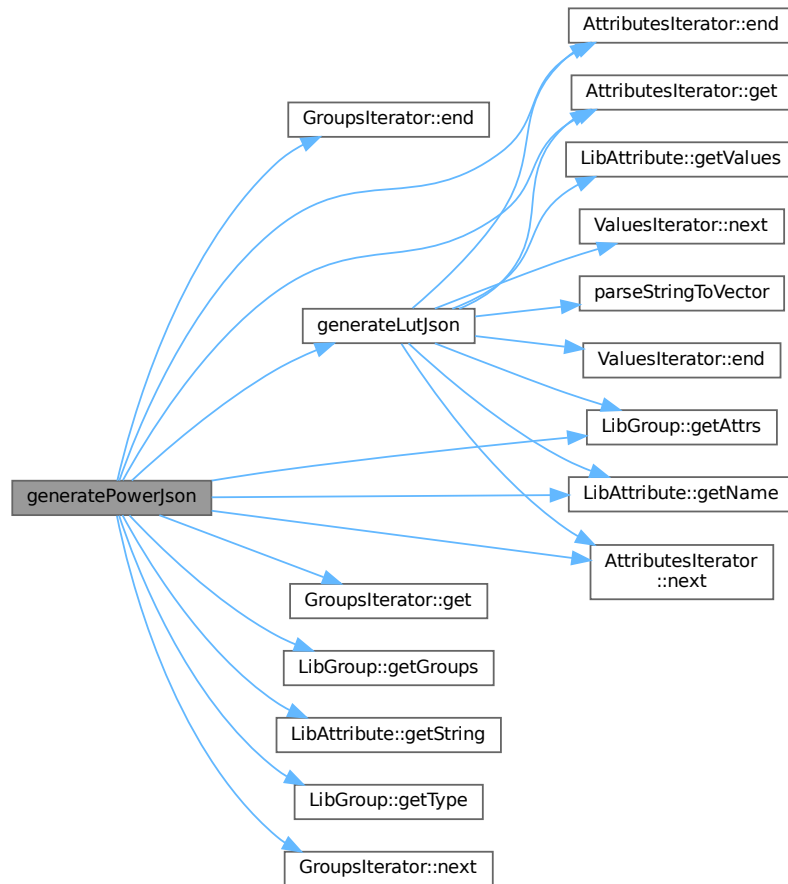
json A JSON object representing the power group data

The function:

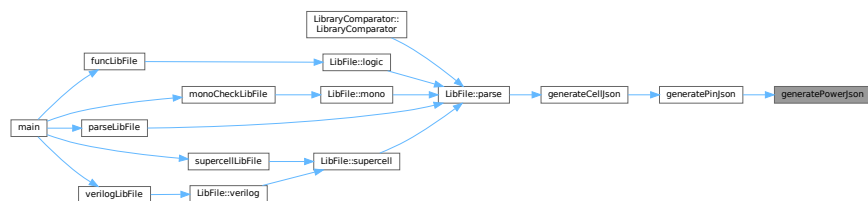
- Extracts string attributes (when, related_pin, related_pg_pin)
- Processes rise_power and fall_power subgroups by converting them to LUT JSON format
- Logs warnings for unknown power subgroup types

Definition at line 154 of file json_utils.cpp.

Here is the call graph for this function:



Here is the caller graph for this function:



8.4 json_utils.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef JSON_UTILS_HPP
00002 #define JSON_UTILS_HPP

```

```

00003
00004 #include <string>
00005 #include <utility>
00006
00007 #include "nlohmann/json.hpp"
00008 #include "si2dr_liberty.h"
00009
00010 #include "LibGroup.hpp"
00011
00012 using json = nlohmann::json;
00013
00014 json generateLutJson(LibGroup &lib_lut_group, si2drErrorT &err);
00015 json generatePowerJson(LibGroup &lib_power_group, si2drErrorT &err);
00016 std::pair<std::string, json> generatePinJson(LibGroup &lib_pin_group, si2drErrorT &err);
00017 json generateCellJson(LibGroup &lib_cell_group, si2drErrorT &err);
00018
00019 #endif // JSON_UTILS_HPP

```

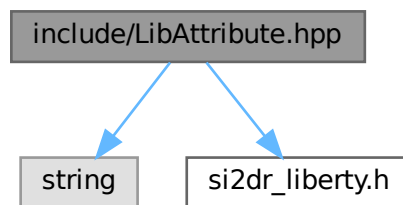
8.5 include/LibAttribute.hpp File Reference

```

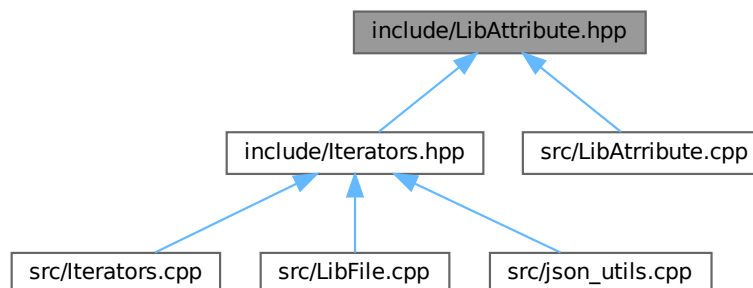
#include <string>
#include "si2dr_liberty.h"

```

Include dependency graph for LibAttribute.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [LibAttribute](#)

8.6 LibAttribute.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef LIB_ATTRIBUTE_H
00002 #define LIB_ATTRIBUTE_H
00003
00004 #include <string>
00005
00006 #include "si2dr_liberty.h"
00007
00008 class LibAttribute {
00009 public:
00010     LibAttribute(si2drAttrIdT attr, si2drErrorT &err);
00011     ~LibAttribute();
00012     std::string getName();
00013     bool isComplex();
00014     si2drValuesIdT getValues();
00015     long int getInt();
00016     double getFloat();
00017     std::string getString();
00018     bool getBoolean();
00019
00020 private:
00021     si2drAttrIdT attr_;
00022     si2drErrorT &err_;
00023 };
00024
00025 #endif // LIB_ATTRIBUTE_H

```

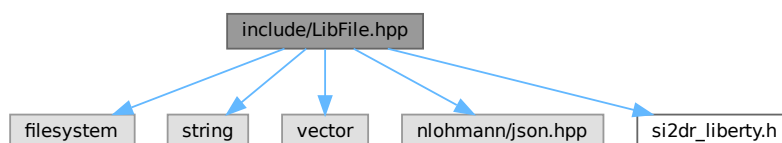
8.7 include/LibFile.hpp File Reference

```

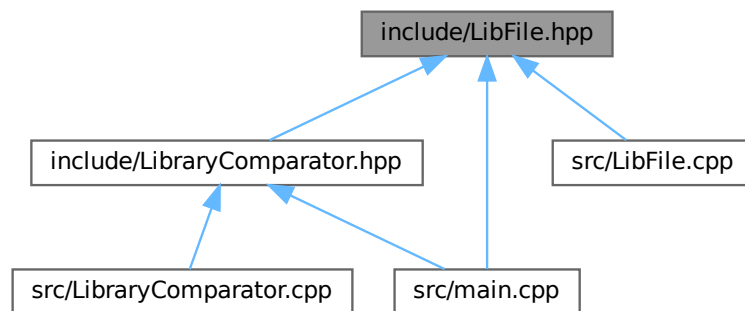
#include <filesystem>
#include <string>
#include <vector>
#include "nlohmann/json.hpp"
#include "si2dr_liberty.h"

```

Include dependency graph for LibFile.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [LibFile](#)

Typedefs

- using [json](#) = `nlohmann::json`

8.7.1 Typedef Documentation

8.7.1.1 json

```
using json = nlohmann::json
```

Definition at line 11 of file [LibFile.hpp](#).

8.8 LibFile.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef LIB_FILE_H
00002 #define LIB_FILE_H
00003
00004 #include <filesystem>
00005 #include <string>
00006 #include <vector>
00007
00008 #include "nlohmann/json.hpp"
00009 #include "si2dr_liberty.h"
00010
00011 using json = nlohmann::json;
00012
00013 class LibFile {
00014 public:
00015     LibFile(const std::string &filepath, const std::string &loggername);
00016     ~LibFile();
00017     std::shared_ptr<spdlog::logger> logger_;
00018     std::filesystem::path filepath_; // full path to the file
00019     std::string basename_;          // file name without extension
00020     std::string filename_;          // full file name with extension
00021     std::string libname_ = "";      // library name obtained through parsing
00022     std::string jsonname_ = "";     // json file name to store parsed data
00023     std::string loggername_ = "";   // log file name
00024     json lib_json_ = json::object();
00025     void writeJsonToFile();
00026     void parse();
00027     void modify();
00028     void mono(const bool is_slew);
00029     void supercell(const int chain_length, const std::vector<std::string> &cell_names);
00030     void verilog(const int chain_length, const std::vector<std::string> &cell_names);
00031     std::map<std::string, std::string> logic(const std::string &cell_name);
00032
00033 private:
00034     si2drErrorT err_;
00035     int process_ = 0;
00036     float voltage_ = 0.0;
00037     int temperature_ = 0;
00038     void read();
00039     bool checkTimingArcMonotonicity(const json &cell, const json &pin, const json &arc,
00040                                     const std::string &type, bool is_slew);
00041 };
00042
00043 #endif // LIB_FILE_H

```

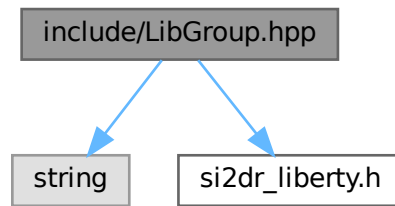
8.9 include/LibGroup.hpp File Reference

```

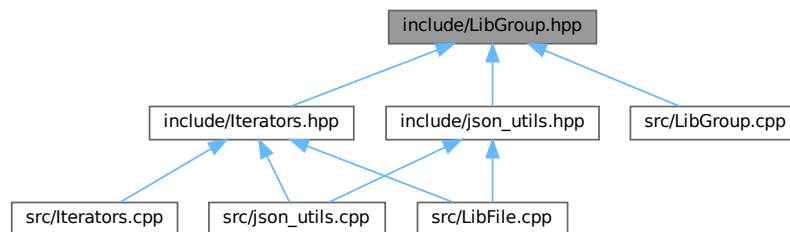
#include <string>
#include "si2dr_liberty.h"

```


Include dependency graph for LibGroup.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [LibGroup](#)

8.10 LibGroup.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef LIB_GROUP_H
00002 #define LIB_GROUP_H
00003
00004 #include <string>
00005
00006 #include "si2dr_liberty.h"
00007
00008 class LibGroup {
00009 public:
00010     LibGroup(si2drGroupIdT group, si2drErrorT &err);
00011     ~LibGroup();
00012     std::string getName();
00013     std::string getType();
00014     si2drAttrIdT getAttrs();
  
```

```

00015     si2drGroupsIdT getGroups();
00016
00017 private:
00018     si2drGroupIdT group_;
00019     si2drErrorT &err_;
00020 };
00021
00022 #endif // LIB_GROUP_H

```

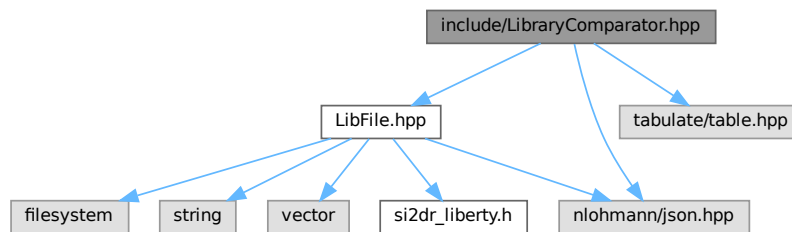
8.11 include/LibraryComparator.hpp File Reference

```

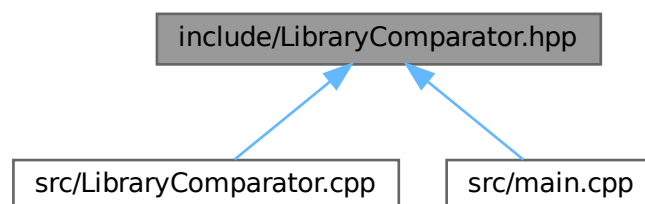
#include "LibFile.hpp"
#include "nlohmann/json.hpp"
#include "tabulate/table.hpp"

```

Include dependency graph for LibraryComparator.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class `LibraryComparator`

Typedefs

- using `json` = `nlohmann::json`

8.11.1 Typedef Documentation

8.11.1.1 json

using `json` = `nlohmann::json`

Definition at line 9 of file [LibraryComparator.hpp](#).

8.12 LibraryComparator.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef COMPARE_HPP
00002 #define COMPARE_HPP
00003
00004 #include "LibFile.hpp"
00005
00006 #include "nlohmann/json.hpp"
00007 #include "tabulate/table.hpp"
00008
00009 using json = nlohmann::json;
00010 using namespace tabulate;
00011
00012 class LibraryComparator {
00013 public:
00014     LibraryComparator(LibFile &ref_libfile, LibFile &comp_libfile, double reltol, double abstol);
00015     std::filesystem::path ref_lib_path_;
00016     std::filesystem::path comp_lib_path_;
00017     void generateReport(const std::string &output_file);
00018
00019 private:
00020     json ref_json_;
00021     json comp_json_;
00022     double reltol_;
00023     double abstol_;
00024
00025     void compareCell(const std::string &cell_name, const json &ref_cell, const json &comp_cell,
00026                     Table &table);
00027     void comparePin(const std::string &cell_name, const std::string &pin_name, const json &ref_pin,
00028                    const json &comp_pin, Table &table);
00029     void compareTimingArc(const std::string &cell_name, const std::string &pin_name,
00030                           const std::string &timing_type, const json &ref_timing_arc,
00031                           const json &comp_timing_arc, Table &table);
00032     void compareLut(const std::string &cell_name, const std::string &pin_name,
00033                     const std::string &timing_type, const std::string &related_pin,
00034                     const std::string &arc_name, const json &ref_lut, const json &comp_lut, Table &table);
00035     // void configureTableFormat(Table &table);
00036 };
00037
00038 #endif // COMPARE_HPP
```

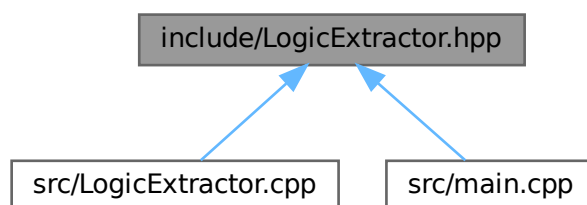
8.13 include/LogicExtractor.hpp File Reference

```
#include <fstream>
#include <iostream>
#include <map>
#include <string>
#include <unordered_map>
#include <unordered_set>
#include <vector>
#include "slang/parsing/Token.h"
#include "slang/syntax/SyntaxPrinter.h"
#include "slang/syntax/SyntaxVisitor.h"
#include "spdlog/spdlog.h"
#include "verilog_utils.hpp"
```

Include dependency graph for LogicExtractor.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- struct [GateInfo](#)
- class [LogicExtractor](#)

Functions

- void [extractAndPrintNetlistInfo](#) (const std::string &verilog_file, const std::string &cell)
- std::map< std::string, std::string > [extractLogicFromVerilog](#) (const std::string &verilog_file, const std::string &cell)

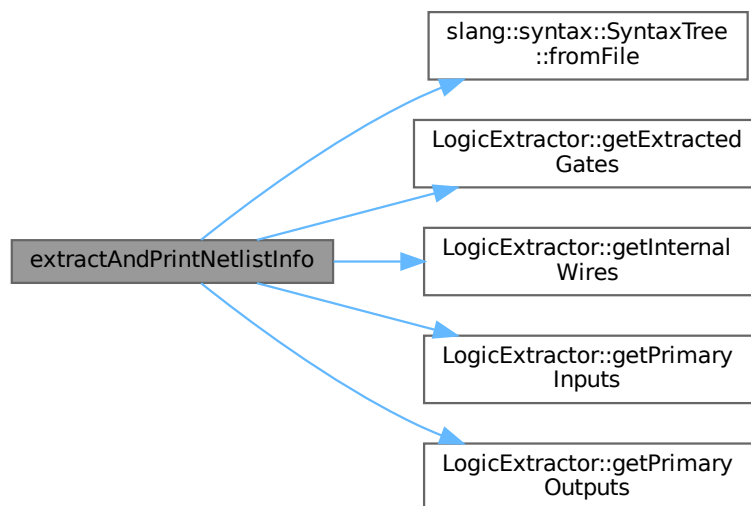
8.13.1 Function Documentation

8.13.1.1 extractAndPrintNetlistInfo()

```
void extractAndPrintNetlistInfo (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 514 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:

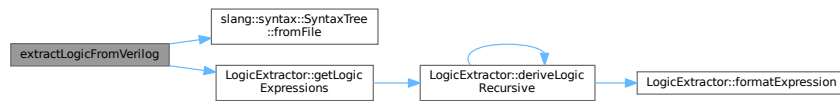


8.13.1.2 extractLogicFromVerilog()

```
std::map< std::string, std::string > extractLogicFromVerilog (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 575 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.14 LogicExtractor.hpp

[Go to the documentation of this file.](#)

```

00001 // include/LogicExtractor.hpp
00002
00003 #ifndef LOGIC_EXTRACTOR_HPP
00004 #define LOGIC_EXTRACTOR_HPP
00005
00006 #include <fstream>
00007 #include <iostream>
00008 #include <map>
00009 #include <string>
00010 #include <unordered_map>
00011 #include <unordered_set>
00012 #include <vector>
00013
00014 #include "slang/parsing/Token.h" // Include Token header for TokenKind
00015 #include "slang/syntax/SyntaxPrinter.h"
00016 #include "slang/syntax/SyntaxVisitor.h"
00017 #include "spdlog/spdlog.h" // Include spdlog
00018
00019 #include "verilog_utils.hpp"
00020
00021 // Structure to represent a gate instance's information
00022 struct GateInfo {
00023     slang::parsing::TokenKind kind; // Store the TokenKind for reliable checks
00024     std::string gateTypeName;       // Store the string name like "and", "xor"
00025     std::vector<std::string> inputSignals;
00026     std::string outputSignal;
00027 };
00028
00029 // Visitor class to extract netlist information and derive logic expressions
00030 class LogicExtractor : public slang::syntax::SyntaxVisitor<LogicExtractor> {
00031 public:
00032     // Constructor takes the target cell name
00033     explicit LogicExtractor(const std::string &targetCell)
00034         : targetCell_(targetCell), inTargetModule_(false), parsingComplete_(false) {}
00035

```

```

00036 // --- Visitor Handlers ---
00037 void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00038 void handle(
00039     const slang::syntax::PortDeclarationSyntax &portDecl); // Handles direction/type declaration
00040 void
00041 handle(const slang::syntax::NonAnsiPortListSyntax &portList); // Handles port names list in header
00042 void
00043 handle(const slang::syntax::NetDeclarationSyntax &netDecl); // To find explicitly declared wires
00044 void handle(const slang::syntax::PrimitiveInstantiationSyntax
00045     &primitiveInst); // Handles gate instantiation
00046 // Potentially handle ContinuousAssignSyntax if needed later:
00047 // void handle(const slang::syntax::ContinuousAssignSyntax& assign);
00048
00049 // --- Access Extracted Info (for Step 1 debugging) ---
00050 const std::unordered_map<std::string, GateInfo> &getExtractedGates()const {
00051     return gateOutputDrivers_;
00052 }
00053 const std::unordered_set<std::string> &getPrimaryInputs()const { return primaryInputs_; }
00054 const std::unordered_set<std::string> &getPrimaryOutputs()const { return primaryOutputs_; }
00055 const std::unordered_set<std::string> &getInternalWires()const { return internalWires_; }
00056
00057 // --- Logic Derivation (Commented out for Step 1) ---
00058 std::map<std::string, std::string> getLogicExpressions();
00059
00060 private:
00061 // --- Internal State ---
00062 const std::string &targetCell_;
00063 bool inTargetModule_;
00064 bool parsingComplete_; // Flag to indicate AST traversal is done
00065
00066 // Netlist Information
00067 std::unordered_set<std::string> primaryInputs_;
00068 std::unordered_set<std::string> primaryOutputs_;
00069 std::unordered_set<std::string> internalWires_; // Includes gate outputs
00070
00071 // Temporary map to store directions found in PortDeclarationSyntax
00072 // Key: port name, Value: direction ("input", "output", "inout", "unknown")
00073 std::unordered_map<std::string, std::string> portDirections_;
00074
00075 // Map: output signal name -> GateInfo driving it
00076 std::unordered_map<std::string, GateInfo> gateOutputDrivers_;
00077
00078 // Map: signal name -> Logic expression string (Memoization Cache) - (Used in Step 2)
00079 std::unordered_map<std::string, std::string> logicCache_;
00080
00081 // --- Helper Methods ---
00082 // Recursive function to derive logic for a given signal (Used in Step 2)
00083 std::string deriveLogicRecursive(const std::string &signalName);
00084 // Helper to format expressions based on gate type (Used in Step 2)
00085 std::string formatExpression(const GateInfo &gateInfo,
00086     const std::vector<std::string> &inputExprs);
00087 };
00088
00089 // --- Function Declaration ---
00090 // For Step 1, this function will just run the visitor and maybe print summary
00091 void extractAndPrintNetlistInfo(const std::string &verilog_file, const std::string &cell);
00092
00093 // Function to be implemented in Step 2
00094 std::map<std::string, std::string> extractLogicFromVerilog(const std::string &verilog_file,
00095     const std::string &cell);
00096
00097 #endif // LOGIC_EXTRACTOR_HPP

```

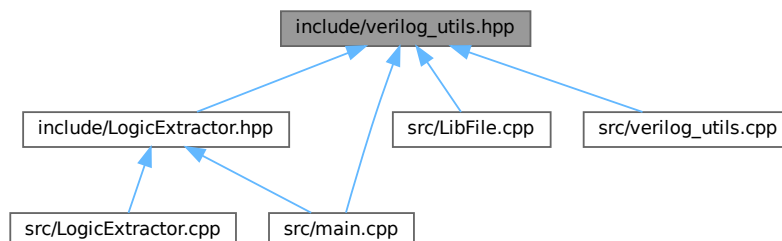
8.15 include/verilog_utils.hpp File Reference

```
#include <algorithm>
#include <fstream>
#include <iostream>
#include <map>
#include <string>
#include <unordered_map>
#include <unordered_set>
#include <vector>
#include "slang/parsing/Token.h"
#include "slang/syntax/SyntaxPrinter.h"
#include "slang/syntax/SyntaxVisitor.h"
#include "spdlog/spdlog.h"
```

Include dependency graph for verilog_utils.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [VerilogVisitor](#)
- class [CellExtractor](#)
- class [CellPrinter](#)
- class [ModuleRewriter](#)

Functions

- void [getAST](#) (const std::string &verilog_file, const std::string &cell)

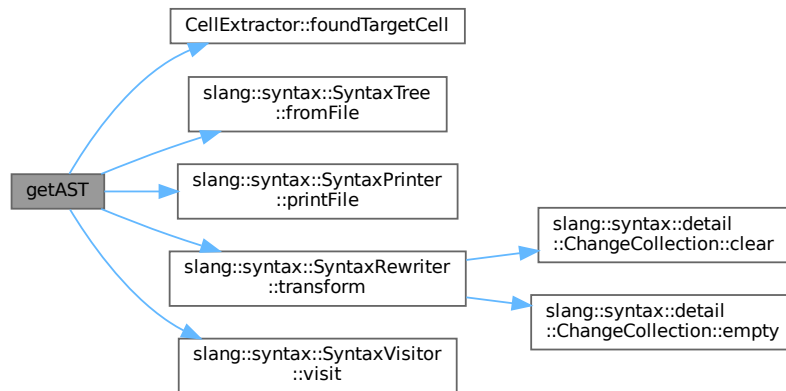
8.15.1 Function Documentation

8.15.1.1 getAST()

```
void getAST (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 544 of file [verilog_utils.cpp](#).

Here is the call graph for this function:



8.16 verilog_utils.hpp

[Go to the documentation of this file.](#)

```
00001 // verilog_utils.hpp
00002
00003 #ifndef VERILOG_UTILS_H
00004 #define VERILOG_UTILS_H
00005
00006 #include <algorithm> // For std::find
00007 #include <fstream>
00008 #include <iostream>
00009 #include <map>
00010 #include <string>
00011 #include <unordered_map>
00012 #include <unordered_set>
00013 #include <vector>
00014
00015 #include "slang/parsing/Token.h" // Include Token header for TokenKind
00016 #include "slang/syntax/SyntaxPrinter.h"
00017 #include "slang/syntax/SyntaxVisitor.h"
```

```

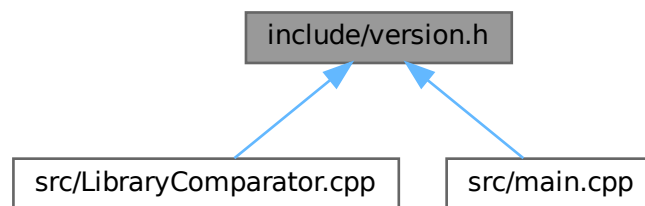
00018 #include "spdlog/spdlog.h" // Include spdlog
00019
00020 // Creating custom visitor class
00021 class VerilogVisitor : public slang::syntax::SyntaxVisitor<VerilogVisitor> {
00022 public:
00023     explicit VerilogVisitor(const std::string &targetCell)
00024         : targetCell_(targetCell), depth_(0), inTargetModule_(false) {}
00025     void handle(const slang::syntax::SyntaxNode &node);
00026     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00027     void handle(const slang::syntax::PortDeclarationSyntax &portDecl);
00028     void handle(const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst);
00029     // void handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst);
00030     void handle(const slang::syntax::SpecifyBlockSyntax &specifyBlock);
00031
00032 private:
00033     const std::string &targetCell_;
00034     int depth_;
00035     bool inTargetModule_;
00036 };
00037
00038 // Creating a Rewriter to extract a specific cell
00039 class CellExtractor : public slang::syntax::SyntaxRewriter<CellExtractor> {
00040 public:
00041     explicit CellExtractor(const std::string &targetCell)
00042         : targetCell_(targetCell), foundTarget_(false) {}
00043     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00044     bool foundTargetCell() const;
00045
00046 private:
00047     const std::string &targetCell_;
00048     bool foundTarget_;
00049 };
00050
00051 // Print specific cell when visiting SyntaxTree
00052 class CellPrinter : public slang::syntax::SyntaxVisitor<CellPrinter> {
00053 public:
00054     explicit CellPrinter(const std::string &targetCell, std::ostream &out)
00055         : targetCell_(targetCell), out_(out), foundTarget_(false) {}
00056     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00057
00058 private:
00059     const std::string &targetCell_;
00060     std::ostream &out_;
00061     bool foundTarget_;
00062 };
00063
00064 // Comprehensive module rewriter for adding ports, instances, and connections
00065 class ModuleRewriter : public slang::syntax::SyntaxRewriter<ModuleRewriter> {
00066 public:
00067     explicit ModuleRewriter(const std::vector<std::string> &inputPins,
00068                             const std::vector<std::string> &outputPins,
00069                             const std::pair<std::string, std::string> &supercell_entry,
00070                             int instance_count, std::shared_ptr<spdlog::logger> logger)
00071         : inputPins_(inputPins), outputPins_(outputPins), cellName_(supercell_entry.first),
00072           moduleName_(supercell_entry.second), logger_(logger), depth_(0),
00073           instance_count_(instance_count) {}
00074     std::shared_ptr<spdlog::logger> logger_;
00075     void handle(const slang::syntax::SyntaxNode &node);
00076     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00077
00078 private:
00079     const std::vector<std::string> &inputPins_;
00080     const std::vector<std::string> &outputPins_;
00081     const std::string cellName_;
00082     const std::string moduleName_;
00083     std::map<std::string, std::string> portInfoMap_; // Map from port name to direction
00084     int depth_;
00085     int instance_count_;
00086 };

```

```
00087
00088 void getAST(const std::string &verilog_file, const std::string &cell);
00089
00090 #endif // VERILOG_UTILS_H
```

8.17 include/version.h File Reference

This graph shows which files directly or indirectly include this file:



Macros

- `#define` [APP_NAME](#) "ZlibValidation"
- `#define` [APP_VERSION_MAJOR](#) 0
- `#define` [APP_VERSION_MINOR](#) 1
- `#define` [APP_VERSION_PATCH](#) 0
- `#define` [APP_VERSION](#) "0.1.0"
- `#define` [APP_AUTHOR](#) "Song Zixuan"
- `#define` [APP_CONTACT](#) "cedar@zju.edu.cn"
- `#define` [BUILD_TIMESTAMP](#) "2025-03-31 22:35:20"

8.17.1 Macro Definition Documentation

8.17.1.1 APP_AUTHOR

```
#define APP_AUTHOR "Song Zixuan"
```

Definition at line 10 of file [version.h](#).

8.17.1.2 APP_CONTACT

```
#define APP_CONTACT "cedar@zju.edu.cn"
```

Definition at line 11 of file [version.h](#).

8.17.1.3 APP_NAME

```
#define APP_NAME "ZlibValidation"
```

Definition at line 5 of file [version.h](#).

8.17.1.4 APP_VERSION

```
#define APP_VERSION "0.1.0"
```

Definition at line 9 of file [version.h](#).

8.17.1.5 APP_VERSION_MAJOR

```
#define APP_VERSION_MAJOR 0
```

Definition at line 6 of file [version.h](#).

8.17.1.6 APP_VERSION_MINOR

```
#define APP_VERSION_MINOR 1
```

Definition at line 7 of file [version.h](#).

8.17.1.7 APP_VERSION_PATCH

```
#define APP_VERSION_PATCH 0
```

Definition at line 8 of file [version.h](#).

8.17.1.8 BUILD_TIMESTAMP

```
#define BUILD_TIMESTAMP "2025-03-31 22:35:20"
```

Definition at line 12 of file [version.h](#).

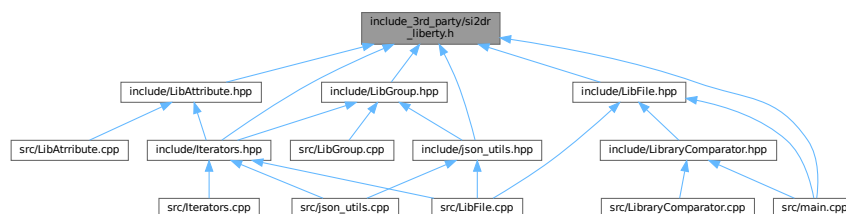
8.18 version.h

[Go to the documentation of this file.](#)

```
00001 // version.h.in
00002 #pragma once
00003
00004 // Auto-generated by CMake - DO NOT EDIT
00005 #define APP_NAME          "ZlibValidation"
00006 #define APP_VERSION_MAJOR 0
00007 #define APP_VERSION_MINOR 1
00008 #define APP_VERSION_PATCH 0
00009 #define APP_VERSION       "0.1.0"
00010 #define APP_AUTHOR        "Song Zixuan"
00011 #define APP_CONTACT       "cedar@zju.edu.cn"
00012 #define BUILD_TIMESTAMP   "2025-03-31 22:35:20"
```

8.19 include_3rd_party/si2dr_liberty.h File Reference

This graph shows which files directly or indirectly include this file:



Classes

- struct [si2OID](#)
- struct [si2drExprT](#)
- struct [liberty_value_data](#)

Macros

- `#define` [SI2_ARGS](#)(args) ()
- `#define` [SI2_LONG_MAX](#) 2147483647
- `#define` [SI2_LONG_MIN](#) -2147483647
- `#define` [SI2_ULONG_MAX](#) 4294967295
- `#define` [SI2DR_MIN_FLOAT32](#) -1E+37
- `#define` [SI2DR_MAX_FLOAT32](#) 1E+37
- `#define` [SI2DR_MIN_FLOAT64](#) -1.797693e+308
- `#define` [SI2DR_MAX_FLOAT64](#) 1.797693e+308
- `#define` [SI2DR_MIN_INT32](#) -2147483647
- `#define` [SI2DR_MAX_INT32](#) 2147483647
- `#define` [SI2DR_MAX_STRING_LEN](#) 1048576 /*1024*1024*/
- `#define` [SI2DR_TRUE](#) SI2_TRUE
- `#define` [SI2DR_FALSE](#) SI2_FALSE
- `#define` [LONG_DOUBLE](#) long double

Typedefs

- typedef enum [si2BooleanEnum](#) [si2BooleanT](#)
- typedef long int [si2LongT](#)
- typedef float [si2FloatT](#)
- typedef double [si2DoubleT](#)
- typedef char * [si2StringT](#)
- typedef void [si2VoidT](#)
- typedef enum [si2TypeEnum](#) [si2TypeT](#)
- typedef struct [si2OID](#) [si2ObjectIdT](#)
- typedef void * [si2IteratorIdT](#)
- typedef [si2BooleanT](#) [si2drBooleanT](#)
- typedef [si2LongT](#) [si2drInt32T](#)
- typedef [si2FloatT](#) [si2drFloat32T](#)
- typedef [si2DoubleT](#) [si2drFloat64T](#)
- typedef [si2StringT](#) [si2drStringT](#)
- typedef [si2VoidT](#) [si2drVoidT](#)
- typedef [si2IteratorIdT](#) [si2drIterIdT](#)

- typedef [si2ObjectIdT](#) [si2drObjectIdT](#)
- typedef [si2drObjectIdT](#) [si2drGroupIdT](#)
- typedef [si2drObjectIdT](#) [si2drAttrIdT](#)
- typedef [si2drObjectIdT](#) [si2drAttrComplexValIdT](#)
- typedef [si2drObjectIdT](#) [si2drDefinIdT](#)
- typedef [si2drIterIdT](#) [si2drObjectsIdT](#)
- typedef [si2drIterIdT](#) [si2drGroupsIdT](#)
- typedef [si2drIterIdT](#) [si2drAttrIdT](#)
- typedef [si2drIterIdT](#) [si2drDefinesIdT](#)
- typedef [si2drIterIdT](#) [si2drNamesIdT](#)
- typedef [si2drIterIdT](#) [si2drValuesIdT](#)
- typedef enum [si2drSeverityT](#) [si2drSeverityT](#)
- typedef enum [si2drObjectTypeT](#) [si2drObjectTypeT](#)
- typedef enum [si2drAttrTypeT](#) [si2drAttrTypeT](#)
- typedef enum [si2drValueTypeT](#) [si2drValueTypeT](#)
- typedef enum [si2drExprTypeT](#) [si2drExprTypeT](#)
- typedef struct [si2drExprT](#) [si2drExprT](#)
- typedef enum [si2drErrorT](#) [si2drErrorT](#)
- typedef [si2drVoidT](#)(* [si2drMessageHandlerT](#)) ([si2drSeverityT](#) sev, [si2drErrorT](#) errToPrint, [si2drStringT](#) auxText, [si2drErrorT](#) *err)

Enumerations

- enum [si2BooleanEnum](#) { [SI2_FALSE](#) = 0 , [SI2_TRUE](#) = 1 }
- enum [si2TypeEnum](#) {
[SI2_UNDEFINED_VALUETYPE](#) = 0 , [SI2_BOOLEAN](#) = 1 , [SI2_LONG](#) = 2 , [SI2_FLOAT](#) = 3
,
[SI2_DOUBLE](#) = 4 , [SI2_STRING](#) = 5 , [SI2_VOID](#) = 6 , [SI2_OID](#) = 7 ,
[SI2_ITER](#) = 8 , [SI2_EXPR](#) = 9 , [SI2_MAX_VALUETYPE](#) = 10 }
- enum [si2drSeverityT](#) { [SI2DR_SEVERITY_NOTE](#) = 0 , [SI2DR_SEVERITY_WARN](#) = 1 ,
[SI2DR_SEVERITY_ERR](#) = 2 }
- enum [si2drObjectTypeT](#) {
[SI2DR_UNDEFINED_OBJECTTYPE](#) = 0 , [SI2DR_GROUP](#) = 1 , [SI2DR_ATTR](#) = 2 ,
[SI2DR_DEFINE](#) = 3 ,
[SI2DR_COMPLEXVAL](#) = 4 , [SI2DR_MAX_OBJECTTYPE](#) = 5 }
- enum [si2drAttrTypeT](#) { [SI2DR_SIMPLE](#) , [SI2DR_COMPLEX](#) }
- enum [si2drValueTypeT](#) {
[SI2DR_UNDEFINED_VALUETYPE](#) = [SI2_UNDEFINED_VALUETYPE](#) , [SI2DR_INT32](#) = [SI2_](#)
[_LONG](#) , [SI2DR_STRING](#) = [SI2_STRING](#) , [SI2DR_FLOAT64](#) = [SI2_DOUBLE](#) ,
[SI2DR_BOOLEAN](#) = [SI2_BOOLEAN](#) , [SI2DR_EXPR](#) = [SI2_EXPR](#) , [SI2DR_MAX_VALUETYPE](#)
= [SI2_MAX_VALUETYPE](#) }

- enum `si2drExprTypeT` {
`SI2DR_EXPR_VAL` = 0 , `SI2DR_EXPR_OP_ADD` = 1 , `SI2DR_EXPR_OP_SUB` = 2 ,
`SI2DR_EXPR_OP_MUL` = 3 ,
`SI2DR_EXPR_OP_DIV` = 4 , `SI2DR_EXPR_OP_PAREN` = 5 , `SI2DR_EXPR_OP_LOG2` = 6
, `SI2DR_EXPR_OP_LOG10` = 7 ,
`SI2DR_EXPR_OP_EXP` = 8 }
- enum `si2drErrorT` {
`SI2DR_NO_ERROR` = 0 , `SI2DR_INTERNAL_SYSTEM_ERROR` = 1 , `SI2DR_INVALID_VALUE`
= 4 , `SI2DR_INVALID_NAME` = 5 ,
`SI2DR_INVALID_OBJECTTYPE` = 6 , `SI2DR_INVALID_ATTRTYPE` = 10 , `SI2DR_UNUSABLE_OID`
= 7 , `SI2DR_OBJECT_ALREADY_EXISTS` = 8 ,
`SI2DR_OBJECT_NOT_FOUND` = 9 , `SI2DR_SYNTAX_ERROR` = 2 , `SI2DR_TRACE_FILES_CANNOT_BE_C`
= 3 , `SI2DR_PIINIT_NOT_CALLED` = 11 ,
`SI2DR_SEMANTIC_ERROR` = 12 , `SI2DR_REFERENCE_ERROR` = 13 , `SI2DR_MAX_ERROR`
= 14 }

Functions

- `si2drVoidT si2drDefaultMessageHandler` (`si2drSeverityT` sev, `si2drErrorT` errToPrint, `si2drStringT` auxText, `si2drErrorT` *err)
- `si2drMessageHandlerT si2drPIGetMessageHandler` (`si2drErrorT` *err)
- `si2drVoidT si2drPISetMessageHandler` (`si2drMessageHandlerT` MsgHandFuncPtr, `si2drErrorT` *err)
- `si2drGroupIdT si2drPICreateGroup` `SI2_ARGS` ((`si2drStringT` name, `si2drStringT` group_type, `si2drErrorT` *err))
- `si2drAttrIdT si2drGroupCreateAttr` `SI2_ARGS` ((`si2drGroupIdT` group, `si2drStringT` name, `si2drAttrTypeT` type, `si2drErrorT` *err))
- `si2drAttrTypeT si2drAttrGetAttrType` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddInt32Value` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drInt32T` integr, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddStringValue` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drStringT` string, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddBooleanValue` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drBooleanT` boolval, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddFloat64Value` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drFloat64T` float64, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddExprValue` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drExprT` *expr, `si2drErrorT` *err))
- `si2drVoidT si2drIterNextComplexValue` `SI2_ARGS` ((`si2drValuesIdT` iter, `si2drValueTypeT` *type, `si2drInt32T` *integr, `si2drFloat64T` *float64, `si2drStringT` *string, `si2drBooleanT` *boolval, `si2drExprT` **expr, `si2drErrorT` *err))
- `si2drAttrComplexValIdT si2drIterNextComplex` `SI2_ARGS` ((`si2drValuesIdT` iter, `si2drErrorT` *err))

- `si2drVoidT` `si2drSimpleAttrSetBooleanValue` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drBooleanT` integr, `si2drErrorT` *err))
- `si2drVoidT` `si2drExprDestroy` `SI2_ARGS` ((`si2drExprT` *expr, `si2drErrorT` *err))
- `si2drExprT` *`si2drCreateExpr` `SI2_ARGS` ((`si2drExprTypeT` type, `si2drErrorT` *err))
- `si2drExprT` *`si2drCreateBooleanValExpr` `SI2_ARGS` ((`si2drBooleanT` b, `si2drErrorT` *err))
- `si2drExprT` *`si2drCreateDoubleValExpr` `SI2_ARGS` ((`si2drFloat64T` d, `si2drErrorT` *err))
- `si2drExprT` *`si2drCreateStringValExpr` `SI2_ARGS` ((`si2drStringT` s, `si2drErrorT` *err))
- `si2drExprT` *`si2drCreateIntValExpr` `SI2_ARGS` ((`si2drInt32T` i, `si2drErrorT` *err))
- `si2drExprT` *`si2drCreateBinaryOpExpr` `SI2_ARGS` ((`si2drExprT` *left, `si2drExprTypeT` optype, `si2drExprT` *right, `si2drErrorT` *err))
- `si2drExprT` *`si2drCreateUnaryOpExpr` `SI2_ARGS` ((`si2drExprTypeT` optype, `si2drExprT` *expr, `si2drErrorT` *err))
- `si2drDefinIdT` `si2drGroupCreateDefine` `SI2_ARGS` ((`si2drGroupIdT` group, `si2drStringT` name, `si2drStringT` allowed_group_name, `si2drValueTypeT` valtype, `si2drErrorT` *err))
- `si2drVoidT` `si2drDefineGetInfo` `SI2_ARGS` ((`si2drDefinIdT` def, `si2drStringT` *name, `si2drStringT` *allowed_group_name, `si2drValueTypeT` *valtype, `si2drErrorT` *err))
- `si2drStringT` `si2drDefineGetName` `SI2_ARGS` ((`si2drDefinIdT` def, `si2drErrorT` *err))
- `si2drGroupIdT` `si2drGroupCreateGroup` `SI2_ARGS` ((`si2drGroupIdT` group, `si2drStringT` name, `si2drStringT` group_type, `si2drErrorT` *err))
- `si2drStringT` `si2drGroupGetGroupType` `SI2_ARGS` ((`si2drGroupIdT` group, `si2drErrorT` *err))
- `si2drVoidT` `si2drGroupSetComment` `SI2_ARGS` ((`si2drGroupIdT` group, `si2drStringT` comment, `si2drErrorT` *err))
- `si2drVoidT` `si2drAttrSetComment` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drStringT` comment, `si2drErrorT` *err))
- `si2drVoidT` `si2drDefineSetComment` `SI2_ARGS` ((`si2drDefinIdT` def, `si2drStringT` comment, `si2drErrorT` *err))
- `si2drVoidT` `si2drGroupAddName` `SI2_ARGS` ((`si2drGroupIdT` group, `si2drStringT` name, `si2drErrorT` *err))
- `si2drGroupIdT` `si2drPIFindGroupByName` `SI2_ARGS` ((`si2drStringT` name, `si2drStringT` type, `si2drErrorT` *err))
- `si2drGroupIdT` `si2drGroupFindGroupByName` `SI2_ARGS` ((`si2drGroupIdT` group, `si2drStringT` name, `si2drStringT` type, `si2drErrorT` *err))
- `si2drDefinIdT` `si2drPIFindDefineByName` `SI2_ARGS` ((`si2drStringT` name, `si2drErrorT` *err))
- `si2drGroupsIdT` `si2drPIGetGroups` `SI2_ARGS` ((`si2drErrorT` *err))
- `si2drVoidT` `si2drObjectDelete` `SI2_ARGS` ((`si2drObjectIdT` object, `si2drErrorT` *err))
- `si2drStringT` `si2drPIGetErrorText` `SI2_ARGS` ((`si2drErrorT` errorCode, `si2drErrorT` *err))
- `si2drBooleanT` `si2drObjectIsSame` `SI2_ARGS` ((`si2drObjectIdT` object1, `si2drObjectIdT` object2, `si2drErrorT` *err))
- `si2drVoidT` `si2drObjectSetFileName` `SI2_ARGS` ((`si2drObjectIdT` object, `si2drStringT` filename, `si2drErrorT` *err))
- `si2drVoidT` `si2drObjectSetLineNo` `SI2_ARGS` ((`si2drObjectIdT` object, `si2drInt32T` lineno, `si2drErrorT` *err))

- `si2drVoidT` `si2drReadLibertyFile` `SI2_ARGS` ((char *filename, `si2drErrorT` *err))
- `si2drVoidT` `si2drWriteLibertyFile` `SI2_ARGS` ((char *filename, `si2drGroupIdT` group, char *cell-name, `si2drErrorT` *err))
- `si2drVoidT` `si2drPISetTraceMode` `SI2_ARGS` ((`si2drStringT` fname, `si2drErrorT` *err))
- `si2drVoidT` `si2drGroupMoveAfter` `SI2_ARGS` ((`si2drGroupIdT` groupToMove, `si2drGroupIdT` targetGroup, `si2drErrorT` *err))
- `LONG_DOUBLE` `liberty_get_element` (struct `liberty_value_data` *vd,...)
- void `liberty_destroy_value_data` (struct `liberty_value_data` *vd)
- struct `liberty_value_data` * `liberty_get_values_data` (`si2drGroupIdT` table_group)
- void `print_version` ()
- void `usage` ()
- int `main_liberty_parser` (int argc, char *argv[])
- void `get_function` (char *filename)

8.19.1 Macro Definition Documentation

8.19.1.1 LONG_DOUBLE

```
#define LONG_DOUBLE long double
```

Definition at line 609 of file `si2dr_liberty.h`.

8.19.1.2 SI2_ARGS

```
#define SI2_ARGS(  
    args ) ()
```

Definition at line 34 of file `si2dr_liberty.h`.

8.19.1.3 SI2_LONG_MAX

```
#define SI2_LONG_MAX 2147483647
```

Definition at line 74 of file `si2dr_liberty.h`.

8.19.1.4 SI2_LONG_MIN

```
#define SI2_LONG_MIN -2147483647
```

Definition at line 75 of file [si2dr_liberty.h](#).

8.19.1.5 SI2_ULONG_MAX

```
#define SI2_ULONG_MAX 4294967295
```

Definition at line 76 of file [si2dr_liberty.h](#).

8.19.1.6 SI2DR_FALSE

```
#define SI2DR_FALSE SI2_FALSE
```

Definition at line 96 of file [si2dr_liberty.h](#).

8.19.1.7 SI2DR_MAX_FLOAT32

```
#define SI2DR_MAX_FLOAT32 1E+37
```

Definition at line 87 of file [si2dr_liberty.h](#).

8.19.1.8 SI2DR_MAX_FLOAT64

```
#define SI2DR_MAX_FLOAT64 1.797693e+308
```

Definition at line 89 of file [si2dr_liberty.h](#).

8.19.1.9 SI2DR_MAX_INT32

```
#define SI2DR_MAX_INT32 2147483647
```

Definition at line 91 of file [si2dr_liberty.h](#).

8.19.1.10 SI2DR_MAX_STRING_LEN

```
#define SI2DR_MAX_STRING_LEN 1048576 /*1024*1024*/
```

Definition at line 93 of file [si2dr_liberty.h](#).

8.19.1.11 SI2DR_MIN_FLOAT32

```
#define SI2DR_MIN_FLOAT32 -1E+37
```

Definition at line 86 of file [si2dr_liberty.h](#).

8.19.1.12 SI2DR_MIN_FLOAT64

```
#define SI2DR_MIN_FLOAT64 -1.797693e+308
```

Definition at line 88 of file [si2dr_liberty.h](#).

8.19.1.13 SI2DR_MIN_INT32

```
#define SI2DR_MIN_INT32 -2147483647
```

Definition at line 90 of file [si2dr_liberty.h](#).

8.19.1.14 SI2DR_TRUE

```
#define SI2DR_TRUE SI2_TRUE
```

Definition at line 95 of file [si2dr_liberty.h](#).

8.19.2 Typedef Documentation

8.19.2.1 si2BooleanT

```
typedef enum si2BooleanEnum si2BooleanT
```

8.19.2.2 si2DoubleT

```
typedef double si2DoubleT
```

Definition at line 52 of file [si2dr_liberty.h](#).

8.19.2.3 si2drAttrComplexValIdT

```
typedef si2drObjectIdT si2drAttrComplexValIdT
```

Definition at line 111 of file [si2dr_liberty.h](#).

8.19.2.4 si2drAttrIdT

```
typedef si2drObjectIdT si2drAttrIdT
```

Definition at line 110 of file [si2dr_liberty.h](#).

8.19.2.5 si2drAttrsIdT

```
typedef si2drIterIdT si2drAttrsIdT
```

Definition at line 116 of file [si2dr_liberty.h](#).

8.19.2.6 si2drAttrTypeT

```
typedef enum si2drAttrTypeT si2drAttrTypeT
```

8.19.2.7 si2drBooleanT

```
typedef si2BooleanT si2drBooleanT
```

Definition at line 98 of file [si2dr_liberty.h](#).

8.19.2.8 si2drDefineIdT

```
typedef si2drObjectIdT si2drDefineIdT
```

Definition at line 112 of file [si2dr_liberty.h](#).

8.19.2.9 si2drDefinesIdT

```
typedef si2drIterIdT si2drDefinesIdT
```

Definition at line 117 of file [si2dr_liberty.h](#).

8.19.2.10 si2drErrorT

```
typedef enum si2drErrorT si2drErrorT
```

8.19.2.11 si2drExprT

```
typedef struct si2drExprT si2drExprT
```

8.19.2.12 si2drExprTypeT

```
typedef enum si2drExprTypeT si2drExprTypeT
```

8.19.2.13 si2drFloat32T

```
typedef si2FloatT si2drFloat32T
```

Definition at line 101 of file [si2dr_liberty.h](#).

8.19.2.14 si2drFloat64T

```
typedef si2DoubleT si2drFloat64T
```

Definition at line 102 of file [si2dr_liberty.h](#).

8.19.2.15 si2drGroupIdT

```
typedef si2drObjectIdT si2drGroupIdT
```

Definition at line 109 of file [si2dr_liberty.h](#).

8.19.2.16 si2drGroupsIdT

```
typedef si2drIterIdT si2drGroupsIdT
```

Definition at line 115 of file [si2dr_liberty.h](#).

8.19.2.17 si2drInt32T

```
typedef si2LongT si2drInt32T
```

Definition at line 100 of file [si2dr_liberty.h](#).

8.19.2.18 si2drIterIdT

```
typedef si2IteratorIdT si2drIterIdT
```

Definition at line 106 of file [si2dr_liberty.h](#).

8.19.2.19 si2drMessageHandlerT

```
typedef si2drVoidT(* si2drMessageHandlerT) (si2drSeverityT sev, si2drErrorT errToPrint, si2drStringT auxText,  
si2drErrorT *err)
```

Definition at line 215 of file [si2dr_liberty.h](#).

8.19.2.20 si2drNamesIdT

```
typedef si2drIterIdT si2drNamesIdT
```

Definition at line 118 of file [si2dr_liberty.h](#).

8.19.2.21 si2drObjectIdT

```
typedef si2ObjectIdT si2drObjectIdT
```

Definition at line 107 of file [si2dr_liberty.h](#).

8.19.2.22 si2drObjectsIdT

```
typedef si2drIterIdT si2drObjectsIdT
```

Definition at line 114 of file [si2dr_liberty.h](#).

8.19.2.23 si2drObjectTypeT

```
typedef enum si2drObjectTypeT si2drObjectTypeT
```

8.19.2.24 si2drSeverityT

```
typedef enum si2drSeverityT si2drSeverityT
```

8.19.2.25 si2drStringT

```
typedef si2StringT si2drStringT
```

Definition at line 103 of file [si2dr_liberty.h](#).

8.19.2.26 si2drValuesIdT

```
typedef si2drIterIdT si2drValuesIdT
```

Definition at line 119 of file [si2dr_liberty.h](#).

8.19.2.27 si2drValueTypeT

```
typedef enum si2drValueTypeT si2drValueTypeT
```

8.19.2.28 si2drVoidT

```
typedef si2VoidT si2drVoidT
```

Definition at line 104 of file [si2dr_liberty.h](#).

8.19.2.29 si2FloatT

```
typedef float si2FloatT
```

Definition at line 51 of file [si2dr_liberty.h](#).

8.19.2.30 si2IteratorIdT

```
typedef void* si2IteratorIdT
```

Definition at line 83 of file [si2dr_liberty.h](#).

8.19.2.31 si2LongT

```
typedef long int si2LongT
```

Definition at line 50 of file [si2dr_liberty.h](#).

8.19.2.32 si2ObjectIdT

```
typedef struct si2OID si2ObjectIdT
```

8.19.2.33 si2StringT

```
typedef char* si2StringT
```

Definition at line 53 of file [si2dr_liberty.h](#).

8.19.2.34 si2TypeT

```
typedef enum si2TypeEnum si2TypeT
```

8.19.2.35 si2VoidT

```
typedef void si2VoidT
```

Definition at line [54](#) of file [si2dr_liberty.h](#).

8.19.3 Enumeration Type Documentation

8.19.3.1 si2BooleanEnum

```
enum si2BooleanEnum
```

Enumerator

SI2_FALSE	
SI2_TRUE	

Definition at line [49](#) of file [si2dr_liberty.h](#).

8.19.3.2 si2drAttrTypeT

```
enum si2drAttrTypeT
```

Enumerator

SI2DR_SIMPLE	
SI2DR_COMPLEX	

Definition at line [139](#) of file [si2dr_liberty.h](#).

8.19.3.3 si2drErrorT

enum [si2drErrorT](#)

Enumerator

SI2DR_NO_ERROR	
SI2DR_INTERNAL_SYSTEM_ERROR	
SI2DR_INVALID_VALUE	
SI2DR_INVALID_NAME	
SI2DR_INVALID_OBJECTTYPE	
SI2DR_INVALID_ATTRTYPE	
SI2DR_UNUSABLE_OID	
SI2DR_OBJECT_ALREADY_EXISTS	
SI2DR_OBJECT_NOT_FOUND	
SI2DR_SYNTAX_ERROR	
SI2DR_TRACE_FILES_CANNOT_BE_OPENED	
SI2DR_PIINIT_NOT_CALLED	
SI2DR_SEMANTIC_ERROR	
SI2DR_REFERENCE_ERROR	
SI2DR_MAX_ERROR	

Definition at line 190 of file [si2dr_liberty.h](#).

8.19.3.4 si2drExprTypeT

enum [si2drExprTypeT](#)

Enumerator

SI2DR_EXPR_VAL	
SI2DR_EXPR_OP_ADD	
SI2DR_EXPR_OP_SUB	
SI2DR_EXPR_OP_MUL	
SI2DR_EXPR_OP_DIV	
SI2DR_EXPR_OP_PAREN	
SI2DR_EXPR_OP_LOG2	
SI2DR_EXPR_OP_LOG10	
SI2DR_EXPR_OP_EXP	

Definition at line 157 of file [si2dr_liberty.h](#).

8.19.3.5 si2drObjectTypeT

enum [si2drObjectTypeT](#)

Enumerator

SI2DR_UNDEFINED_OBJECTTYPE	
SI2DR_GROUP	
SI2DR_ATTR	
SI2DR_DEFINE	
SI2DR_COMPLEXVAL	
SI2DR_MAX_OBJECTTYPE	

Definition at line 129 of file [si2dr_liberty.h](#).

8.19.3.6 si2drSeverityT

enum [si2drSeverityT](#)

Enumerator

SI2DR_SEVERITY_NOTE	
SI2DR_SEVERITY_WARN	
SI2DR_SEVERITY_ERR	

Definition at line 121 of file [si2dr_liberty.h](#).

8.19.3.7 si2drValueTypeT

enum [si2drValueTypeT](#)

Enumerator

SI2DR_UNDEFINED_VALUETYPE	
---------------------------	--

Enumerator

SI2DR_INT32	
SI2DR_STRING	
SI2DR_FLOAT64	
SI2DR_BOOLEAN	
SI2DR_EXPR	
SI2DR_MAX_VALUETYPE	

Definition at line 146 of file [si2dr_liberty.h](#).

8.19.3.8 si2TypeEnum

enum [si2TypeEnum](#)

Enumerator

SI2_UNDEFINED_VALUETYPE	
SI2_BOOLEAN	
SI2_LONG	
SI2_FLOAT	
SI2_DOUBLE	
SI2_STRING	
SI2_VOID	
SI2_OID	
SI2_ITER	
SI2_EXPR	
SI2_MAX_VALUETYPE	

Definition at line 56 of file [si2dr_liberty.h](#).

8.19.4 Function Documentation

8.19.4.1 get_function()

```
void get_function (
    char * filename )
```

8.19.4.2 liberty_destroy_value_data()

```
void liberty_destroy_value_data (
    struct liberty_value_data * vd )
```

8.19.4.3 liberty_get_element()

```
LONG_DOUBLE liberty_get_element (
    struct liberty_value_data * vd,
    ... )
```

8.19.4.4 liberty_get_values_data()

```
struct liberty_value_data * liberty_get_values_data (
    si2drGroupIdT table_group )
```

8.19.4.5 main_liberty_parser()

```
int main_liberty_parser (
    int argc,
    char * argv[] )
```

8.19.4.6 print_version()

```
void print_version ( )
```

8.19.4.7 SI2_ARGS() [1/41]

```
si2drVoidT si2drReadLibertyFile SI2_ARGS (
    (char *filename, si2drErrorT *err) )
```

8.19.4.8 SI2_ARGS() [2/41]

```
si2drVoidT si2drWriteLibertyFile SI2_ARGS (  
    (char *filename, si2drGroupIdT group, char *cellname, si2drErrorT *err) )
```

8.19.4.9 SI2_ARGS() [3/41]

```
si2drVoidT si2drComplexAttrAddBooleanValue SI2_ARGS (  
    (si2drAttrIdT attr, si2drBooleanT boolval, si2drErrorT *err) )
```

8.19.4.10 SI2_ARGS() [4/41]

```
si2drVoidT si2drSimpleAttrSetBooleanValue SI2_ARGS (  
    (si2drAttrIdT attr, si2drBooleanT intgr, si2drErrorT *err) )
```

8.19.4.11 SI2_ARGS() [5/41]

```
si2drStringT si2drAttrGetComment SI2_ARGS (  
    (si2drAttrIdT attr, si2drErrorT *err) )
```

8.19.4.12 SI2_ARGS() [6/41]

```
si2drVoidT si2drSimpleAttrSetExprValue SI2_ARGS (  
    (si2drAttrIdT attr, si2drExprT *expr, si2drErrorT *err) )
```

8.19.4.13 SI2_ARGS() [7/41]

```
si2drVoidT si2drSimpleAttrSetFloat64Value SI2_ARGS (  
    (si2drAttrIdT attr, si2drFloat64T float64, si2drErrorT *err) )
```


8.19.4.14 SI2_ARGS() [8/41]

```
si2drVoidT si2drSimpleAttrSetInt32Value SI2_ARGS (  
    (si2drAttrIdT attr, si2drInt32T intgr, si2drErrorT *err) )
```

8.19.4.15 SI2_ARGS() [9/41]

```
si2drVoidT si2drAttrSetComment SI2_ARGS (  
    (si2drAttrIdT attr, si2drStringT comment, si2drErrorT *err) )
```

8.19.4.16 SI2_ARGS() [10/41]

```
si2drVoidT si2drSimpleAttrSetStringValue SI2_ARGS (  
    (si2drAttrIdT attr, si2drStringT string, si2drErrorT *err) )
```

8.19.4.17 SI2_ARGS() [11/41]

```
si2drExprT *si2drCreateBooleanValExpr SI2_ARGS (  
    (si2drBooleanT b, si2drErrorT *err) )
```

8.19.4.18 SI2_ARGS() [12/41]

```
si2drStringT si2drDefineGetComment SI2_ARGS (  
    (si2drDefineIdT def, si2drErrorT *err) )
```

8.19.4.19 SI2_ARGS() [13/41]

```
si2drVoidT si2drDefineGetInfo SI2_ARGS (  
    (si2drDefineIdT def, si2drStringT *name, si2drStringT *allowed_group_name, si2drValueType *valtype,  
    si2drErrorT *err) )
```

8.19.4.20 SI2_ARGS() [14/41]

```
si2drVoidT si2drDefineSetComment SI2_ARGS (  
    (si2drDefineIdT def, si2drStringT comment, si2drErrorT *err) )
```

8.19.4.21 SI2_ARGS() [15/41]

```
si2drBooleanT si2drPIGetNocheckMode SI2_ARGS (  
    (si2drErrorT *err) )
```

8.19.4.22 SI2_ARGS() [16/41]

```
si2drStringT si2drPIGetErrorText SI2_ARGS (  
    (si2drErrorT errorCode, si2drErrorT *err) )
```

8.19.4.23 SI2_ARGS() [17/41]

```
si2drExprT *si2drOpExprGetRightExpr SI2_ARGS (  
    (si2drExprT *expr, si2drErrorT *err) )
```

8.19.4.24 SI2_ARGS() [18/41]

```
si2drExprT *si2drCreateBinaryOpExpr SI2_ARGS (  
    (si2drExprT *left, si2drExprTypeT optype, si2drExprT *right, si2drErrorT *err) )
```

8.19.4.25 SI2_ARGS() [19/41]

```
si2drExprT *si2drCreateUnaryOpExpr SI2_ARGS (  
    (si2drExprTypeT optype, si2drExprT *expr, si2drErrorT *err) )
```

8.19.4.26 SI2_ARGS() [20/41]

```
si2drExprT *si2drCreateExpr SI2_ARGS (  
    (si2drExprTypeT type, si2drErrorT *err) )
```

8.19.4.27 SI2_ARGS() [21/41]

```
si2drExprT *si2drCreateDoubleValExpr SI2_ARGS (  
    (si2drFloat64T d, si2drErrorT *err) )
```

8.19.4.28 SI2_ARGS() [22/41]

```
si2drVoidT si2drCheckLibertyLibrary SI2_ARGS (  
    (si2drGroupIdT group, si2drErrorT *err) )
```

8.19.4.29 SI2_ARGS() [23/41]

```
si2drVoidT si2drGroupSetComment SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT comment, si2drErrorT *err) )
```

8.19.4.30 SI2_ARGS() [24/41]

```
si2drAttrIdT si2drGroupCreateAttr SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drAttrTypeT type, si2drErrorT *err) )
```

8.19.4.31 SI2_ARGS() [25/41]

```
si2drDefineIdT si2drGroupFindDefineByName SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drErrorT *err) )
```

8.19.4.32 SI2_ARGS() [26/41]

```
si2drDefineIdT si2drGroupCreateDefine SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drStringT allowed_group_name, si2drValueTypeT valtype,  
    si2drErrorT *err) )
```

8.19.4.33 SI2_ARGS() [27/41]

```
si2drGroupIdT si2drGroupCreateGroup SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drStringT group_type, si2drErrorT *err) )
```

8.19.4.34 SI2_ARGS() [28/41]

```
si2drGroupIdT si2drGroupFindGroupByName SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drStringT type, si2drErrorT *err) )
```

8.19.4.35 SI2_ARGS() [29/41]

```
si2drVoidT si2drGroupMoveBefore SI2_ARGS (  
    (si2drGroupIdT groupToMove, si2drGroupIdT targetGroup, si2drErrorT *err) )
```

8.19.4.36 SI2_ARGS() [30/41]

```
si2drExprT *si2drCreateIntValExpr SI2_ARGS (  
    (si2drInt32T i, si2drErrorT *err) )
```

8.19.4.37 SI2_ARGS() [31/41]

```
si2drStringT si2drObjectGetFileName SI2_ARGS (  
    (si2drObjectIdT object, si2drErrorT *err) )
```

8.19.4.38 SI2_ARGS() [32/41]

```
si2drVoidT si2drObjectSetLineNo SI2_ARGS (  
    (si2drObjectIdT object, si2drInt32T lineno, si2drErrorT *err) )
```

8.19.4.39 SI2_ARGS() [33/41]

```
si2drVoidT si2drObjectSetFileName SI2_ARGS (  
    (si2drObjectIdT object, si2drStringT filename, si2drErrorT *err) )
```

8.19.4.40 SI2_ARGS() [34/41]

```
si2drBooleanT si2drObjectIsSame SI2_ARGS (  
    (si2drObjectIdT object1, si2drObjectIdT object2, si2drErrorT *err) )
```

8.19.4.41 SI2_ARGS() [35/41]

```
si2drVoidT si2drPISetTraceMode SI2_ARGS (  
    (si2drStringT fname, si2drErrorT *err) )
```

8.19.4.42 SI2_ARGS() [36/41]

```
si2drDefineIdT si2drPIFindDefineByName SI2_ARGS (  
    (si2drStringT name, si2drErrorT *err) )
```

8.19.4.43 SI2_ARGS() [37/41]

```
si2drGroupIdT si2drPICreateGroup SI2_ARGS (  
    (si2drStringT name, si2drStringT group_type, si2drErrorT *err) )
```

8.19.4.44 SI2_ARGS() [38/41]

```
si2drGroupIdT si2drPIFindGroupByName SI2_ARGS (  
    (si2drStringT name, si2drStringT type, si2drErrorT *err) )
```

8.19.4.45 SI2_ARGS() [39/41]

```
si2drExprT *si2drCreateStringValExpr SI2_ARGS (  
    (si2drStringT s, si2drErrorT *err) )
```

8.19.4.46 SI2_ARGS() [40/41]

```
si2drVoidT si2drIterQuit SI2_ARGS (  
    (si2drValuesIdT iter, si2drErrorT *err) )
```

8.19.4.47 SI2_ARGS() [41/41]

```
si2drVoidT si2drIterNextComplexValue SI2_ARGS (  
    (si2drValuesIdT iter, si2drValueTypeT *type, si2drInt32T *intgr, si2drFloat64T *float64, si2drStringT  
    *string, si2drBooleanT *boolval, si2drExprT **expr, si2drErrorT *err) )
```

8.19.4.48 si2drDefaultMessageHandler()

```
si2drVoidT si2drDefaultMessageHandler (  
    si2drSeverityT sev,  
    si2drErrorT errToPrint,  
    si2drStringT auxText,  
    si2drErrorT * err )
```

8.19.4.49 si2drPIGetMessageHandler()

```
si2drMessageHandlerT si2drPIGetMessageHandler (
    si2drErrorT * err )
```

8.19.4.50 si2drPISetMessageHandler()

```
si2drVoidT si2drPISetMessageHandler (
    si2drMessageHandlerT MsgHandFuncPtr,
    si2drErrorT * err )
```

8.19.4.51 usage()

```
void usage ( )
```

8.20 si2dr_liberty.h

[Go to the documentation of this file.](#)

```
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00016
00017 *****/
00018
00019 /* define a PI for manipulating the liberty data */
00020
00021
00022 /* basic data types */
00023
00024
00025 #ifndef SI2DR_H
00026 #define SI2DR_H
00027
00028 #ifdef SI2_ARGS
00029 #undef SI2_ARGS
00030 #endif
```

```

00031 #if defined(__STDC__) || defined(HPPA) || defined(ibmpoweraix) || defined(_MSC_VER) || defined(__cplusplus)
00032 #define SI2_ARGS(args)  args
00033 #else
00034 #define SI2_ARGS(args)  ()
00035 #endif
00036
00037
00038 /* *****
00039 File contents : LIBERTY-DR type definitions.
00040 ***** */
00041 #ifdef __cplusplus
00042 extern "C" {
00043 #endif
00044
00045 /*
00046 * These primitive types are compatible with the ANSI C Standard (ANSI/
00047 * X3.159-1990 Programming Language C Standard.)
00048 */
00049 typedef enum si2BooleanEnum { SI2_FALSE = 0, SI2_TRUE = 1 } si2BooleanT;
00050 typedef long int    si2LongT;
00051 typedef float       si2FloatT;
00052 typedef double      si2DoubleT;
00053 typedef char*       si2StringT;
00054 typedef void        si2VoidT;
00055
00056 typedef enum si2TypeEnum {
00057     SI2_UNDEFINED_VALUETYPE = 0,
00058     SI2_BOOLEAN = 1,
00059     SI2_LONG = 2,
00060     SI2_FLOAT = 3,
00061     SI2_DOUBLE = 4,
00062     SI2_STRING = 5,
00063     SI2_VOID = 6,
00064     SI2_OID = 7,
00065     SI2_ITER = 8,
00066     SI2_EXPR = 9,
00067     SI2_MAX_VALUETYPE = 10
00068 } si2TypeT;
00069
00070 /*
00071 * Size limits
00072 */
00073
00074 #define SI2_LONG_MAX    2147483647
00075 #define SI2_LONG_MIN    -2147483647
00076 #define SI2_ULONG_MAX   4294967295
00077
00078
00079 typedef struct si2OID {void *v1, *v2;} si2ObjectIdT;
00080 /*
00081 * Iterators are blind handles the size of a single pointer.
00082 */
00083 typedef void *si2IteratorIdT;
00084
00085
00086 #define SI2DR_MIN_FLOAT32  -1E+37
00087 #define SI2DR_MAX_FLOAT32   1E+37
00088 #define SI2DR_MIN_FLOAT64  -1.797693e+308
00089 #define SI2DR_MAX_FLOAT64   1.797693e+308
00090 #define SI2DR_MIN_INT32    -2147483647
00091 #define SI2DR_MAX_INT32     2147483647
00092
00093 #define SI2DR_MAX_STRING_LEN  1048576 /*1024*1024*/
00094
00095 #define SI2DR_TRUE SI2_TRUE
00096 #define SI2DR_FALSE SI2_FALSE
00097
00098 typedef si2BooleanT si2drBooleanT;
00099

```



```

00100 typedef si2LongT      si2drInt32T;
00101 typedef si2FloatT     si2drFloat32T;
00102 typedef si2DoubleT     si2drFloat64T;
00103 typedef si2StringT     si2drStringT;
00104 typedef si2VoidT       si2drVoidT;
00105
00106 typedef si2IteratorIdT  si2drIterIdT;
00107 typedef si2ObjectIdT    si2drObjectIdT;
00108
00109 typedef si2drObjectIdT  si2drGroupIdT;
00110 typedef si2drObjectIdT  si2drAttrIdT;
00111 typedef si2drObjectIdT  si2drAttrComplexValIdT;
00112 typedef si2drObjectIdT  si2drDefineIdT;
00113
00114 typedef si2drIterIdT    si2drObjectsIdT;
00115 typedef si2drIterIdT    si2drGroupsIdT;
00116 typedef si2drIterIdT    si2drAttrsIdT;
00117 typedef si2drIterIdT    si2drDefinesIdT;
00118 typedef si2drIterIdT    si2drNamesIdT;
00119 typedef si2drIterIdT    si2drValuesIdT;
00120
00121 typedef enum si2drSeverityT
00122 {
00123     SI2DR_SEVERITY_NOTE = 0,
00124     SI2DR_SEVERITY_WARN = 1,
00125     SI2DR_SEVERITY_ERR  = 2
00126 } si2drSeverityT;
00127
00128
00129 typedef enum si2drObjectTypeT
00130 {
00131     SI2DR_UNDEFINED_OBJECTTYPE = 0,
00132     SI2DR_GROUP                 = 1,
00133     SI2DR_ATTR                  = 2,
00134     SI2DR_DEFINE                = 3,
00135     SI2DR_COMPLEXVAL            = 4,
00136     SI2DR_MAX_OBJECTTYPE        = 5
00137 } si2drObjectTypeT;
00138
00139 typedef enum si2drAttrTypeT
00140 {
00141     SI2DR_SIMPLE,
00142     SI2DR_COMPLEX
00143 } si2drAttrTypeT;
00144
00145
00146 typedef enum si2drValueTypeT
00147 {
00148     SI2DR_UNDEFINED_VALUETYPE = SI2_UNDEFINED_VALUETYPE,
00149     SI2DR_INT32                = SI2_LONG,
00150     SI2DR_STRING               = SI2_STRING,
00151     SI2DR_FLOAT64              = SI2_DOUBLE,
00152     SI2DR_BOOLEAN              = SI2_BOOLEAN,
00153     SI2DR_EXPR                 = SI2_EXPR,
00154     SI2DR_MAX_VALUETYPE        = SI2_MAX_VALUETYPE
00155 } si2drValueTypeT;
00156
00157 typedef enum si2drExprTypeT
00158 {
00159     SI2DR_EXPR_VAL = 0,      /* unary ops just have left arg set: add, sub, logs, paren */
00160     SI2DR_EXPR_OP_ADD = 1,
00161     SI2DR_EXPR_OP_SUB = 2,
00162     SI2DR_EXPR_OP_MUL = 3,
00163     SI2DR_EXPR_OP_DIV = 4,
00164     SI2DR_EXPR_OP_PAREN = 5,
00165     SI2DR_EXPR_OP_LOG2 = 6, /* pretty much flight of fancy from here on? */
00166     SI2DR_EXPR_OP_LOG10 = 7,
00167     SI2DR_EXPR_OP_EXP = 8, /* exponent? */
00168

```

```

00169 } si2drExprTypeT;
00170
00171
00172 typedef struct si2drExprT
00173 {
00174     si2drExprTypeT type;
00175     union
00176     {
00177         si2drInt32T i;
00178         si2drFloat64T d;
00179         si2drStringT s; /* most likely an identifier */
00180         si2drBooleanT b;
00181     } u;
00182
00183     si2drValueTypeT valuetype; /* if the type is a fixed value */
00184
00185     struct si2drExprT *left; /* the exprs form a classic binary tree rep of an arithmetic expression */
00186     struct si2drExprT *right;
00187 } si2drExprT;
00188
00189
00190 typedef enum si2drErrorT
00191 {
00192     SI2DR_NO_ERROR = 0,
00193     SI2DR_INTERNAL_SYSTEM_ERROR = 1,
00194     SI2DR_INVALID_VALUE = 4,
00195     SI2DR_INVALID_NAME = 5,
00196     SI2DR_INVALID_OBJECTTYPE = 6,
00197     SI2DR_INVALID_ATTRTYPE = 10,
00198     SI2DR_UNUSABLE_OID = 7,
00199     SI2DR_OBJECT_ALREADY_EXISTS = 8,
00200     SI2DR_OBJECT_NOT_FOUND = 9,
00201     SI2DR_SYNTAX_ERROR = 2,
00202     SI2DR_TRACE_FILES_CANNOT_BE_OPENED = 3,
00203     SI2DR_PIINIT_NOT_CALLED = 11,
00204     SI2DR_SEMANTIC_ERROR = 12,
00205     SI2DR_REFERENCE_ERROR = 13,
00206     SI2DR_MAX_ERROR = 14
00207 } si2drErrorT;
00208
00209 si2drVoidT si2drDefaultMessageHandler(si2drSeverityT sev,
00210                                       si2drErrorT errToPrint,
00211                                       si2drStringT auxText,
00212                                       si2drErrorT *err);
00213
00214
00215 typedef si2drVoidT (*si2drMessageHandlerT)(si2drSeverityT sev,
00216                                             si2drErrorT errToPrint,
00217                                             si2drStringT auxText,
00218                                             si2drErrorT *err);
00219
00220
00221 si2drMessageHandlerT si2drPIMessageHandler( si2drErrorT *err);
00222
00223
00224 si2drVoidT si2drPISetMessageHandler( si2drMessageHandlerT MsgHandFuncPtr,
00225                                       si2drErrorT *err);
00226
00227
00228 si2drGroupIdT si2drPICreateGroup SI2_ARGS(( si2drStringT name,
00229                                              si2drStringT group_type,
00230                                              si2drErrorT *err));
00231
00232
00233 si2drAttrIdT si2drGroupCreateAttr SI2_ARGS(( si2drGroupIdT group,
00234                                              si2drStringT name,
00235                                              si2drAttrTypeT type,
00236                                              si2drErrorT *err));
00237

```

```

00238     si2drAttrTypeT si2drAttrGetAttrType    SI2_ARGS(( si2drAttrIdT attr,
00239                                                         si2drErrorT *err));
00240
00241     si2drStringT   si2drAttrGetName         SI2_ARGS(( si2drAttrIdT attr,
00242                                                         si2drErrorT *err));
00243
00244     si2drVoidT     si2drComplexAttrAddInt32Value    SI2_ARGS(( si2drAttrIdT attr,
00245                                                         si2drInt32T   intgr,
00246                                                         si2drErrorT   *err ));
00247
00248     si2drVoidT     si2drComplexAttrAddStringValue    SI2_ARGS(( si2drAttrIdT attr,
00249                                                         si2drStringT   string,
00250                                                         si2drErrorT   *err ));
00251
00252     si2drVoidT     si2drComplexAttrAddBooleanValue    SI2_ARGS(( si2drAttrIdT attr,
00253                                                         si2drBooleanT   boolval,
00254                                                         si2drErrorT   *err ));
00255
00256     si2drVoidT     si2drComplexAttrAddFloat64Value    SI2_ARGS(( si2drAttrIdT attr,
00257                                                         si2drFloat64T   float64,
00258                                                         si2drErrorT   *err ));
00259
00260     si2drVoidT     si2drComplexAttrAddExprValue    SI2_ARGS(( si2drAttrIdT attr,
00261                                                         si2drExprT      *expr,
00262                                                         si2drErrorT   *err ));
00263
00264     si2drValuesIdT si2drComplexAttrGetValues    SI2_ARGS(( si2drAttrIdT attr,
00265                                                         si2drErrorT   *err ));
00266
00267     si2drVoidT     si2drIterNextComplexValue    SI2_ARGS(( si2drValuesIdT iter,
00268                                                         si2drValueTypeT *type,
00269                                                         si2drInt32T     *intgr,
00270                                                         si2drFloat64T   *float64,
00271                                                         si2drStringT     *string,
00272                                                         si2drBooleanT   *boolval,
00273                                                         si2drExprT      **expr,
00274                                                         si2drErrorT   *err ));
00275
00276     si2drAttrComplexValIdT si2drIterNextComplex    SI2_ARGS(( si2drValuesIdT iter,
00277                                                         si2drErrorT   *err ));
00278
00279     si2drValueTypeT si2drComplexValGetValueType    SI2_ARGS(( si2drAttrComplexValIdT attr,
00280                                                         si2drErrorT   *err ));
00281
00282     si2drInt32T     si2drComplexValGetInt32Value    SI2_ARGS(( si2drAttrComplexValIdT attr,
00283                                                         si2drErrorT   *err ));
00284
00285     si2drFloat64T   si2drComplexValGetFloat64Value    SI2_ARGS(( si2drAttrComplexValIdT attr,
00286                                                         si2drErrorT   *err ));
00287
00288     si2drStringT     si2drComplexValGetStringValue    SI2_ARGS(( si2drAttrComplexValIdT attr,
00289                                                         si2drErrorT   *err ));
00290
00291     si2drBooleanT    si2drComplexValGetBooleanValue    SI2_ARGS(( si2drAttrComplexValIdT attr,
00292                                                         si2drErrorT   *err ));
00293
00294     si2drExprT      *si2drComplexValGetExprValue    SI2_ARGS(( si2drAttrComplexValIdT attr,
00295                                                         si2drErrorT   *err ));
00296
00297
00298
00299
00300
00301
00302     si2drValueTypeT si2drSimpleAttrGetValueType    SI2_ARGS(( si2drAttrIdT attr,
00303                                                         si2drErrorT   *err ));
00304
00305     si2drInt32T     si2drSimpleAttrGetInt32Value    SI2_ARGS(( si2drAttrIdT attr,
00306                                                         si2drErrorT   *err ));

```

```

00307
00308     si2drFloat64T    si2drSimpleAttrGetFloat64Value    SI2_ARGS(( si2drAttrIdT attr,
00309                                                     si2drErrorT    *err ));
00310
00311     si2drStringT     si2drSimpleAttrGetStringValue     SI2_ARGS(( si2drAttrIdT attr,
00312                                                     si2drErrorT    *err ));
00313
00314     si2drBooleanT    si2drSimpleAttrGetBooleanValue    SI2_ARGS(( si2drAttrIdT attr,
00315                                                     si2drErrorT    *err ));
00316
00317     si2drExprT       *si2drSimpleAttrGetExprValue      SI2_ARGS(( si2drAttrIdT attr,
00318                                                     si2drErrorT    *err ));
00319
00320
00321
00322     si2drVoidT       si2drSimpleAttrSetInt32Value      SI2_ARGS(( si2drAttrIdT attr,
00323                                                     si2drInt32T    intgr,
00324                                                     si2drErrorT    *err ));
00325
00326     si2drVoidT       si2drSimpleAttrSetBooleanValue    SI2_ARGS(( si2drAttrIdT attr,
00327                                                     si2drBooleanT  intgr,
00328                                                     si2drErrorT    *err ));
00329
00330     si2drVoidT       si2drSimpleAttrSetFloat64Value    SI2_ARGS(( si2drAttrIdT attr,
00331                                                     si2drFloat64T float64,
00332                                                     si2drErrorT    *err ));
00333
00334     si2drVoidT       si2drSimpleAttrSetStringValue     SI2_ARGS(( si2drAttrIdT attr,
00335                                                     si2drStringT   string,
00336                                                     si2drErrorT    *err ));
00337
00338     si2drVoidT       si2drSimpleAttrSetExprValue      SI2_ARGS(( si2drAttrIdT attr,
00339                                                     si2drExprT     *expr,
00340                                                     si2drErrorT    *err ));
00341
00342     si2drBooleanT    si2drSimpleAttrGetIsVar           SI2_ARGS(( si2drAttrIdT attr,
00343                                                     si2drErrorT    *err ));
00344
00345     si2drVoidT       si2drSimpleAttrSetIsVar           SI2_ARGS(( si2drAttrIdT attr,
00346                                                     si2drErrorT    *err ));
00347
00348 /* helper functions for expr creation, building, and destruction */
00349
00350     si2drVoidT       si2drExprDestroy                  SI2_ARGS(( si2drExprT *expr, /* recursively free the structures */
00351                                                     si2drErrorT    *err));
00352
00353     si2drExprT       *si2drCreateExpr                  SI2_ARGS(( si2drExprTypeT type, /* malloc an Expr and return it */
00354                                                     si2drErrorT    *err));
00355
00356     si2drExprT       *si2drCreateBooleanValExpr        SI2_ARGS(( si2drBooleanT b,
00357                                                     si2drErrorT    *err ));
00358
00359     si2drExprT       *si2drCreateDoubleValExpr         SI2_ARGS(( si2drFloat64T d,
00360                                                     si2drErrorT    *err ));
00361
00362     si2drExprT       *si2drCreateStringValExpr         SI2_ARGS(( si2drStringT s,
00363                                                     si2drErrorT    *err ));
00364
00365     si2drExprT       *si2drCreateIntValExpr            SI2_ARGS(( si2drInt32T i,
00366                                                     si2drErrorT    *err ));
00367
00368     si2drExprT       *si2drCreateBinaryOpExpr          SI2_ARGS(( si2drExprT *left,
00369                                                     si2drExprTypeT optype,
00370                                                     si2drExprT *right,
00371                                                     si2drErrorT    *err ));
00372
00373     si2drExprT       *si2drCreateUnaryOpExpr           SI2_ARGS(( si2drExprTypeT optype,
00374                                                     si2drExprT *expr,
00375                                                     si2drErrorT    *err ));

```

```

00376
00377 si2drStringT    si2drExprToString          SI2_ARGS(( si2drExprT  *expr,
00378                                                         si2drErrorT  *err ));
00379
00380 si2drExprTypeT  si2drExprGetType           SI2_ARGS(( si2drExprT  *expr,
00381                                                         si2drErrorT  *err ));
00382
00383 si2drValueTypeT si2drValExprGetValueType   SI2_ARGS(( si2drExprT  *expr,
00384                                                         si2drErrorT  *err ));
00385
00386 si2drInt32T     si2drIntValExprGetInt      SI2_ARGS(( si2drExprT  *expr,
00387                                                         si2drErrorT  *err ));
00388
00389 si2drFloat64T   si2drDoubleValExprGetDouble SI2_ARGS(( si2drExprT  *expr,
00390                                                         si2drErrorT  *err ));
00391
00392 si2drBooleanT   si2drBooleanValExprGetBoolean SI2_ARGS(( si2drExprT  *expr,
00393                                                         si2drErrorT  *err ));
00394
00395 si2drStringT     si2drStringValExprGetString SI2_ARGS(( si2drExprT  *expr,
00396                                                         si2drErrorT  *err ));
00397
00398 si2drExprT      *si2drOpExprGetLeftExpr    SI2_ARGS(( si2drExprT  *expr, /* will apply to Unary & binary Ops */
00399                                                         si2drErrorT  *err ));
00400
00401 si2drExprT      *si2drOpExprGetRightExpr    SI2_ARGS(( si2drExprT  *expr,
00402                                                         si2drErrorT  *err ));
00403
00404 /* Define related funcs */
00405
00406 si2drDefineIdT  si2drGroupCreateDefine      SI2_ARGS(( si2drGroupIdT group,
00407                                                         si2drStringT  name,
00408                                                         si2drStringT  allowed_group_name,
00409                                                         si2drValueTypeT valtype,
00410                                                         si2drErrorT  *err));
00411
00412 si2drVoidT      si2drDefineGetInfo          SI2_ARGS(( si2drDefineIdT def,
00413                                                         si2drStringT  *name,
00414                                                         si2drStringT  *allowed_group_name,
00415                                                         si2drValueTypeT *valtype,
00416                                                         si2drErrorT  *err));
00417
00418 si2drStringT    si2drDefineGetName          SI2_ARGS(( si2drDefineIdT def,
00419                                                         si2drErrorT  *err));
00420
00421 si2drStringT    si2drDefineGetAllowedGroupName SI2_ARGS(( si2drDefineIdT def,
00422                                                         si2drErrorT  *err));
00423
00424 si2drValueTypeT si2drDefineGetValueType     SI2_ARGS(( si2drDefineIdT def,
00425                                                         si2drErrorT  *err));
00426
00427 si2drGroupIdT   si2drGroupCreateGroup       SI2_ARGS(( si2drGroupIdT group,
00428                                                         si2drStringT  name,
00429                                                         si2drStringT  group_type,
00430                                                         si2drErrorT  *err));
00431
00432 si2drStringT    si2drGroupGetGroupType      SI2_ARGS(( si2drGroupIdT group,
00433                                                         si2drErrorT  *err));
00434
00435 si2drStringT    si2drGroupGetComment        SI2_ARGS(( si2drGroupIdT group,
00436                                                         si2drErrorT  *err));
00437
00438 si2drVoidT      si2drGroupSetComment        SI2_ARGS(( si2drGroupIdT group,
00439                                                         si2drStringT  comment,
00440                                                         si2drErrorT  *err));
00441
00442 si2drStringT    si2drAttrGetComment         SI2_ARGS(( si2drAttrIdT attr,
00443                                                         si2drErrorT  *err));
00444

```

```

00445     si2drVoidT      si2drAttrSetComment      SI2_ARGS(( si2drAttrIdT attr,
00446                                                    si2drStringT comment,
00447                                                    si2drErrorT *err));
00448
00449     si2drStringT    si2drDefineGetComment      SI2_ARGS(( si2drDefineIdT def,
00450                                                    si2drErrorT *err));
00451
00452     si2drVoidT      si2drDefineSetComment      SI2_ARGS(( si2drDefineIdT def,
00453                                                    si2drStringT comment,
00454                                                    si2drErrorT *err));
00455
00456     si2drVoidT      si2drGroupAddName          SI2_ARGS(( si2drGroupIdT group,
00457                                                    si2drStringT name,
00458                                                    si2drErrorT *err));
00459
00460     si2drVoidT      si2drGroupDeleteName       SI2_ARGS(( si2drGroupIdT group,
00461                                                    si2drStringT name,
00462                                                    si2drErrorT *err));
00463
00464
00465     si2drGroupIdT    si2drPIFindGroupByName     SI2_ARGS(( si2drStringT name,
00466                                                    si2drStringT type,
00467                                                    si2drErrorT *err));
00468
00469     si2drGroupIdT    si2drGroupFindGroupByName SI2_ARGS(( si2drGroupIdT group,
00470                                                    si2drStringT name,
00471                                                    si2drStringT type,
00472                                                    si2drErrorT *err));
00473
00474     si2drAttrIdT     si2drGroupFindAttrByName  SI2_ARGS(( si2drGroupIdT group,
00475                                                    si2drStringT name,
00476                                                    si2drErrorT *err));
00477
00478     si2drDefineIdT   si2drGroupFindDefineByName SI2_ARGS(( si2drGroupIdT group,
00479                                                    si2drStringT name,
00480                                                    si2drErrorT *err));
00481
00482
00483     si2drDefineIdT   si2drPIFindDefineByName   SI2_ARGS(( si2drStringT name,
00484                                                    si2drErrorT *err));
00485
00486
00487
00488     si2drGroupsIdT   si2drPIGetGroups          SI2_ARGS(( si2drErrorT *err));
00489
00490
00491     si2drGroupsIdT   si2drGroupGetGroups       SI2_ARGS(( si2drGroupIdT group,
00492                                                    si2drErrorT *err));
00493
00494     si2drNamesIdT    si2drGroupGetNames        SI2_ARGS(( si2drGroupIdT group,
00495                                                    si2drErrorT *err));
00496
00497     si2drAttrsIdT    si2drGroupGetAttrs        SI2_ARGS(( si2drGroupIdT group,
00498                                                    si2drErrorT *err));
00499
00500     si2drDefinesIdT  si2drGroupGetDefines      SI2_ARGS(( si2drGroupIdT group,
00501                                                    si2drErrorT *err));
00502
00503     si2drGroupIdT    si2drIterNextGroup        SI2_ARGS(( si2drGroupsIdT iter,
00504                                                    si2drErrorT *err));
00505     si2drStringT     si2drIterNextName         SI2_ARGS(( si2drNamesIdT iter,
00506                                                    si2drErrorT *err));
00507     si2drAttrIdT     si2drIterNextAttr         SI2_ARGS(( si2drAttrsIdT iter,
00508                                                    si2drErrorT *err));
00509     si2drDefineIdT   si2drIterNextDefine       SI2_ARGS(( si2drDefinesIdT iter,
00510                                                    si2drErrorT *err));
00511     si2drVoidT       si2drIterQuit             SI2_ARGS(( si2drIterIdT iter,
00512                                                    si2drErrorT *err));
00513

```

```

00514
00515
00516     si2drVoidT      si2drObjectDelete    SI2_ARGS(( si2drObjectIdT object,
00517                                           si2drErrorT  *err));
00518
00519
00520
00521     si2drStringT    si2drPIGetErrorText    SI2_ARGS(( si2drErrorT errorCode,
00522                                           si2drErrorT  *err));
00523
00524     si2drObjectIdT  si2drPIGetNullId      SI2_ARGS(( si2drErrorT  *err));
00525
00526     si2drVoidT      si2drPIInit          SI2_ARGS(( si2drErrorT  *err));
00527
00528     si2drVoidT      si2drPIQuit          SI2_ARGS(( si2drErrorT  *err));
00529
00530     si2drObjectTypeT si2drObjectGetType    SI2_ARGS(( si2drObjectIdT object,
00531                                           si2drErrorT  *err));
00532
00533     si2drObjectIdT  si2drObjectGetOwner    SI2_ARGS(( si2drObjectIdT object,
00534                                           si2drErrorT  *err));
00535
00536     si2drBooleanT   si2drObjectIsNull      SI2_ARGS(( si2drObjectIdT object,
00537                                           si2drErrorT  *err));
00538
00539     si2drBooleanT   si2drObjectIsSame      SI2_ARGS(( si2drObjectIdT object1,
00540                                           si2drObjectIdT object2,
00541                                           si2drErrorT  *err));
00542
00543     si2drBooleanT   si2drObjectIsUsable    SI2_ARGS(( si2drObjectIdT object,
00544                                           si2drErrorT  *err));
00545
00546     si2drVoidT      si2drObjectSetFileName SI2_ARGS(( si2drObjectIdT object,
00547                                           si2drStringT  filename,
00548                                           si2drErrorT  *err));
00549
00550     si2drVoidT      si2drObjectSetLineNo   SI2_ARGS(( si2drObjectIdT object,
00551                                           si2drInt32T   lineno,
00552                                           si2drErrorT  *err));
00553
00554     si2drInt32T     si2drObjectGetLineNo   SI2_ARGS(( si2drObjectIdT object,
00555                                           si2drErrorT  *err));
00556
00557     si2drStringT    si2drObjectGetFileName SI2_ARGS(( si2drObjectIdT object,
00558                                           si2drErrorT  *err));
00559
00560     si2drVoidT      si2drReadLibertyFile   SI2_ARGS(( char *filename,
00561                                           si2drErrorT  *err));
00562
00563     si2drVoidT      si2drWriteLibertyFile  SI2_ARGS(( char *filename,
00564                                           si2drGroupIdT group,
00565                                           char* cellname,
00566                                           si2drErrorT  *err ));
00567
00568     si2drVoidT      si2drCheckLibertyLibrary SI2_ARGS(( si2drGroupIdT group,
00569                                           si2drErrorT  *err));
00570
00571
00572     si2drBooleanT   si2drPIGetTraceMode    SI2_ARGS((si2drErrorT  *err));
00573
00574     si2drVoidT      si2drPIUnSetTraceMode  SI2_ARGS((si2drErrorT  *err));
00575
00576     si2drVoidT      si2drPISetTraceMode    SI2_ARGS((si2drStringT fname,
00577                                           si2drErrorT  *err));
00578
00579
00580     si2drVoidT      si2drPISetDebugMode    SI2_ARGS((si2drErrorT  *err));
00581
00582     si2drVoidT      si2drPIUnSetDebugMode  SI2_ARGS((si2drErrorT  *err));

```

```

00583
00584     si2drBooleanT    si2drPIGetDebugMode    SI2_ARGS((si2drErrorT *err));
00585
00586
00587     si2drVoidT       si2drPISetNocheckMode    SI2_ARGS((si2drErrorT *err));
00588
00589     si2drVoidT       si2drPIUnSetNocheckMode    SI2_ARGS((si2drErrorT *err));
00590
00591     si2drBooleanT    si2drPIGetNocheckMode    SI2_ARGS((si2drErrorT *err));
00592
00593
00594     si2drVoidT       si2drGroupMoveAfter SI2_ARGS((si2drGroupIdT groupToMove,
00595                                                    si2drGroupIdT targetGroup,
00596                                                    si2drErrorT *err));
00597
00598     si2drVoidT       si2drGroupMoveBefore SI2_ARGS((si2drGroupIdT groupToMove,
00599                                                    si2drGroupIdT targetGroup,
00600                                                    si2drErrorT *err));
00601
00602
00603 /* HELPER FUNCTIONS/STRUCTURES */
00604
00605 #ifdef NO_LONG_DOUBLE
00606 #define LONG_DOUBLE double
00607 #define strtold strtod
00608 #else
00609 #define LONG_DOUBLE long double
00610 #endif
00611
00612 struct liberty_value_data
00613 {
00614     int dimensions;
00615     int *dim_sizes; /* a malloc'd array of <dimensions> numbers,
00616 each number is the size of the array in that dim.*/
00617     LONG_DOUBLE **index_info; /* a Ptr to a malloc'd array of size <dimensions>, ptrs to
00618 malloc'd arrays of long double index values.
00619 Each array of long double is of length set by
00620 the corresponding dim_sizes array values */
00621     LONG_DOUBLE *values; /* a ptr to a malloc'd array of long doubles,
00622 starting with [0,0,...,0,0], and ending with
00623 [z-1,y-1,...,b-1,a-1], where a-z are the max
00624 number of elements in each dimension. */
00625 };
00626
00627 LONG_DOUBLE liberty_get_element(struct liberty_value_data *vd, ...); /* returns NaN if bounds exceeded */
00628
00629 void liberty_destroy_value_data(struct liberty_value_data *vd);
00630
00631 struct liberty_value_data *liberty_get_values_data( si2drGroupIdT table_group);
00632
00633 // directly call the main function of liberty parser
00634 extern void print_version();
00635 extern void usage();
00636 extern int main_liberty_parser(int argc, char* argv[]);
00637 extern void get_function(char* filename);
00638
00639 #ifdef __cplusplus
00640 }
00641 #endif
00642
00643 #endif
00644

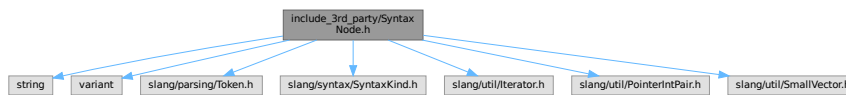
```


8.21 include_3rd_party/SyntaxNode.h File Reference

Base class and utilities for syntax nodes.

```
#include <string>
#include <variant>
#include "slang/parsing/Token.h"
#include "slang/syntax/SyntaxKind.h"
#include "slang/util/Iterator.h"
#include "slang/util/PointerIntPair.h"
#include "slang/util/SmallVector.h"
```

Include dependency graph for SyntaxNode.h:



Classes

- struct [slang::syntax::PtrTokenOrSyntax](#)
A token pointer or a syntax node.
- struct [slang::syntax::ConstTokenOrSyntax](#)
A token or a constant syntax node.
- struct [slang::syntax::TokenOrSyntax](#)
A token or a syntax node.
- class [slang::syntax::SyntaxNode](#)
Base class for all syntax nodes.
- class [slang::syntax::DeferredSourceRange](#)
- class [slang::syntax::SyntaxListBase](#)
A base class for syntax nodes that represent a list of items.
- class [slang::syntax::SyntaxList< T >](#)
A syntax node that represents a list of child syntax nodes.
- class [slang::syntax::TokenList](#)
A syntax node that represents a list of child tokens.
- class [slang::syntax::SeparatedSyntaxList< T >](#)
- class [slang::syntax::SeparatedSyntaxList< T >::iterator_base< U >](#)

Namespaces

- namespace [slang](#)
- namespace [slang::syntax](#)

Functions

- SLANG_EXPORT SyntaxNode * [slang::syntax::clone](#) (const SyntaxNode &node, BumpAllocator &alloc)
Performs a shallow clone of the given syntax node.
- SLANG_EXPORT SyntaxNode * [slang::syntax::deepClone](#) (const SyntaxNode &node, BumpAllocator &alloc)
Performs a deep clone of the given syntax node.
- template<std::derived_from< SyntaxNode > T>
T * [slang::syntax::clone](#) (const T &node, BumpAllocator &alloc)
Performs a shallow clone of the given syntax node.
- template<std::derived_from< SyntaxNode > T>
T * [slang::syntax::deepClone](#) (const T &node, BumpAllocator &alloc)
Performs a deep clone of the given syntax node.
- template<typename T >
SyntaxList< T > * [slang::syntax::deepClone](#) (const SyntaxList< T > &node, BumpAllocator &alloc)
- template<typename T >
SeparatedSyntaxList< T > * [slang::syntax::deepClone](#) (const SeparatedSyntaxList< T > &node, BumpAllocator &alloc)
- TokenList * [slang::syntax::deepClone](#) (const TokenList &node, BumpAllocator &alloc)

8.21.1 Detailed Description

Base class and utilities for syntax nodes.

Definition in file [SyntaxNode.h](#).

8.22 SyntaxNode.h

[Go to the documentation of this file.](#)

```
00001 //-----
00004 //
00005 // SPDX-FileCopyrightText: Michael Popoloski
00006 // SPDX-License-Identifier: MIT
00007 //-----
00008 #pragma once
00009
00010 #include <string>
00011 #include <variant>
00012
00013 #include "slang/parsing/Token.h"
00014 #include "slang/syntax/SyntaxKind.h"
00015 #include "slang/util/Iterator.h"
00016 #include "slang/util/PointerIntPair.h"
00017 #include "slang/util/SmallVector.h"
```

```

00018
00019 namespace slang::syntax {
00020
00021 class SyntaxNode;
00022
00024 struct SLANG_EXPORT PtrTokenOrSyntax : public std::variant<parsing::Token*, SyntaxNode*> {
00025     using Base = std::variant<parsing::Token*, SyntaxNode*>;
00026     PtrTokenOrSyntax(parsing::Token* token) : Base(token) {}
00027     PtrTokenOrSyntax(SyntaxNode* node) : Base(node) {}
00028     PtrTokenOrSyntax(nullptr_t) : Base((parsing::Token*)nullptr) {}
00029
00031     bool isToken()const { return this->index() == 0; }
00032
00034     bool isNode()const { return this->index() == 1; }
00035
00038     parsing::Token* token()const { return std::get<0>(*this); }
00039
00042     SyntaxNode* node()const { return std::get<1>(*this); }
00043
00046     SourceRange range() const;
00047 };
00048
00050 struct SLANG_EXPORT ConstTokenOrSyntax : public std::variant<parsing::Token, const SyntaxNode*> {
00051     using Base = std::variant<parsing::Token, const SyntaxNode*>;
00052     ConstTokenOrSyntax(parsing::Token token) : Base(token) {}
00053     ConstTokenOrSyntax(const SyntaxNode* node) : Base(node) {}
00054     ConstTokenOrSyntax(nullptr_t) : Base(parsing::Token()) {}
00055
00057     bool isToken()const { return this->index() == 0; }
00058
00060     bool isNode()const { return this->index() == 1; }
00061
00064     parsing::Token token()const { return std::get<0>(*this); }
00065
00068     const SyntaxNode* node()const { return std::get<1>(*this); }
00069
00071     SourceRange range() const;
00072 };
00073
00075 struct SLANG_EXPORT TokenOrSyntax : public ConstTokenOrSyntax {
00076     TokenOrSyntax(parsing::Token token) : ConstTokenOrSyntax(token) {}
00077     TokenOrSyntax(SyntaxNode* node) : ConstTokenOrSyntax(node) {}
00078     TokenOrSyntax(nullptr_t) : ConstTokenOrSyntax(parsing::Token()) {}
00079
00082     SyntaxNode* node()const {
00083         // const_cast is safe because we only could have constructed this
00084         // object with a non-const pointer.
00085         return const_cast<SyntaxNode*>(std::get<1>(*this));
00086     }
00087 };
00088
00090 class SLANG_EXPORT SyntaxNode {
00091 public:
00092     using Token = parsing::Token;
00093
00095     SyntaxKind kind;
00096
00099     SyntaxNode* parent = nullptr;
00100
00108     const SyntaxNode* previewNode = nullptr;
00109
00111     std::string toString() const;
00112
00114     Token getFirstToken() const;
00115
00117     Token getLastToken() const;
00118
00120     Token* getFirstTokenPtr();
00121

```

```

00123     Token* getLastTokenPtr();
00124
00126     SourceRange sourceRange() const;
00127
00130     const SyntaxNode* childNode(size_t index) const;
00131
00134     SyntaxNode* childNode(size_t index);
00135
00139     Token childToken(size_t index) const;
00140
00144     Token* childTokenPtr(size_t index);
00145
00147     size_t getChildCount() const; // Note: implemented in AllSyntax.cpp
00148
00153     bool isEquivalentTo(const SyntaxNode& other) const;
00154
00157     template<typename T>
00158     T& as() {
00159         SLANG_ASSERT(T::isKind(kind));
00160         return *static_cast<T*>(this);
00161     }
00162
00165     template<typename T>
00166     const T& as()const {
00167         SLANG_ASSERT(T::isKind(kind));
00168         return *static_cast<const T*>(this);
00169     }
00170
00171     // Reinterprets this node as being of type T, if it is not belong type T, return nullptr
00172     template<typename T>
00173     T* as_if() {
00174         if (!T::isKind(kind))
00175             return nullptr;
00176         return static_cast<T*>(this);
00177     }
00178
00179     // Reinterprets this node as being of type T, if it is not belong type T, return nullptr
00180     template<typename T>
00181     const T* as_if()const {
00182         if (!T::isKind(kind))
00183             return nullptr;
00184         return static_cast<const T*>(this);
00185     }
00186
00189     template<typename TVisitor, typename... Args>
00190     decltype(auto) visit(TVisitor& visitor, Args&&... args);
00191
00194     template<typename TVisitor, typename... Args>
00195     decltype(auto) visit(TVisitor& visitor, Args&&... args) const;
00196
00200     static bool isKind(SyntaxKind) { return true; }
00201
00204     static bool isChildOptional(size_t) { return true; }
00205
00206 protected:
00207     explicit SyntaxNode(SyntaxKind kind) : kind(kind) {}
00208
00209 private:
00210     ConstTokenOrSyntax getChild(size_t index) const;
00211     TokenOrSyntax getChild(size_t index);
00212     PtrTokenOrSyntax getChildPtr(size_t index);
00213 };
00214
00219 SLANG_EXPORT SyntaxNode* clone(const SyntaxNode& node, BumpAllocator& alloc);
00220
00226 SLANG_EXPORT SyntaxNode* deepClone(const SyntaxNode& node, BumpAllocator& alloc);
00227
00232 template<std::derived_from<SyntaxNode> T>
00233 T* clone(const T& node, BumpAllocator& alloc) {

```

```

00234     return static_cast<T*>(clone(static_cast<const SyntaxNode*>(node), alloc));
00235 }
00236
00242 template<std::derived_from<SyntaxNode> T>
00243 T* deepClone(const T& node, BumpAllocator& alloc) {
00244     return static_cast<T*>(deepClone(static_cast<const SyntaxNode*>(node), alloc));
00245 }
00246
00250 class DeferredSourceRange {
00251 public:
00252     DeferredSourceRange() = default;
00253     DeferredSourceRange(SourceRange range) : range(range), node(nullptr, true) {}
00254     DeferredSourceRange(const SyntaxNode& node) : node(&node, false) {}
00255     DeferredSourceRange(const SyntaxNode& node, SourceRange range) :
00256         range(range), node(&node, true) {}
00257
00258     SourceRange get()const {
00259         if (!node.getInt()) {
00260             if (auto ptr = node.getPointer())
00261                 range = ptr->sourceRange();
00262             node.setInt(true);
00263         }
00264         return range;
00265     }
00266
00267     SourceRange operator*()const { return get(); }
00268
00269     const SyntaxNode* syntax()const { return node.getPointer(); }
00270
00271 private:
00272     mutable SourceRange range;
00273     mutable PointerIntPair<const SyntaxNode*, 1, 1, bool> node;
00274 };
00275
00277 class SLANG_EXPORT SyntaxListBase : public SyntaxNode {
00278 public:
00280     size_t getChildCount()const { return childCount; }
00281
00283     virtual TokenOrSyntax getChild(size_t index) = 0;
00284
00286     virtual ConstTokenOrSyntax getChild(size_t index) const = 0;
00287
00288     // Gets the child pointer (token or node) at given index.
00289     virtual PtrTokenOrSyntax getChildPtr(size_t index) = 0;
00290
00292     virtual void setChild(size_t index, TokenOrSyntax child) = 0;
00293
00295     virtual SyntaxListBase* clone(BumpAllocator& alloc) const = 0;
00296
00299     virtual void resetAll(BumpAllocator& alloc, std::span<const TokenOrSyntax> children) = 0;
00300
00301     static bool isKind(SyntaxKind kind);
00302
00303     static bool isChildOptional(size_t index);
00304
00305 protected:
00306     SyntaxListBase(SyntaxKind kind, size_t childCount) : SyntaxNode(kind), childCount(childCount) {}
00307
00308     size_t childCount;
00309 };
00310
00312 template<typename T>
00313 class SLANG_EXPORT SyntaxList : public SyntaxListBase, public std::span<T*> {
00314 public:
00315     SyntaxList(nullptr_t) : SyntaxList(std::span<T*>()) {}
00316     SyntaxList(std::span<T*> elements) :
00317         SyntaxListBase(SyntaxKind::SyntaxList, elements.size(), std::span<T*>(elements)) {}
00318
00319 private:

```

```

00320 TokenOrSyntax getChild(size_t index) final { return (*this)[index]; }
00321 ConstTokenOrSyntax getChild(size_t index) const final { return (*this)[index]; }
00322 PtrTokenOrSyntax getChildPtr(size_t index) final { return (*this)[index]; };
00323
00324 void setChild(size_t index, TokenOrSyntax child) final {
00325     (*this)[index] = &child.node()->as<T>();
00326 }
00327
00328 SyntaxListBase* clone(BumpAllocator& alloc) const final {
00329     return alloc.emplace<SyntaxList<T>>>(*this);
00330 }
00331
00332 void resetAll(BumpAllocator& alloc, std::span<const TokenOrSyntax> children) final {
00333     SmallVector<T*> buffer(children.size(), UninitializedTag());
00334     for (auto& t : children)
00335         buffer.push_back(&t.node()->as<T>());
00336
00337     *this = buffer.copy(alloc);
00338     childCount = buffer.size();
00339 }
00340 };
00341
00342 template<typename T>
00343 SyntaxList<T>* deepClone(const SyntaxList<T>& node, BumpAllocator& alloc) {
00344     SmallVector<T*> buffer(node.size(), UninitializedTag());
00345     for (auto& t : node)
00346         buffer.push_back(deepClone(*t, alloc));
00347
00348     return alloc.emplace<SyntaxList<T>>(buffer.copy(alloc));
00349 }
00350
00352 class SLANG_EXPORT TokenList : public SyntaxListBase, public std::span<parsing::Token> {
00353 public:
00354     TokenList(nullptr_t) : TokenList(std::span<Token>()) {}
00355     TokenList(std::span<Token> elements) :
00356         SyntaxListBase(SyntaxKind::TokenList, elements.size(), std::span<Token>(elements)) {}
00357 private:
00358     TokenOrSyntax getChild(size_t index) final { return (*this)[index]; }
00359     ConstTokenOrSyntax getChild(size_t index) const final { return (*this)[index]; }
00360     PtrTokenOrSyntax getChildPtr(size_t index) final { return &(*this)[index]; };
00361     void setChild(size_t index, TokenOrSyntax child) final { (*this)[index] = child.token(); }
00362
00363     SyntaxListBase* clone(BumpAllocator& alloc) const final {
00364         return alloc.emplace<TokenList>(*this);
00365     }
00366
00367     void resetAll(BumpAllocator& alloc, std::span<const TokenOrSyntax> children) final {
00368         SmallVector<Token> buffer(children.size(), UninitializedTag());
00369         for (auto& t : children)
00370             buffer.push_back(t.token());
00371
00372         *this = buffer.copy(alloc);
00373         childCount = buffer.size();
00374     }
00375 }
00376 };
00377
00380 template<typename T>
00381 class SLANG_EXPORT SeparatedSyntaxList : public SyntaxListBase {
00382 public:
00383     template<typename U>
00384     class iterator_base : public iterator_facade<iterator_base<U>> {
00385     public:
00386         using ParentList = std::conditional_t<std::is_const_v<std::remove_pointer_t<U>>,
00387             const SeparatedSyntaxList, SeparatedSyntaxList>;
00388
00389         iterator_base() : list(nullptr), index(0) {}
00390         iterator_base(ParentList& list, size_t index) : list(&list), index(index) {}
00391     };
00392 }

```

```

00394         U dereference()const { return (*list)[index]; }
00395
00396         bool equals(const iterator_base& other)const {
00397             return list == other.list && index == other.index;
00398         }
00399
00400         ptrdiff_t distance_to(const iterator_base& other)const {
00401             return ptrdiff_t(other.index) - ptrdiff_t(index);
00402         }
00403
00404         void advance(ptrdiff_t n) { index = size_t(ptrdiff_t(index) + n); }
00405
00406     private:
00407         ParentList* list;
00408         size_t index;
00409     };
00410
00411     using iterator = iterator_base<T*>;
00412     using const_iterator = iterator_base<const T*>;
00413
00414     SeparatedSyntaxList(nullptr_t) : SeparatedSyntaxList(std::span<TokenOrSyntax>()) {}
00415     SeparatedSyntaxList(std::span<TokenOrSyntax> elements) :
00416         SyntaxListBase(SyntaxKind::SeparatedList, elements.size(), elements(elements)) {}
00417
00419     [[nodiscard]] std::span<const ConstTokenOrSyntax> elems()const {
00420         return std::span<const ConstTokenOrSyntax>(
00421             static_cast<ConstTokenOrSyntax*>(elements.data()), elements.size());
00422     }
00423
00425     [[nodiscard]] std::span<TokenOrSyntax> elems() { return elements; }
00426
00428     [[nodiscard]] bool empty()const { return elements.empty(); }
00429
00431     [[nodiscard]] size_t size() const noexcept { return (elements.size() + 1) / 2; }
00432
00433     const T* operator[](size_t index)const {
00434         index <= 1;
00435         return &elements[index].node()->template as<T>();
00436     }
00437
00438     T* operator[](size_t index) {
00439         index <= 1;
00440         return &elements[index].node()->template as<T>();
00441     }
00442
00443     iterator begin() { return iterator(*this, 0); }
00444     iterator end() { return iterator(*this, size()); }
00445     const_iterator begin()const { return cbegin(); }
00446     const_iterator end()const { return cend(); }
00447     const_iterator cbegin()const { return const_iterator(*this, 0); }
00448     const_iterator cend()const { return const_iterator(*this, size()); }
00449
00450     private:
00451     TokenOrSyntax getChild(size_t index) final { return elements[index]; }
00452     ConstTokenOrSyntax getChild(size_t index) const final { return elements[index]; }
00453     PtrTokenOrSyntax getChildPtr(size_t index) final {
00454         if (elements[index].isNode())
00455             return elements[index].node();
00456         else
00457             return &(std::get<parsing::Token>(elements[index]));
00458     }
00459     void setChild(size_t index, TokenOrSyntax child) final { elements[index] = child; }
00460
00461     SyntaxListBase* clone(BumpAllocator& alloc) const final {
00462         return alloc.emplace<SeparatedSyntaxList<T*>>(*this);
00463     }
00464
00465     void resetAll(BumpAllocator& alloc, std::span<const TokenOrSyntax> children) final {
00466         SmallVector<TokenOrSyntax> buffer(children.size(), UninitializedTag());

```

```

00467         buffer.append_range(children);
00468
00469         elements = buffer.copy(alloc);
00470         childCount = buffer.size();
00471     }
00472
00473     std::span<TokenOrSyntax> elements;
00474 };
00475
00476 template<typename T>
00477 SeparatedSyntaxList<T>* deepClone(const SeparatedSyntaxList<T>& node, BumpAllocator& alloc) {
00478     SmallVector<TokenOrSyntax> buffer(node.size(), UninitializedTag());
00479     for (const auto& ele : node.elms()) {
00480         if (ele.isToken())
00481             buffer.push_back(ele.token().deepClone(alloc));
00482         else
00483             buffer.push_back(static_cast<T*>(deepClone(*ele.node(), alloc)));
00484     }
00485     return alloc.emplace<SeparatedSyntaxList<T>>(buffer.copy(alloc));
00486 }
00487
00488 inline TokenList* deepClone(const TokenList& node, BumpAllocator& alloc) {
00489     SmallVector<parsing::Token> buffer(node.size(), UninitializedTag());
00490     for (const auto& ele : node) {
00491         buffer.push_back(ele.deepClone(alloc));
00492     }
00493     return alloc.emplace<TokenList>(buffer.copy(alloc));
00494 }
00495
00496 } // namespace slang::syntax

```

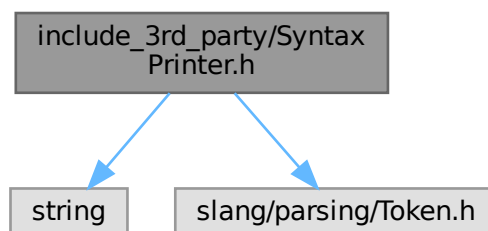
8.23 include_3rd_party/SyntaxPrinter.h File Reference

Support for printing syntax nodes and tokens.

```
#include <string>
```

```
#include "slang/parsing/Token.h"
```

Include dependency graph for SyntaxPrinter.h:



Classes

- class [slang::syntax::SyntaxPrinter](#)

Namespaces

- namespace `slang`
- namespace `slang::syntax`

8.23.1 Detailed Description

Support for printing syntax nodes and tokens.

Definition in file [SyntaxPrinter.h](#).

8.24 SyntaxPrinter.h

[Go to the documentation of this file.](#)

```
00001 -----
00004 //
00005 // SPDX-FileCopyrightText: Michael Popoloski
00006 // SPDX-License-Identifier: MIT
00007 -----
00008 #pragma once
00009
00010 #include <string>
00011
00012 #include "slang/parsing/Token.h"
00013
00014 namespace slang {
00015     class SourceManager;
00016 }
00017
00018 namespace slang::syntax {
00019
00020     class SyntaxNode;
00021     class SyntaxTree;
00022
00025     class SLANG_EXPORT SyntaxPrinter {
00026     public:
00027         SyntaxPrinter() = default;
00028         explicit SyntaxPrinter(const SourceManager& sourceManager);
00029
00032         SyntaxPrinter& append(std::string_view text);
00033
00036         SyntaxPrinter& print(parsing::Trivia trivia);
00037
00040         SyntaxPrinter& print(parsing::Token token);
00041
00044         SyntaxPrinter& print(const SyntaxNode& node);
00045
00048         SyntaxPrinter& print(const SyntaxTree& tree);
00049
00052         SyntaxPrinter& setIncludeTrivia(bool include) {
00053             includeTrivia = include;
00054             return *this;
00055         }
00056
00059         SyntaxPrinter& setIncludeMissing(bool include) {
00060             includeMissing = include;
00061             return *this;
00062         }
00063     }
```

```

00063
00066     SyntaxPrinter& setIncludeSkipped(bool include) {
00067         includeSkipped = include;
00068         return *this;
00069     }
00070
00073     SyntaxPrinter& setIncludeDirectives(bool include) {
00074         includeDirectives = include;
00075         return *this;
00076     }
00077
00080     SyntaxPrinter& setIncludePreprocessed(bool include) {
00081         includePreprocessed = include;
00082         return *this;
00083     }
00084
00087     SyntaxPrinter& setIncludeComments(bool include) {
00088         includeComments = include;
00089         return *this;
00090     }
00091
00094     SyntaxPrinter& setSquashNewlines(bool include) {
00095         squashNewlines = include;
00096         return *this;
00097     }
00098
00100     std::string str()const { return buffer; }
00101
00105     static std::string printFile(const SyntaxTree& tree);
00106
00107 private:
00108     std::string buffer;
00109     const SourceManager* sourceManager = nullptr;
00110     bool includeTrivia = true;
00111     bool includeMissing = false;
00112     bool includeSkipped = false;
00113     bool includeDirectives = false;
00114     bool includePreprocessed = true;
00115     bool includeComments = true;
00116     bool squashNewlines = true;
00117 };
00118
00119 } // namespace slang::syntax

```

8.25 include_3rd_party/SyntaxTree.h File Reference

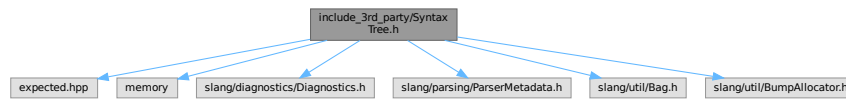
Top-level parser interface.

```

#include <expected.hpp>
#include <memory>
#include "slang/diagnostics/Diagnostics.h"
#include "slang/parsing/ParserMetadata.h"
#include "slang/util/Bag.h"
#include "slang/util/BumpAllocator.h"

```

Include dependency graph for SyntaxTree.h:



Classes

- class [slang::syntax::SyntaxTree](#)

Namespaces

- namespace [slang](#)
- namespace [slang::parsing](#)
- namespace [slang::syntax](#)

8.25.1 Detailed Description

Top-level parser interface.

Definition in file [SyntaxTree.h](#).

8.26 SyntaxTree.h

[Go to the documentation of this file.](#)

```

00001 //-----
00004 //
00005 // SPDX-FileCopyrightText: Michael Popoloski
00006 // SPDX-License-Identifier: MIT
00007 //-----
00008 #pragma once
00009
00010 #include <expected.hpp>
00011 #include <memory>
00012
00013 #include "slang/diagnostics/Diagnostics.h"
00014 #include "slang/parsing/ParserMetadata.h"
00015 #include "slang/util/Bag.h"
00016 #include "slang/util/BumpAllocator.h"
00017
00018 namespace slang {
00019
00020 class SourceManager;
00021 struct SourceBuffer;
00022
00023 } // namespace slang
00024

```

```

00025 namespace slang::parsing {
00026 struct ParserMetadata;
00027 }
00028
00029 namespace slang::syntax {
00030
00031 class SyntaxNode;
00032 struct DefineDirectiveSyntax;
00033
00039 class SLANG_EXPORT SyntaxTree {
00040 public:
00041     using TreeOrError =
00042         nonstd::expected<std::shared_ptr<SyntaxTree>, std::pair<std::error_code, std::string_view>>;
00043     using MacroList = std::span<const DefineDirectiveSyntax* const>;
00044
00047     bool isLibraryUnit = false;
00048
00049     SyntaxTree(SyntaxNode* root, SourceManager& sourceManager, BumpAllocator&& alloc,
00050         const SourceLibrary* library, const std::shared_ptr<SyntaxTree>& parent = nullptr);
00051
00052     SyntaxTree(SyntaxTree&& other) = default;
00053     ~SyntaxTree();
00054
00058     static TreeOrError fromFile(std::string_view path);
00059
00066     static TreeOrError fromFile(std::string_view path, SourceManager& sourceManager,
00067         const Bag& options = {});
00068
00073     static TreeOrError fromFiles(std::span<const std::string_view> paths);
00074
00081     static TreeOrError fromFiles(std::span<const std::string_view> paths,
00082         SourceManager& sourceManager, const Bag& options = {});
00083
00089     static std::shared_ptr<SyntaxTree> fromText(std::string_view text,
00090         std::string_view name = "source",
00091         std::string_view path = "");
00092
00099     static std::shared_ptr<SyntaxTree> fromText(std::string_view text, const Bag& options,
00100         std::string_view name = "source"sv,
00101         std::string_view path = "");
00102
00111     static std::shared_ptr<SyntaxTree> fromText(std::string_view text, SourceManager& sourceManager,
00112         std::string_view name = "source"sv,
00113         std::string_view path = "", const Bag& options = {},
00114         const SourceLibrary* library = nullptr);
00115
00123     static std::shared_ptr<SyntaxTree> fromFileInMemory(std::string_view text,
00124         SourceManager& sourceManager,
00125         std::string_view name = "source"sv,
00126         std::string_view path = "",
00127         const Bag& options = {});
00128
00135     static std::shared_ptr<SyntaxTree> fromBuffer(const SourceBuffer& buffer,
00136         SourceManager& sourceManager,
00137         const Bag& options = {},
00138         MacroList inheritedMacros = {});
00139
00147     static std::shared_ptr<SyntaxTree> fromBuffers(std::span<const SourceBuffer> buffers,
00148         SourceManager& sourceManager,
00149         const Bag& options = {},
00150         MacroList inheritedMacros = {});
00151
00157     static std::shared_ptr<SyntaxTree> fromLibraryMapFile(std::string_view path,
00158         SourceManager& sourceManager,
00159         const Bag& options = {});
00160
00168     static std::shared_ptr<SyntaxTree> fromLibraryMapText(std::string_view text,
00169         SourceManager& sourceManager,
00170         std::string_view name = "source"sv,

```

```

00171         std::string_view path = "",
00172         const Bag& options = {});
00173
00179     static std::shared_ptr<SyntaxTree> fromLibraryMapBuffer(const SourceBuffer& buffer,
00180         SourceManager& sourceManager,
00181         const Bag& options = {});
00182
00184     Diagnostics& diagnostics() { return diagnosticsBuffer; }
00185
00187     BumpAllocator& allocator() { return alloc; }
00188
00190     SourceManager& sourceManager() { return sourceMan; }
00191
00193     const SourceManager& sourceManager()const { return sourceMan; }
00194
00196     const SourceLibrary* getSourceLibrary()const { return library; }
00197
00199     SyntaxNode& root() { return *rootNode; }
00200
00202     const SyntaxNode& root()const { return *rootNode; }
00203
00205     const Bag& options()const { return options_; }
00206
00208     const parsing::ParserMetadata& getMetadata()const { return *metadata; }
00209
00211     MacroList getDefinedMacros()const { return macros; }
00212
00217     static SourceManager& getDefaultSourceManager();
00218
00219 private:
00220     SyntaxTree(SyntaxNode* root, const SourceLibrary* library, SourceManager& sourceManager,
00221         BumpAllocator&& alloc, Diagnostics&& diagnostics, parsing::ParserMetadata&& metadata,
00222         std::vector<const DefineDirectiveSyntax*>&& macros, Bag options);
00223
00224     static std::shared_ptr<SyntaxTree> create(SourceManager& sourceManager,
00225         std::span<const SourceBuffer> source,
00226         const Bag& options, MacroList inheritedMacros,
00227         bool guess);
00228
00229     SyntaxNode* rootNode;
00230     const SourceLibrary* library;
00231     SourceManager& sourceMan;
00232     BumpAllocator alloc;
00233     Diagnostics diagnosticsBuffer;
00234     Bag options_;
00235     std::unique_ptr<parsing::ParserMetadata> metadata;
00236     std::vector<const DefineDirectiveSyntax*> macros;
00237 };
00238
00239 } // namespace slang::syntax

```

8.27 include_3rd_party/SyntaxVisitor.h File Reference

Syntax tree visitor support.

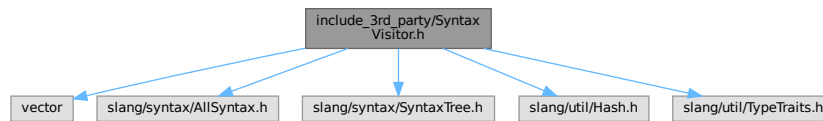
```

#include <vector>
#include "slang/syntax/AllSyntax.h"
#include "slang/syntax/SyntaxTree.h"
#include "slang/util/Hash.h"

```

```
#include "slang/util/TypeTraits.h"
```

Include dependency graph for SyntaxVisitor.h:



Classes

- class [slang::syntax::SyntaxVisitor< TDerived >](#)
- struct [slang::syntax::detail::SyntaxChange](#)
- struct [slang::syntax::detail::RemoveChange](#)
- struct [slang::syntax::detail::ReplaceChange](#)
- struct [slang::syntax::detail::ChangeCollection](#)
- class [slang::syntax::SyntaxRewriter< TDerived >](#)

Namespaces

- namespace [slang](#)
- namespace [slang::syntax](#)
- namespace [slang::syntax::detail](#)

Macros

- `#define` [DERIVED](#) `static_cast<TDerived*>(this)`

Typedefs

- using [slang::syntax::detail::InsertChangeMap](#) = `flat_hash_map< const SyntaxNode *, std::vector< SyntaxChange > >`
- using [slang::syntax::detail::ModifyChangeMap](#) = `flat_hash_map< const SyntaxNode *, std::variant< RemoveChange, ReplaceChange > >`
- using [slang::syntax::detail::ListChangeMap](#) = `flat_hash_map< const SyntaxNode *, std::vector< SyntaxChange > >`

Functions

- SLANG_EXPORT std::shared_ptr< SyntaxTree > [slang::syntax::detail::transformTree](#) (BumpAllocator &&alloc, const std::shared_ptr< SyntaxTree > &tree, const ChangeCollection &commits, const std::vector< std::shared_ptr< SyntaxTree > > &tempTrees, const SourceLibrary *library)

8.27.1 Detailed Description

Syntax tree visitor support.

Definition in file [SyntaxVisitor.h](#).

8.27.2 Macro Definition Documentation

8.27.2.1 DERIVED

```
#define DERIVED static_cast<TDerived*>(this)
```

Definition at line 19 of file [SyntaxVisitor.h](#).

8.28 SyntaxVisitor.h

[Go to the documentation of this file.](#)

```
00001 //-----
00004 //
00005 // SPDX-FileCopyrightText: Michael Popoloski
00006 // SPDX-License-Identifier: MIT
00007 //-----
00008 #pragma once
00009
00010 #include <vector>
00011
00012 #include "slang/syntax/AllSyntax.h"
00013 #include "slang/syntax/SyntaxTree.h"
00014 #include "slang/util/Hash.h"
00015 #include "slang/util/TypeTraits.h"
00016
00017 namespace slang::syntax {
00018
00019 #define DERIVED static_cast<TDerived*>(this)
00020
00024 template<typename TDerived>
00025 class SyntaxVisitor {
00026 public:
00028     template<typename T>
```

```

00029     void visit(T&& t) {
00030         if constexpr (requires { DERIVED->handle(t); })
00031             DERIVED->handle(t);
00032         else
00033             DERIVED->visitDefault(t);
00034     }
00035
00036     template<typename T>
00037     void visitDefault(T&& node) {
00038         for (uint32_t i = 0; i < node.getChildCount(); i++) {
00039             auto child = node.childNode(i);
00040             if (child)
00041                 child->visit(*DERIVED);
00042             else {
00043                 auto token = node.childToken(i);
00044                 if (token)
00045                     DERIVED->visitToken(token);
00046             }
00047         }
00048     }
00049 }
00050
00051 private:
00052     // This is to make things compile if the derived class doesn't provide an implementation.
00053     void visitToken(parsing::Token) {}
00054 };
00055
00056 namespace detail {
00057     struct SLANG_EXPORT SyntaxChange {
00058         const SyntaxNode* first = nullptr;
00059         SyntaxNode* second = nullptr;
00060         parsing::Token separator = {};
00061     };
00062
00063     struct SLANG_EXPORT RemoveChange : SyntaxChange {};
00064     struct SLANG_EXPORT ReplaceChange : SyntaxChange {};
00065
00066     using InsertChangeMap = flat_hash_map<const SyntaxNode*, std::vector<SyntaxChange>>;
00067     using ModifyChangeMap = flat_hash_map<const SyntaxNode*, std::variant<RemoveChange, ReplaceChange>>;
00068     using ListChangeMap = flat_hash_map<const SyntaxNode*, std::vector<SyntaxChange>>;
00069
00070     struct SLANG_EXPORT ChangeCollection {
00071         InsertChangeMap insertBefore;
00072         InsertChangeMap insertAfter;
00073         ModifyChangeMap removeOrReplace;
00074         ListChangeMap listInsertAtFront;
00075         ListChangeMap listInsertAtBack;
00076
00077         void clear() {
00078             insertBefore.clear();
00079             insertAfter.clear();
00080             removeOrReplace.clear();
00081             listInsertAtFront.clear();
00082             listInsertAtBack.clear();
00083         }
00084
00085         bool empty() const {
00086             return insertBefore.empty() && insertAfter.empty() && removeOrReplace.empty() &&
00087                 listInsertAtFront.empty() && listInsertAtBack.empty();
00088         }
00089     };
00090
00091     SLANG_EXPORT std::shared_ptr<SyntaxTree> transformTree(
00092         BumpAllocator&& alloc, const std::shared_ptr<SyntaxTree>& tree, const ChangeCollection& commits,
00093         const std::vector<std::shared_ptr<SyntaxTree>>& tempTrees, const SourceLibrary* library);
00094 } // namespace detail
00095
00096 template<typename TDerived>

```



```

00102 class SyntaxRewriter : public SyntaxVisitor<TDerived> {
00103 public:
00104     SyntaxRewriter() : factory(alloc) {}
00105
00116     std::shared_ptr<SyntaxTree> transform(const std::shared_ptr<SyntaxTree>& tree,
00117                                         const SourceLibrary* library = nullptr) {
00118         sourceManager = &tree->sourceManager();
00119         commits.clear();
00120         tempTrees.clear();
00121
00122         tree->root().visit(*DERIVED);
00123
00124         if (commits.empty())
00125             return tree;
00126
00127         return transformTree(std::move(alloc), tree, commits, tempTrees, library);
00128     }
00129
00130 protected:
00131     using Token = parsing::Token;
00132
00133     SyntaxNode& parse(std::string_view text) {
00134         tempTrees.emplace_back(SyntaxTree::fromText(text, *sourceManager));
00135         return tempTrees.back()->root();
00136     }
00137
00138     void remove(const SyntaxNode& oldNode) {
00139         if (auto [_ , ok] = commits.removeOrReplace.emplace(&oldNode,
00140                                                         detail::RemoveChange{&oldNode, nullptr});
00141             !ok) {
00142             SLANG_THROW(std::logic_error("Node only permit one remove/replace operation"));
00143         }
00144     }
00145
00146     void replace(const SyntaxNode& oldNode, SyntaxNode& newNode) {
00147         if (auto [_ , ok] = commits.removeOrReplace.emplace(
00148             &oldNode, detail::ReplaceChange{&oldNode, &newNode});
00149             !ok) {
00150             SLANG_THROW(std::logic_error("Node only permit one remove/replace operation"));
00151         }
00152     }
00153
00154     void insertBefore(const SyntaxNode& oldNode, SyntaxNode& newNode) {
00155         commits.insertBefore[&oldNode].push_back({&oldNode, &newNode});
00156     }
00157
00158     void insertAfter(const SyntaxNode& oldNode, SyntaxNode& newNode) {
00159         commits.insertAfter[&oldNode].push_back({&oldNode, &newNode});
00160     }
00161
00162     void insertAtFront(const SyntaxListBase& list, SyntaxNode& newNode, Token separator = {}) {
00163         commits.listInsertAtFront[&list].push_back({&list, &newNode, separator});
00164     }
00165
00166     void insertAtBack(const SyntaxListBase& list, SyntaxNode& newNode, Token separator = {}) {
00167         commits.listInsertAtBack[&list].push_back({&list, &newNode, separator});
00168     }
00169
00170     Token makeToken(parsing::TokenKind kind, std::string_view text,
00171                    std::span<const parsing::Trivia> trivia = {}) {
00172         return Token(alloc, kind, trivia, text, SourceLocation::NoLocation);
00173     }
00174
00175     Token makeId(std::string_view text, std::span<const parsing::Trivia> trivia = {}) {
00176         return makeToken(parsing::TokenKind::Identifier, text, trivia);
00177     }
00178
00179     Token makeId(std::string_view text, parsing::Trivia trivia) {
00180         auto triviaPtr = alloc.emplace<parsing::Trivia>(trivia);

```

```

00188         return makeId(text, {triviaPtr, 1});
00189     }
00190
00191     Token makeComma() { return makeToken(parsing::TokenKind::Comma, ", "sv); }
00192
00193     BumpAllocator alloc;
00194     SyntaxFactory factory;
00195
00196     static const parsing::Trivia SingleSpace;
00197
00198 private:
00199     SourceManager* sourceManager = nullptr;
00200     detail::ChangeCollection commits;
00201     std::vector<std::shared_ptr<SyntaxTree>> tempTrees;
00202 };
00203
00204 template<typename TDerived>
00205 const parsing::Trivia SyntaxRewriter<TDerived>::SingleSpace{parsing::TriviaKind::Whitespace, " "sv};
00206
00207 #undef DERIVED
00208
00209 } // namespace slang::syntax

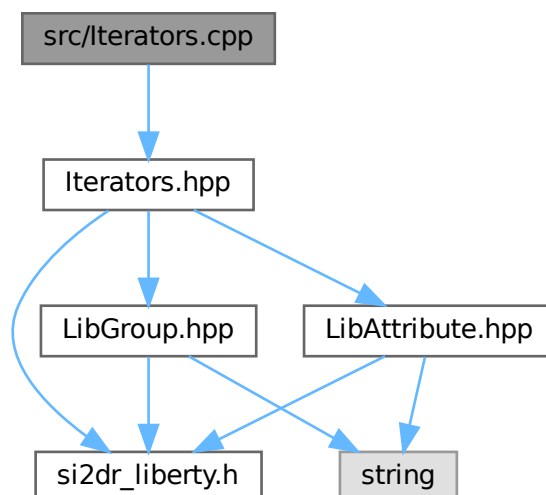
```

8.29 README.md File Reference

8.30 src/Iterators.cpp File Reference

#include "Iterators.hpp"

Include dependency graph for Iterators.cpp:



8.31 Iterators.cpp

[Go to the documentation of this file.](#)

```

00001 #include "Iterators.hpp"
00002
00003 GroupsIterator::GroupsIterator(si2drGroupsIdT groups, si2drErrorT &err) : groups_(groups), err_(err) {
00004     group_ = si2drIterNextGroup(groups_, &err_);
00005 }
00006 GroupsIterator::~GroupsIterator() { si2drIterQuit(groups_, &err_); }
00007
00008 void GroupsIterator::next() { group_ = si2drIterNextGroup(groups_, &err_); }
00009 bool GroupsIterator::end() { return si2drObjectIsNull(group_, &err_); }
00010
00011 LibGroup GroupsIterator::get() { return LibGroup(group_, err_); }
00012
00013 AttributesIterator::AttributesIterator(si2drAttrsIdT attrs, si2drErrorT &err)
00014     : attrs_(attrs), err_(err) {
00015     attr_ = si2drIterNextAttr(attrs_, &err_);
00016 }
00017 AttributesIterator::~AttributesIterator() { si2drIterQuit(attrs_, &err_); }
00018
00019 void AttributesIterator::next() { attr_ = si2drIterNextAttr(attrs_, &err_); }
00020 bool AttributesIterator::end() { return si2drObjectIsNull(attr_, &err_); }
00021
00022 LibAttribute AttributesIterator::get() { return LibAttribute(attr_, err_); }
00023
00024 ValuesIterator::ValuesIterator(si2drValuesIdT values, si2drErrorT &err) : values_(values), err_(err) {
00025     si2drIterNextComplexValue(values_, &vtype_, &int_, &float_, &str_, &bool_, &exprp_, &err_);
00026 }
00027 ValuesIterator::~ValuesIterator() { si2drIterQuit(values_, &err_); }
00028
00029 void ValuesIterator::next() {
00030     si2drIterNextComplexValue(values_, &vtype_, &int_, &float_, &str_, &bool_, &exprp_, &err_);
00031 }
00032 bool ValuesIterator::end() { return vtype_ == SI2DR_UNDEFINED_VALUETYPE; }

```

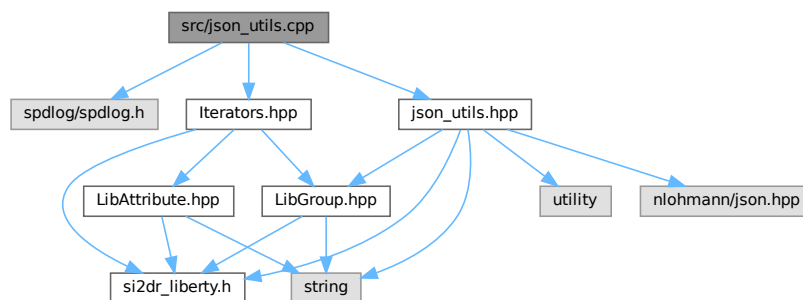
8.32 src/json_utils.cpp File Reference

```
#include "spdlog/spdlog.h"
```

```
#include "Iterators.hpp"
```

```
#include "json_utils.hpp"
```

Include dependency graph for json_utils.cpp:



Functions

- `std::vector< double > parseStringToVector (const std::string &str)`
Parses a string representation of a vector of doubles into a vector of doubles.
- `json generateLutJson (LibGroup &lib_lut_group, si2drErrorT &err)`
Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.
- `json generateTimingJson (LibGroup &lib_timing_group, si2drErrorT &err)`
Generates a JSON representation of timing information from a library timing group.
- `json generatePowerJson (LibGroup &lib_power_group, si2drErrorT &err)`
Converts a Liberty power group into a JSON representation.
- `std::pair< std::string, json > generatePinJson (LibGroup &lib_pin_group, si2drErrorT &err)`
Generates a JSON representation of a Liberty pin group.
- `json generateCellJson (LibGroup &lib_cell_group, si2drErrorT &err)`
Generates a JSON representation of a cell from a [LibGroup](#) object.

8.32.1 Function Documentation

8.32.1.1 generateCellJson()

```
json generateCellJson (
    LibGroup & lib_cell_group,
    si2drErrorT & err )
```

Generates a JSON representation of a cell from a [LibGroup](#) object.

This function processes a [LibGroup](#) representing a cell and converts it into a JSON object. It extracts the cell name and processes specific attributes such as `area`, `cell_leakage_power`, and `cell_footprint`. It also processes pins within the cell by categorizing them based on their direction (input, output, internal, inout).

Parameters

<code>lib_cell_group</code>	The LibGroup object representing the cell
<code>err</code>	Reference to a <code>si2drErrorT</code> object for error handling

Returns

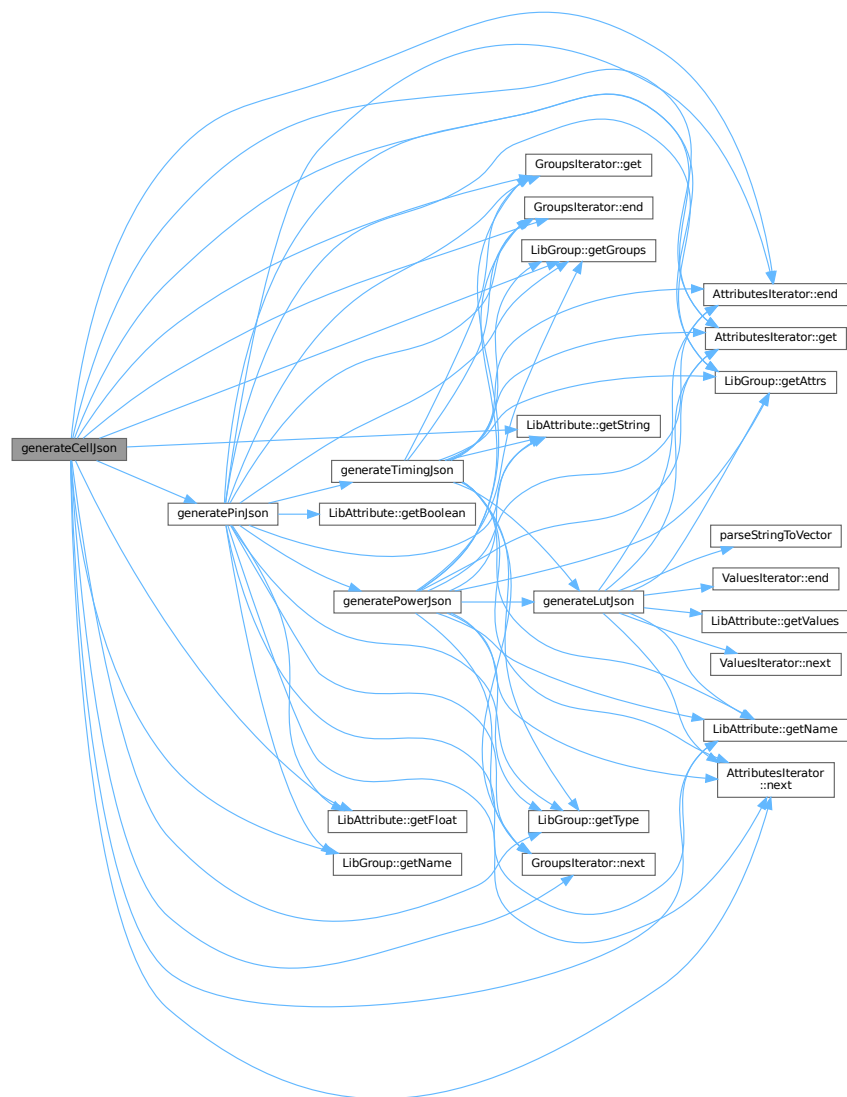
json A JSON object containing the cell's information with the following structure:

- `"cell_name"`: string - name of the cell

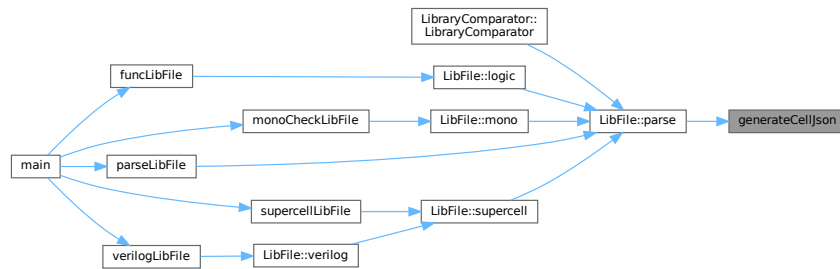
- "area": float (optional) - cell area if present
- "cell_leakage_power": float (optional) - cell leakage power if present
- "cell_footprint": string (optional) - cell footprint if present
- "input_pins": array - list of input pins
- "output_pins": array - list of output pins
- "internal_pins": array - list of internal pins
- "inout_pins": array - list of inout pins

Definition at line 272 of file `json_utils.cpp`.

Here is the call graph for this function:



Here is the caller graph for this function:



8.32.1.2 generateLutJson()

```

json generateLutJson (
    LibGroup & lib_lut_group,
    si2drErrorT & err )

```

Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.

This function iterates through the attributes of the provided [LibGroup](#) object representing a LUT and converts them into a JSON structure. It specifically handles the following attributes:

- "index_1": Converted to a vector and stored in the JSON
- "index_2": Converted to a vector and stored in the JSON
- "values": Each value is parsed into a vector and added to an array in the JSON

Any other attributes encountered will trigger a warning message.

Parameters

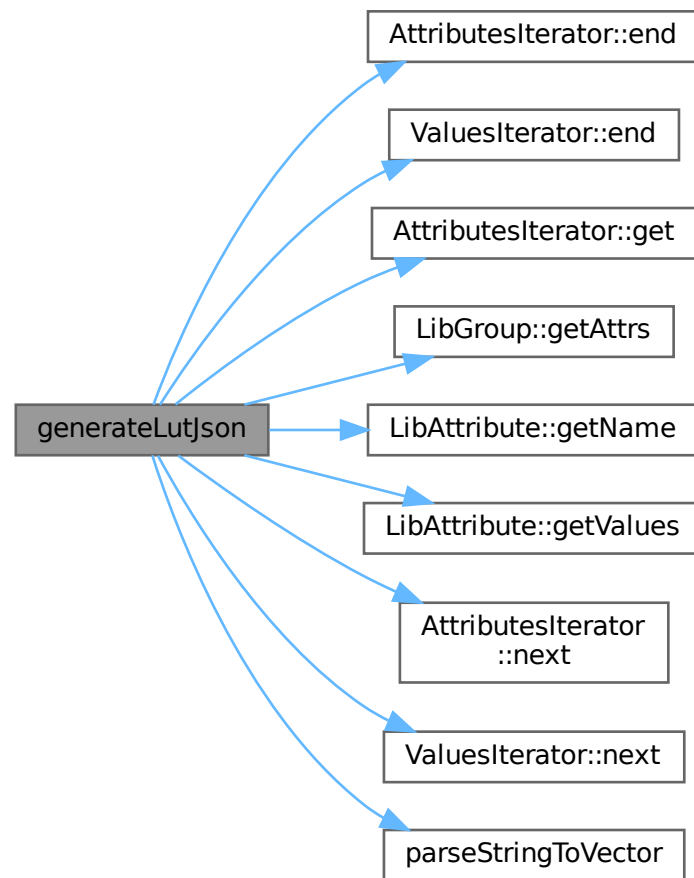
<i>lib_lut_group</i>	The LibGroup object containing the LUT data to be converted
<i>err</i>	Reference to an <code>si2drErrorT</code> object to track any errors during processing

Returns

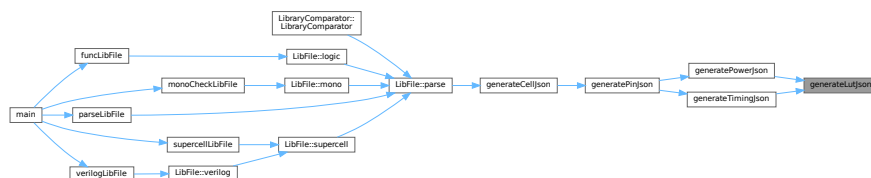
json A JSON object representing the LUT data with keys for "index_1", "index_2", and "values"

Definition at line 62 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.32.1.3 generatePinJson()

```
std::pair< std::string, json > generatePinJson (
```

```
LibGroup & lib_pin_group,  
si2drErrorT & err )
```

Generates a JSON representation of a Liberty pin group.

This function traverses a Liberty pin group, extracts relevant attributes and sub-groups, and converts them into a JSON object. It also determines the pin's direction.

Processes the following pin attributes:

- direction
- max_transition, capacitance, rise_capacitance, fall_capacitance, max_capacitance (float types)
- function, power_down_function, related_ground_pin, related_power_pin, three_state (string types)
- clock (boolean type)

Handles the following sub-groups:

- internal_power: Converted using [generatePowerJson\(\)](#)
- timing: Converted using [generateTimingJson\(\)](#)

Parameters

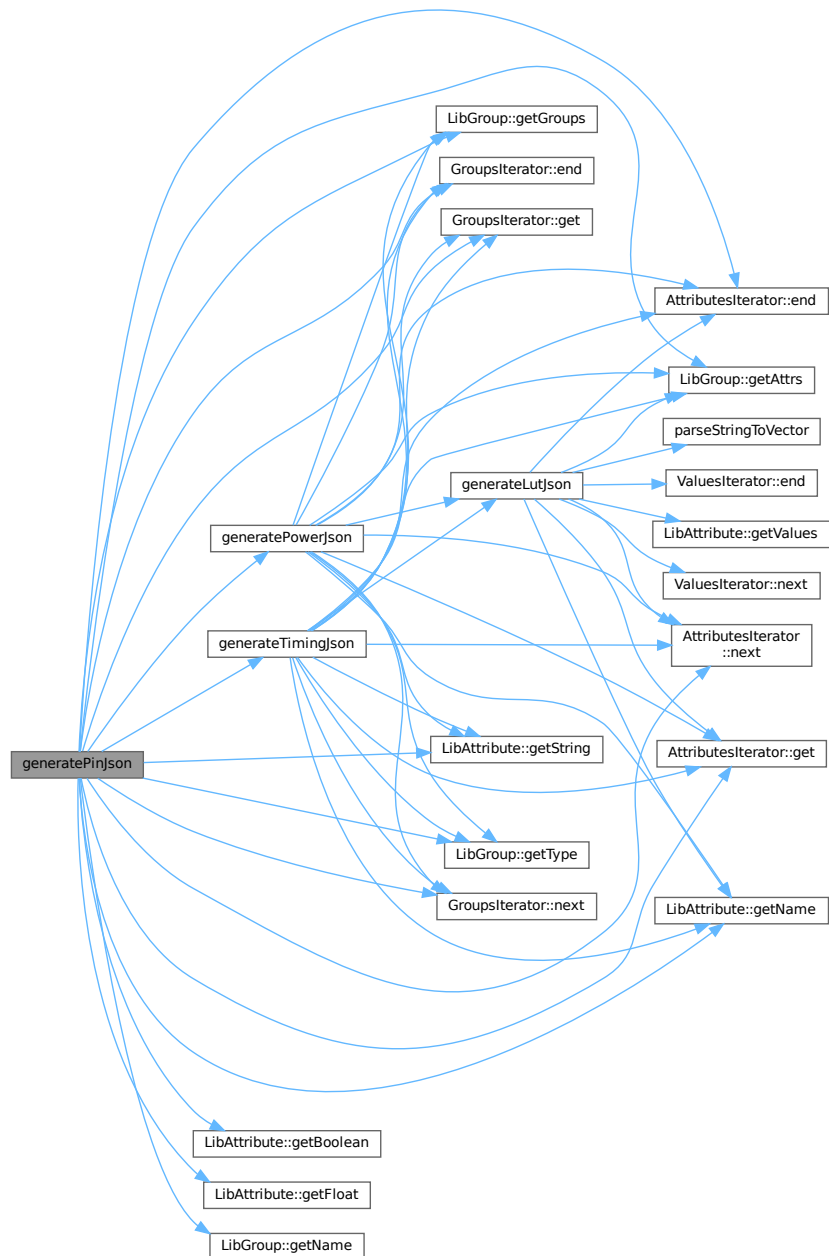
<i>lib_pin_group</i>	The Liberty pin group to process
<i>err</i>	Reference to an error object for tracking Liberty API errors

Returns

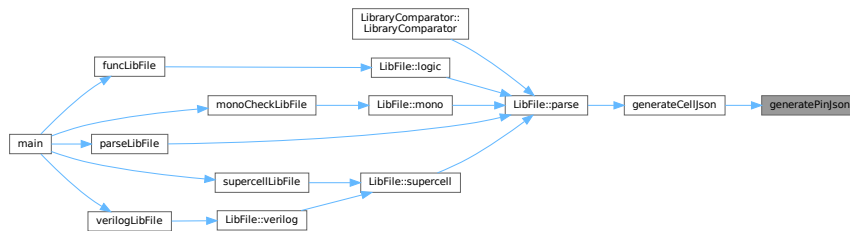
A pair containing the pin direction (string) and the JSON representation of the pin

Definition at line 206 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.32.1.4 generatePowerJson()

```

json generatePowerJson (
    LibGroup & lib_power_group,
    si2drErrorT & err )

```

Converts a Liberty power group into a JSON representation.

This function processes a Liberty power group and converts its attributes and subgroups into a JSON object. It handles attributes like "when", "related_pin", and "related_pg_pin", as well as "rise_power" and "fall_power" subgroups.

Parameters

<i>lib_power_group</i>	The Liberty power group to convert
<i>err</i>	Reference to an si2drErrorT object for error handling

Returns

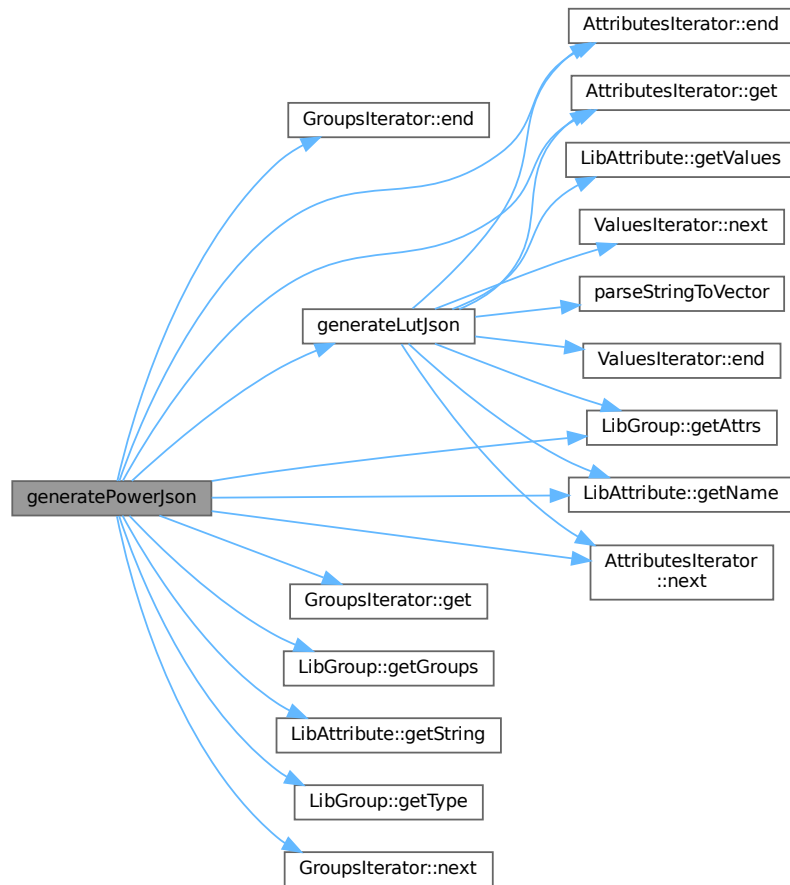
json A JSON object representing the power group data

The function:

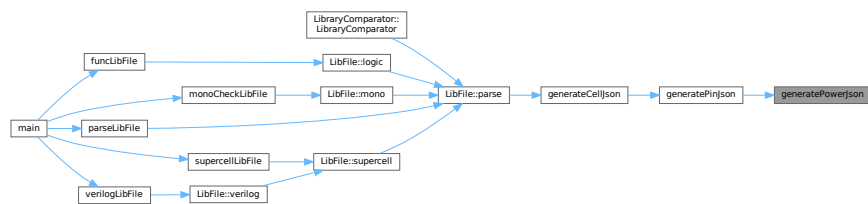
- Extracts string attributes (when, related_pin, related_pg_pin)
- Processes rise_power and fall_power subgroups by converting them to LUT JSON format
- Logs warnings for unknown power subgroup types

Definition at line 154 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.32.1.5 generateTimingJson()

```

json generateTimingJson (
    LibGroup & lib_timing_group,
    si2drErrorT & err )

```

Generates a JSON representation of timing information from a library timing group.

This function extracts timing attributes and sub-groups from a Liberty timing group and organizes them into a JSON object. It processes standard timing attributes like 'related_pin', 'timing_type', 'timing_sense', and 'when', as well as timing tables such as 'cell_fall', 'cell_rise', 'fall_transition', 'rise_transition', 'fall_constraint', and 'rise_constraint'.

Parameters

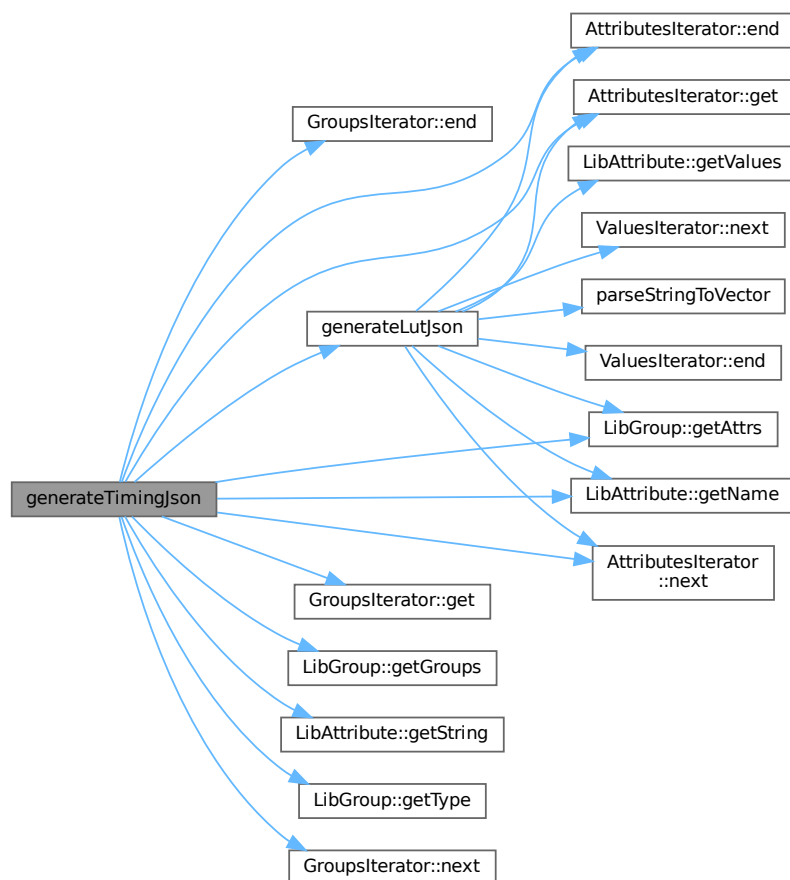
<i>lib_timing_group</i>	The Liberty timing group to process
<i>err</i>	Reference to an si2drErrorT object for error reporting

Returns

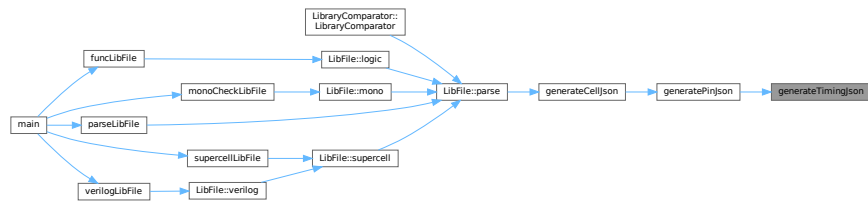
json A JSON object containing the extracted timing information

Definition at line 106 of file [json_utils.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.32.1.6 parseStringToVector()

```
std::vector< double > parseStringToVector (
    const std::string & str )
```

Parses a string representation of a vector of doubles into a vector of doubles.

This function takes a string containing comma-separated numbers, cleans it by removing backslashes and newline characters, and then converts each comma-separated value into a double that is added to the resulting vector.

Parameters

<i>str</i>	The input string to be parsed, containing comma-separated numbers
------------	---

Returns

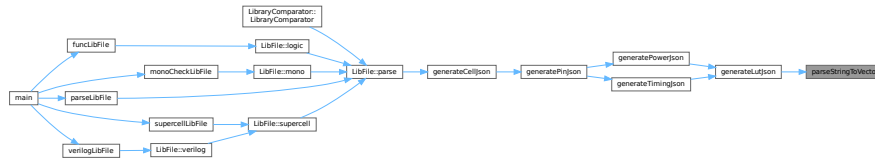
`std::vector<double>` A vector of doubles parsed from the input string

Exceptions

<code>std::invalid_argument</code>	If the string contains values that cannot be converted to double
<code>std::out_of_range</code>	If the string contains values that are out of range for double

Definition at line 28 of file [json_utils.cpp](#).

Here is the caller graph for this function:



8.33 json_utils.cpp

[Go to the documentation of this file.](#)

```

00001 #include "spdlog/spdlog.h"
00002
00003 #include "Iterators.hpp"
00004 #include "json_utils.hpp"
00005
00006 /*
00007 json 类型的值可以是以下几种类型之一：
00008 方括号 []：用于包含一组有序的值（数组）。
00009 数组中的每个值可以是任意类型的 JSON 值，包括对象、数组、字符串、数字、布尔值或 null。
00010 花括号 {}：用于包含一组键值对（对象）。
00011 对象中的每个键都是一个字符串，值可以是任意类型的 JSON 值，包括对象、数组、字符串、数字、布尔值或
00012 null。
00013 */
00014
00028 std::vector<double> parseStringToVector(const std::string &str) {
00029     std::vector<double> result;
00030     std::string cleaned_str;
00031
00032     // Remove backslashes and newline characters
00033     for (char c : str) {
00034         if (c != '\\' && c != '\n') {
00035             cleaned_str += c;
00036         }
00037     }
00038
00039     std::stringstream ss(cleaned_str);
00040     std::string item;
00041     while (std::getline(ss, item, ',')) {
00042         result.push_back(std::stod(item));
00043     }
00044     return result;
00045 }
00046
00062 json generateLutJson(LibGroup &lib_lut_group, si2drErrorT &err) {
00063     json lut_json;
00064
00065     AttributesIterator attr_iter(lib_lut_group.getAttrs(), err);
00066     for (; !attr_iter.end(); attr_iter.next()) {
00067         LibAttribute lib_attr = attr_iter.get();
00068         std::string lut_attr_name = lib_attr.getName();
00069
00070         // spdlog::debug("LUT Attribute Name: {}", lib_attr.getName());
00071         // spdlog::debug("Is Complex? {}", lib_attr.isComplex());
00072
00073         if (lut_attr_name == "index_1" || lut_attr_name == "index_2") {
00074             ValuesIterator values_iter(lib_attr.getValues(), err);
00075             for (; !values_iter.end(); values_iter.next()) {
00076                 // spdlog::debug("Type: {}", int(values_iter.vtype_)); // 5 is string
00077                 // spdlog::debug("Str: {}", values_iter.str_);
00078             }
00079         }
00080     }
00081     return lut_json;
00082 }

```

```

00078         lut_json[lut_attr_name] = parseStringToVector(std::string(values_iter.str_));
00079     }
00080 } else if (lut_attr_name == "values") {
00081     ValuesIterator values_iter(lib_attr.getValues(), err);
00082     for (; !values_iter.end(); values_iter.next()) {
00083         // spdlog::debug("{}", values_iter.str_);
00084         lut_json["values"].push_back(parseStringToVector(std::string(values_iter.str_)));
00085     }
00086 } else {
00087     spdlog::warn("Unknown LUT attribute name: {}", lut_attr_name);
00088 }
00089 }
00090 return lut_json;
00091 }
00092
00106 json generateTimingJson(LibGroup &lib_timing_group, si2drErrorT &err) {
00107     json timing_json;
00108
00109     AttributesIterator attr_iter(lib_timing_group.getAttrs(), err);
00110     for (; !attr_iter.end(); attr_iter.next()) {
00111         LibAttribute lib_attr = attr_iter.get();
00112
00113         std::string attr_name = lib_attr.getName();
00114         if (attr_name == "related_pin" || attr_name == "timing_type" || attr_name == "timing_sense" ||
00115             attr_name == "when") {
00116             timing_json[attr_name] = lib_attr.getString();
00117         }
00118         // More timing attributes can be added here
00119     }
00120
00121     GroupsIterator timing_sub_group_iter(lib_timing_group.getGroups(), err);
00122     for (; !timing_sub_group_iter.end(); timing_sub_group_iter.next()) {
00123         LibGroup lib_timing_sub_group = timing_sub_group_iter.get();
00124
00125         std::string timing_sub_group_type = lib_timing_sub_group.getType();
00126         // std::string timing_sub_group_name = lib_timing_sub_group.getName();
00127         if (timing_sub_group_type == "cell_fall" || timing_sub_group_type == "cell_rise" ||
00128             timing_sub_group_type == "fall_transition" || timing_sub_group_type == "rise_transition" ||
00129             timing_sub_group_type == "fall_constraint" || timing_sub_group_type == "rise_constraint") {
00130             timing_json[timing_sub_group_type] = generateLutJson(lib_timing_sub_group, err);
00131         } else {
00132             spdlog::warn("Unknown timing sub group type: {}", timing_sub_group_type);
00133         }
00134     }
00135     return timing_json;
00136 }
00137
00154 json generatePowerJson(LibGroup &lib_power_group, si2drErrorT &err) {
00155     json power_json;
00156
00157     AttributesIterator attr_iter(lib_power_group.getAttrs(), err);
00158     for (; !attr_iter.end(); attr_iter.next()) {
00159         LibAttribute lib_attr = attr_iter.get();
00160
00161         std::string attr_name = lib_attr.getName();
00162         if (attr_name == "when" || attr_name == "related_pin" || attr_name == "related_pg_pin") {
00163             power_json[attr_name] = lib_attr.getString();
00164         }
00165         // More power attributes can be added here
00166     }
00167
00168     GroupsIterator power_sub_group_iter(lib_power_group.getGroups(), err);
00169     for (; !power_sub_group_iter.end(); power_sub_group_iter.next()) {
00170         LibGroup lib_power_sub_group = power_sub_group_iter.get();
00171
00172         std::string power_sub_group_type = lib_power_sub_group.getType();
00173         // std::string power_sub_group_name = lib_power_sub_group.getName();
00174         if (power_sub_group_type == "rise_power") {
00175             // spdlog::debug("Rise Power: {}", power_sub_group_name);

```

```

00176     power_json["rise_power"] = generateLutJson(lib_power_sub_group, err);
00177 } else if (power_sub_group_type == "fall_power") {
00178     power_json["fall_power"] = generateLutJson(lib_power_sub_group, err);
00179 } else {
00180     spdlog::warn("Unknown power sub group type: {}", power_sub_group_type);
00181 }
00182 }
00183 return power_json;
00184 }
00185
00206 std::pair<std::string, json> generatePinJson(LibGroup &lib_pin_group, si2drErrorT &err) {
00207     json pin_json;
00208     std::string direction;
00209     pin_json["pin_name"] = lib_pin_group.getName();
00210
00211     AttributesIterator attr_iter(lib_pin_group.getAttrs(), err);
00212     for (; !attr_iter.end(); attr_iter.next()) {
00213         LibAttribute lib_attr = attr_iter.get();
00214
00215         std::string attr_name = lib_attr.getName();
00216         if (attr_name == "direction") {
00217             direction = lib_attr.getString();
00218         } else if (attr_name == "max_transition" || attr_name == "capacitance" ||
00219                 attr_name == "rise_capacitance" || attr_name == "fall_capacitance" ||
00220                 attr_name == "max_capacitance") {
00221             pin_json[attr_name] = lib_attr.getFloat();
00222         } else if (attr_name == "function" || attr_name == "power_down_function" ||
00223                 attr_name == "related_ground_pin" || attr_name == "related_power_pin" ||
00224                 attr_name == "three_state") {
00225             pin_json[attr_name] = lib_attr.getString();
00226         } else if (attr_name == "clock") {
00227             pin_json[attr_name] = lib_attr.getBoolean();
00228         }
00229         // More pin attributes can be added here
00230     }
00231
00232     GroupsIterator pin_sub_group_iter(lib_pin_group.getGroups(), err);
00233     for (; !pin_sub_group_iter.end(); pin_sub_group_iter.next()) {
00234         LibGroup lib_pin_sub_group = pin_sub_group_iter.get();
00235
00236         std::string pin_sub_group_type = lib_pin_sub_group.getType();
00237         // std::string pin_sub_group_name = lib_pin_sub_group.getName();
00238         if (pin_sub_group_type == "internal_power") {
00239             json power_json = generatePowerJson(lib_pin_sub_group, err);
00240             pin_json["power_arcs"].push_back(power_json);
00241         } else if (pin_sub_group_type == "timing") {
00242             // spdlog::debug("Has Timing: {}", lib_pin_group.getName());
00243             json timing_json = generateTimingJson(lib_pin_sub_group, err);
00244             pin_json["timing_arcs"].push_back(timing_json);
00245         } else {
00246             spdlog::warn("Unknown pin sub group type: {}", pin_sub_group_type);
00247         }
00248     }
00249     return std::make_pair(direction, pin_json);
00250 }
00251
00272 json generateCellJson(LibGroup &lib_cell_group, si2drErrorT &err) {
00273     json cell_json;
00274     cell_json["cell_name"] = lib_cell_group.getName();
00275
00276     AttributesIterator attr_iter(lib_cell_group.getAttrs(), err);
00277     for (; !attr_iter.end(); attr_iter.next()) {
00278         LibAttribute lib_attr = attr_iter.get();
00279
00280         std::string attr_name = lib_attr.getName();
00281         if (attr_name == "area" || attr_name == "cell_leakage_power") {
00282             cell_json[attr_name] = lib_attr.getFloat();
00283         } else if (attr_name == "cell_footprint") {
00284             cell_json[attr_name] = lib_attr.getString();

```



```

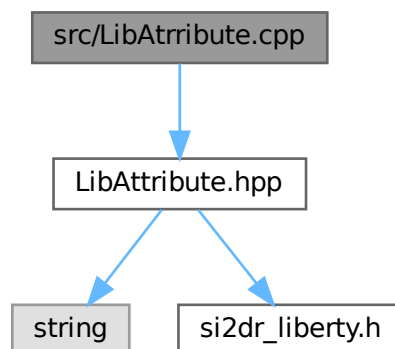
00285     }
00286     // More cell attributes can be added here
00287 }
00288
00289 GroupsIterator cell_sub_group_iter(lib_cell_group.getGroups(), err);
00290 for (; !cell_sub_group_iter.end(); cell_sub_group_iter.next()) {
00291     LibGroup lib_cell_sub_group = cell_sub_group_iter.get();
00292
00293     std::string cell_sub_group_type = lib_cell_sub_group.getType();
00294     // std::string cell_sub_group_name = lib_cell_sub_group.getName();
00295
00296     // spdlog::debug("Cell Sub Group Type: {}", cell_sub_group_type);
00297     // spdlog::debug("Cell Sub Group Name: {}", cell_sub_group_name);
00298
00299     if (cell_sub_group_type == "pin") {
00300         auto [direction, pin_json] = generatePinJson(lib_cell_sub_group, err);
00301         if (direction == "input") {
00302             cell_json["input_pins"].push_back(pin_json);
00303         } else if (direction == "output") {
00304             cell_json["output_pins"].push_back(pin_json);
00305         } else if (direction == "internal") {
00306             cell_json["internal_pins"].push_back(pin_json);
00307         } else if (direction == "inout") {
00308             cell_json["inout_pins"].push_back(pin_json);
00309         } else {
00310             spdlog::warn("Unknown direction: {}", direction);
00311         }
00312     }
00313 }
00314 return cell_json;
00315 }

```

8.34 src/LibAttribute.cpp File Reference

```
#include "LibAttribute.hpp"
```

Include dependency graph for LibAttribute.cpp:



8.35 LibAttribute.cpp

[Go to the documentation of this file.](#)

```
00001 #include "LibAttribute.hpp"
00002
00003 LibAttribute::LibAttribute(si2drAttrIdT attr, si2drErrorT &err) : attr_(attr), err_(err) {}
00004
00005 LibAttribute::~LibAttribute() {}
00006
00007 std::string LibAttribute::getName() {
00008     si2drStringT name = si2drAttrGetName(attr_, &err_);
00009     return name ? std::string(name) : std::string();
00010 }
00011
00012 bool LibAttribute::isComplex() { return si2drAttrGetAttrType(attr_, &err_) ? 1 : 0; }
00013
00014 si2drValuesIdT LibAttribute::getValues() { return si2drComplexAttrGetValues(attr_, &err_); }
00015
00016 long int LibAttribute::getInt() { return si2drSimpleAttrGetInt32Value(attr_, &err_); }
00017
00018 double LibAttribute::getFloat() { return si2drSimpleAttrGetFloat64Value(attr_, &err_); }
00019
00020 std::string LibAttribute::getString() {
00021     si2drStringT str = si2drSimpleAttrGetStringValue(attr_, &err_);
00022     return str ? std::string(str) : std::string();
00023 }
00024
00025 bool LibAttribute::getBoolean() { return si2drSimpleAttrGetBooleanValue(attr_, &err_) ? 1 : 0; }
```

8.36 src/LibFile.cpp File Reference

```
#include <chrono>
#include <filesystem>
#include <fstream>
#include <sstream>
#include <unordered_set>
#include "spdlog/sinks/basic_file_sink.h"
#include "spdlog/sinks/stdout_color_sinks.h"
#include "spdlog/spdlog.h"
#include "Iterators.hpp"
#include "json_utils.hpp"
#include "LibFile.hpp"
#include "verilog_utils.hpp"
```

Include dependency graph for LibFile.cpp:



8.37 LibFile.cpp

[Go to the documentation of this file.](#)

```

00001 #include <chrono>
00002 #include <filesystem>
00003 #include <fstream>
00004 #include <sstream>
00005 #include <unordered_set>
00006
00007 #include "spdlog/sinks/basic_file_sink.h"
00008 #include "spdlog/sinks/stdout_color_sinks.h"
00009 #include "spdlog/spdlog.h"
00010
00011 #include "Iterators.hpp"
00012 #include "json_utils.hpp"
00013 #include "LibFile.hpp"
00014 #include "verilog_utils.hpp"
00015
00028 LibFile::LibFile(const std::string &filepath, const std::string &loggername)
00029     : filepath_(filepath), loggername_(loggername) {
00030     filename_ = filepath_.string();
00031     basename_ = filepath_.stem().string();
00032     jsonname_ = basename_ + ".json";
00033
00034     auto console_sink = std::make_shared<spdlog::sinks::stdout_color_sink_mt>();
00035     console_sink->set_level(spdlog::level::info);
00036
00037     auto file_sink = std::make_shared<spdlog::sinks::basic_file_sink_mt>(loggername_, true);
00038     file_sink->set_level(spdlog::level::debug);
00039
00040     std::vector<spdlog::sink_ptr> sinks{console_sink, file_sink};
00041     logger_ = std::make_shared<spdlog::logger>(loggername_, sinks.begin(), sinks.end());
00042     logger_->set_level(spdlog::level::debug);
00043
00044     logger_->info("Created LibFile object for '{}'", filename_);
00045     logger_->info("Debug log in: '{}', loggername_");
00046 }
00047
00048 LibFile::~LibFile() { logger_->info("Closing file: '{}', filename_"); }
00049
00059 void LibFile::writeJsonToFile() {
00060     std::ofstream out(jsonname_);
00061     if (!out.is_open()) {
00062         logger_->error("Could not open file '{}' for writing", jsonname_);
00063         return;
00064     }
00065     out << lib_json_.dump(2);
00066     out.close();
00067     logger_->info("JSON data written to '{}', jsonname_");
00068 }
00069
00090 void LibFile::read() {
00091     logger_->info("Reading '{} ...', filename_");
00092
00093     auto start = std::chrono::high_resolution_clock::now();
00094     si2drReadLibertyFile(const_cast<char *>(filepath_.c_str()), &err_);
00095     auto end = std::chrono::high_resolution_clock::now();
00096     std::chrono::duration<double> duration = end - start;
00097
00098     if (err_ == SI2DR_INVALID_NAME) {
00099         logger_->error("Could not open file '{}' for parsing, quitting...", filename_);
00100         exit(301);
00101     } else if (err_ == SI2DR_SYNTAX_ERROR) {
00102         logger_->error("Syntax Errors were detected in the input file!");
00103         exit(401);
00104     } else {
00105         logger_->info("Done. Read time: {:.2f} seconds", duration.count());

```

```

00106 }
00107 }
00108
00129 void LibFile::parse() {
00130     si2drPIInit(&err_); // Initialize private error handler
00131     this->read();       // Read the Liberty file
00132
00133     logger_>info("Parsing '{}' ...", filename_);
00134     // Create a scope for the top-level groups iteration
00135     {
00136         GroupsIterator group_iter(si2drPIGetGroups(&err_), err_);
00137         for (; !group_iter.end(); group_iter.next()) {
00138             LibGroup lib_group = group_iter.get();
00139
00140             if (lib_group.getName() != "") {
00141                 libname_ = lib_group.getName();
00142                 lib_json["library_name"] = this->libname_;
00143                 logger_>info("Library Name: {}", libname_);
00144             } else {
00145                 logger_>warn("Library Name: <NONAME>");
00146             }
00147             // Second level groups
00148             GroupsIterator sub_group_iter(lib_group.getGroups(), err_);
00149             for (; !sub_group_iter.end(); sub_group_iter.next()) {
00150                 LibGroup lib_sub_group = sub_group_iter.get();
00151
00152                 std::string sub_group_type = lib_sub_group.getType();
00153                 std::string sub_group_name = lib_sub_group.getName();
00154
00155                 if (sub_group_type == "cell") {
00156                     logger_>debug("Cell Name: {}", sub_group_name);
00157                     // if (sub_group_name != "AN2D0") {
00158                     //     continue;
00159                     // }
00160                     // Handle each cell
00161                     json cell_json = generateCellJson(lib_sub_group, err_);
00162                     lib_json["cells"].push_back(cell_json);
00163                 } else if (sub_group_type == "operating_conditions") {
00164                     logger_>info("Operating Conditions: {}", sub_group_name);
00165                     // Get PVT from Operating Conditions
00166                     AttributesIterator attr_iter(lib_sub_group.getAttrs(), err_);
00167                     for (; !attr_iter.end(); attr_iter.next()) {
00168                         LibAttribute lib_attr = attr_iter.get();
00169                         logger_>debug("Attribute Name: {}", lib_attr.getName());
00170                         logger_>debug("Int Value: {}", lib_attr.getInt());
00171                         logger_>debug("Float Value: {}", lib_attr.getFloat());
00172                         logger_>debug("String Value: {}", lib_attr.getString());
00173                         if (lib_attr.getName() == "process") {
00174                             process_ = lib_attr.getInt();
00175                             lib_json["process"] = process_;
00176                         } else if (lib_attr.getName() == "voltage") {
00177                             voltage_ = lib_attr.getFloat();
00178                             // Round to 2 decimal places to avoid floating-point precision issues
00179                             lib_json["voltage"] = std::round(voltage_ * 100) / 100.0;
00180                         } else if (lib_attr.getName() == "temperature") {
00181                             temperature_ = lib_attr.getInt();
00182                             lib_json["temperature"] = temperature_;
00183                         }
00184                     }
00185                     logger_>info("P: {}, V: {}, T: {}", process_, voltage_, temperature_);
00186                 }
00187                 // sub_group_iter's lifetime ends here, and the destructor is called
00188             }
00189         }
00190         // group_iter's lifetime ends here, and the destructor is called
00191     }
00192     si2drPIQuit(&err_);
00193 }
00194

```

```

00195 void LibFile::modify() { logger_>info("Modifying the file..."); }
00196
00218 bool LibFile::checkTimingArcMonotonicity(const json &cell, const json &pin, const json &arc,
00219                                           const std::string &timing_arc_name, const bool is_slew) {
00220     if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00221         logger_>debug(
00222             "Checking cell: '{}', pin: '{}', related_pin: '{}', timing_arc: '{}', when: '{}'",
00223             cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00224             arc["related_pin"].get<std::string>(), timing_arc_name, arc["when"].get<std::string>());
00225     } else {
00226         logger_>debug("Checking cell: '{}', pin: '{}', related_pin: '{}', timing_arc: '{}'",
00227             cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00228             arc["related_pin"].get<std::string>(), timing_arc_name);
00229     }
00230     bool is_monotonic = true; // Assume the values are monotonic initially
00231     if (arc.contains(timing_arc_name) && arc[timing_arc_name].contains("values")) {
00232         std::vector<std::vector<double>> value_matrix;
00233         // Generate a matrix of values from the JSON array
00234         for (const auto &row_val : arc[timing_arc_name]["values"]) {
00235             std::vector<double> row_data;
00236             // Check if the value is an array
00237             if (!row_val.is_array()) {
00238                 logger_>error("Invalid format: '{}' values should be an array of arrays in cell '{}', pin "
00239                     "'{}', related_pin '{}'",
00240                     timing_arc_name, cell["cell_name"].get<std::string>(),
00241                     pin["pin_name"].get<std::string>(), arc["related_pin"].get<std::string>());
00242                 return false;
00243             }
00244             // Check for non-numeric values
00245             for (const auto &val : row_val) {
00246                 if (val.is_number()) {
00247                     row_data.push_back(val.get<double>());
00248                 } else {
00249                     logger_>warn("Non-numeric value found in '{}'.values, skipping value: {} in cell '{}', "
00250                         "pin '{}', related_pin '{}'",
00251                         timing_arc_name, val.dump(), cell["cell_name"].get<std::string>(),
00252                         pin["pin_name"].get<std::string>(), arc["related_pin"].get<std::string>());
00253                 }
00254             }
00255             value_matrix.push_back(row_data);
00256         }
00257         // Check the matrix for monotonicity
00258         if (!value_matrix.empty() && !value_matrix[0].empty()) {
00259             // Check the monotonic incrementality of rows (by output load capacitance, index 2)
00260             for (size_t i = 0; i < value_matrix.size(); ++i) {
00261                 for (size_t j = 1; j < value_matrix[i].size(); ++j) {
00262                     if ((value_matrix[i][j] < value_matrix[i][j - 1] &&
00263                         pin["pin_name"] != arc["related_pin"]) ||
00264                         (value_matrix[i][j] == 0 && value_matrix[i][j - 1] == 0)) {
00265                         // If contains "when" key, log the when value
00266                         if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00267                             logger_>warn("Non-monotonic (by load) '{}' values: ({}), ({}), {} < ({}), ({}), {} "
00268                                 "for Cell: {} Pin: {}->{} when: '{}'",
00269                                 timing_arc_name, i, j, value_matrix[i][j], i, j - 1,
00270                                 value_matrix[i][j - 1], cell["cell_name"].get<std::string>(),
00271                                 arc["related_pin"].get<std::string>(),
00272                                 pin["pin_name"].get<std::string>(), arc["when"].get<std::string>());
00273                         } else {
00274                             logger_>warn("Non-monotonic (by load) '{}' values: ({}), ({}), {} < ({}), ({}), {} "
00275                                 "for Cell: {} Pin: {}->{}",
00276                                 timing_arc_name, i, j, value_matrix[i][j], i, j - 1,
00277                                 value_matrix[i][j - 1], cell["cell_name"].get<std::string>(),
00278                                 arc["related_pin"].get<std::string>(),
00279                                 pin["pin_name"].get<std::string>());
00280                         }
00281                         is_monotonic = false;
00282                     }
00283                 }
00284             }

```

```

00285     if (is_slew) {
00286         // Check the monotonic incrementality of columns (by input slew, index 1)
00287         for (size_t j = 0; j < value_matrix[0].size(); ++j) {
00288             for (size_t i = 1; i < value_matrix.size(); ++i) {
00289                 if ((value_matrix[i][j] < value_matrix[i - 1][j] &&
00290                     pin["pin_name"] != arc["related_pin"]) ||
00291                     (value_matrix[i][j] == 0 && value_matrix[i - 1][j] == 0)) {
00292                     // If contains "when" key, log the when value
00293                     if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00294                         logger_>warn("Non-monotonic (by slew) '{}' values: ({} , {}) {} < ({} , {}) {} "
00295                                     "for Cell: {} Pin: {}->{} when: \"{}\"",
00296                                     timing_arc_name, i, j, value_matrix[i][j], i - 1, j,
00297                                     value_matrix[i - 1][j], cell["cell_name"].get<std::string>(),
00298                                     arc["related_pin"].get<std::string>(),
00299                                     pin["pin_name"].get<std::string>(), arc["when"].get<std::string>());
00300                     } else {
00301                         logger_>warn("Non-monotonic (by slew) '{}' values: ({} , {}) {} < ({} , {}) {} "
00302                                     "for Cell: {} Pin: {}->{}",
00303                                     timing_arc_name, i, j, value_matrix[i][j], i - 1, j,
00304                                     value_matrix[i - 1][j], cell["cell_name"].get<std::string>(),
00305                                     arc["related_pin"].get<std::string>(),
00306                                     pin["pin_name"].get<std::string>());
00307                     }
00308                     is_monotonic = false;
00309                 }
00310             }
00311         }
00312     }
00313 } else {
00314     logger_>warn(
00315         "Empty or invalid 'values' array found for cell: '{}', pin: '{}', related_pin: '{}', "
00316         "timing_arc: '{}'",
00317         cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00318         arc["related_pin"].get<std::string>(), timing_arc_name);
00319 }
00320 }
00321 return is_monotonic;
00322 }
00323
00344 void LibFile::mono(const bool is_slew) {
00345     /*
00346     * Use this command to check the following data in the current library to ensure
00347     * the tables are monotonically increasing with respect to output load:
00348     * cell_rise retaining_rise
00349     * cell_fall retaining_fall
00350     * rise_transition retain_rise_slew
00351     * fall_transition retain_fall_slew
00352     * mpw
00353     */
00354     logger_>info("Monotonicity check of '{}' starting ...", filename_);
00355
00356     if (!std::filesystem::exists(jsonname_)) {
00357         logger_>info("JSON file not found. Parsing Liberty file first.");
00358         this->parse();
00359         this->writeJsonToFile();
00360     } else {
00361         // Read the JSON file into json object
00362         std::ifstream in(jsonname_);
00363         if (!in.is_open()) {
00364             logger_>error("Could not open file '{}' for reading", jsonname_);
00365             return;
00366         }
00367         try {
00368             lib_json_ = json::parse(in);
00369         } catch (const json::parse_error &e) {
00370             logger_>error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00371             in.close();
00372             return;
00373         }

```

```

00374     in.close();
00375 }
00376
00377 std::map<std::string, bool> cell_monotonicity_status; // Track pass/fail status for each cell
00378 std::vector<std::string> failed_cells;                // List of failed cell names
00379 int total_cells = 0;
00380 int passed_cells = 0;
00381
00382 // Check the monotonicity of delay values
00383 // The index 1 (time values) must be monotonically increasing and >= 0
00384 for (const auto &cell : lib_json_["cells"]) {
00385     total_cells++;
00386     bool cell_is_monotonic = true; // Assume cell is monotonic initially
00387     std::string cell_name = cell["cell_name"].get<std::string>();
00388     cell_monotonicity_status[cell_name] =
00389         true; // Initialize to pass, will be set to false if any check fails
00390
00391     if (cell.contains("output_pins")) {
00392         for (const auto &pin : cell["output_pins"]) {
00393             if (pin.contains("timing_arcs")) {
00394                 for (const auto &arc : pin["timing_arcs"]) {
00395                     // Check 4 timing arc names and accumulate monotonicity status
00396                     for (const auto &name :
00397                         {"cell_rise", "cell_fall", "rise_transition", "fall_transition"}) {
00398                         if (!checkTimingArcMonotonicity(cell, pin, arc, name, is_slew)) {
00399                             cell_is_monotonic = false;
00400                         }
00401                     }
00402                 }
00403             }
00404         }
00405     }
00406     if (cell.contains("input_pins")) {
00407         for (const auto &pin : cell["input_pins"]) {
00408             if (!pin.contains("clock")) {
00409                 continue; // Skip non-clock pins
00410             }
00411             // logger->debug("Clock pin: {}", pin["pin_name"].get<std::string>());
00412             if (pin.contains("timing_arcs")) {
00413                 for (const auto &arc : pin["timing_arcs"]) {
00414                     if (arc.contains("timing_type")) {
00415                         // logger->debug("Timing type: {}", arc["timing_type"].get<std::string>());
00416                         if (arc["timing_type"].get<std::string>() == "min_pulse_width") {
00417                             for (const auto &name : {"rise_constraint", "fall_constraint"}) {
00418                                 if (!checkTimingArcMonotonicity(cell, pin, arc, name, is_slew)) {
00419                                     cell_is_monotonic = false;
00420                                 }
00421                             }
00422                         }
00423                     }
00424                 }
00425             }
00426         }
00427     }
00428     // Update cell status based on all checks
00429     cell_monotonicity_status[cell_name] = cell_is_monotonic;
00430     if (cell_is_monotonic) {
00431         passed_cells++;
00432     } else {
00433         failed_cells.push_back(cell_name);
00434     }
00435 }
00436 logger->info("Monotonicity check of '{}' completed.", filename_);
00437
00438 // Output summary statistics
00439 logger->info("validate_monotonicity : {} out of {} cells passed", passed_cells, total_cells);
00440 logger->info("validate_monotonicity : {} out of {} cells failed", total_cells - passed_cells,
00441             total_cells);
00442 if (!failed_cells.empty()) {

```

```

00443     std::stringstream failed_cells_ss;
00444     for (size_t i = 0; i < failed_cells.size(); ++i) {
00445         failed_cells_ss << failed_cells[i];
00446         if (i < failed_cells.size() - 1) {
00447             failed_cells_ss << ", "; // Add comma if not the last element
00448         }
00449     }
00450     logger_>info("Failed cell list : {}", failed_cells_ss.str());
00451 }
00452 }
00453
00473 void LibFile::supercell(const int chain_length, const std::vector<std::string> &cell_names) {
00474     /*
00475     * Supercells are named as follows:
00476     * <cellname>_X<chain_length>_<input_pin>_<output_pin>
00477     */
00478     logger_>info("Creating supercells for '{}'", filename_);
00479
00480     if (!std::filesystem::exists(jsonname_)) {
00481         logger_>info("JSON file not found. Parsing Liberty file first.");
00482         this->parse();
00483         this->writeJsonToFile();
00484     } else {
00485         // Read the JSON file into json object
00486         std::ifstream in(jsonname_);
00487         if (!in.is_open()) {
00488             logger_>error("Could not open file '{}' for reading", jsonname_);
00489             return;
00490         }
00491         try {
00492             lib_json_ = json::parse(in);
00493         } catch (const json::parse_error &e) {
00494             logger_>error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00495             in.close();
00496             return;
00497         }
00498         in.close();
00499     }
00500
00501     // write to .map file
00502     std::ofstream out(basename_ + ".map");
00503     if (!out.is_open()) {
00504         logger_>error("Could not open file '{}' for writing", basename_ + ".map");
00505         return;
00506     }
00507     // if cell_names is empty, create supercells for all cells
00508     // else create supercells for only the specified cells
00509
00510     // Check if we're processing specific cells or all cells
00511     bool process_all_cells = cell_names.empty();
00512
00513     if (process_all_cells) {
00514         logger_>info("Creating supercells for ALL cells in '{}'", filename_);
00515     } else {
00516         logger_>info("Creating supercells for {} specified cells in '{}', cell_names.size(),
00517                     filename_);
00518     }
00519
00520     // Create a set for faster lookups if we have specific cell names
00521     std::unordered_set<std::string> cell_set;
00522     std::unordered_set<std::string> found_cells;
00523     if (!process_all_cells) {
00524         cell_set.insert(cell_names.begin(), cell_names.end());
00525     }
00526
00527     for (const auto &cell : lib_json_["cells"]) {
00528         bool is_sequential = false;
00529         std::string cell_name = cell["cell_name"].get<std::string>();
00530

```



```

00531     // Skip cells not in the specified list
00532     if (!process_all_cells && cell_set.find(cell_name) == cell_set.end()) {
00533         continue;
00534     }
00535
00536     // Mark this cell as found
00537     if (!process_all_cells) {
00538         found_cells.insert(cell_name);
00539     }
00540
00541     logger_>debug("Creating supercells for cell: '{}'", cell_name);
00542
00543     std::vector<std::string> output_pins;
00544     std::vector<std::string> input_pins;
00545     // Extract input and output pins
00546     if (cell.contains("output_pins")) {
00547         for (const auto &pin : cell["output_pins"]) {
00548             output_pins.push_back(pin["pin_name"].get<std::string>());
00549         }
00550     }
00551     if (cell.contains("input_pins")) {
00552         for (const auto &pin : cell["input_pins"]) {
00553             if (pin.contains("clock")) {
00554                 is_sequential = true;
00555                 // clock pin not use for supercell
00556                 continue;
00557             }
00558             input_pins.push_back(pin["pin_name"].get<std::string>());
00559         }
00560     }
00561
00562     // create supercells for all combinations of input and output pins
00563     for (const auto &output_pin : output_pins) {
00564         for (const auto &input_pin : input_pins) {
00565             if (is_sequential) {
00566                 const int chain_len = 1;
00567                 std::string supercell_name =
00568                     cell_name + "__X" + std::to_string(chain_len) + "__" + input_pin + "__" + output_pin;
00569                 out << cell_name << " " << supercell_name << std::endl;
00570             } else {
00571                 std::string supercell_name = cell_name + "__X" + std::to_string(chain_length) + "__" +
00572                     input_pin + "__" + output_pin;
00573                 out << cell_name << " " << supercell_name << std::endl;
00574             }
00575         }
00576     }
00577 }
00578
00579 // Check for cells that were specified but not found
00580 if (!process_all_cells) {
00581     std::vector<std::string> not_found_cells;
00582     for (const auto &requested_cell : cell_names) {
00583         if (found_cells.find(requested_cell) == found_cells.end()) {
00584             not_found_cells.push_back(requested_cell);
00585         }
00586     }
00587
00588     // Output warning for cells that weren't found
00589     if (!not_found_cells.empty()) {
00590         std::string missing_cells = not_found_cells[0];
00591         for (size_t i = 1; i < not_found_cells.size(); ++i) {
00592             missing_cells += ", " + not_found_cells[i];
00593         }
00594         logger_>warn("{} specified cell{} not found in the library: {}", not_found_cells.size(),
00595             not_found_cells.size() > 1 ? "s were" : " was", missing_cells);
00596     }
00597 }
00598
00599 out.close();

```

```

00600 logger_>info("Supercell creation complete in '{}'", basename_ + ".map");
00601 }
00602
00627 void LibFile::verilog(const int chain_length, const std::vector<std::string> &cell_names) {
00628 logger_>info("Creating Verilog for '{}'", filename_);
00629
00630 this->supercell(chain_length, cell_names);
00631
00632 // Create a temporary file for individual modules
00633 std::string temp_file = basename_ + "_temp.v";
00634 std::ofstream out(temp_file);
00635 if (!out.is_open()) {
00636 logger_>error("Could not open temp file '{}' for writing", temp_file);
00637 return;
00638 }
00639
00640 // Read the .map file into a vector to preserve all entries and their order
00641 std::vector<std::pair<std::string, std::string>> supercell_entries;
00642 std::ifstream in(basename_ + ".map");
00643 if (!in.is_open()) {
00644 logger_>error("Could not open file '{}' for reading", basename_ + ".map");
00645 return;
00646 }
00647
00648 std::string line;
00649 while (std::getline(in, line)) {
00650 std::istringstream iss(line);
00651 std::string cell_name, supercell_name;
00652 if (!(iss >> cell_name >> supercell_name)) {
00653 logger_>warn("Invalid line format in map file: '{}', line);
00654 continue;
00655 }
00656 supercell_entries.push_back({cell_name, supercell_name});
00657 }
00658 in.close();
00659 logger_>info("Read {} supercells from '{}'", supercell_entries.size(), basename_ + ".map");
00660
00661 // Store module information for top-level creation
00662 std::vector<std::string> module_names;
00663 std::map<std::string, std::vector<std::string>> module_inputs;
00664 std::map<std::string, std::vector<std::string>> module_outputs;
00665
00666 // Generate verilog module for each supercell
00667 for (const auto &supercell_entry : supercell_entries) {
00668 // Check if the cell is sequential
00669 bool is_sequential = false;
00670 // Count the number of instances will be created in verilog
00671 int instance_count = 0;
00672 // Get original cell name from pair
00673 const std::string &cell_name = supercell_entry.first;
00674 // Get supercell name(as well as module name) from pair
00675 const std::string &module_name = supercell_entry.second;
00676
00677 // Store module name for top-level creation
00678 module_names.push_back(module_name);
00679
00680 logger_>info("Creating Module: '{}' from Cell '{}', module_name, cell_name);
00681
00682 // Get the cell JSON object
00683 json cell_json;
00684 for (const auto &cell : lib_json_["cells"]) {
00685 if (cell["cell_name"].get<std::string>() == cell_name) {
00686 cell_json = cell;
00687 break;
00688 }
00689 }
00690
00691 // Get input/output pins from the cell JSON object
00692 std::vector<std::string> input_pins;

```

```

00693     std::vector<std::string> output_pins;
00694     std::stringstream input_pins_ss;
00695     std::stringstream output_pins_ss;
00696     if (cell_json.contains("input_pins")) {
00697         for (const auto &pin : cell_json["input_pins"]) {
00698             if (pin.contains("clock")) {
00699                 is_sequential = true;
00700             }
00701             std::string pin_name = pin["pin_name"].get<std::string>();
00702             input_pins.push_back(pin_name);
00703             input_pins_ss << pin_name << ", ";
00704
00705             // Store input pin name for top-level creation
00706             module_inputs[module_name].push_back(pin_name);
00707         }
00708     }
00709     if (cell_json.contains("output_pins")) {
00710         for (const auto &pin : cell_json["output_pins"]) {
00711             std::string pin_name = pin["pin_name"].get<std::string>();
00712             output_pins.push_back(pin_name);
00713             output_pins_ss << pin_name << ", ";
00714
00715             // Store output pin name for top-level creation
00716             module_outputs[module_name].push_back(pin_name);
00717         }
00718     }
00719     logger_>debug("Input pins set: {}", input_pins_ss.str());
00720     logger_>debug("Output pins set: {}", output_pins_ss.str());
00721
00722     // Sequential cells will have only one instance
00723     // Combinational cells will have instances equal to chain length
00724     if (is_sequential) {
00725         instance_count = 1;
00726     } else {
00727         instance_count = chain_length;
00728     }
00729
00730     // Create ANSI port list
00731     std::string port_list_text = "(";
00732     // Add input ports
00733     bool isFirstPort = true;
00734     for (const auto &input_pin : input_pins) {
00735         if (!isFirstPort) {
00736             port_list_text += ", ";
00737         }
00738         port_list_text += "input " + input_pin;
00739         isFirstPort = false;
00740     }
00741     // Add output ports
00742     for (const auto &output_pin : output_pins) {
00743         if (!isFirstPort) {
00744             port_list_text += ", ";
00745         }
00746         port_list_text += "output " + output_pin;
00747         isFirstPort = false;
00748     }
00749     port_list_text += ")";
00750     logger_>debug("ANSI Port list: {}", port_list_text);
00751
00752     // Create full module header text
00753     std::string fullModuleText = "module " + module_name + port_list_text + ";\nendmodule";
00754     logger_>debug("Full module text: {}", fullModuleText);
00755
00756     // Generate the syntax tree for the module using slang library
00757     auto tree = slang::syntax::SyntaxTree::fromText(fullModuleText);
00758     if (tree) {
00759         // Add ports to the syntax tree
00760         ModuleRewriter rewriter(input_pins, output_pins, supercell_entry, instance_count, logger_);
00761     }

```

```

00762     tree = rewriter.transform(tree);
00763     // Revisit the syntax tree to check architecture
00764     rewriter.visit(tree->root());
00765
00766     // Output the transformed syntax tree
00767     out << "`timescale 1ns/10ps" << std::endl;
00768     out << slang::syntax::SyntaxPrinter::printFile(*tree) << std::endl << std::endl;
00769 } else {
00770     logger->error("Failed to create syntax tree for module '{}'", module_name);
00771     continue;
00772 }
00773 }
00774
00775 out.close();
00776
00777 // Create the top-level module
00778 std::ofstream final_out(basename_ + ".v");
00779 if (!final_out.is_open()) {
00780     logger->error("Could not open file '{}' for writing", basename_ + ".v");
00781     return;
00782 }
00783
00784 // Collect all port names separately for inputs and outputs
00785 std::vector<std::string> all_input_ports;
00786 std::vector<std::string> all_output_ports;
00787 std::stringstream top_instances;
00788
00789 // First pass: collect all port names
00790 for (const auto &module_name : module_names) {
00791     // Collect input port names
00792     for (const auto &pin : module_inputs[module_name]) {
00793         all_input_ports.push_back(module_name + "__" + pin);
00794     }
00795
00796     // Collect output port names
00797     for (const auto &pin : module_outputs[module_name]) {
00798         all_output_ports.push_back(module_name + "__" + pin);
00799     }
00800
00801     // Create instance
00802     std::string instance_name = "I_" + module_name.substr(0, module_name.find("__X"));
00803     for (size_t i = module_name.find("__X") + 3; i < module_name.length(); i++) {
00804         if (module_name[i] == '_' && module_name[i + 1] == '_') {
00805             instance_name += "__" + module_name.substr(i + 2);
00806             break;
00807         }
00808     }
00809
00810     // Generate port connections for this instance
00811     std::stringstream instance_ports;
00812     for (const auto &pin : module_inputs[module_name]) {
00813         instance_ports << "." << pin << "(" << module_name << "__" << pin << ")", ";";
00814     }
00815     for (const auto &pin : module_outputs[module_name]) {
00816         instance_ports << "." << pin << "(" << module_name << "__" << pin << ")", ";";
00817     }
00818
00819     // Remove last comma and space
00820     std::string instance_ports_str = instance_ports.str();
00821     if (!instance_ports_str.empty()) {
00822         instance_ports_str = instance_ports_str.substr(0, instance_ports_str.length() - 2);
00823     }
00824
00825     top_instances << " " << module_name << " " << instance_name << " (" << instance_ports_str
00826         << ");\n";
00827 }
00828
00829 // Now generate the port list with all inputs first, then all outputs
00830 std::stringstream top_ports;

```

```

00831
00832 // Add all input ports first
00833 for (size_t i = 0; i < all_input_ports.size(); ++i) {
00834     top_ports << all_input_ports[i];
00835     if (i < all_input_ports.size() - 1 || !all_output_ports.empty()) {
00836         top_ports << ", ";
00837     }
00838 }
00839
00840 // Then add all output ports
00841 for (size_t i = 0; i < all_output_ports.size(); ++i) {
00842     top_ports << all_output_ports[i];
00843     if (i < all_output_ports.size() - 1) {
00844         top_ports << ", ";
00845     }
00846 }
00847
00848 // Generate input and output declarations
00849 std::stringstream top_inputs;
00850 std::stringstream top_outputs;
00851
00852 // Generate input declarations
00853 for (const auto &port : all_input_ports) {
00854     top_inputs << " input " << port << ";\n";
00855 }
00856
00857 // Generate output declarations
00858 for (const auto &port : all_output_ports) {
00859     top_outputs << " output " << port << ";\n";
00860 }
00861
00862 // Write the top module
00863 final_out << "timescale 1ns/10ps" << std::endl;
00864 final_out << "module validate_top ( " << top_ports.str() << " );" << std::endl;
00865 final_out << std::endl; // Add newline before port declarations
00866 final_out << top_inputs.str();
00867 final_out << top_outputs.str();
00868 final_out << std::endl;
00869 final_out << top_instances.str();
00870 final_out << "endmodule" << std::endl << std::endl;
00871
00872 // Append the module definitions from the temporary file
00873 std::ifstream temp_in(temp_file);
00874 if (temp_in.is_open()) {
00875     final_out << temp_in.rdbuf();
00876     temp_in.close();
00877 } else {
00878     logger_>error("Could not open temp file '{}'" for reading", temp_file);
00879 }
00880
00881 final_out.close();
00882
00883 // Remove temporary file
00884 // if (std::filesystem::exists(temp_file)) {
00885 //     std::filesystem::remove(temp_file);
00886 // }
00887
00888 logger_>info("Verilog creation complete in '{}'", basename_ + ".v");
00889 }
00890
00891 std::map<std::string, std::string> LibFile::logic(const std::string &cell_name) {
00892     std::map<std::string, std::string> logic_map = {};
00893
00894     logger_>info("Getting output pin logic function for '{}'", filename_);
00895
00896     if (!std::filesystem::exists(jsonname_)) {
00897         logger_>info("JSON file not found. Parsing Liberty file first.");
00898         this->parse();
00899         this->writeJsonToFile();

```

```

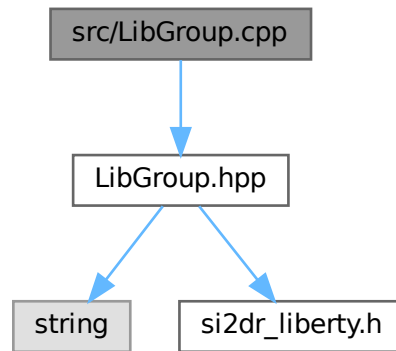
00900 } else {
00901     // Read the JSON file into json object
00902     std::ifstream in(jsonname_);
00903     if (!in.is_open()) {
00904         logger_>error("Could not open file '{}' for reading", jsonname_);
00905         return logic_map;
00906     }
00907     try {
00908         lib_json_ = json::parse(in);
00909     } catch (const json::parse_error &e) {
00910         logger_>error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00911         in.close();
00912         return logic_map;
00913     }
00914     in.close();
00915 }
00916
00917 // Check if the cell name exists in the JSON object
00918 for (const auto &cell : lib_json_["cells"]) {
00919     if (cell["cell_name"].get<std::string>() == cell_name) {
00920         logger_>debug("Found cell: '{}'", cell_name);
00921
00922         // Tranverse all output pins
00923         if (cell.contains("output_pins")) {
00924             for (const auto &pin : cell["output_pins"]) {
00925                 std::string pin_name = pin["pin_name"].get<std::string>();
00926                 // Check if the pin has a "timing_arcs" attribute
00927                 if (pin.contains("function")) {
00928                     std::string function = pin["function"].get<std::string>();
00929                     logic_map[pin_name] = function;
00930                     logger_>debug("Pin: '{}' Logic Function: '{}'", pin_name, function);
00931                 } else {
00932                     logger_>warn("Pin: '{}' does not have a 'function' attribute", pin_name);
00933                 }
00934             }
00935         }
00936     }
00937 }
00938 if (logic_map.empty()) {
00939     logger_>warn("No logic functions found for cell: '{}'", cell_name);
00940 } else {
00941     logger_>info("Found {} Logic functions for cell: '{}', logic_map.size(), cell_name);
00942     // Print the logic functions
00943     for (const auto &pair : logic_map) {
00944         logger_>info("Pin -> Logic Function: {} -> {}", pair.first, pair.second);
00945     }
00946 }
00947 return logic_map;
00948 }

```

8.38 src/LibGroup.cpp File Reference

```
#include "LibGroup.hpp"
```

Include dependency graph for LibGroup.cpp:



8.39 LibGroup.cpp

[Go to the documentation of this file.](#)

```

00001 #include "LibGroup.hpp"
00002
00003 LibGroup::LibGroup(si2drGroupIdT group, si2drErrorT &err) : group_(group), err_(err) {}
00004
00005 LibGroup::~LibGroup() {}
00006
00007 std::string LibGroup::getName() {
00008     si2drNamesIdT names = si2drGroupGetNames(group_, &err_);
00009     si2drStringT name = si2drIterNextName(names, &err_);
00010     si2drIterQuit(names, &err_);
00011     return name ? std::string(name) : std::string();
00012 }
00013
00014 std::string LibGroup::getType() {
00015     si2drStringT type = si2drGroupGetGroupType(group_, &err_);
00016     return type ? std::string(type) : std::string();
00017 }
00018
00019 si2drAttrIdT LibGroup::getAttrs() { return si2drGroupGetAttrs(group_, &err_); }
00020
00021 si2drGroupsIdT LibGroup::getGroups() { return si2drGroupGetGroups(group_, &err_); }
  
```

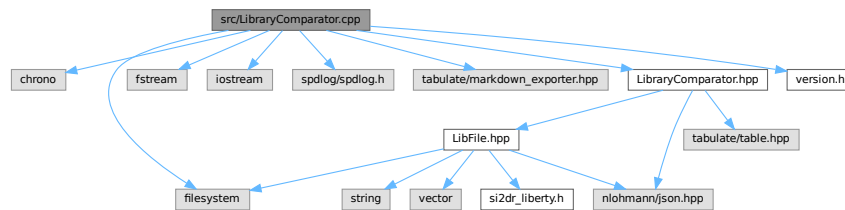
8.40 src/LibraryComparator.cpp File Reference

```
#include <chrono>
```

```
#include <filesystem>
```

```
#include <fstream>
#include <iostream>
#include "spdlog/spdlog.h"
#include <tabulate/markdown_exporter.hpp>
#include "LibraryComparator.hpp"
#include "version.h"
```

Include dependency graph for LibraryComparator.cpp:



8.41 LibraryComparator.cpp

[Go to the documentation of this file.](#)

```
00001 #include <chrono>
00002 #include <filesystem>
00003 #include <fstream>
00004 #include <iostream>
00005
00006 #include "spdlog/spdlog.h"
00007 #include <tabulate/markdown_exporter.hpp>
00008
00009 #include "LibraryComparator.hpp"
00010 #include "version.h"
00011
00012
00031 LibraryComparator::LibraryComparator(LibFile &ref_libfile, LibFile &comp_libfile, double reltol,
00032                                     double abstol)
00033 : reltol_(reltol), abstol_(abstol) {
00034     std::string ref_json_name = ref_libfile.basename_ + ".json";
00035     if (!std::filesystem::exists(ref_json_name)) {
00036         spdlog::info("Reference JSON file {} not found. Parsing first.", ref_json_name);
00037         ref_libfile.parse();
00038         ref_libfile.writeJsonToFile();
00039         ref_json_ = ref_libfile.lib_json_;
00040     } else {
00041         // Read the JSON file into json object
00042         std::ifstream ref_in(ref_json_name);
00043         if (!ref_in.is_open()) {
00044             spdlog::error("Could not open file '{}' for reading", ref_json_name);
00045             return;
00046         }
00047         try {
00048             ref_json_ = json::parse(ref_in);
00049         } catch (const json::parse_error &e) {
00050             spdlog::error("Error parsing reference JSON file '{}': {}", ref_json_name, e.what());
00051             return;
00052         }
00053     }
00054     std::string comp_json_name = comp_libfile.basename_ + ".json";
```



```

00055 if (!std::filesystem::exists(comp_json_name)) {
00056     spdlog::info("Comparison JSON file {} not found. Parsing first.", comp_json_name);
00057     comp_libfile.parse();
00058     comp_libfile.writeJsonToFile();
00059     comp_json_ = comp_libfile.lib_json_;
00060 } else {
00061     // Read the JSON file into json object
00062     std::ifstream comp_in(comp_json_name);
00063     if (!comp_in.is_open()) {
00064         spdlog::error("Could not open file '{}' for reading", comp_json_name);
00065         return;
00066     }
00067     try {
00068         comp_json_ = json::parse(comp_in);
00069     } catch (const json::parse_error &e) {
00070         spdlog::error("Error parsing comparison JSON file '{}': {}", comp_json_name, e.what());
00071         return;
00072     }
00073 }
00074
00075 ref_lib_path_ = ref_libfile.filepath_;
00076 comp_lib_path_ = comp_libfile.filepath_;
00077 spdlog::info("Successfully loaded JSON files for comparison");
00078 }
00079
00107 void LibraryComparator::compareLut(const std::string &cell_name, const std::string &pin_name,
00108                                     const std::string &timing_type, const std::string &related_pin,
00109                                     const std::string &arc_name, const json &ref_lut,
00110                                     const json &comp_lut, Table &table) {
00111     spdlog::info("Comparing LUT of '{}' ...", arc_name);
00112     std::vector<double> ref_index_1 = ref_lut["index_1"].get<std::vector<double>>();
00113     std::vector<double> ref_index_2 = ref_lut["index_2"].get<std::vector<double>>();
00114     std::vector<double> comp_index_1 = comp_lut["index_1"].get<std::vector<double>>();
00115     std::vector<double> comp_index_2 = comp_lut["index_2"].get<std::vector<double>>();
00116     if (ref_index_1 != comp_index_1 || ref_index_2 != comp_index_2) {
00117         spdlog::warn("Mismatch in LUT indices for '{}' in cell: '{}', pin: '{}', related_pin: '{}', "
00118                     "timing_type: '{}'",
00119                     arc_name, cell_name, pin_name, related_pin, timing_type);
00120         return;
00121     } else {
00122         spdlog::debug("LUT indices match for '{}'", arc_name);
00123
00124         std::vector<std::vector<double>> comp_value_matrix;
00125         // Generate a matrix of values from the JSON array for comparison library
00126         for (const auto &row_val : comp_lut["values"]) {
00127             std::vector<double> row_data;
00128             // Check if the row value is an array
00129             if (!row_val.is_array()) {
00130                 spdlog::error("Invalid LUT format: '{}' values should be an array of arrays in comp_lib '{}', "
00131                             "cell '{}', pin '{}', "
00132                             "related_pin '{}', timing_type '{}'",
00133                             arc_name, this->comp_lib_path_.string(), cell_name, pin_name, related_pin,
00134                             timing_type);
00135                 return;
00136             }
00137             // Check if non-numeric values
00138             for (const auto &val : row_val) {
00139                 if (val.is_number()) {
00140                     row_data.push_back(val.get<double>());
00141                 } else {
00142                     spdlog::error(
00143                         "Non-numeric value found in '{}' values, skipping value: {} in comp_lib '{}', cell '{}', "
00144                         "pin '{}', related_pin '{}', timing_type '{}'",
00145                         arc_name, val.dump(), this->comp_lib_path_.string(), cell_name, pin_name, related_pin,
00146                         timing_type);
00147                     return;
00148                 }
00149             }
00150             comp_value_matrix.push_back(row_data);

```

```

00151     }
00152     // Generate a matrix of values from the JSON array for reference library
00153     std::vector<std::vector<double>> ref_value_matrix;
00154     for (const auto &row_val : ref_lut["values"]) {
00155         std::vector<double> row_data;
00156         // Check if the row value is an array
00157         if (!row_val.is_array()) {
00158             spdlog::error("Invalid LUT format: '{}' values should be an array of arrays in ref_lib '{}', "
00159                 "cell '{}', pin '{}', "
00160                 "related_pin '{}', timing_type '{}'",
00161                 arc_name, this->ref_lib_path_.string(), cell_name, pin_name, related_pin,
00162                 timing_type);
00163             return;
00164         }
00165         // Check if non-numeric values
00166         for (const auto &val : row_val) {
00167             if (val.is_number()) {
00168                 row_data.push_back(val.get<double>());
00169             } else {
00170                 spdlog::error(
00171                     "Non-numeric value found in '{}'.values, skipping value: {} in ref_lib '{}', cell '{}', "
00172                     "pin '{}', related_pin '{}', timing_type '{}'",
00173                     arc_name, val.dump(), this->ref_lib_path_.string(), cell_name, pin_name, related_pin,
00174                     timing_type);
00175                 return;
00176             }
00177         }
00178         ref_value_matrix.push_back(row_data);
00179     }
00180     // Compare the matrix for relative tolerance
00181     if (!comp_value_matrix.empty() && !comp_value_matrix[0].empty()) {
00182         for (size_t i = 0; i < comp_value_matrix.size(); ++i) {
00183             for (size_t j = 0; j < comp_value_matrix[i].size(); ++j) {
00184                 double ref_val = ref_value_matrix[i][j];
00185                 double comp_val = comp_value_matrix[i][j];
00186                 if (std::abs(ref_val - comp_val) > reltol * std::abs(ref_val) &&
00187                     std::abs(ref_val - comp_val) > abstol) {
00188                     spdlog::debug("LUT value mismatch for '{}', index_1: {}, index_2: {}", arc_name,
00189                         ref_index_1[i], ref_index_2[j]);
00190                     table.add_row({related_pin + "->" + pin_name, std::to_string(ref_val),
00191                         std::to_string(comp_val), std::to_string(comp_val - ref_val),
00192                         std::to_string((comp_val - ref_val) / ref_val * 100), timing_type, arc_name,
00193                         std::to_string(i + 1), std::to_string(ref_index_1[i]), std::to_string(j + 1),
00194                         std::to_string(ref_index_2[j]), "<"});
00195                     spdlog::debug("table size: {}", table.size());
00196                 } else {
00197                     spdlog::debug("LUT value match for '{}', index_1: {}, index_2: {}", arc_name, ref_index_1[i],
00198                         ref_index_2[j]);
00199                 }
00200             }
00201         }
00202     } else {
00203         spdlog::error(
00204             "Empty or invalid 'values' array found for cell: '{}', pin: '{}', related_pin: '{}', "
00205             "timing_type: '{}', arc: '{}'",
00206             cell_name, pin_name, related_pin, timing_type, arc_name);
00207     }
00208 }
00209 }
00210
00229 void LibraryComparator::compareTimingArc(const std::string &cell_name, const std::string &pin_name,
00230     const std::string &timing_type, const json &ref_timing_arc,
00231     const json &comp_timing_arc, Table &table) {
00232     spdlog::info("Comparing timing type '{}' ...", timing_type);
00233
00234     std::string related_pin = ref_timing_arc["related_pin"].get<std::string>();
00235     spdlog::debug("Related pin: '{}'", related_pin);
00236
00237     std::vector<std::string> arc_names = {"cell_rise", "cell_fall", "rise_transition", "fall_transition"};

```

```

00238 for (auto arc_name : arc_names) {
00239     if (comp_timing_arc.contains(arc_name)) {
00240         spdlog::debug("Comparing timing arc: '{}'", arc_name);
00241         // Look for the arc in the reference JSON
00242         if (ref_timing_arc.contains(arc_name)) {
00243             // Found this arc
00244             spdlog::debug("Found timing arc: '{}'", arc_name);
00245             compareLut(cell_name, pin_name, timing_type, related_pin, arc_name, ref_timing_arc[arc_name],
00246                 comp_timing_arc[arc_name], table);
00247         } else {
00248             // arc not found in reference JSON
00249             spdlog::warn("Timing arc: '{}' not found in reference JSON", arc_name);
00250         }
00251     }
00252 }
00253 }
00254
00269 void LibraryComparator::comparePin(const std::string &cell_name, const std::string &pin_name,
00270     const json &ref_pin, const json &comp_pin, Table &table) {
00271     spdlog::info("Comparing pin '{}' ...", pin_name);
00272
00273     if (comp_pin.contains("timing_arcs")) {
00274         for (const auto &comp_arc : comp_pin["timing_arcs"]) {
00275             std::string timing_type = comp_arc["timing_type"].get<std::string>();
00276             spdlog::debug("Comparing timing type: '{}'", timing_type);
00277             // Look for the timing type in the reference JSON
00278             auto ref_arc_it = std::find_if(
00279                 ref_pin["timing_arcs"].begin(), ref_pin["timing_arcs"].end(),
00280                 [&timing_type](const json &ref_arc) { return ref_arc["timing_type"] == timing_type; });
00281             if (ref_arc_it != ref_pin["timing_arcs"].end()) {
00282                 // Found this timing type
00283                 spdlog::debug("Found timing type: '{}'", timing_type);
00284                 compareTimingArc(cell_name, pin_name, timing_type, *ref_arc_it, comp_arc, table);
00285             } else {
00286                 // timing arc not found in reference JSON
00287                 spdlog::warn("Timing arc: '{}' not found in reference JSON", timing_type);
00288             }
00289         }
00290     } else {
00291         spdlog::info("No timing arcs found for pin: '{}'", comp_pin["pin_name"].get<std::string>());
00292     }
00293 }
00294
00308 void LibraryComparator::compareCell(const std::string &cell_name, const json &ref_cell,
00309     const json &comp_cell, Table &table) {
00310     spdlog::info("Comparing cell '{}' ...", cell_name);
00311
00312     if (comp_cell.contains("output_pins")) {
00313         for (const auto &comp_pin : comp_cell["output_pins"]) {
00314             std::string pin_name = comp_pin["pin_name"].get<std::string>();
00315             spdlog::debug("Comparing pin: '{}'", pin_name);
00316             // Look for the pin in the reference JSON
00317             auto ref_pin_it =
00318                 std::find_if(ref_cell["output_pins"].begin(), ref_cell["output_pins"].end(),
00319                     [&pin_name](const json &ref_pin) { return ref_pin["pin_name"] == pin_name; });
00320             if (ref_pin_it != ref_cell["output_pins"].end()) {
00321                 // Found this pin
00322                 spdlog::debug("Found pin: '{}'", pin_name);
00323                 comparePin(cell_name, pin_name, *ref_pin_it, comp_pin, table);
00324             } else {
00325                 // pin not found in reference JSON
00326                 spdlog::warn("Pin: '{}' not found in reference JSON", pin_name);
00327             }
00328         }
00329     } else {
00330         spdlog::info("No output pins found for cell: '{}'", cell_name);
00331     }
00332 }
00333

```

```

00350 void LibraryComparator::generateReport(const std::string &output_file) {
00351     std::ofstream outfile(output_file);
00352     outfile << "# LIBRARY comparison\n" << std::endl;
00353     outfile << "**Reference library: " << ref_lib_path_ << "**\n" << std::endl;
00354     outfile << "**Comparison library: " << comp_lib_path_ << "**\n" << std::endl;
00355     outfile << "**Absolute tolerance: " << abstol_ << "**\n" << std::endl;
00356     outfile << "**Relative tolerance: " << reltol_ << "**\n" << std::endl;
00357     outfile << "**Performed by " << APP_NAME << " v" << APP_VERSION << " from " << APP_AUTHOR;
00358     auto now = std::chrono::system_clock::now();
00359     std::time_t now_time_t = std::chrono::system_clock::to_time_t(now);
00360     std::tm *now_tm = std::localtime(&now_time_t);
00361     outfile << ". on: " << std::put_time(now_tm, "%c") << "**\n" << std::endl;
00362     outfile << "> Legend: < outlier, * scaled, ! indices switched, ^ slews extrapolated, - loads "
00363         "extrapolated,\n";
00364     outfile << "> \n";
00365     outfile
00366         << "> Legend: + padding added, /0 divide by zero, / slews interpolated, # loads interpolated\n";
00367     outfile << "> \n";
00368     outfile << "> Legend: << value is less but unknown, >> value is more but unknown\n\n";
00369
00370     spdlog::info("Starting comparison report generation ...");
00371     for (const auto &comp_cell : comp_json_["cells"]) {
00372         std::string cell_name = comp_cell["cell_name"].get<std::string>();
00373         spdlog::debug("Comparing cell: '{}'", cell_name);
00374         // Look for the cell in the reference JSON
00375         auto ref_cell_it =
00376             std::find_if(ref_json_["cells"].begin(), ref_json_["cells"].end(),
00377                 [&cell_name](const json &ref_cell) { return ref_cell["cell_name"] == cell_name; });
00378         if (ref_cell_it != ref_json_["cells"].end()) {
00379             // Found this cell
00380             spdlog::debug("Found cell: '{}'", cell_name);
00381             Table table;
00382             table.add_row({"Pin Name", "Reference", "Comparison", "Diff", "Diff %", "Type", "Arc Name",
00383                 "Row #", "Index_1", "Col #", "Index_2", "Note"});
00384             // Header formatting
00385             for (size_t i = 0; i < table[0].size(); ++i) {
00386                 table[0][i].format().font_color(Color::yellow).font_style({FontStyle::bold});
00387             }
00388             compareCell(cell_name, *ref_cell_it, comp_cell, table);
00389
00390             if (table.size() > 1) {
00391                 // Data starts from row 1, row 0 is header
00392                 size_t failed_count = table.size() - 1;
00393                 double sum_diff = 0.0;
00394                 double sum_diff_percent = 0.0;
00395                 size_t outlier_count = 0;
00396                 double max_diff = -1e9;
00397                 double max_diff_percent = -1e9;
00398                 std::vector<double> diff_values; // Store diff values for std deviation calculation
00399
00400                 // Index of columns
00401                 const int diff_index = 3;
00402                 const int diff_percent_index = 4;
00403                 const int note_index = 11;
00404
00405                 for (size_t i = 1; i < table.size(); ++i) {
00406                     try {
00407                         double diff = std::stod(table[i][diff_index].get_text());
00408                         double diff_percent = std::stod(table[i][diff_percent_index].get_text());
00409
00410                         sum_diff += diff;
00411                         sum_diff_percent += diff_percent;
00412                         diff_values.push_back(diff_percent);
00413
00414                         if (table[i][note_index].get_text() == "<") {
00415                             outlier_count++;
00416                         }
00417                         max_diff = std::max(max_diff, diff);
00418                         max_diff_percent = std::max(max_diff_percent, diff_percent);

```

```

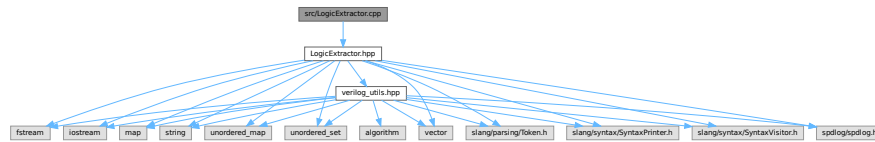
00419
00420     } catch (const std::invalid_argument &e) {
00421         spdlog::error("Could not convert value to double: {}", e.what());
00422         continue; // Skip this row if there's an error
00423     }
00424 }
00425
00426 double avg_diff = sum_diff / failed_count;
00427 double avg_diff_percent = sum_diff_percent / failed_count;
00428
00429 outfile << "## " << cell_name << "\n\n";
00430 outfile << "Delay Comparison in ns\n\n";
00431 if (output_file.substr(output_file.size() - 3) == ".md") {
00432     MarkdownExporter exporter;
00433     auto markdown = exporter.dump(table);
00434     outfile << markdown << std::endl;
00435 } else {
00436     outfile << table << std::endl;
00437 }
00438 outfile << std::endl;
00439 std::cout << table << std::endl;
00440 outfile << cell_name << " delay SUMMARY (abstol: " << abstol_ << ", reltol: " << reltol_
00441     << ")\n";
00442
00443 // Create summary table
00444 Table summary_table;
00445 summary_table.add_row({"Cell Name", "Data Type", "Failed Count", "Avg Diff", "Avg Diff%",
00446     "Max Diff", "Max Diff%", "Outliers"});
00447 summary_table.add_row({cell_name, "delay(ns)", std::to_string(failed_count),
00448     std::to_string(avg_diff), std::to_string(avg_diff_percent) + "%",
00449     std::to_string(max_diff), std::to_string(max_diff_percent) + "%",
00450     std::to_string(outlier_count)});
00451
00452 // Output summary table
00453 if (output_file.substr(output_file.size() - 3) == ".md") {
00454     MarkdownExporter exporter;
00455     auto markdown = exporter.dump(summary_table);
00456     outfile << markdown << std::endl;
00457 } else {
00458     outfile << summary_table << std::endl;
00459 }
00460 outfile << "Worst delay outlier: Max Abs: " << max_diff << ", Max Rel: " << max_diff_percent
00461     << "%, Outliers: " << outlier_count << "\n\n";
00462 }
00463 } else {
00464     // cell not found in reference JSON
00465     spdlog::warn("Cell: '{} not found in reference JSON", cell_name);
00466 }
00467 }
00468
00469 outfile.close();
00470 }

```

8.42 src/LogicExtractor.cpp File Reference

```
#include "LogicExtractor.hpp"
```

Include dependency graph for LogicExtractor.cpp:



Functions

- void [extractAndPrintNetlistInfo](#) (const std::string &verilog_file, const std::string &cell)
- std::map< std::string, std::string > [extractLogicFromVerilog](#) (const std::string &verilog_file, const std::string &cell)

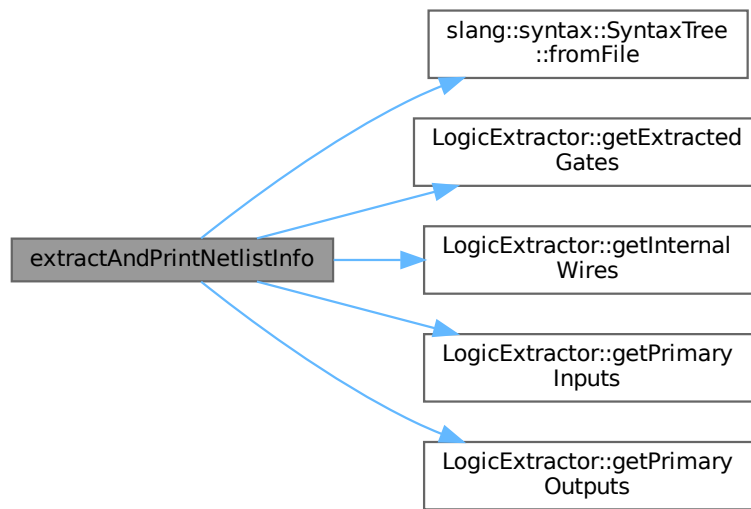
8.42.1 Function Documentation

8.42.1.1 extractAndPrintNetlistInfo()

```
void extractAndPrintNetlistInfo (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line [514](#) of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



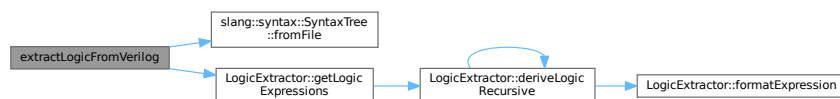
8.42.1.2 extractLogicFromVerilog()

```

std::map< std::string, std::string > extractLogicFromVerilog (
    const std::string & verilog_file,
    const std::string & cell )
  
```

Definition at line 575 of file [LogicExtractor.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.43 LogicExtractor.cpp

[Go to the documentation of this file.](#)

```

00001 #include "LogicExtractor.hpp"
00002
00003 // --- Implementation for LogicExtractor ---
00004
00005 void LogicExtractor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00006     if (parsingComplete_)
00007         return; // Don't re-process if called again
00008
00009     if (module.header && module.header->name.valid()) {
00010         std::string_view moduleName = module.header->name.valueText();
00011         if (!targetCell_.empty() && moduleName == targetCell_) {
00012             spdlog::info("LogicExtractor: Found target module: {}", targetCell_);
00013             inTargetModule_ = true;
00014
00015             // Reset state for the target module
00016             primaryInputs_.clear();
00017             primaryOutputs_.clear();
00018             internalWires_.clear();
00019             portDirections_.clear(); // Clear temporary direction map
00020             gateOutputDrivers_.clear();
00021             // logicCache_.clear(); // For Step 2
00022
00023             // Visit children (ports, declarations, instances) IN ORDER
00024             // It's often better to visit declarations first, then the port list
00025             // Slang's default visitDefault might handle this, but be aware.
00026             visitDefault(module);
00027
00028             // --- Finalize Ports after visiting declarations and port list ---
00029             // This part might be better placed *after* visiting NonAnsiPortList,
00030             // assuming the list visit confirms the names from the header.
00031             // Let's move the finalization logic to handle(NonAnsiPortListSyntax)
00032
00033             // Mark that we finished processing the target module
00034             inTargetModule_ = false; // Important to prevent processing other modules
00035             parsingComplete_ = true; // Stop processing after finding the target
00036             spdlog::info("LogicExtractor: Finished visiting target module '{}'.", targetCell_);
00037
00038         } else if (!parsingComplete_) {
00039             // If not the target module yet, continue searching
00040             visitDefault(module);
00041         }
00042     } else if (!parsingComplete_) {
00043         // Handle modules without valid headers if necessary, or just traverse
00044         visitDefault(module);
00045     }
00046 }
00047
00048 // Handle Port Declarations (defines direction and type, usually inside module body)
00049 void LogicExtractor::handle(const slang::syntax::PortDeclarationSyntax &portDecl) {
00050     if (!inTargetModule_)
00051         return;
00052     spdlog::debug("LogicExtractor: Handling Port Declaration");
00053
00054     std::string direction = "unknown";
00055     if (portDecl.header) {
00056         // Determine direction (input/output/inout)
00057         // Important: Use valueText() as Slang typically represents keywords as text tokens
00058         if (auto varHeader = portDecl.header->as_if<slang::syntax::VariablePortHeaderSyntax>()) {
00059             if (varHeader->direction.valid()) {
00060                 direction = std::string(varHeader->direction.valueText());
00061             }
00062         } else if (auto netHeader = portDecl.header->as_if<slang::syntax::NetPortHeaderSyntax>()) {
00063             if (netHeader->direction.valid()) {
00064                 direction = std::string(netHeader->direction.valueText());

```



```

00065     }
00066 }
00067 // Add InterfacePortHeaderSyntax if needed
00068 } else {
00069     spdlog::warn("LogicExtractor: Port declaration without header found.");
00070 }
00071
00072 for (const auto &declarator : portDecl.declarators) {
00073     if (declarator->name.valid()) {
00074         std::string portName = std::string(declarator->name.valueText());
00075         if (portDirections_.count(portName) && portDirections_[portName] != "unknown") {
00076             spdlog::warn("LogicExtractor: Multiple direction declarations for port '{}'. Keeping first "
00077                 "one '{}'. New direction: '{}'",
00078                 portName, portDirections_[portName], direction);
00079         } else {
00080             spdlog::info("LogicExtractor: Storing direction '{}' for port '{}'", direction, portName);
00081             portDirections_[portName] = direction;
00082             if (direction == "input") {
00083                 primaryInputs_.insert(portName);
00084                 internalWires_.insert(portName); // Inputs are also technically 'wires' usable internally
00085             } else if (direction == "output") {
00086                 primaryOutputs_.insert(portName);
00087                 internalWires_.insert(portName); // Outputs are also wires driven by something
00088             } else if (direction == "inout") {
00089                 spdlog::info("LogicExtractor: Inout port '{}' found. Adding to both inputs and outputs.",
00090                     portName);
00091                 primaryInputs_.insert(portName);
00092                 primaryOutputs_.insert(portName);
00093                 internalWires_.insert(portName);
00094             } else {
00095                 spdlog::warn("LogicExtractor: Port '{}' found in declaration but has unknown or missing "
00096                     "direction '{}'. Treating as internal wire.",
00097                     portName, direction);
00098                 internalWires_.insert(portName);
00099             }
00100             // logicCache_[portName] = portName; // For Step 2
00101         }
00102     } else {
00103         spdlog::warn(
00104             "LogicExtractor: Port declarator without a name found in PortDeclarationSyntax.");
00105     }
00106 }
00107 // Don't call visitDefault here, as it might revisit things unexpectedly.
00108 // Let the main module visitor handle traversing into children if necessary.
00109 }
00110
00111 // Handle Non-ANSI Port List (defines names, usually in module header)
00112 void LogicExtractor::handle(const slang::syntax::NonAnsiPortListSyntax &portList) {
00113     if (!inTargetModule_)
00114         return;
00115     spdlog::debug("LogicExtractor: Handling NonAnsi Port List");
00116
00117     for (const auto portSyntax : portList.ports) {
00118         if (!portSyntax)
00119             continue;
00120
00121         std::string portName = "";
00122         // Most common case: ImplicitNonAnsiPortSyntax contains the expression (usually just the name)
00123         if (auto implicitPort = portSyntax->as_if<slang::syntax::ImplicitNonAnsiPortSyntax>()) {
00124             if (implicitPort->expr) {
00125                 // The expression itself might be complex, try to get the simple name
00126                 if (auto portRef = implicitPort->expr->as_if<slang::syntax::PortReferenceSyntax>()) {
00127                     if (portRef->name.valid()) {
00128                         portName = std::string(portRef->name.valueText());
00129                     }
00130                 } else if (auto portConcat =
00131                     implicitPort->expr->as_if<slang::syntax::PortConcatenationSyntax>()) {
00132                     // Handle concatenations if necessary - complex
00133                     spdlog::warn("LogicExtractor: Port concatenation found in NonAnsi port list - currently "

```

```

00134         "not fully handled for logic extraction. Port: {}",
00135         implicitPort->toString());
00136     continue; // Skip complex ports for now
00137 } else {
00138     // Fallback: try getting name from the expression directly (might be just identifier)
00139     portName = implicitPort->toString();
00140 }
00141 }
00142 }
00143 // Handle ExplicitNonAnsiPortSyntax (e.g., .A(A)) if needed
00144 else if (auto explicitPort = portSyntax->as_if<slang::syntax::ExplicitNonAnsiPortSyntax>()) {
00145     if (explicitPort->name.valid()) {
00146         portName = std::string(explicitPort->name.valueText());
00147         // Note: explicitPort->expr is the internal signal it connects to.
00148         // We are primarily interested in the port's own name (explicitPort->name) here.
00149     }
00150 }
00151 // Handle EmptyNonAnsiPortSyntax (commas for placeholders) if necessary
00152 else if (portSyntax->kind == slang::syntax::SyntaxKind::EmptyNonAnsiPort) {
00153     spdlog::debug("LogicExtractor: Skipping empty non-ANSI port placeholder.");
00154     continue;
00155 } else {
00156     spdlog::warn("LogicExtractor: Unhandled NonAnsi port syntax kind: {}",
00157         slang::syntax::toString(portSyntax->kind));
00158     continue;
00159 }
00160
00161 // Now we (hopefully) have the portName.
00162 if (!portName.empty()) {
00163     if (portDirections_.count(portName)) {
00164         const std::string &direction = portDirections_[portName];
00165         spdlog::debug("LogicExtractor: Finalizing port '{}' with direction '{}'", portName,
00166             direction);
00167
00168         if (direction == "input") {
00169             primaryInputs_.insert(portName);
00170             internalWires_.insert(portName); // Inputs are also technically 'wires' usable internally
00171             // logicCache_[portName] = portName; // For Step 2
00172         } else if (direction == "output") {
00173             primaryOutputs_.insert(portName);
00174             internalWires_.insert(portName); // Outputs are also wires driven by something
00175         } else if (direction == "inout") {
00176             spdlog::info("LogicExtractor: Inout port '{}' found. Adding to both inputs and outputs.",
00177                 portName);
00178             primaryInputs_.insert(portName);
00179             primaryOutputs_.insert(portName);
00180             internalWires_.insert(portName);
00181             // logicCache_[portName] = portName; // For Step 2
00182         } else {
00183             spdlog::warn("LogicExtractor: Port '{}' found in list but has unknown or missing "
00184                 "direction '{}'. Treating as internal wire.",
00185                 portName, direction);
00186             internalWires_.insert(portName);
00187         }
00188     } else {
00189         spdlog::debug("LogicExtractor: Port '{}' found in NonAnsi list, no direction declaration ",
00190             portName);
00191     }
00192 } else {
00193     spdlog::warn("LogicExtractor: Could not determine port name from NonAnsi port list item: {}",
00194         portSyntax->toString());
00195 }
00196 }
00197 // Don't call visitDefault here
00198 }
00199
00200 // Handle explicit wire declarations
00201 void LogicExtractor::handle(const slang::syntax::NetDeclarationSyntax &netDecl) {
00202     if (!inTargetModule_)

```

```

00203     return;
00204
00205     for (const auto &declarator : netDecl.declarators) {
00206         if (declarator->name.valid()) {
00207             std::string wireName = std::string(declarator->name.valueText());
00208             spdlog::debug("LogicExtractor: Found wire declaration: {}", wireName);
00209             // Avoid adding ports again if they were also declared as nets (common)
00210             if (!primaryInputs_.count(wireName) && !primaryOutputs_.count(wireName)) {
00211                 internalWires_.insert(wireName);
00212             } else {
00213                 spdlog::trace("LogicExtractor: Wire '{}' is already known as a port.", wireName);
00214             }
00215         }
00216     }
00217     // Don't call visitDefault here
00218 }
00219
00220 // Handle primitive gate instantiations (MOST IMPORTANT PART)
00221 void LogicExtractor::handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst) {
00222     if (!inTargetModule_)
00223         return;
00224
00225     if (!primitiveInst.type.valid()) {
00226         spdlog::warn("LogicExtractor: Primitive instance without a type token found.");
00227         return;
00228     }
00229
00230     // Use token kind for reliable checking, text for name storage
00231     slang::parsing::TokenKind gateKind = primitiveInst.type.kind;
00232     std::string gateTypeName = std::string(primitiveInst.type.valueText());
00233
00234     spdlog::debug("LogicExtractor: Found Primitive Instance of Type: {} (Kind: {})", gateTypeName,
00235                 slang::parsing::toString(gateKind));
00236
00237     for (const auto &instance : primitiveInst.instances) {
00238         // if (!instance || !instance->connections.isInitialized()) { // Check if connections are valid
00239         //     spdlog::warn("LogicExtractor: Skipping primitive instance of type {} due to null pointer or
00240         //         "
00241         //             "uninitialized connections.",
00242         //             gateTypeName);
00243         //     continue;
00244         // }
00245
00246         GateInfo currentGateInfo;
00247         currentGateInfo.gateTypeName = gateTypeName;
00248         currentGateInfo.kind = gateKind;
00249
00250         // Primitives usually have ordered connections.
00251         // The FIRST connection is typically the OUTPUT.
00252         // The REST are INPUTS.
00253         if (instance->connections.empty()) {
00254             spdlog::warn("LogicExtractor: Gate instance of type {} has no connections.", gateTypeName);
00255             continue;
00256         }
00257
00258         // Extract Output Signal
00259         // Ensure the connection itself is not null
00260         if (!instance->connections[0]) {
00261             spdlog::error("LogicExtractor: First connection (output) is null for primitive {}",
00262                         gateTypeName);
00263             continue;
00264         }
00265         if (auto firstConn =
00266             instance->connections[0]->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00267             currentGateInfo.outputSignal =
00268                 firstConn->expr->toString(); // Get the output signal name from the connection
00269                 // remove extra spaces in the signal name
00270             currentGateInfo.outputSignal.erase(
00271                 std::remove_if(currentGateInfo.outputSignal.begin(), currentGateInfo.outputSignal.end(),

```

```

00272         [](unsigned char x) { return std::isspace(x); }},
00273         currentGateInfo.outputSignal.end());
00274     if (!currentGateInfo.outputSignal.empty()) {
00275         spdlog::debug(" Output Signal: {}", currentGateInfo.outputSignal);
00276         // Gate outputs are internal signals (unless they are module outputs)
00277         // Add to internal wires if not already a primary output.
00278         if (!primaryOutputs_.count(currentGateInfo.outputSignal)) {
00279             internalWires_.insert(currentGateInfo.outputSignal);
00280         }
00281     } else {
00282         spdlog::error(
00283             "LogicExtractor: Could not determine output signal name for gate instance of type {}",
00284             gateTypeName);
00285         continue; // Skip this instance if output is unknown
00286     }
00287 } else {
00288     spdlog::error("LogicExtractor: Expected OrderedPortConnectionSyntax for output of primitive "
00289         "{}", but got kind: {}. Conn: {}",
00290         gateTypeName, slang::syntax::toString(instance->connections[0]->kind),
00291         instance->connections[0]->toString());
00292     continue; // Skip this instance
00293 }
00294
00295 // Extract Input Signals
00296 for (size_t i = 1; i < instance->connections.size(); ++i) {
00297     // Ensure the connection itself is not null
00298     if (!instance->connections[i]) {
00299         spdlog::warn("LogicExtractor: Input connection {} is null for primitive {}", i,
00300             gateTypeName);
00301         continue;
00302     }
00303     if (auto conn =
00304         instance->connections[i]->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00305         std::string inputSig =
00306             conn->expr->toString(); // Get the input signal name from the connection
00307         // remove extra spaces in the signal name
00308         inputSig.erase(std::remove_if(inputSig.begin(), inputSig.end(),
00309             [](unsigned char x) { return std::isspace(x); }},
00310             inputSig.end());
00311         if (!inputSig.empty()) {
00312             currentGateInfo.inputSignals.push_back(inputSig);
00313             spdlog::debug(" Input Signal {}: {}", i, inputSig);
00314         } else {
00315             spdlog::warn("LogicExtractor: Could not determine input signal name for input {} of gate "
00316                 "instance type {}. Conn: {}",
00317                 i, gateTypeName, conn->toString());
00318             // Decide whether to skip or continue with partial inputs
00319         }
00320     } else {
00321         spdlog::warn("LogicExtractor: Expected OrderedPortConnectionSyntax for input {} of "
00322             "primitive {}, but got kind: {}. Conn: {}",
00323             i, gateTypeName, slang::syntax::toString(instance->connections[i]->kind),
00324             instance->connections[i]->toString());
00325     }
00326 }
00327
00328 // Store the gate information, mapping the output signal to its driving gate
00329 if (!currentGateInfo.outputSignal.empty()) {
00330     if (gateOutputDrivers_.count(currentGateInfo.outputSignal)) {
00331         // This is a critical warning - indicates multiple drivers for the same net!
00332         spdlog::error("LogicExtractor: Multiple drivers found for signal '{}'. Previous driver: "
00333             "{}", New driver: {}. Netlist is likely invalid.",
00334             currentGateInfo.outputSignal,
00335             gateOutputDrivers_[currentGateInfo.outputSignal].gateTypeName, gateTypeName);
00336         // Keep the first one found for now, or decide on error handling
00337     } else {
00338         spdlog::info(" Storing driver for '{}': Gate Type '{}'", currentGateInfo.outputSignal,
00339             gateTypeName);
00340         gateOutputDrivers_[currentGateInfo.outputSignal] = currentGateInfo;

```

```

00341     }
00342 }
00343 }
00344 // Don't call visitDefault here
00345 }
00346
00347 // --- Logic Derivation Implementation ---
00348
00349 // Public method called after visiting the tree
00350 std::map<std::string, std::string> LogicExtractor::getLogicExpressions() {
00351     std::map<std::string, std::string> result_map;
00352     if (!parsingComplete_) {
00353         spdlog::error("LogicExtractor: AST parsing did not complete or target module '{}' not found. "
00354             "Cannot extract logic.",
00355             targetCell_);
00356         return result_map; // Return empty map
00357     }
00358
00359     spdlog::info("LogicExtractor: Deriving logic expressions for {} output ports...",
00360         primaryOutputs_.size());
00361
00362     for (const std::string &outputPort : primaryOutputs_) {
00363         spdlog::debug("LogicExtractor: Deriving logic for output: {}", outputPort);
00364         try {
00365             result_map[outputPort] = deriveLogicRecursive(outputPort);
00366             spdlog::info("  Output: {} => {}", outputPort, result_map[outputPort]);
00367         } catch (const std::runtime_error &e) {
00368             spdlog::error("LogicExtractor: Error deriving logic for output '{}': {}", outputPort,
00369                 e.what());
00370             result_map[outputPort] = "/* Error deriving logic */";
00371         } catch (...) {
00372             spdlog::error("LogicExtractor: Unknown error deriving logic for output '{}'", outputPort);
00373             result_map[outputPort] = "/* Unknown error deriving logic */";
00374         }
00375     }
00376     return result_map;
00377 }
00378
00379 // Recursive function with memoization
00380 std::string LogicExtractor::deriveLogicRecursive(const std::string &signalName) {
00381     // 1. Check Cache (Memoization)
00382     if (logicCache_.count(signalName)) {
00383         return logicCache_.at(signalName);
00384     }
00385
00386     // 2. Base Case: Is it a primary input?
00387     if (primaryInputs_.count(signalName)) {
00388         // Already cached during port handling, but double-check
00389         if (!logicCache_.count(signalName)) {
00390             logicCache_[signalName] = signalName;
00391         }
00392         return signalName;
00393     }
00394
00395     // 3. Recursive Step: Is it driven by a gate?
00396     if (gateOutputDrivers_.count(signalName)) {
00397         const GateInfo &driverGate = gateOutputDrivers_.at(signalName);
00398         std::vector<std::string> inputExpressions;
00399         spdlog::debug("  Tracing signal '{}', driven by {} gate", signalName,
00400             driverGate.gateTypeName());
00401
00402         // Recursively find expressions for all inputs of this gate
00403         for (const std::string &inputSig : driverGate.inputSignals) {
00404             if (inputSig.empty()) {
00405                 throw std::runtime_error("Empty input signal name encountered for gate driving " +
00406                     signalName);
00407             }
00408             spdlog::debug("    Recursing for input: {}", inputSig);
00409             inputExpressions.push_back(deriveLogicRecursive(inputSig));

```

```

00410     }
00411
00412     // Format the expression based on gate type and input expressions
00413     std::string currentExpr = formatExpression(driverGate, inputExpressions);
00414
00415     // Cache the result
00416     logicCache_[signalName] = currentExpr;
00417     return currentExpr;
00418 }
00419
00420 // 4. Handle Assign statements (if implemented)
00421 // if (assignDrivers_.count(signalName)) { ... }
00422
00423 // 5. Error Case: Signal not found or not driven by known element
00424 // Check if it's just an internal wire that wasn't driven?
00425 if (internalWires_.count(signalName)) {
00426     throw std::runtime_error(
00427         "Signal '" + signalName +
00428         "' is an internal wire but has no identified driver (gate or assign).");
00429 } else {
00430     throw std::runtime_error(
00431         "Signal '" + signalName +
00432         "' is not a primary input, known wire, or driven by a recognized gate/assignment.");
00433 }
00434 }
00435
00436 // Helper to format the expression string based on gate type
00437 std::string LogicExtractor::formatExpression(const GateInfo &gateInfo,
00438                                             const std::vector<std::string> &inputExprs) {
00439     if (inputExprs.empty() && gateInfo.kind != slang::parsing::TokenKind::NotKeyword &&
00440         gateInfo.kind != slang::parsing::TokenKind::BufKeyword) {
00441         // Gates like AND/OR/XOR need inputs
00442         spdlog::warn("Gate type {} requires inputs, but none were provided/derived for output {}",
00443             gateInfo.gateTypeName, gateInfo.outputSignal);
00444         return "/*Error: Missing Inputs for " + gateInfo.gateTypeName + ">*/";
00445     }
00446
00447     std::string result = "";
00448
00449     // Use gateInfo.type (enum) for reliable checking
00450     switch (gateInfo.kind) {
00451     case slang::parsing::TokenKind::AndKeyword:
00452         result = "(" + inputExprs[0];
00453         for (size_t i = 1; i < inputExprs.size(); ++i)
00454             result += " * " + inputExprs[i]; // Use * for AND
00455         result += ")";
00456         break;
00457     case slang::parsing::TokenKind::NandKeyword:
00458         result = "!(" + inputExprs[0];
00459         for (size_t i = 1; i < inputExprs.size(); ++i)
00460             result += " * " + inputExprs[i];
00461         result += ")";
00462         break;
00463     case slang::parsing::TokenKind::OrKeyword:
00464         result = "(" + inputExprs[0];
00465         for (size_t i = 1; i < inputExprs.size(); ++i)
00466             result += " + " + inputExprs[i]; // Use + for OR
00467         result += ")";
00468         break;
00469     case slang::parsing::TokenKind::NorKeyword:
00470         result = "!(" + inputExprs[0];
00471         for (size_t i = 1; i < inputExprs.size(); ++i)
00472             result += " + " + inputExprs[i];
00473         result += ")";
00474         break;
00475     case slang::parsing::TokenKind::XorKeyword:
00476         result = "(" + inputExprs[0];
00477         for (size_t i = 1; i < inputExprs.size(); ++i)
00478             result += " ^ " + inputExprs[i]; // Use ^ for XOR

```

```

00479     result += ")";
00480     break;
00481 case slang::parsing::TokenKind::XnorKeyword:
00482     result = "!(" + inputExprs[0];
00483     for (size_t i = 1; i < inputExprs.size(); ++i)
00484         result += " ^ " + inputExprs[i];
00485     result += ")";
00486     break;
00487 case slang::parsing::TokenKind::NotKeyword: // NOT gate
00488     if (inputExprs.size() != 1) {
00489         spdlog::warn("NOT gate expects 1 input, got {}", inputExprs.size());
00490         return "/*<Error: Incorrect Inputs for NOT>*/";
00491     }
00492     result = "!" + inputExprs[0]; // Use ! for NOT
00493     break;
00494 case slang::parsing::TokenKind::BufKeyword: // BUF gate
00495     if (inputExprs.size() != 1) {
00496         spdlog::warn("BUF gate expects 1 input, got {}", inputExprs.size());
00497         return "/*<Error: Incorrect Inputs for BUF>*/";
00498     }
00499     result = inputExprs[0]; // Output is the same as input
00500     break;
00501 // Add cases for bufif0, bufif1, notif0, notif1, pullup, pulldown, cmos, nmos, pmos, tran etc. if
00502 // needed
00503 default:
00504     spdlog::warn("Unsupported primitive gate type for logic expression generation: {}",
00505                 gateInfo.gateTypeName);
00506     result = "/*<Unsupported Gate: " + gateInfo.gateTypeName + ">*/";
00507     break;
00508 }
00509 return result;
00510 }
00511
00512 // --- New Function Implementation (Step 1: Visit and Print) ---
00513
00514 void extractAndPrintNetlistInfo(const std::string &verilog_file, const std::string &cell) {
00515     spdlog::info("--- Step 1: Starting Netlist Info Extraction ---");
00516     spdlog::info("Verilog file: '{}', Target cell: '{}'", verilog_file, cell);
00517
00518     try {
00519         auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00520         if (!result) { // Check if result is valid
00521             spdlog::error("Error parsing Verilog file '{}'. Cannot extract info.", verilog_file);
00522             return;
00523         }
00524
00525         spdlog::info("Successfully parsed Verilog file.");
00526         std::shared_ptr<slang::syntax::SyntaxTree> tree = result.value();
00527
00528         LogicExtractor extractor(cell);
00529         tree->root().visit(extractor); // Populate extractor's internal state
00530
00531         // --- Print Summary (Debug for Step 1) ---
00532         spdlog::info("--- Extraction Summary for Cell '{}': ---", cell);
00533
00534         const auto &inputs = extractor.getPrimaryInputs();
00535         spdlog::info("Found {} Primary Inputs:", inputs.size());
00536         for (const auto &name : inputs) {
00537             spdlog::info("  - {}", name);
00538         }
00539
00540         const auto &outputs = extractor.getPrimaryOutputs();
00541         spdlog::info("Found {} Primary Outputs:", outputs.size());
00542         for (const auto &name : outputs) {
00543             spdlog::info("  - {}", name);
00544         }
00545
00546         const auto &wires = extractor.getInternalWires();
00547         spdlog::info("Found {} Internal Wires:", wires.size());

```

```

00548     for (const auto &name : wires) {
00549         spdlog::info(" - {}", name);
00550     }
00551
00552     const auto &gates = extractor.getExtractedGates();
00553     spdlog::info("Found {} Gate Drivers:", gates.size());
00554     for (const auto &pair : gates) {
00555         const std::string &outputNet = pair.first;
00556         const GateInfo &info = pair.second;
00557         std::string inputsStr = "";
00558         for (size_t i = 0; i < info.inputSignals.size(); ++i) {
00559             inputsStr += info.inputSignals[i] + (i == info.inputSignals.size() - 1 ? " : " : ", ");
00560         }
00561         spdlog::info(" - Output: {} <= Driven by: {} ({} Inputs: [{}]", outputNet,
00562             info.gateTypeName, slang::parsing::toString(info.kind), inputsStr);
00563     }
00564     spdlog::info("--- End Netlist Info Extraction ---");
00565
00566 } catch (const std::exception &e) {
00567     spdlog::error("Exception during Verilog parsing or info extraction: {}", e.what());
00568 } catch (...) {
00569     spdlog::error("Unknown exception during Verilog parsing or info extraction.");
00570 }
00571 }
00572
00573 // --- (Step 2: Extract Logic Expressions) ---
00574
00575 std::map<std::string, std::string> extractLogicFromVerilog(const std::string &verilog_file,
00576     const std::string &cell) {
00577     spdlog::info("--- Starting Logic Expression Extraction for cell: '{}' ---", cell);
00578     std::map<std::string, std::string> logicMap; // Default empty map
00579
00580     try {
00581         auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00582         if (!result) {
00583             spdlog::error("Error parsing Verilog file '{}'. Cannot extract logic.", verilog_file);
00584             return logicMap;
00585         }
00586
00587         spdlog::info("Successfully parsed Verilog file.");
00588         std::shared_ptr<slang::syntax::SyntaxTree> tree = result.value();
00589
00590         LogicExtractor extractor(cell);
00591         tree->root().visit(extractor); // Populate extractor's internal state
00592
00593         // Now, call the method to derive and get the expressions
00594         logicMap = extractor.getLogicExpressions();
00595
00596     } catch (const std::exception &e) {
00597         spdlog::error("Exception during Verilog parsing or logic extraction: {}", e.what());
00598     } catch (...) {
00599         spdlog::error("Unknown exception during Verilog parsing or logic extraction.");
00600     }
00601
00602     if (logicMap.empty()) {
00603         spdlog::warn("Logic extraction finished, but no expressions were derived for cell '{}'. Check "
00604             "if cell exists and is correctly defined.",
00605             cell);
00606     } else {
00607         spdlog::info("Logic extraction completed for cell '{}'. Found {} output expression strings",
00608             cell, logicMap.size());
00609     }
00610
00611     spdlog::info("--- End Logic Expression Extraction ---");
00612     return logicMap;
00613 }

```


8.44 src/main.cpp File Reference

```
#include <filesystem>
#include <iostream>
#include <string>
#include <thread>
#include <vector>
#include "CLI/CLI.hpp"
#include "si2dr_liberty.h"
#include "spdlog/sinks/basic_file_sink.h"
#include "spdlog/sinks/stdout_color_sinks.h"
#include "spdlog/spdlog.h"
#include "LibraryComparator.hpp"
#include "LibFile.hpp"
#include "LogicExtractor.hpp"
#include "verilog_utils.hpp"
#include "version.h"
```

Include dependency graph for main.cpp:



Functions

- void [printInfo](#) ()
Sets up and configures the global logger and prints application information.
- void [parseLibFile](#) (const std::string &library_path, const std::string log_file_name)
Parses a library file and generates corresponding output.
- void [monoCheckLibFile](#) (const std::string &library_path, const std::string log_file_name, bool is↵_slew)
Performs monotonicity check on a library file.
- void [supercellLibFile](#) (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names)
Creates supercell map structures from a Liberty library file.
- void [verilogLibFile](#) (const std::string &library_path, const std::string &log_file_name, int chain↵_length, const std::vector< std::string > &cell_names)
Generates Verilog files from a library file for specified cells.
- void [compareLibFiles](#) (const std::string &ref_lib, const std::string &comp_lib, const double reltol, const double abstol, std::string &report_file_name)

Compares two library files and generates a detailed comparison report.

- void [funcLibFile](#) (const std::string &ref_file, const std::string &comp_file, const std::vector< std::string > &cell_names, std::string &report_file_name)
- int [main](#) (int argc, char *argv[])

8.44.1 Function Documentation

8.44.1.1 compareLibFiles()

```
void compareLibFiles (
    const std::string & ref_lib,
    const std::string & comp_lib,
    const double reltol,
    const double abstol,
    std::string & report_file_name )
```

Compares two library files and generates a detailed comparison report.

This function compares a reference library file with another library file, performing validation checks based on specified tolerance parameters. The comparison results are written to a report file in markdown or text format.

Parameters

<i>ref_lib</i>	Path to the reference library file to use as the baseline
<i>comp_lib</i>	Path to the library file to compare against the reference
<i>reltol</i>	Relative tolerance for numerical comparisons (must be ≥ 0.0)
<i>abstol</i>	Absolute tolerance for numerical comparisons
<i>report_file_name</i>	[in,out] Name of the file to write the comparison report to. If empty, defaults to [comp_lib_basename].cmp.md. If provided but doesn't end with .txt or .md, .md will be appended.

Note

The function will log an error and return without comparing if reltol is invalid.

Log files will be created with the naming pattern [library_basename].cmp.log

The function uses the [LibraryComparator](#) class to perform the actual comparison.

Definition at line 220 of file [main.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.44.1.2 funcLibFile()

```
void funcLibFile (  
    const std::string & ref_file,  
    const std::string & comp_file,  
    const std::vector< std::string > & cell_names,  
    std::string & report_file_name )
```

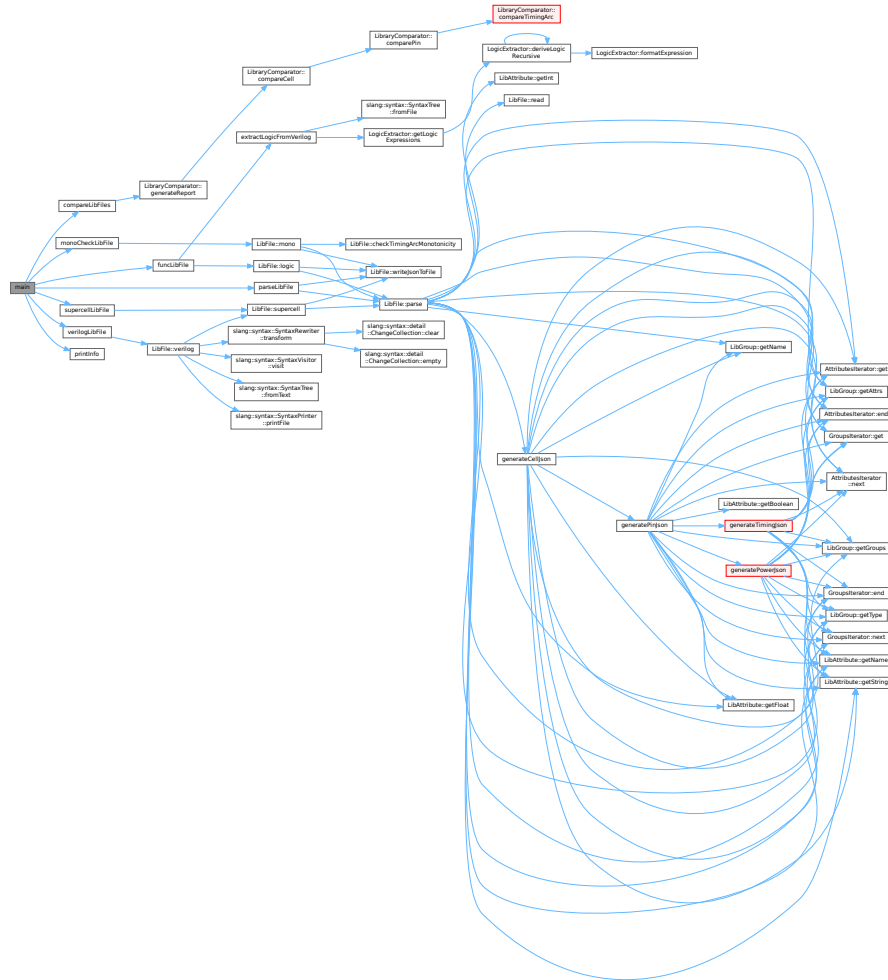
Definition at line 248 of file [main.cpp](#).

```
graph LR; main[main] --> funcLibFile[funcLibFile]
```

```
int main (
```

Generated on Mon Mar 31 2025 22:47:15 for ZlibValidation by Doxygen

Here is the call graph for this function:



```
void monoCheckLibFile (
    const std::string & library_path,
    const std::string log_file_name,
    bool is_slew )
```

This function validates the monotonicity of timing data in a library file. It creates a log file to record the results of the check and handles any exceptions that occur during the process.

Parameters

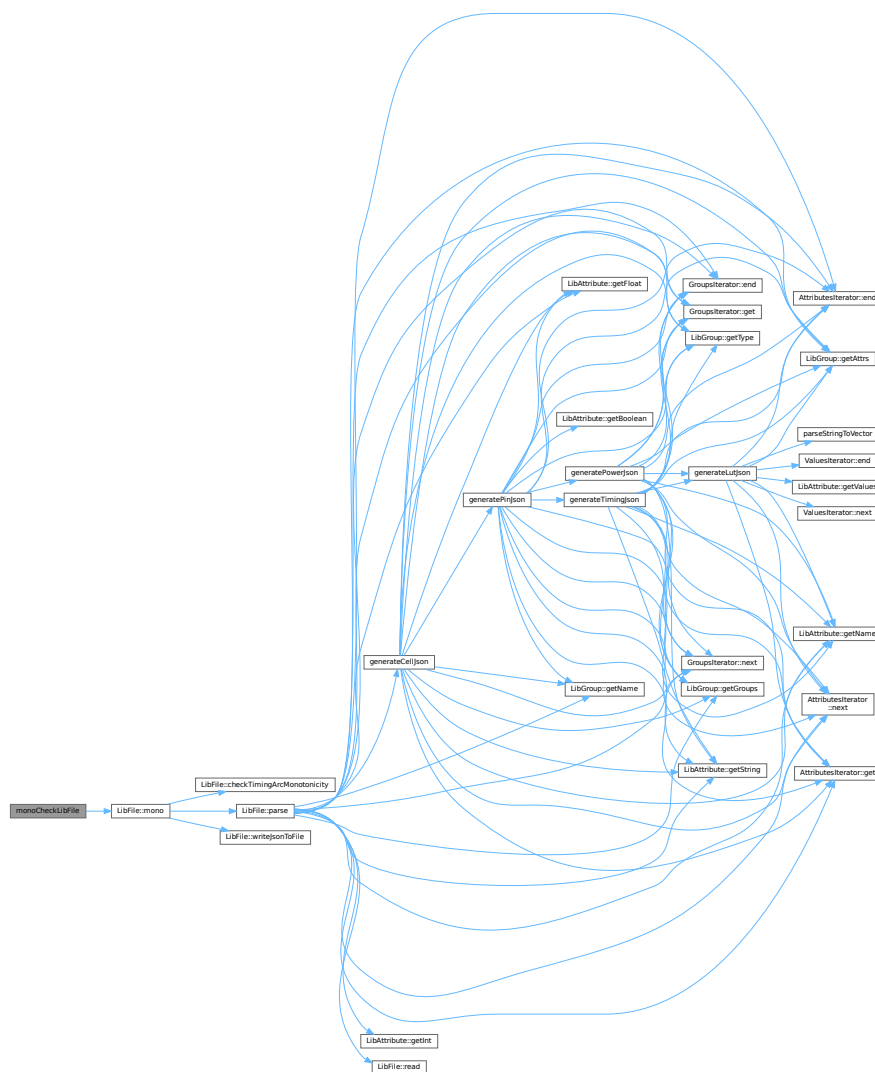
<i>library_path</i>	Path to the library file to check
<i>log_file_name</i>	Name of the log file to create (optional). If empty, a default name will be generated from the library file name
<i>is_slew</i>	Flag indicating whether to check input slew monotonicity (true) or output load monotonicity (false)

Exceptions

<i>The</i>	function catches and logs any exceptions but does not rethrow them
------------	--

Definition at line 98 of file [main.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.44.1.5 parseLibFile()

```
void parseLibFile (  
    const std::string & library_path,  
    const std::string log_file_name )
```

Parses a library file and generates corresponding output.

This function processes the given library file, parsing its contents and generating a JSON output. It also logs the parsing process to a specified log file or creates a default log file if none is provided.

Parameters

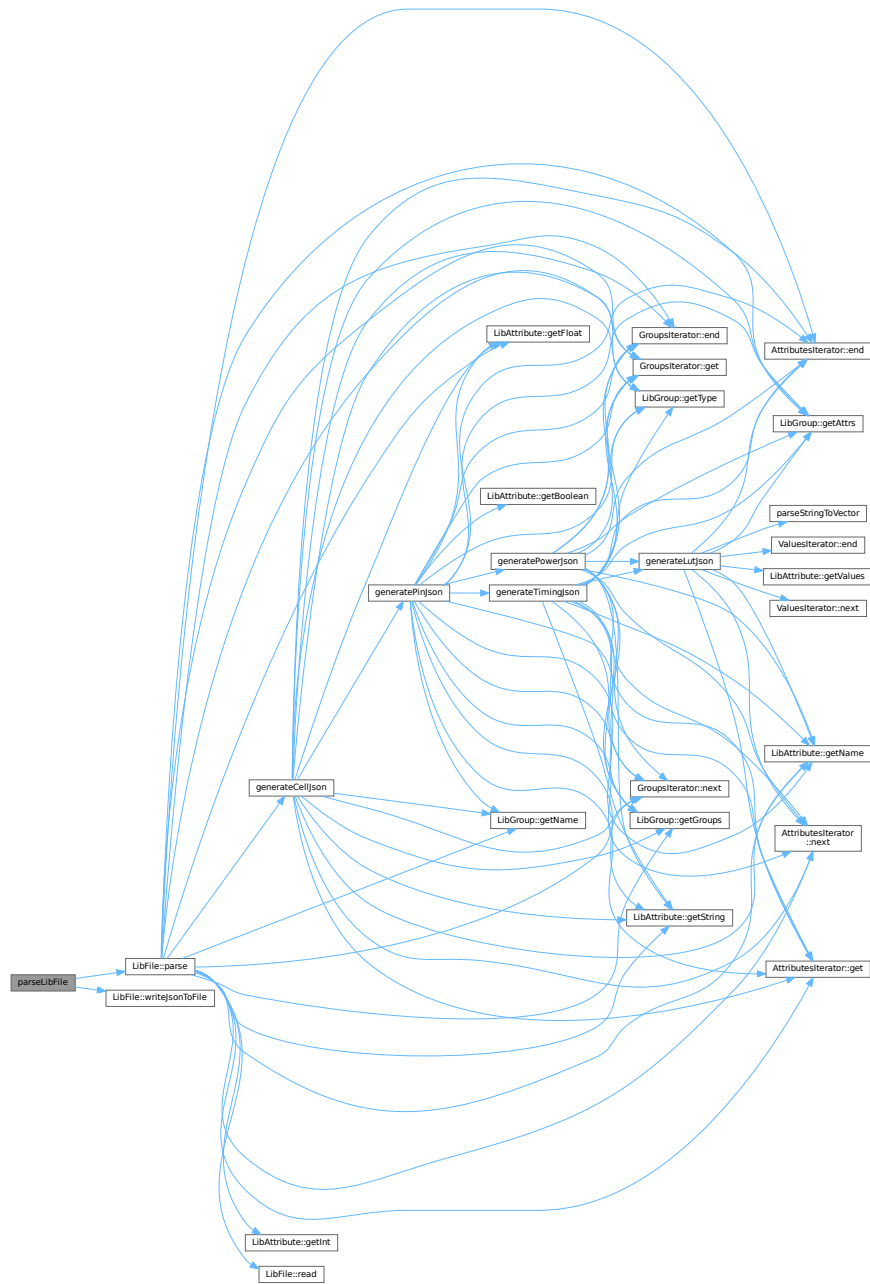
<i>library_path</i>	Path to the library file that needs to be parsed
<i>log_file_name</i>	Optional name for the log file. If empty, a default name is generated based on the library filename with ".parse.log" extension

Exceptions

<i>The</i>	function catches and logs any exceptions but doesn't propagate them
------------	---

Definition at line 66 of file [main.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.44.1.6 printInfo()

```
void printInfo ( )
```

Sets up and configures the global logger and prints application information.

This function performs the following operations:

1. Creates a console sink for logging with level set to INFO
2. Creates a file sink for logging with level set to DEBUG, saving to [APP_NAME].log
3. Configures a logger with both sinks and sets it as the default logger
4. Outputs basic application information:
 - Version and build timestamp
 - Author information
 - Log file location

Note

Uses spdlog library for logging functionality

Depends on APP_NAME, APP_VERSION, BUILD_TIMESTAMP, APP_AUTHOR, and APP↵_CONTACT macros

Definition at line 35 of file [main.cpp](#).

Here is the caller graph for this function:



8.44.1.7 supercellLibFile()

```
void supercellLibFile (  
    const std::string & library_path,  
    const std::string & log_file_name,  
    int chain_length,  
    const std::vector< std::string > & cell_names )
```

Creates supercell map structures from a Liberty library file.

This function reads a Liberty file, creates supercell structures based on the specified chain length and cell names, and logs the process to a file.

Parameters

<i>library_path</i>	Path to the Liberty file to process
<i>log_file_name</i>	Name of the log file (if empty, defaults to "[library_name].supercell.log")
<i>chain_length</i>	The length of chains to create (must be ≥ 1)
<i>cell_names</i>	Vector of cell names to process for supercell generation

Exceptions

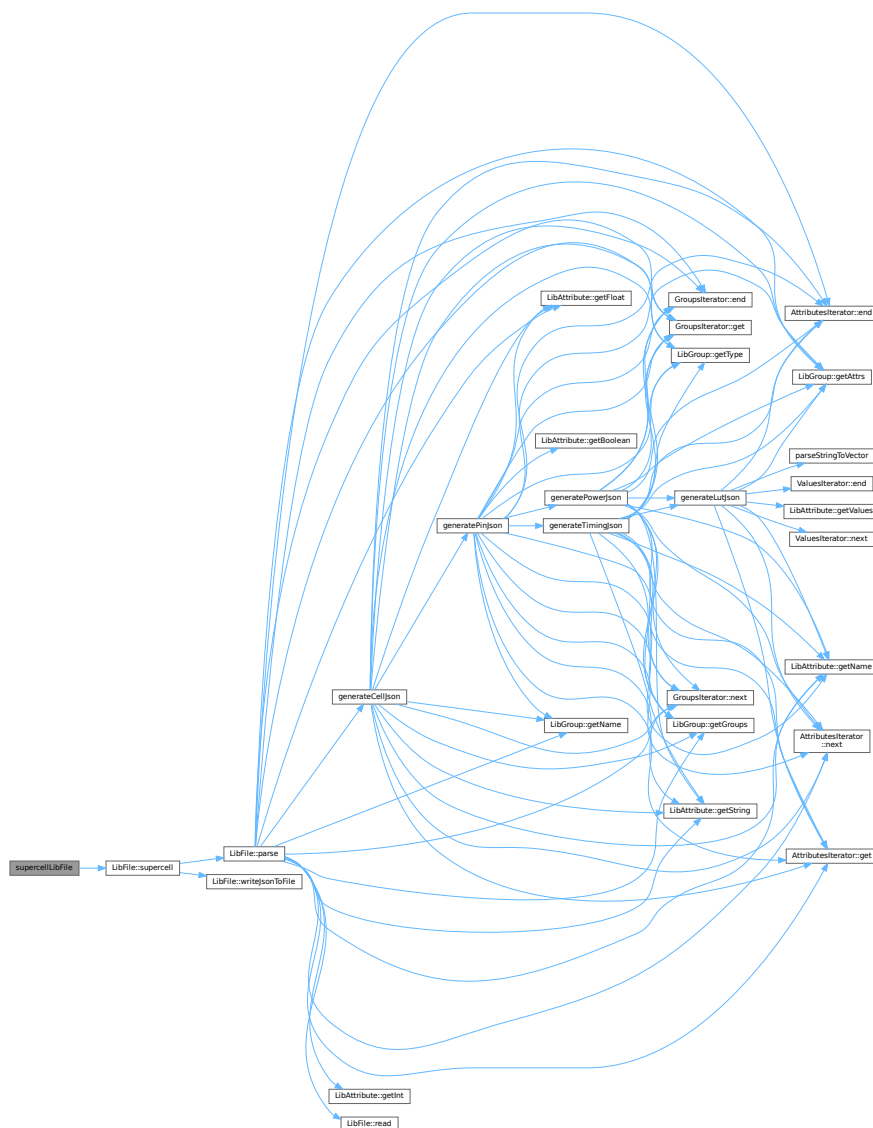
May	pass through exceptions from the LibFile::supercell method
-----	--

Note

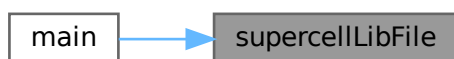
The function validates the chain length and logs all activities including errors that might occur during processing

Definition at line [133](#) of file [main.cpp](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.44.1.8 verilogLibFile()

```
void verilogLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names )
```

Generates Verilog files from a library file for specified cells.

This function processes a library file and generates Verilog representation for the specified cell names with a given chain length. The operation results are logged to a specified or default log file.

Parameters

<i>library_path</i>	Path to the library file to process
<i>log_file_name</i>	Name for the log file (if empty, a default name will be generated)
<i>chain_length</i>	Number of cells to chain together, must be ≥ 1
<i>cell_names</i>	Vector of cell names to generate Verilog for

Exceptions

<i>Catches</i>	any exceptions from the verilog generation process and logs them
----------------	--

Definition at line 174 of file [main.cpp](#).


```

00004 #include <string>
00005 #include <thread>
00006 #include <vector>
00007
00008 #include "CLI/CLI.hpp"
00009 #include "si2dr_liberty.h"
00010 #include "spdlog/sinks/basic_file_sink.h"
00011 #include "spdlog/sinks/stdout_color_sinks.h"
00012 #include "spdlog/spdlog.h"
00013
00014 #include "LibraryComparator.hpp"
00015 #include "LibFile.hpp"
00016 #include "LogicExtractor.hpp"
00017 #include "verilog_utils.hpp"
00018 #include "version.h"
00019
00035 void printInfo() {
00036     // Set Global Logger
00037     auto console_sink = std::make_shared<spdlog::sinks::stdout_color_sink_mt>();
00038     console_sink->set_level(spdlog::level::info);
00039     std::string app_name = APP_NAME;
00040     auto file_sink = std::make_shared<spdlog::sinks::basic_file_sink_mt>(app_name + ".log", true);
00041     file_sink->set_level(spdlog::level::trace);
00042
00043     std::vector<spdlog::sink_ptr> sinks{console_sink, file_sink};
00044     auto logger = std::make_shared<spdlog::logger>(APP_NAME, sinks.begin(), sinks.end());
00045     logger->set_level(spdlog::level::debug);
00046     spdlog::set_default_logger(logger);
00047
00048     spdlog::info("Version {}, Built: {}", APP_VERSION, BUILD_TIMESTAMP);
00049     spdlog::info("Author: {}, Email: {}", APP_AUTHOR, APP_CONTACT);
00050     spdlog::info("Global log file in: '{}', app_name + ".log");
00051 }
00052
00066 void parseLibFile(const std::string &library_path, const std::string log_file_name) {
00067     std::string logname = log_file_name.empty()
00068         ? std::filesystem::path(library_path).stem().string() + ".parse.log"
00069         : log_file_name;
00070     LibFile libfile(library_path, logname);
00071
00072     libfile.logger->info("Starting parse for file: '{} ...', library_path);
00073
00074     try {
00075         libfile.parse();
00076         libfile.writeJsonToFile();
00077         libfile.logger->info("Successfully parsed file: '{}', library_path);
00078     } catch (const std::exception &e) {
00079         libfile.logger->error("Error parsing file '{}': {}", library_path, e.what());
00080     }
00081 }
00082
00098 void monoCheckLibFile(const std::string &library_path, const std::string log_file_name,
00099     bool is_slew) {
00100     std::string logname = log_file_name.empty()
00101         ? std::filesystem::path(library_path).stem().string() + ".mono.log"
00102         : log_file_name;
00103     LibFile libfile(library_path, logname);
00104
00105     libfile.logger->info("Starting monotonicity check for file: '{}', input slew: {}", library_path,
00106         is_slew);
00107
00108     try {
00109         libfile.mono(is_slew);
00110         libfile.logger->info("Successfully completed monotonicity check for file: '{}', library_path);
00111     } catch (const std::exception &e) {
00112         libfile.logger->error("Error during monotonicity check for file '{}': {}", library_path,
00113             e.what());
00114     }
00115 }

```

```

00116
00133 void supercellLibFile(const std::string &library_path, const std::string &log_file_name,
00134                      int chain_length, const std::vector<std::string> &cell_names) {
00135     std::string logname = log_file_name.empty()
00136         ? std::filesystem::path(library_path).stem().string() + ".supercell.log"
00137         : log_file_name;
00138     LibFile libfile(library_path, logname);
00139
00140     // Check chain length validity
00141     if (chain_length < 1) {
00142         libfile.logger_>error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00143         return;
00144     }
00145     std::stringstream ss;
00146     for (const auto &cell : cell_names) {
00147         ss << cell << ", ";
00148     }
00149     libfile.logger_>info("Starting supercell generation for file: '{}', chain length: {}, cells: {}",
00150                         library_path, chain_length, ss.str());
00151     try {
00152         libfile.supercell(chain_length, cell_names);
00153         libfile.logger_>info("Successfully generated supercells for file: '{}', library_path);
00154     } catch (const std::exception &e) {
00155         libfile.logger_>error("Error during supercell generation for file '{}': {}", library_path,
00156                               e.what());
00157     }
00158 }
00159
00174 void verilogLibFile(const std::string &library_path, const std::string &log_file_name,
00175                   int chain_length, const std::vector<std::string> &cell_names) {
00176     std::string logname = log_file_name.empty()
00177         ? std::filesystem::path(library_path).stem().string() + ".verilog.log"
00178         : log_file_name;
00179     LibFile libfile(library_path, logname);
00180
00181     // Check chain length validity
00182     if (chain_length < 1) {
00183         libfile.logger_>error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00184         return;
00185     }
00186     std::stringstream ss;
00187     for (const auto &cell : cell_names) {
00188         ss << cell << ", ";
00189     }
00190     libfile.logger_>info("Starting Verilog generation for file: '{}', chain length: {}, cells: {}",
00191                         library_path, chain_length, ss.str());
00192     try {
00193         libfile.verilog(chain_length, cell_names);
00194         libfile.logger_>info("Successfully generated Verilog for file: '{}', library_path);
00195     } catch (const std::exception &e) {
00196         libfile.logger_>error("Error during Verilog generation for file '{}': {}", library_path,
00197                               e.what());
00198     }
00199 }
00200
00220 void compareLibFiles(const std::string &ref_lib, const std::string &comp_lib, const double reltol,
00221                    const double abstol, std::string &report_file_name) {
00222     std::string ref_logname = std::filesystem::path(ref_lib).stem().string() + ".cmp.log";
00223     std::string comp_logname = std::filesystem::path(comp_lib).stem().string() + ".cmp.log";
00224     LibFile ref_libfile(ref_lib, ref_logname), comp_libfile(comp_lib, comp_logname);
00225
00226     // Check relative tolerance validity
00227     if (reltol < 0.0) {
00228         spdlog::error("Invalid relative tolerance: {}. Relative tolerance must be >= 0.0.", reltol);
00229         return;
00230     }
00231
00232     // Check the report file name
00233     if (report_file_name.empty()) {

```

```

00234     report_file_name = comp_libfile.basename_ + ".cmp.md";
00235 } else if (std::filesystem::path(report_file_name).extension() != ".txt" &&
00236           std::filesystem::path(report_file_name).extension() != ".md") {
00237     // report file name not end in .txt or .md, warn user and append .md
00238     report_file_name = report_file_name + ".md";
00239     spdlog::warn("Report file name does not end in .txt or .md. Report will be written to '{}'",
00240                 report_file_name);
00241 }
00242
00243 LibraryComparator comparator(ref_libfile, comp_libfile, reltol, abstol);
00244 comparator.generateReport(report_file_name);
00245 spdlog::info("Comparison completed. Report written to: '{}'", report_file_name);
00246 }
00247
00248 void funcLibFile(const std::string &ref_file, const std::string &comp_file,
00249                 const std::vector<std::string> &cell_names, std::string &report_file_name) {
00250     // Check if the files are Liberty or Verilog
00251     std::string ref_ext = std::filesystem::path(ref_file).extension();
00252     std::string comp_ext = std::filesystem::path(comp_file).extension();
00253     std::string ref_logname = std::filesystem::path(ref_file).stem().string() + ".cmp.log";
00254     std::string comp_logname = std::filesystem::path(comp_file).stem().string() + ".cmp.log";
00255
00256     // Pre cell names check
00257     if (cell_names.empty()) {
00258         spdlog::error("No cell names provided for functional equivalence check.");
00259         return;
00260     }
00261
00262     // Check the report file name
00263     if (report_file_name.empty()) {
00264         report_file_name = std::filesystem::path(comp_file).stem().string() + ".cmp.md";
00265     } else if (std::filesystem::path(report_file_name).extension() != ".txt" &&
00266               std::filesystem::path(report_file_name).extension() != ".md") {
00267         // report file name not end in .txt or .md, warn user and append .md
00268         report_file_name = report_file_name + ".md";
00269         spdlog::warn("Report file name does not end in .txt or .md. Report will be written to '{}'",
00270                     report_file_name);
00271     }
00272
00273     // Declare LibFile objects as pointers, initialized to nullptr
00274     LibFile *ref_libfile = nullptr;
00275     LibFile *comp_libfile = nullptr;
00276
00277     // Check the reference file extension
00278     if (ref_ext == ".v") {
00279         spdlog::info("Reference file in Verilog format.");
00280     } else if (ref_ext == ".lib") {
00281         spdlog::info("Reference file in Liberty format.");
00282         ref_libfile = new LibFile(ref_file, ref_logname);
00283     } else {
00284         spdlog::error("Unsupported reference file format: '{}'", ref_ext);
00285         return;
00286     }
00287
00288     // Check the comparison file extension
00289     if (comp_ext == ".v") {
00290         spdlog::info("Comparison file in Verilog format.");
00291     } else if (comp_ext == ".lib") {
00292         spdlog::info("Comparison file in Liberty format.");
00293         comp_libfile = new LibFile(comp_file, comp_logname);
00294     } else {
00295         spdlog::error("Unsupported comparison file format: '{}'", comp_ext);
00296         return;
00297     }
00298
00299     // Tranverse the cell names and check functional equivalence
00300     for (const auto &cell : cell_names) {
00301         spdlog::info("Checking functional equivalence for cell: '{}'", cell);
00302         std::map<std::string, std::string> ref_outpin_map;
00303         std::map<std::string, std::string> comp_outpin_map;

```



```

00303
00304     if (ref_ext == ".v") {
00305         ref_outpin_map = extractLogicFromVerilog(ref_file, cell);
00306     } else if (ref_ext == ".lib") {
00307         ref_outpin_map = ref_libfile->logic(cell);
00308     }
00309
00310     if (comp_ext == ".v") {
00311         // getAST(comp_file, cell);
00312         // extractAndPrintNetlistInfo(comp_file, cell);
00313         comp_outpin_map = extractLogicFromVerilog(comp_file, cell);
00314     } else if (comp_ext == ".lib") {
00315         comp_outpin_map = comp_libfile->logic(cell);
00316     }
00317
00318     spdlog::info("Logic function expressions collected for cell: '{}'", cell);
00319     spdlog::info("Starting logic comparison ...");
00320
00321     // TODO Perform functional equivalence check
00322     // LogicComparator comparator(ref_outpin_map, comp_outpin_map, cell);
00323     // comparator.generateReport(report_file_name);
00324     spdlog::info("Functional equivalence check completed for cell: '{}'", cell);
00325 }
00326 spdlog::info("Functional equivalence check completed for all cells,");
00327 spdlog::info("Report written to: '{}'", report_file_name);
00328 }
00329
00330 int main(int argc, char *argv[]) {
00331     // Parse command line arguments
00332     std::vector<std::string> library_paths; // Support multiple files
00333     std::string log_file_name = "";
00334
00335     // Command line parameter parsing using CLI11
00336     CLI::App app{APP_NAME};
00337     app.set_version_flag("-v,--version", APP_VERSION);
00338
00339     // Add subcommand for parse mode
00340     CLI::App *parse_cmd =
00341         app.add_subcommand("parse", "Parse the Liberty file and write JSON to a file");
00342     parse_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00343         ->check(CLI::ExistingFile)
00344         ->required();
00345     parse_cmd->add_option("-l,--log", log_file_name,
00346         "Specify the log file name. Default: <basename>.parse.log");
00347     parse_cmd->callback([&] {
00348         printInfo();
00349         // Check if multi files
00350         if (library_paths.size() > 1) {
00351             spdlog::info("Running sequential parsing for {} files.", library_paths.size());
00352             spdlog::info("Each library will write to its own log file.");
00353             // Sequential parsing
00354             for (const auto &library_path : library_paths) {
00355                 parseLibFile(library_path, log_file_name = "");
00356             }
00357         } else {
00358             parseLibFile(library_paths[0], log_file_name);
00359         }
00360     });
00361
00362     // Add subcommand for mono check mode
00363     bool is_slew = false;
00364     CLI::App *mono_cmd = app.add_subcommand("mono", "Check the monotonicity of timing arc values");
00365     mono_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00366         ->check(CLI::ExistingFile)
00367         ->required();
00368     mono_cmd->add_option("-l,--log", log_file_name,
00369         "Specify the log file name. Default: <basename>.mono.log");
00370     mono_cmd->add_flag("-s,--slew", is_slew,
00371         "Specify that monotonicity checks also include input slew.");

```

```

00372 mono_cmd->callback([&] {
00373     printInfo();
00374     // Check if multi files
00375     if (library_paths.size() > 1) {
00376         spdlog::info("Running multi-threaded monotonicity check for {} files.", library_paths.size());
00377         spdlog::info("Each thread will write to its own log file.");
00378
00379         // Sequential json file check
00380         for (const auto &library_path : library_paths) {
00381             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00382                                     ".json")) {
00383                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00384                 parseLibFile(library_path, log_file_name = "");
00385             }
00386         }
00387         spdlog::info("All JSON files prepared.");
00388         // Parallel monotonicity check
00389         std::vector<std::thread> threads;
00390         for (const auto &library_path : library_paths) {
00391             threads.emplace_back(monoCheckLibFile, library_path, log_file_name = "", is_slew);
00392         }
00393         for (auto &thread : threads) {
00394             thread.join();
00395         }
00396         spdlog::info("All threads completed.");
00397     } else {
00398         monoCheckLibFile(library_paths[0], log_file_name, is_slew);
00399     }
00400 });
00401
00402 // Add subcommand for compare mode
00403 std::string ref_lib, comp_lib, report_file_name;
00404 CLI::App *compare_cmd = app.add_subcommand(
00405     "compare",
00406     "Compare the comparison library against the reference library and report differences");
00407 compare_cmd->add_option("--ref", ref_lib, "Specify the reference library file")
00408     ->check(CLI::ExistingFile)
00409     ->required();
00410 compare_cmd->add_option("--comp", comp_lib, "Specify the comparison library file")
00411     ->check(CLI::ExistingFile)
00412     ->required();
00413 double abstol = 0.002;
00414 compare_cmd->add_option("--abstol", abstol,
00415     "Specify the absolute tolerance for comparison. Default: 0.002ns");
00416 double reltol = 0.02;
00417 compare_cmd->add_option("--reltol", reltol,
00418     "Specify the relative tolerance for comparison. Default: 0.02/2.0%");
00419 compare_cmd->add_option("--report", report_file_name,
00420     "Specify the report file name. Default: <comp_lib>.cmp.md");
00421 compare_cmd->callback([&] {
00422     printInfo();
00423     compareLibFiles(ref_lib, comp_lib, reltol, abstol, report_file_name);
00424 });
00425
00426 // Add subcommand for supercell generation
00427 int chain_length = 1;
00428 CLI::App *supercell_cmd =
00429     app.add_subcommand("supercell", "Generate supercells for the given Liberty file");
00430 supercell_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00431     ->check(CLI::ExistingFile)
00432     ->required();
00433 supercell_cmd->add_option("-l,--log", log_file_name,
00434     "Specify the log file name. Default: <basename>.supercell.log");
00435 supercell_cmd->add_option("-c,--chain", chain_length,
00436     "Specify the chain length for supercell generation. Default: 1");
00437 std::vector<std::string> cell_names = {}; // "CMPE42D1" "AN2D0", "DFQD1"
00438 supercell_cmd->add_option("--cells", cell_names,
00439     "Specify the cell names to generate supercells for");
00440 supercell_cmd->callback([&] {

```

```

00441     printInfo();
00442     // Check if multi files
00443     if (library_paths.size() > 1) {
00444         spdlog::info("Running multi-threaded supercell generation for {} files.",
00445             library_paths.size());
00446         spdlog::info("Each library will write to its own log file.");
00447
00448         // Sequential json file check
00449         for (const auto &library_path : library_paths) {
00450             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00451                 ".json")) {
00452                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00453                 parseLibFile(library_path, log_file_name = "");
00454             }
00455         }
00456         spdlog::info("All JSON files prepared.");
00457         // Parallel supercell generation
00458         std::vector<std::thread> threads;
00459         for (const auto &library_path : library_paths) {
00460             threads.emplace_back(supercellLibFile, library_path, log_file_name = "", chain_length,
00461                 cell_names);
00462         }
00463         for (auto &thread : threads) {
00464             thread.join();
00465         }
00466         spdlog::info("All threads completed.");
00467     } else {
00468         supercellLibFile(library_paths[0], log_file_name, chain_length, cell_names);
00469     }
00470 });
00471
00472 // Add subcommand for zlibboost
00473 CLI::App *zlibboost_cmd =
00474     app.add_subcommand("zlibboost", "ZlibBoost - Multi-threaded Library Processing Tool");
00475 std::string config_dir = "/home/songzx/Projects/zlibboost/config.tcl";
00476 std::string python_dir = "/home/guocj/anaconda3/envs/myenv/bin/python";
00477 std::string main_py_dir = "/home/songzx/Projects/zlibboost/zlibboost.py";
00478 zlibboost_cmd->add_option(
00479     "-c, --config", config_dir,
00480     "Specify the configuration TCL file. Default: /home/songzx/Projects/zlibboost/config.tcl");
00481 zlibboost_cmd->add_option(
00482     "--python", python_dir,
00483     "Specify the python directory. Default: /home/guocj/anaconda3/envs/myenv/bin/python");
00484 zlibboost_cmd->add_option("--main", main_py_dir,
00485     "Specify the main python script directory. Default: "
00486     "/home/songzx/Projects/zlibboost/zlibboost.py");
00487 zlibboost_cmd->callback([&] {
00488     printInfo();
00489     // Run the ZlibBoost tool
00490     std::string command = python_dir + " " + main_py_dir + " -c " + config_dir;
00491     spdlog::info("Running ZlibBoost with command: '{}'", command);
00492     int ret = std::system(command.c_str());
00493     if (ret == 0) {
00494         spdlog::info("ZlibBoost completed successfully.");
00495     } else {
00496         spdlog::error("ZlibBoost failed with return code: {}", ret);
00497     }
00498 });
00499
00500 // Add subcommand for clear
00501 CLI::App *clear_cmd =
00502     app.add_subcommand("clear", "Clear the log, JSON, map, markdown files in this directory");
00503 clear_cmd->callback([&] {
00504     printInfo();
00505     std::filesystem::path current_dir = std::filesystem::current_path();
00506     for (const auto &entry : std::filesystem::directory_iterator(current_dir)) {
00507         if (entry.path().extension() == ".log" || entry.path().extension() == ".json" ||
00508             entry.path().extension() == ".map" || entry.path().extension() == ".md") {
00509             spdlog::info("Removing file: '{}', entry.path().string());

```

```

00510         std::filesystem::remove(entry.path());
00511     }
00512 }
00513 spdlog::info("All log, JSON, map, markdown files cleared.");
00514 });
00515
00516 // Add subcommand for Verilog generation
00517 CLI::App *verilog_cmd =
00518     app.add_subcommand("verilog", "Generate Verilog file for given Liberty file");
00519 verilog_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00520     ->check(CLI::ExistingFile)
00521     ->required();
00522 verilog_cmd->add_option("-l,--log", log_file_name,
00523     "Specify the log file name. Default: <basename>.verilog.log");
00524 verilog_cmd->add_option("-c,--chain", chain_length,
00525     "Specify the chain length for verilog generation. Default: 1");
00526 verilog_cmd->add_option("--cells", cell_names, "Specify the cell names to generate Verilog for");
00527 verilog_cmd->callback([&] {
00528     printInfo();
00529     // Check if multi files
00530     if (library_paths.size() > 1) {
00531         spdlog::info("Running multi-threaded Verilog generation for {} files.", library_paths.size());
00532         spdlog::info("Each library will write to its own log file.");
00533
00534         // Sequential json file check
00535         for (const auto &library_path : library_paths) {
00536             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00537                 ".json")) {
00538                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00539                 parseLibFile(library_path, log_file_name = "");
00540             }
00541         }
00542         spdlog::info("All JSON files prepared.");
00543         // Parallel Verilog generation
00544         std::vector<std::thread> threads;
00545         for (const auto &library_path : library_paths) {
00546             threads.emplace_back(verilogLibFile, library_path, log_file_name = "", chain_length,
00547                 cell_names);
00548         }
00549         for (auto &thread : threads) {
00550             thread.join();
00551         }
00552         spdlog::info("All threads completed.");
00553     } else {
00554         verilogLibFile(library_paths[0], log_file_name, chain_length, cell_names);
00555     }
00556 });
00557
00558 // Add subcommand for SPICE generation
00559 CLI::App *spice_cmd = app.add_subcommand("spice", "Generate SPICE file for given Liberty file");
00560 spice_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00561     ->check(CLI::ExistingFile)
00562     ->required();
00563 spice_cmd->add_option("-l,--log", log_file_name,
00564     "Specify the log file name. Default: <basename>.spice.log");
00565 spice_cmd->add_option("--cells", cell_names, "Specify the cell names to generate SPICE for");
00566 spice_cmd->callback([&] {
00567     printInfo();
00568     // Check if multi files
00569     if (library_paths.size() > 1) {
00570         spdlog::info("Running multi-threaded SPICE generation for {} files.", library_paths.size());
00571         spdlog::info("Each library will write to its own log file.");
00572
00573         // Sequential json file check
00574         for (const auto &library_path : library_paths) {
00575             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() +
00576                 ".json")) {
00577                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00578                 parseLibFile(library_path, log_file_name = "");

```

```

00579     }
00580 }
00581 spdlog::info("All JSON files prepared.");
00582 // Parallel SPICE generation
00583 std::vector<std::thread> threads;
00584 for (const auto &library_path : library_paths) {
00585     // threads.emplace_back(spiceLibFile, library_path, log_file_name = "", cell_names);
00586 }
00587 for (auto &thread : threads) {
00588     thread.join();
00589 }
00590 spdlog::info("All threads completed.");
00591 } else {
00592     // spiceLibFile(library_paths[0], log_file_name, cell_names);
00593 }
00594 });
00595
00596 // Add subcommand for funtional equivalence check
00597 CLI::App *func_cmd = app.add_subcommand(
00598     "func", "Check functional equivalence of two Liberty files or Verilog files");
00599 std::string ref_file, comp_file;
00600 func_cmd->add_option("--ref", ref_file, "Specify the reference Liberty or Verilog file")
00601     ->check(CLI::ExistingFile)
00602     ->required();
00603 func_cmd->add_option("--comp", comp_file, "Specify the comparison Liberty or Verilog file")
00604     ->check(CLI::ExistingFile)
00605     ->required();
00606 func_cmd->add_option("--cells", cell_names,
00607     "Specify the cell names to check functional equivalence for");
00608 func_cmd->add_option("--report", report_file_name,
00609     "Specify the report file name. Default: <comp_lib>.cmp.md");
00610
00611 func_cmd->callback([&] {
00612     printInfo();
00613     funcLibFile(ref_file, comp_file, cell_names, report_file_name);
00614 });
00615
00616 CLI11_PARSE(app, argc, argv);
00617
00618 // End of program
00619 char hostname[256];
00620 gethostname(hostname, sizeof(hostname));
00621
00622 auto now = std::chrono::system_clock::now();
00623 auto time_now = std::chrono::system_clock::to_time_t(now);
00624 std::stringstream ss;
00625 ss << std::put_time(std::localtime(&time_now), "%c");
00626
00627 spdlog::info("ZlibValidation exited on '{}' at {}", hostname, ss.str());
00628 return 0;
00629 }

```

8.46 src/verilog_utils.cpp File Reference

#include "verilog_utils.hpp"

Include dependency graph for verilog_utils.cpp:



Functions

- void `getAST` (const std::string &verilog_file, const std::string &cell)

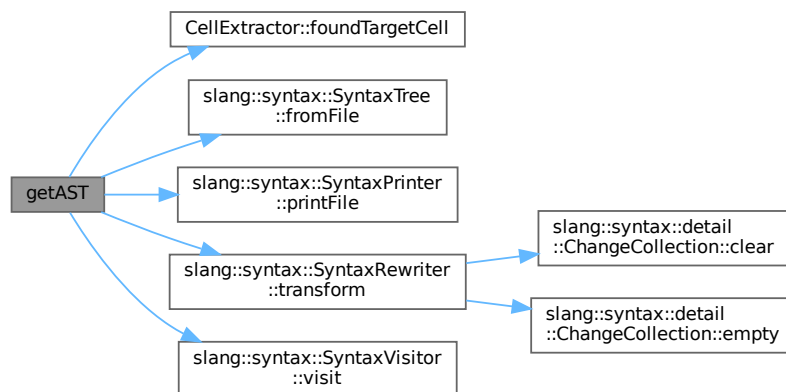
8.46.1 Function Documentation

8.46.1.1 `getAST()`

```
void getAST (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 544 of file `verilog_utils.cpp`.

Here is the call graph for this function:



8.47 `verilog_utils.cpp`

[Go to the documentation of this file.](#)

```
00001 // verilog_utils.cpp
00002
00003 #include "verilog_utils.hpp"
00004
00005 // Adding a generic handler method to record all visited nodes and increment the depth counter
00006 void VerilogVisitor::handle(const slang::syntax::SyntaxNode &node) {
00007     std::string indent(depth_ * 2, ' ');
00008     spdlog::debug("{}Node: {}", indent, slang::syntax::toString(node.kind));
00009
00010     // Increase depth, visit child nodes, then decrease depth
```

```

00011     depth_++;
00012     visitDefault(node);
00013     depth--;
00014 }
00015
00016 // Handle module declarations
00017 void VerilogVisitor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00018     std::string indent(depth_ * 2, ' ');
00019
00020     if (module.header && module.header->name.valid()) {
00021         std::string_view moduleName = module.header->name.valueText();
00022         spdlog::debug("{}Module Name:{}", indent, moduleName);
00023
00024         inTargetModule_ = !targetCell_.empty() && moduleName == targetCell_;
00025         if (!inTargetModule_) {
00026             return;
00027         }
00028
00029         // If a specific module name is specified, check if it matches
00030         if (inTargetModule_) {
00031             spdlog::info("{}Found target module: {}", indent, targetCell_);
00032             // Print the module ports
00033             if (module.header->ports) {
00034                 spdlog::info("{}Ports: {}", indent, module.header->ports->toString());
00035             }
00036         }
00037     }
00038
00039     // Continue processing child nodes
00040     depth_++;
00041     visitDefault(module);
00042     depth--;
00043 }
00044
00045 // Handle port declarations
00046 void VerilogVisitor::handle(const slang::syntax::PortDeclarationSyntax &portDecl) {
00047     if (!inTargetModule_) {
00048         return;
00049     }
00050
00051     std::string indent = std::string(depth_ * 2, ' ');
00052
00053     // Get port direction
00054     std::string direction = "unknown";
00055     if (portDecl.header) {
00056         // Try to convert the header to different types of port headers
00057         if (auto *varPort = portDecl.header->as_if<slang::syntax::VariablePortHeaderSyntax>()) {
00058             if (varPort->direction) {
00059                 direction = varPort->direction.valueText();
00060             }
00061         } else if (auto *netPort = portDecl.header->as_if<slang::syntax::NetPortHeaderSyntax>()) {
00062             if (netPort->direction) {
00063                 direction = netPort->direction.valueText();
00064             }
00065         }
00066
00067         // Get port name
00068         for (const auto &declarator : portDecl.declarators) {
00069             if (declarator->name.valid()) {
00070                 std::string_view portName = declarator->name.valueText();
00071                 spdlog::info("{}Port Name: {} ({})", indent, portName, direction);
00072             }
00073         }
00074     }
00075 }
00076
00077 // Handle hierarchical instantiation (module instance)
00078 void VerilogVisitor::handle(const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst) {
00079     if (!inTargetModule_) {

```

```

00080     return;
00081 }
00082
00083 std::string indent(depth_ * 2, ' ');
00084
00085 if (hierarchyInst.type.valid()) {
00086     std::string_view instType = hierarchyInst.type.valueText();
00087     spdlog::info("{}Hierarchy Instance Type: {}", indent, instType);
00088
00089     // Safely print parameter values (if any)
00090     try {
00091         if (hierarchyInst.parameters) {
00092             spdlog::info("{}Parameters:", indent);
00093             for (const auto &param : hierarchyInst.parameters->parameters) {
00094                 if (!param)
00095                     continue; // Null pointer check
00096
00097                 if (auto *orderedParam = param->as_if<slang::syntax::OrderedParamAssignmentSyntax>()) {
00098                     if (orderedParam->expr) {
00099                         spdlog::info("{} Parameter: {}", indent, orderedParam->expr->toString());
00100                     }
00101                 } else if (auto *namedParam = param->as_if<slang::syntax::NamedParamAssignmentSyntax>()) {
00102                     if (namedParam->name.valid() && namedParam->expr) {
00103                         spdlog::info("{} Parameter {}: {}", indent, namedParam->name.valueText(),
00104                                     namedParam->expr->toString());
00105                     }
00106                 }
00107             }
00108         }
00109     } catch (const std::exception &e) {
00110         spdlog::warn("{}Exception while processing parameters: {}", indent, e.what());
00111     }
00112
00113     // Safely handle instances
00114     try {
00115         for (const auto &instance : hierarchyInst.instances) {
00116             if (!instance) {
00117                 spdlog::warn("{}Invalid instance", indent);
00118                 continue;
00119             } else if (!instance->decl) {
00120                 spdlog::warn("{}Invalid instance declaration", indent);
00121                 // continue;
00122             } else {
00123                 try {
00124                     if (instance->decl->name.valid()) {
00125                         std::string_view instName = instance->decl->name.valueText();
00126                         spdlog::info("{}Instance Name: {}", indent, instName);
00127                     }
00128                 } catch (...) {
00129                     spdlog::debug("{}Error printing name", indent);
00130                 }
00131
00132                 // Safely print dimensions
00133                 try {
00134                     if (!instance->decl->dimensions.empty()) {
00135                         spdlog::info("{} Dimensions: {}", indent, instance->decl->dimensions.toString());
00136                     }
00137                 } catch (...) {
00138                     spdlog::debug("{}Error printing dimensions", indent);
00139                 }
00140             }
00141
00142             // Safely print port connections
00143             spdlog::info("{} Port connections:", indent);
00144             for (const auto &conn : instance->connections) {
00145                 if (!conn) {
00146                     spdlog::warn("{} Null connection", indent);
00147                     continue; // Null pointer check
00148                 }

```



```

00149
00150     try {
00151         // Use as_if method for safe type conversion
00152         if (auto *ordered = conn->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00153             if (ordered->expr) {
00154                 spdlog::info("{}    Ordered connection: {}", indent, ordered->expr->toString());
00155             } else {
00156                 spdlog::info("{}    Ordered connection: <empty>", indent);
00157             }
00158         } else if (auto *named = conn->as_if<slang::syntax::NamedPortConnectionSyntax>()) {
00159             if (named->name.valid()) {
00160                 std::string exprStr = named->expr ? named->expr->toString() : "<empty>";
00161                 spdlog::info("{}    .{}({})", indent, named->name.valueText(), exprStr);
00162             }
00163         } else if (conn->as_if<slang::syntax::EmptyPortConnectionSyntax>()) {
00164             spdlog::info("{}    <empty connection>", indent);
00165         } else if (conn->as_if<slang::syntax::WildcardPortConnectionSyntax>()) {
00166             spdlog::info("{}    .* (wildcard connection)", indent);
00167         } else {
00168             spdlog::info("{}    Unknown connection type: {}", indent,
00169                 slang::syntax::toString(conn->kind));
00170         }
00171     } catch (const std::exception &e) {
00172         spdlog::warn("{}Exception processing connection: {}", indent, e.what());
00173     } catch (...) {
00174         spdlog::warn("{}Unknown exception processing connection", indent);
00175     }
00176 }
00177 }
00178 } catch (const std::exception &e) {
00179     spdlog::warn("{}Exception while processing instances: {}", indent, e.what());
00180 }
00181
00182 // Continue processing child nodes
00183 depth_++;
00184 visitDefault(hierarchyInst);
00185 depth--;
00186 }
00187 }
00188 /*
00189 // Handle primitive gate instantiation
00190 void VerilogVisitor::handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst) {
00191     if (!inTargetModule_) {
00192         return;
00193     }
00194
00195     std::string indent(depth_ * 2, ' ');
00196
00197     // Get primitive gate type
00198     if (primitiveInst.type.valid()) {
00199         std::string_view gateType = primitiveInst.type.valueText();
00200         spdlog::info("{}Primitive Gate: {}", indent, gateType);
00201
00202         // Print delay information (if any)
00203         if (primitiveInst.delay) {
00204             spdlog::info("{}Delay: {}", indent, primitiveInst.delay->toString());
00205         }
00206
00207         // Print strength information (if any)
00208         if (primitiveInst.strength) {
00209             spdlog::info("{}Strength: {}", indent, primitiveInst.strength->toString());
00210         }
00211
00212         // Print each instance
00213         for (const auto &instance : primitiveInst.instances) {
00214             // Primitive gates usually don't have names, but if they do, print them
00215             if (instance->decl && instance->decl->name.valid()) {
00216                 std::string_view instName = instance->decl->name.valueText();
00217                 spdlog::info("{} Gate instance: {}", indent, instName);

```

```

00218 }
00219
00220 // Print connections
00221 spdlog::info("{} Gate connections:", indent);
00222 for (size_t i = 0; i < instance->connections.size(); ++i) {
00223     const auto &conn = instance->connections[i];
00224     // For gate-level instantiation, connections are usually ordered
00225     if (auto *ordered = conn->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00226         if (ordered->expr) {
00227             // The first is usually the output, the rest are inputs
00228             std::string portType = (i == 0) ? "output" : "input";
00229             spdlog::info("{} {} {}: {}", indent, portType, i, ordered->expr->toString());
00230         }
00231     }
00232 }
00233 }
00234
00235 // Continue to traverse deeper when specific standard gate type
00236 // Create a set of safe standard gate types
00237 static const std::unordered_set<std::string_view> safeGateTypes = {
00238     "and", "or", "nand", "nor", "xor", "xnor", "not",
00239     "buf", "bufif0", "bufif1", "notif0", "notif1", "pullup", "pulldown",
00240     "cmos", "rcmos", "nmos", "pmos", "rnmos", "rpmos", "tran",
00241     "rtran", "tranif0", "tranif1", "rtranif0", "rtranif1"};
00242
00243 if (safeGateTypes.find(gateType) != safeGateTypes.end()) {
00244     // Continue processing child nodes
00245     depth++;
00246     visitDefault(primitiveInst);
00247     depth--;
00248 } else {
00249     spdlog::warn("{}Skipping deeper traversal of primitive: {}", indent, gateType);
00250 }
00251
00252 } else {
00253     spdlog::warn("{}Primitive instantiation missing gate type", indent);
00254 }
00255 }
00256 */
00257 // Handle specify block
00258 void VerilogVisitor::handle(const slang::syntax::SpecifyBlockSyntax &specifyBlock) {
00259     if (!inTargetModule_) {
00260         return;
00261     }
00262
00263     std::string indent(depth_ * 2, ' ');
00264     spdlog::info("{}Specify Block:", indent);
00265
00266     // Iterate through all path declarations
00267     for (const auto &item : specifyBlock.items) {
00268         if (auto *pathDecl = item->as_if<slang::syntax::PathDeclarationSyntax>()) {
00269             if (pathDecl->desc) {
00270                 std::string pathSrc;
00271                 std::string pathDst;
00272
00273                 // Get path source
00274                 if (!pathDecl->desc->inputs.empty()) {
00275                     // Get the first input
00276                     if (auto *identifier =
00277                         pathDecl->desc->inputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00278                         if (identifier->identifier.valid()) {
00279                             pathSrc = identifier->identifier.valueText();
00280                         }
00281                     }
00282                 }
00283
00284                 // Get path destination
00285                 if (pathDecl->desc->suffix) {
00286                     if (auto *simpleSuffix =

```

```

00287         pathDecl->desc->suffix->as_if<slang::syntax::SimplePathSuffixSyntax>()) {
00288     if (!simpleSuffix->outputs.empty()) {
00289         if (auto *identifier =
00290             simpleSuffix->outputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00291             if (identifier->identifier.valid()) {
00292                 pathDst = identifier->identifier.valueText();
00293             }
00294         }
00295     } else if (auto *edgeSuffix =
00296         pathDecl->desc->suffix
00297         ->as_if<slang::syntax::EdgeSensitivePathSuffixSyntax>()) {
00298         if (!edgeSuffix->outputs.empty()) {
00299             if (auto *identifier =
00300                 edgeSuffix->outputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00301                 if (identifier->identifier.valid()) {
00302                     pathDst = identifier->identifier.valueText();
00303                 }
00304             }
00305         }
00306     }
00307 }
00308
00309 // Construct path string
00310 std::string pathStr = pathSrc + " => " + pathDst;
00311
00312 // Get path delay
00313 std::string delayStr = "(";
00314 for (size_t i = 0; i < pathDecl->delays.size(); ++i) {
00315     if (i > 0)
00316         delayStr += ", ";
00317     delayStr += pathDecl->delays[i]->toString();
00318 }
00319 delayStr += ")";
00320
00321 spdlog::info("{} Path: {} = {}", indent, pathStr, delayStr);
00322 }
00323 }
00324 // Can add handling for ConditionalPathDeclarationSyntax and IfNonePathDeclarationSyntax
00325 else if (auto *condPath = item->as_if<slang::syntax::ConditionalPathDeclarationSyntax>()) {
00326     if (condPath->path && condPath->predicate) {
00327         spdlog::info("{} Conditional Path (if {})", indent, condPath->predicate->toString());
00328         // Recursively handle internal path
00329         handle(*condPath->path);
00330     }
00331 } else if (auto *ifNonePath = item->as_if<slang::syntax::IfNonePathDeclarationSyntax>()) {
00332     if (ifNonePath->path) {
00333         spdlog::info("{} If-None Path", indent);
00334         // Recursively handle internal path
00335         handle(*ifNonePath->path);
00336     }
00337 }
00338 }
00339
00340 // Continue processing child nodes
00341 depth++;
00342 visitDefault(specifyBlock);
00343 depth--;
00344 }
00345 }
00346
00347 // Handle module declarations, only keep the target module
00348 void CellExtractor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00349     if (module.header && module.header->name.valid()) {
00350         std::string_view moduleName = module.header->name.valueText();
00351
00352         // If it is the target module, mark as found, otherwise remove it
00353         if (!targetCell_.empty() && moduleName == targetCell_) {
00354             foundTarget_ = true;
00355             // Do not modify, keep this module

```

```

00356     } else {
00357         // Remove non-target modules
00358         remove(module);
00359     }
00360 }
00361 }
00362
00363 // Get result
00364 bool CellExtractor::foundTargetCell()const { return foundTarget_; }
00365
00366 // Handle module declarations
00367 void CellPrinter::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00368     if (module.header && module.header->name.valid()) {
00369         std::string_view moduleName = module.header->name.valueText();
00370
00371         // Check if it is the target module
00372         if (!targetCell_.empty() && moduleName == targetCell_) {
00373             foundTarget_ = true;
00374
00375             // Print module definition
00376             out_ << "`timescale 1ns/10ps\n";
00377             out_ << module.toString();
00378
00379             return; // Do not traverse further
00380         }
00381     }
00382
00383     // Continue traversing other modules
00384     visitDefault(module);
00385 }
00386
00396 void ModuleRewriter::handle(const slang::syntax::SyntaxNode &node) {
00397     std::string indent(depth_ * 2, ' ');
00398     logger_>debug("{}Node: {}", indent, slang::syntax::toString(node.kind));
00399
00400     // Continue processing child nodes
00401     depth_++;
00402     visitDefault(node);
00403     depth_--;
00404 }
00405
00406 void ModuleRewriter::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00407     logger_>debug("Processing module: {}", module.header->name.valueText());
00408
00409     // Create intermediate wires for connections between instances
00410     for (int i = 0; i < instance_count_ - 1; i++) {
00411         auto &newNetNode = parse("\n wire OP_" + std::to_string(i) + ";");
00412         insertAtBack(module.members, newNetNode);
00413         logger_>debug("Added intermediate wire: {}", newNetNode.toString());
00414     }
00415
00416     // Get critical input port & output port from
00417     // the module name pattern: CELLNAME_X#_CRITICALPORT_OUTPUTPORT
00418     std::string criticalInputPort = "";
00419     std::string criticalOutputPort = "";
00420     if (!this->moduleName_.empty()) {
00421         size_t firstSep = this->moduleName_.find("__");
00422         if (firstSep != std::string::npos) {
00423             size_t secondSep = this->moduleName_.find("__", firstSep + 2);
00424             if (secondSep != std::string::npos) {
00425                 size_t thirdSep = this->moduleName_.find("__", secondSep + 2);
00426                 if (thirdSep != std::string::npos) {
00427                     criticalInputPort = this->moduleName_.substr(secondSep + 2, thirdSep - (secondSep + 2));
00428                     criticalOutputPort = this->moduleName_.substr(thirdSep + 2);
00429                     logger_>debug("Extracted critical input port: {}", criticalInputPort);
00430                     logger_>debug("Extracted output port: {}", criticalOutputPort);
00431                 } else {
00432                     logger_>warn("Can't find thirdSep. Invalid module name pattern: {}", this->moduleName_);
00433                     return;

```

```

00434     }
00435   } else {
00436     logger_>warn("Can't find secondSep. Invalid module name pattern: {}", this->moduleName_);
00437     return;
00438   }
00439 } else {
00440   logger_>warn("Can't find firstSep. Invalid module name pattern: {}", this->moduleName_);
00441   return;
00442 }
00443 } else {
00444   logger_>warn("Module name is empty. Can't extract critical input port.");
00445   return;
00446 }
00447
00448 // Add additional wires for intermediate outputs that aren't part of the chain
00449 if (this->outputPins_.size() > 1) { // Only add if there are multiple output pins
00450   for (int i = 0; i < instance_count_ - 1; i++) { // Skip the first and last instances
00451     for (const auto &outputPin : this->outputPins_) {
00452       if (outputPin != criticalOutputPort) { // Found an intermediate output pin
00453         auto &newNetNode = parse("\n wire P_" + std::to_string(i) + "_" + outputPin + ";");
00454         insertAtBack(module.members, newNetNode);
00455         logger_>debug("Added intermediate wire: {}", newNetNode.toString());
00456       }
00457     }
00458   }
00459 }
00460
00461 std::string portList_str = module.header->ports->toString();
00462 logger_>debug("Port list: {}", portList_str);
00463
00464 std::vector<std::string> allPorts;
00465 auto ansiPortList = module.header->ports->as_if<slang::syntax::AnsiPortListSyntax>();
00466 if (!ansiPortList) {
00467   logger_>warn("Port list is not ANSI style.");
00468   return;
00469 }
00470 portInfoMap_.clear(); // Clear the port info map
00471 for (const auto portMember : ansiPortList->ports) {
00472   if (portMember->kind == slang::syntax::SyntaxKind::ImplicitAnsiPort) {
00473     const auto implicitPort = portMember->as_if<slang::syntax::ImplicitAnsiPortSyntax>();
00474     const auto &directionToken =
00475       implicitPort->header->as_if<slang::syntax::VariablePortHeaderSyntax>()->direction;
00476     const auto &nameToken = implicitPort->declarator->name;
00477
00478     std::string_view portName = nameToken.valueText();
00479     std::string_view direction = directionToken.valueText();
00480     portInfoMap_[std::string(portName)] = std::string(direction); // Store port info in the map
00481     logger_>debug("Port Name: {}, Direction: {}", portName, direction);
00482   } else {
00483     logger_>warn("Port member is not ImplicitAnsiPort.");
00484   }
00485 }
00486
00487 for (int i = 0; i < instance_count_; i++) {
00488   std::string instanceName = "I_" + this->cellName_ + "__X" + std::to_string(i) + "__" +
00489     criticalInputPort + "_" + criticalOutputPort;
00490   std::string instanceCode = "\n " + this->cellName_ + " " + instanceName + " (";
00491   std::vector<std::string> portConnections;
00492
00493   for (const auto &pair : portInfoMap_) {
00494     std::string portName = pair.first;
00495     std::string direction = pair.second;
00496     std::string connectionName;
00497
00498     if (portName == criticalInputPort) {
00499       if (i == 0) {
00500         connectionName = portName; // First instance: connect to module input port
00501       } else {
00502         connectionName =

```

```

00503         "OP_" + std::to_string(i - 1); // Intermediate instances: connect to previous OP wire
00504     }
00505 } else if (portName == criticalOutputPort) {
00506     if (i == instance_count_ - 1) {
00507         connectionName = portName; // Last instance: connect to module output port
00508     } else {
00509         connectionName = "OP_" + std::to_string(i); // Intermediate instances: connect to OP wire
00510     }
00511 } else if (direction == "output") { // Handle other output ports
00512     if (i < instance_count_ - 1) {
00513         connectionName = "P_" + std::to_string(i) + "__" +
00514             portName; // Connect to intermediate P_i__portName wire
00515     } else {
00516         connectionName = portName; // Last instance: connect to module output port
00517     }
00518 } else { // For input ports (and inout?), connect directly to module port
00519     connectionName = portName;
00520 }
00521 portConnections.push_back("." + portName + "(" + connectionName + ")");
00522 }
00523
00524 // Connect all ports with comma and space
00525 if (!portConnections.empty()) {
00526     instanceCode += portConnections[0];
00527     for (size_t i = 1; i < portConnections.size(); ++i) {
00528         instanceCode += ", " + portConnections[i];
00529     }
00530 }
00531 instanceCode += ");";
00532
00533 // Blank line before the first instance
00534 if (i == 0) {
00535     instanceCode = "\n" + instanceCode;
00536 }
00537
00538 // Insert the instance code
00539 auto &instanceNode = parse(instanceCode);
00540 insertAtBack(module.members, instanceNode);
00541 }
00542 }
00543
00544 void getAST(const std::string &verilog_file, const std::string &cell) {
00545     try {
00546         spdlog::info("Starting get AST from Verilog file: '{}'", verilog_file);
00547         auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00548         if (result) {
00549             spdlog::info("Successfully parsed Verilog file.");
00550
00551             try {
00552                 // First, use the visitor to print basic information
00553                 VerilogVisitor visitor(cell);
00554                 visitor.visit(result.value()->root());
00555
00556                 // If a target cell is specified, use the rewriter to extract the relevant code
00557                 if (!cell.empty()) {
00558                     // Create a rewriter to only keep the target cell related code
00559                     CellExtractor extractor(cell);
00560                     auto extractedTree = extractor.transform(result.value());
00561
00562                     if (extractor.foundTargetCell()) {
00563                         // Use SyntaxPrinter to print the extracted code
00564                         std::string extractedCode = slang::syntax::SyntaxPrinter::printFile(*extractedTree);
00565
00566                         // Save to a file named after the cell name
00567                         std::string outputFile = cell + ".v";
00568                         std::ofstream cellOut(outputFile);
00569                         if (cellOut) {
00570                             cellOut << extractedCode;
00571                             cellOut.close();

```

```
00572         spdlog::info("Extracted '{}' cell code to '{}'", cell, outputFile);
00573     } else {
00574         spdlog::error("Failed to write extracted cell code to '{}'", outputFile);
00575     }
00576 } else {
00577     spdlog::warn("Target cell '{}' not found in the Verilog file", cell);
00578 }
00579 }
00580
00581 // spdlog::info("Print full source code to 'full_source_code.v'");
00582 // // Optionally save the entire syntax tree
00583 // std::string fullOutput = slang::syntax::SyntaxPrinter::printFile(*result.value());
00584 // std::ofstream out("full_source_code.v");
00585 // out << fullOutput;
00586 // out.close();
00587
00588 spdlog::info("Print target cell code to 'cell_code.v using toString()'");
00589 // Use CellPrinter to print the target cell's code
00590 std::ofstream cellOut("cell_code.v");
00591 CellPrinter cellPrinter(cell, cellOut);
00592 cellPrinter.visit(result.value()->root());
00593 cellOut.close();
00594
00595 } catch (const std::exception &e) {
00596     spdlog::error("Exception during AST traversal: {}", e.what());
00597 } catch (...) {
00598     spdlog::error("Unknown exception during AST traversal");
00599 }
00600 } else {
00601     spdlog::error("Error parsing Verilog file.");
00602 }
00603 } catch (const std::exception &e) {
00604     spdlog::error("Exception during Verilog parsing: {}", e.what());
00605 } catch (...) {
00606     spdlog::error("Unknown exception during Verilog parsing");
00607 }
00608 }
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